

Addis Ababa University  
College of Health Sciences  
School of Pharmacy



Bachelor of Pharmacy (BPharm)  
Curriculum

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Ethiopia

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## **1. Background**

In the past few decades, Ethiopia went through numerous changes that influenced the different sectors in the economy. These changes have positively impacted the way the health system operates and improved major indicators of health care in the country. However, the country still remains to be characterized by its low socioeconomic status and health service coverage. As a result, there is low ratio and improper mix of health professional to the population. The Ethiopian health policy is based on health promotion and disease prevention with focus on ensuring equity. The policy gives priority to ensuring access for the rural and unprivileged urban population. Successful implementation of this policy is highly dependent on availability of well-trained health professionals, both in quality and quantity. In this regard, the pharmacy service as an essential component of the health care delivery system requires properly trained professionals.

While pharmacists' vital role remains to be dispensing of medications and devices and ensuring appropriate therapy and outcomes, they also work in the areas of health promotion and disease prevention, pharmaceutical production and health systems management. In this regard, pharmacists provide their services in a variety of settings in response to a dynamic and evolving set of primarily local health care priorities and needs. There are also regional, national and international policies and factors, which dictate the need for developments in pharmacy practice. Within this context, pharmacists are medication experts in the treatment of disease and in health promotion. This expertise, in its broadest sense, encompasses the preparation, supply and control of medicinal products and assurance of desired outcomes of treatment by medication. It thus begins with the medicine development process and continues through to medication's ultimate benefit to the individual and to society generally. This expertise has its foundations in the pharmaceutical sciences and related research, and has its focus on the individual and populations.

The needs of modern health care systems require that the role of the pharmacist develop rapidly to meet its demands. It is also expected that pharmacists' roles may evolve with time, changes in societal needs and technological progress. Thus, the pharmacy curriculum should change with such developments in order to address the needs of the society. Generally, the pre-service education is designed to ensure that the newly qualified pharmacist has the necessary knowledge and skills to commence practicing competently in a variety of settings including community and hospital pharmacy and the pharmaceutical industry. As such, the curriculum used in pre-service education needs to be responsive to changes in the practice setting, the needs of the society and knowledge

and technological innovations. That is why a robust evaluation of the current performance of the curriculum and a constant formative evaluation of progresses in the field of pharmacy are highly recommended. In line with such recommendations, this curriculum document was prepared after a comprehensive assessment of the performance of the previous curriculum and needs for revision.

## **2. Rationale**

The global landscape of pharmacy practice and thus the education of pharmacists have evolved significantly in the past three decades. The main driver of such changes has been the advent of patient-centered pharmaceutical care as a philosophy of pharmacy practice. As such, universities in Ethiopia have been trying to increasingly incorporate more clinical courses in the existing pharmacy curriculum throughout the past decade. First, Ethiopian universities started offering a clinical oriented curriculum with significantly higher number of clinical-leaning courses in 2008, which started graduating students in 2013. In this endeavour the universities have tried very hard to maintain a balance between the clinical and non-clinical course contents in a way that reflects the Ethiopian context. The same year, the graduates were deployed to the different practice setting in Ethiopia, there was a move to modularize the curriculum in a nationally harmonized manner. This was done by clustering courses into general umbrellas of modules, without the significant modification of course contents in the previous curriculum.

It is generally recommended that higher education curriculums should undergo major revisions at least every five years. As it is a little more than five years since the implementation of the current curriculum, it was imperative to initiate its revision. In addition, the then Ministry of Science and Higher Education (MOSHE), which has since been remerged with the Ministry of Education (MOE), has prescribed certain common courses to be included in all undergraduate curriculums as per the national Educational roadmap spanning from 2020 to 2030. MOSHE also changed the direct placement of candidates, who pass the preparatory school leaving examination, to pharmacy schools. Now students have to take freshman courses in the first semester and join pharmacy schools in the second semester of the first year. Moreover, a nationwide assessment of the performance of the current curriculum including all stakeholders from current students and instructors to graduates and employees indicated the need for revision of the curriculum. Thus, the preparation of this curriculum document was initiated in a bid to respond to these demands for revision.

### **3. Objectives**

#### **General Objectives:**

The general objective of the B.Pharm Program is to train highly qualified pharmacists who fulfil the essential, minimum common expectations of health care systems worldwide while fulfilling local needs.

Graduates would be expected to have the required knowledge, skill, attitude and behaviour to support the health care system and to respond to patient needs working in hospital and community pharmacies; drug manufacturing, import and wholesale companies; public health services; academic institutions; pharmaceutical research; drug regulatory body and quality control organizations.

#### **Specific Objectives:**

- To train manpower that is more patient-oriented while still having a broad pharmaceutical knowledge to be able to easily adapt to working in any of the settings in the country's pharmaceutical sector.
- To provide practice-based training so that future pharmacists acquire problem-solving skills.
- To facilitate and assist in the transfer and adaptation of pharmaceutical knowledge

#### **Demand for the program**

In Ethiopia, health institutions (hospitals, health centers, NGOs, and others), higher institutions (government and private), research institutions and pharmaceutical industries that require competent pharmacy professionals at different level of training and expertise are progressively growing. As a result, ample opportunities are created for pharmacy practitioners to play a vital role in the country's development. However, the input of these professionals to a great extent depends on the quality of education (training) they receive in the higher learning institutions or universities.

The curriculum review has taken into consideration the existing socio-economic demands, government policies and global experience in training pharmacists. During the revision of the curriculum feedback from stakeholders, former graduates of different universities, and internal and

external assessors was given serious consideration. The global experience in pharmacy curriculum design and implementation was also considered.

## **Professional Profile**

To be effective health care team members, pharmacists need skills and attitudes enabling them to assume many different functions. The concept of the “seven-star pharmacist” was introduced by WHO and taken up by FIP in 2000 in its policy statement on Good Pharmacy Education Practice to cover these roles: caregiver, decision-maker, communicator, manager, life-long learner, teacher and leader. The function of the pharmacist as a researcher was later on added.

These roles of the pharmacist are described below and include the following functions:

- **Caregiver:** Pharmacists provide caring services. They must view their practice as integrated and continuous with those of the health care system and other health professionals. Services must be of the highest quality.
- **Decision-maker:** The appropriate, efficacious, safe and cost-effective use of resources (e.g., personnel, medicines, chemicals, equipment, procedures, and practices) should be the foundation of the pharmacist’s work. At the local and national levels, pharmacists play a role in setting medicines policy. Achieving this goal requires the ability to evaluate, synthesize data and information and decide upon the most appropriate course of action.
- **Communicator:** The pharmacist is in an ideal position to provide a link between prescriber and patient, and to communicate information on health and medicines to the public. He or she must be knowledgeable and confident while interacting with other health professionals and the public. Communication involves verbal, non-verbal, listening and writing skills.
- **Manager:** Pharmacists must be able to manage resources (human, physical and financial) and information effectively; they must also be comfortable being managed by others, whether by an employer or the manager/leader of a health care team. More and more, information and its related technology will provide challenges as pharmacists assume greater responsibility for sharing information about medicines and related products and ensuring their quality.
- **Life-long-learner:** It is impossible to acquire in pharmacy school all the knowledge and experience needed to pursue a life-long career as a pharmacist. The concepts, principles and commitment to life-long learning must begin while attending pharmacy school and

must be supported throughout the pharmacist's career. Pharmacists should learn how to keep their knowledge and skills up to date.

- **Teacher:** The pharmacist has a responsibility to assist with the education and training of future generations of pharmacists and the public. Participating as a teacher not only imparts knowledge to others, it offers an opportunity for the practitioner to gain new knowledge and to fine-tune existing skills.
- **Leader:** In multidisciplinary (e.g., team) caring situations or in areas where other health care providers are in short supply or non-existent the pharmacist is obligated to assume a leadership position in the overall welfare of the patient and the community. Leadership involves compassion and empathy as well as vision and the ability to make decisions, communicate, and manage effectively. A pharmacist whose leadership role is to be recognized must have vision and the ability to lead.
- **Researcher:** The pharmacist must be able to use the evidence base (e.g., scientific, pharmacy practice, health system) effectively in order to advise on the rational use of medicines in the health care team. By sharing and documenting experiences, the pharmacist can also contribute to the evidence base with the goal of optimizing patient care and outcomes. As a researcher, the pharmacist is able to increase the accessibility of unbiased health and medicines-related information to the public and other health care professionals

#### **4. Graduate Profile**

It is envisaged that pharmacy graduates with the B.Pharm. Degree will be capable to assume the following responsibilities and attributes:

##### **Core graduate competencies**

- Organize and control the manufacturing, compounding and packaging of pharmaceutical products;
- Organize the selection, procurement, storage, and distribution of pharmaceutical materials and products;
- Provide Pharmaceutical Care and Dispense and ensure the optimal use of medicines by the patient;
- Provide pharmacist-initiated care to patients and ensure the optimal use of medicines;
- Provide education and information on health care and medicines;

- Promote community health and provide related information and advice; and
- Conduct in research to ensure the optimal use of medicines.
- Demonstrate a high level of professional ethics in order to satisfy the pharmaceutical needs of the society.
- Maintain and expand knowledge through self-directed learning.
- Be able to work as a member of the health team.
- Possess the necessary background to pursue further advanced study in the pharmaceutical sciences.

The core competencies have the following knowledge, attitude and skill attitude components in the various domains of pharmacy practice.

| <b>Domains</b> | <b>Competencies</b>                                                                                                                                                                                                                                                                                                                         |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Patient care   | Knowledge:                                                                                                                                                                                                                                                                                                                                  |
|                | <ul style="list-style-type: none"> <li>• Possess the necessary background to pursue further advanced study in the pharmaceutical sciences</li> <li>• Provide pharmacist-initiated care to patients and ensure the optimal use of medicines</li> <li>• Provide education and information on health care and medicines</li> </ul>             |
|                | Attitude:                                                                                                                                                                                                                                                                                                                                   |
|                | <ul style="list-style-type: none"> <li>• Advocate the proper use of necessary materials by screening</li> <li>• Recognize, adhere to and promote established safety rules</li> <li>• Respectful and compassionate to patients, their relatives and other professionals</li> <li>• Be able to work as a member of the health team</li> </ul> |
|                | Skill:                                                                                                                                                                                                                                                                                                                                      |
|                | <ul style="list-style-type: none"> <li>• Provide Pharmaceutical Care and Dispense and ensure the optimal use of medicines by the patient</li> <li>• Communicate effectively both verbally and in writing</li> <li>• Collect, document, retrieve and interpret data related to all their activities clearly and safely</li> </ul>            |
|                | Knowledge:                                                                                                                                                                                                                                                                                                                                  |



|                           |                                                                                                                                                                                                                                                                                                |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pharmaceutical technology | <ul style="list-style-type: none"> <li>• Possess the necessary background to pursue further advanced study in the pharmaceutical sciences</li> </ul>                                                                                                                                           |
|                           | Attitude:                                                                                                                                                                                                                                                                                      |
|                           | <ul style="list-style-type: none"> <li>• Advocate the proper use of necessary materials by screening</li> <li>• Recognize, adhere to and promote established safety rules</li> <li>• Institute and promote safety, quality control and quality assurance in their allowed work area</li> </ul> |
|                           | Skill:                                                                                                                                                                                                                                                                                         |
|                           | <ul style="list-style-type: none"> <li>• Organize and control the manufacturing, compounding and packaging of pharmaceutical products</li> <li>• Monitor and maintain proper functioning of necessary equipment/reagents</li> <li>• Develop and modify laboratory procedures</li> </ul>        |
| Health promotion          | Knowledge:                                                                                                                                                                                                                                                                                     |
|                           | <ul style="list-style-type: none"> <li>• Provide education and information on health care and medicines</li> <li>• Promote community health and provide related information and advice</li> </ul>                                                                                              |
|                           | Attitude:                                                                                                                                                                                                                                                                                      |
|                           | <ul style="list-style-type: none"> <li>• Advocate the proper use of necessary materials by screening</li> </ul>                                                                                                                                                                                |
|                           | Skill:                                                                                                                                                                                                                                                                                         |
|                           | <ul style="list-style-type: none"> <li>• Prepare educational materials to promote the rational use of medicines and medical devices</li> <li>• Screen for drug abuse and refer patients to appropriate care centre</li> </ul>                                                                  |
| Professionalism           | Knowledge:                                                                                                                                                                                                                                                                                     |
|                           | <ul style="list-style-type: none"> <li>• Demonstrate a high level of professional ethics in order to satisfy the pharmaceutical needs of the society</li> <li>• Maintain and expand knowledge through self-directed learning</li> </ul>                                                        |
|                           | Attitude:                                                                                                                                                                                                                                                                                      |
|                           | <ul style="list-style-type: none"> <li>• Maintain the pharmacy ethical code of conduct standards</li> <li>• Contribute to stewardship of their profession</li> </ul>                                                                                                                           |

|                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                     | <ul style="list-style-type: none"> <li>• Recognize, adhere to and promote established safety rules</li> <li>• Pursue graduate training in pharmacy and other health related disciplines</li> <li>• Participate in policy, professional standards, continuing professional development issues pertaining to pharmacy profession</li> <li>• Respectful and compassionate to patients, their relatives and other professionals</li> </ul> |
|                                     | Skill:                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                     | <ul style="list-style-type: none"> <li>• Communicate effectively both verbally and in writing</li> <li>• Engage in policy, professional standards, and continuing professional development issues pertaining to pharmacy profession</li> </ul>                                                                                                                                                                                         |
| Pharmacy law and regulatory affairs | Knowledge:                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                                     | <ul style="list-style-type: none"> <li>• Familiarity with the latest laws, directives and guidelines governing the pharmaceutical sector</li> </ul>                                                                                                                                                                                                                                                                                    |
|                                     | Attitude:                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                     | <ul style="list-style-type: none"> <li>• Maintain the pharmacy ethical code of conduct standards</li> <li>• Recognize, adhere to and promote established safety rules</li> </ul>                                                                                                                                                                                                                                                       |
|                                     | Skill:                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                     | <ul style="list-style-type: none"> <li>• Store and use laboratory supplies and dispose expired drugs safely according to the rules and regulations</li> <li>• Develop and modify laboratory procedures</li> </ul>                                                                                                                                                                                                                      |
| Leadership and management           | Knowledge:                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                                     | <ul style="list-style-type: none"> <li>• Use his/her critical thinking to improve the pharmacy working environment</li> <li>• Familiarize him/her with latest scientific findings to improve the quality of services rendered to the society</li> </ul>                                                                                                                                                                                |
|                                     | Attitude:                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                     | <ul style="list-style-type: none"> <li>• Institute and promote safety, quality control and quality assurance in their allowed work area</li> </ul>                                                                                                                                                                                                                                                                                     |

|                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                | <ul style="list-style-type: none"> <li>• Participate in policy, professional standards, continuing professional development issues pertaining to pharmacy profession</li> </ul>                                                                                                                                                                                                                                                                                                    |
|                                                | Skill: <ul style="list-style-type: none"> <li>• Organize the selection, procurement, storage, and distribution of pharmaceutical materials and products</li> <li>• Plan drugs and equipment logistic procurement, evaluation, setup and auditing</li> <li>• Monitor the inventory in storage, work and laboratory area</li> <li>• Provide professional services, leadership and quality assurance in work areas</li> <li>• Demonstrate leadership and management skills</li> </ul> |
| Scholar (Research and Evidence Based Practice) | Knowledge:                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                                | <ul style="list-style-type: none"> <li>• Familiarize him/her with latest scientific findings to improve the quality of services rendered to the society</li> <li>• Maintain and expand knowledge through self-directed learning</li> <li>• Participate in research to ensure the optimal use of medicines</li> </ul>                                                                                                                                                               |
|                                                | Attitude:                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                                                | <ul style="list-style-type: none"> <li>• Pursue graduate training in pharmacy and other health related disciplines</li> </ul>                                                                                                                                                                                                                                                                                                                                                      |
|                                                | Skill:                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                                                | <ul style="list-style-type: none"> <li>• Participate and/or conduct research and development of new drugs discovery technologies</li> <li>• Develop and modify laboratory procedures</li> </ul>                                                                                                                                                                                                                                                                                    |

## **5. Admission Requirements**

### **Regular Program**

Admission requirements will be as per the University Senate Legislation. The University shall admit for undergraduate studies:

- Students who have completed Ethiopian high school education and satisfy its entrance qualification assessments.
- Applicants with foreign high school education of equivalent level to the Ethiopian high school education, as determined by the Ministry of Science and Higher Education, and who satisfy entrance requirements
- Considering the extensive practical attachments of the program, students with major disabilities (vision, mobility and hearing-related problems or those who cannot effectively use both hands) shall not be admitted
- Requests for transfer from other departments/disciplines shall be handled based on the relevant university regulation/senate legislation

### **Special Admission**

Special admissions may be granted to attract potentially resourceful candidates such as mature students. Such special admissions shall be decided based on relevant university specific legislations and guidelines in line with the National direction.

## **6. Duration of the Program**

- The bachelor degree in pharmacy requires 5 years or 10 semesters out of which the last 2 semesters (Year V) are dedicated to clerkships and community-based team training program.
- The curriculum contains 32 modules with a total of **355 ECTS**. The total ECTS is excluding two non-credited courses.

## 7. Graduation Requirement

Graduation requirement will be according to the rules and regulations of each university. Moreover, the following requirements must be fulfilled for a student enrolled in the B.Pharm program to be eligible for graduation:

- Has taken all the required modules for the program
- Obtained a minimum cGPA of 2.00
- Has not scored 'F' grade in any course,
- Has not scored less than 'C' grade in any of the major professional courses (course code starting with 'Phar...')
- Has not scored 'D' or lesser on any of the supportive courses/modules
- Has successfully completed and passed clerkship/professional practice program
- Has carried out a student research project on a selected and agreed topic of research problem and scored a minimum of 'C' grade
- Has passed a written comprehensive exit examination to be set for the program. After completion of all deductive courses, research and clerkships in good academic standing, students must take the comprehensive exit exam and score 50% or higher to be eligible for graduation. The exit exam shall be graded as Pass/Fail and a pass mark is considered as a prerequisite to sit for the national licensure exam.
- A student who fails to score a passing mark (50%) in the first comprehensive exit exam will be allowed to re-sit for the next exam after 3 months preparation with an appropriate academic support. The detailed conditions for remediation shall be in line with the National Remediation Guidelines to be introduced by the Ministry of Science and Higher Education and the Ministry of Health.

**Attendance:** Students should attend 100% of both lecture classes and practical attachments. The case of students who fail such requirements will be entertained as per senate legislation of the university and preceptor manual of the respective department.

## 8. Degree Nomenclature

- Up on successful completion of this program the graduate will be awarded “The Degree of Bachelor of Pharmacy (B.Pharm)” in English and “የባዕችለር ዲግሪ በፋርማሲ” in Amharic.

## 9. List of Modules/Courses

### 9.1. Module and course code assignment In

#### the module coding:

- The alphabets indicate to which program/department the module belongs
- The numbers should be four digits and
- The first number indicates year of study (starting from 1- for 1<sup>st</sup> year, 2- for 2<sup>nd</sup> year, etc.)
- The 2<sup>nd</sup> and 3<sup>rd</sup> numbers indicate module number (a two digit code starting from 01, 02...etc.)
- The last number indicates module category (core=1, elective=2, basic=3, general=4)

#### In the course coding:

- The alphabets indicate the course hosting department/program
- The numbers should be four digits
- The first number indicates year of study (starting from 1- for 1<sup>st</sup> year, 2- for 2<sup>nd</sup> year, etc.)
- The 2<sup>nd</sup> and 3<sup>rd</sup> numbers indicate module number (a two digit code starting from 01, 02...etc.)
- The last number indicates the course number (starting from 1, 2...etc.) within the module

### 9.2 List of Modules/Courses

During the allocation of credit points and sequencing of modules/courses assumption was made that all semesters other than Year I, Semester I, shall have duration of 20 weeks for the effective implementation of the curriculum.

The following tables show the list of modules and courses.

#### Major Compulsory (227 ECTS).

| Module Number | Module Category | Module title                             | Module code | Course Code | Course Title                  | Course ECTS | Pre-requisite/ Co-requisite                     |
|---------------|-----------------|------------------------------------------|-------------|-------------|-------------------------------|-------------|-------------------------------------------------|
| 7             | Core            | Introductory Pharmacy Module             | Phar-M2071  | Phar 2071   | Introduction to pharmacy      | 2           | None                                            |
|               |                 |                                          |             | Phar2072    | Pharmaceutical calculations   | 2           | None                                            |
| 8             | Core            | Pharmacognosy and Alternative Medicine I | Phar-M2081  | Phar2081    | Chemistry of Natural products | 5           | Organic Chemistry, Organic Chemistry Laboratory |
|               |                 |                                          |             | Phar2082    | Pharmacognosy                 | 7           | Chemistry of Natural Products                   |

|    |      |                                                |            |           |                                                             |   |                                                                                |
|----|------|------------------------------------------------|------------|-----------|-------------------------------------------------------------|---|--------------------------------------------------------------------------------|
| 9  | Core | Dosage form Sciences I                         | Phar-M2091 | Phar2091  | Integrated physical pharmacy and pharmaceutics-I            | 7 | Pharmaceutical Calculations                                                    |
|    |      |                                                | Phar-M2091 | Phar2092  | Practical Integrated physical pharmacy and pharmaceutics-I  | 2 | Pharmaceutical Calculations & Integrated Physical Pharmacy and Pharmaceutics I |
| 10 | Core | Pharmacology module I                          | Phar-M2101 | Phar2101  | Pharmacology-I                                              | 7 | Biochemistry I and II<br>Physiology I and II<br>Human Anatomy                  |
| 11 | Core | Medicinal Chemistry module I                   | Phar-M2111 | Phar2111  | Medicinal chemistry-I                                       | 7 | Organic Chemistry/<br>Pharmacology I                                           |
| 12 | Core | Dosage form Sciences II                        | Phar-M3121 | Phar3121  | Integrated physical pharmacy and pharmaceutics-II           | 7 | Integrated Physical Pharmacy and Pharmaceutics I                               |
|    |      |                                                |            | Phar3122  | Practical Integrated physical pharmacy and pharmaceutics-II | 2 | Integrated Physical Pharmacy and Pharmaceutics II                              |
| 13 | Core | Pharmacology module II                         | Phar-M3131 | Phar3131  | Pharmacology II                                             | 7 | Pharmacology I                                                                 |
|    |      |                                                |            | Phar 3132 | Clinical toxicology                                         | 3 | Pharmacology I and II                                                          |
| 14 | Core | Medicinal Chemistry module II                  | Phar-M3141 | Phar3141  | Medicinal chemistry-II                                      | 5 | Medicinal Chemistry I<br>Pharmacology II                                       |
| 15 | Core | Pharmaceutical Analysis                        | Phar-M3151 | Phar3151  | Pharmaceutical analysis-I                                   | 7 | None                                                                           |
|    |      |                                                |            | Phar3152  | Pharmaceutical analysis-II                                  | 7 | Pharmaceutical Analysis I                                                      |
| 16 | Core | Pharmacotherapeutics Module I                  | Phar-M3161 | Phar3162  | Integrated therapeutics-I                                   | 7 | Pharmacology I                                                                 |
|    |      |                                                |            | Intm 3161 | Physical assessment                                         | 2 | Integrated Therapeutics I                                                      |
|    |      |                                                |            | Phar3163  | Integrated therapeutics-II                                  | 7 | Integrated Therapeutics I                                                      |
| 17 | Core | Pharmaceutical Technology I                    | Phar-M3171 | Phar3171  | Immunological and biological products                       | 3 | Integrated Physical Pharmacy and Pharmaceutics I and II                        |
| 18 | Core | Social and administrative pharmacy module I    | Phar-M3181 | Comh3181  | Health service management and policies                      | 5 | None                                                                           |
| 19 | Core | Biopharmaceutics and Clinical Pharmacokinetics | Phar-M3191 | Phar3191  | Biopharmaceutics and Clinical Pharmacokinetics              | 7 | Physiology II and Pharmacology II                                              |

| Module Number | Module Category | Module title                                 | Module code | Course Code | Course Title                                         | Course ECTS | Pre-requisite/ Co-requisite                             |
|---------------|-----------------|----------------------------------------------|-------------|-------------|------------------------------------------------------|-------------|---------------------------------------------------------|
| 20            | Core            | Pharmaceutical Technology II                 | Phar-M4201  | Phar4201    | Industrial pharmacy                                  | 7           | Integrated Physical Pharmacy and Pharmaceutics I and II |
| 21            | Core            | Social and administrative pharmacy module II | Phar-M4211  | Phar4211    | Introduction to Pharmacoeconomics                    | 3           | None                                                    |
|               |                 |                                              |             | Phar4212    | Pharmaceutical Supply Chain management               | 7           | Health Service Management and Policies (PUBH3181)       |
|               |                 |                                              |             | Phar4213    | Medical supplies, equipment and reagents             | 3           | None                                                    |
|               |                 |                                              |             | Phar4214    | Pharmaceutical Marketing and promotion               | 3           | None                                                    |
| 22            | Core            | Pharmacotherapeutics Module II               | Phar-M4221  | Phar4221    | Integrated therapeutics-III                          | 7           | Integrated Therapeutics I                               |
|               |                 |                                              |             | Phar4222    | Integrated therapeutics-IV                           | 7           | Integrated Therapeutics I                               |
| 23            | Core            | Pharmacognosy and Alternative Medicine II    | Phar-M4231  | Phar4231    | Complementary and alternative medicine               | 3           | Pharmacognosy                                           |
| 24            | Core            | Pharmacy practice module                     | Phar-M4241  | Phar4241    | Drug informatics                                     | 3           | None                                                    |
|               |                 |                                              |             | Phar4242    | Communication skills for pharmacists                 | 3           | None                                                    |
|               |                 |                                              |             | Phar4243    | Pharmacy law and ethics                              | 3           | None                                                    |
|               |                 |                                              |             | Phar4244    | Pharmacy practice                                    | 7           | None                                                    |
|               |                 |                                              |             | Nurs4245    | First aid                                            | 3           | Course on Anatomy and Physiology                        |
|               |                 |                                              |             | PUBH4246    | Nutrition                                            | 3           | None                                                    |
| 26            | Core            | Pharmaceutical Research I                    | Phar-M4261  | Phar4261    | Research Methods                                     | 3           | Epidemiology and biostatistics courses                  |
| 27            | Core            | Pharmacy clerkship I                         | Phar-M5271  | Phar5271    | Ambulatory care clerkship                            | 5           | Successful completion of all course work                |
|               |                 |                                              |             | Phar5272    | Drug information service clerkship                   | 3           | Successful completion of all course work                |
|               |                 |                                              |             | Phar5273    | Internal medicine clerkship                          | 7           | Successful completion of all course work                |
|               |                 |                                              |             | Phar5274    | Hospital pharmacy clerkship                          | 7           | Successful completion of all course work                |
| 28            | Core            | Pharmacy clerkship II                        | Phar-M5281  | Phar5281    | Pediatric clerkship                                  | 7           | Successful completion of all course work                |
|               |                 |                                              |             | Phar5282    | Gynecology, obstetrics and family planning clerkship | 3           | Successful completion of all course work                |



|    |      |                            |            |          |                                        |   |                                                        |
|----|------|----------------------------|------------|----------|----------------------------------------|---|--------------------------------------------------------|
|    |      |                            |            | Phar5283 | Pharmaceutical Manufacturing clerkship | 5 | Successful completion of all course work               |
|    |      |                            |            | Phar5284 | Community pharmacy clerkship           | 5 | Successful completion of all course work               |
| 30 | Core | Pharmaceutical Research II | Phar-M5301 | Phar5301 | Directed study                         | 5 | Research methods                                       |
| 31 | Core | Team Training Program      | PUBH M5311 | ComH5311 | Team training program                  | 7 | Successful completion of all course work               |
| 32 | Core | Comprehensive Exit Exam    | Phar-M5321 | Phar5321 | Comprehensive exit exam                | 0 | Successful completion of all course work and clerkship |

### Major Elective (10 ECTS)

| Module Number | Module Category | Module title                    | Module code | Course Code | Course Title                                         | Course ECTS | Pre-requisite/ Co-requisite                     |
|---------------|-----------------|---------------------------------|-------------|-------------|------------------------------------------------------|-------------|-------------------------------------------------|
| 25            | Elective        | Professional elective module    | Phar-M4252  | Phar4251    | Introduction to Pharmacoepidemiology                 | 5           | None                                            |
|               |                 |                                 |             | Phar4252    | Phytochemistry                                       | 5           | -                                               |
|               |                 |                                 |             | Phar4253    | Pharmaceutical Manufacturing                         | 5           | Pharmaceutical Technology Module                |
|               |                 |                                 |             | Phar4254    | Pharmacogenetics                                     | 5           | Pharmacology I and II                           |
|               |                 |                                 |             | Phar4255    | Pharmaceutical Quality control and quality assurance | 5           | Pharmaceutical Analysis I & II                  |
|               |                 |                                 |             | Phar4256    | Drug design and synthesis                            | 5           | Organic Chemistry, Medicinal Chemistry I and II |
|               |                 |                                 |             | Phar4257    | Warehouse management                                 | 5           | Pharmaceutical supply chain management          |
|               |                 |                                 |             | Phar4258    | Research in pharmacology                             | 5           | Pharmacology I and II, Clinical toxicology      |
| 29            | Elective        | Professional elective clerkship | Phar-M5292  | Phar5291    | Psychiatry clerkship                                 | 5           | Successful completion of all course work        |
|               |                 |                                 |             | Phar5292    | Surgery clerkship                                    | 5           | Successful completion of all course work        |
|               |                 |                                 |             | Phar5293    | Oncology & Hematology clerkship                      | 5           | Successful completion of all course work        |
|               |                 |                                 |             | Phar5294    | Ophthalmology & ENT clerkship                        | 5           | Successful completion of all course work        |
|               |                 |                                 |             | Phar5295    | Emergency Medicine Clerkship                         | 5           | Successful completion of all course work        |
|               |                 |                                 |             | Phar5296    | Dermatology clerkship                                | 5           | Successful completion of all course work        |

|  |  |  |  |          |                                      |   |                                          |
|--|--|--|--|----------|--------------------------------------|---|------------------------------------------|
|  |  |  |  | Phar5297 | Pharmaceutical quality control       | 5 | Successful completion of all course work |
|  |  |  |  | Phar5298 | Pharmaceutical regulatory affairs    | 5 | Successful completion of all course work |
|  |  |  |  | Phar5299 | Pharmaceutical wholesale & promotion | 5 | Successful completion of all course work |

### Supportive (52ECTS)

| Module Number | Module Category | Module title                   | Module code | Course Code | Course Title                              | Course ECTS | Pre-requisite/ Co-requisite                                     |
|---------------|-----------------|--------------------------------|-------------|-------------|-------------------------------------------|-------------|-----------------------------------------------------------------|
| 3             | Basic           | Biomedical sciences-I          | Biom-M1033  | Anat1031    | Human Anatomy and Histology               | 7           | None                                                            |
|               |                 |                                |             | Phyl1032    | Human Physiology-I                        | 5           | None                                                            |
| 4             | Basic           | Chemistry                      | Chem-M1043  | Chem1041    | Organic chemistry                         | 5           | None                                                            |
|               |                 |                                |             | Chem1042    | Organic chemistry laboratory              | 2           | None                                                            |
| 5             | Basic           | Biomedical sciences-II         | Biom-M2053  | Path2054    | Pathology                                 | 5           | Physiology I&II , Biochemistry I&II                             |
|               |                 |                                |             | Phyl2055    | Human Physiology-II                       | 5           | Human Physiology I                                              |
|               |                 |                                |             | Bioc2051    | Biochemistry-I                            | 5           | Organic Chemistry                                               |
|               |                 |                                |             | Bioc2052    | Biochemistry-II                           | 5           | Organic Chemistry                                               |
|               |                 |                                |             | Mbio2053    | Microbiology, Immunology and Parasitology | 7           | Physiology I&II , Biochemistry I&II Human Anatomy and Histology |
| 6             | Basic           | Biostatistics and Epidemiology | PUBH2063    | ComH2061    | Biostatistics                             | 3           | None                                                            |
|               |                 |                                |             | ComH2062    | Epidemiology                              | 3           | None                                                            |

## General Education (66 ECTS)

| Module Number | Module Category | Module title                   | Module code | Course Code | Course Title                          | Course ECTS | Pre-requisite/ Co-requisite |
|---------------|-----------------|--------------------------------|-------------|-------------|---------------------------------------|-------------|-----------------------------|
| 1             | General         | English Language Skills        | Enla-M1014  | FLEn1011    | Communicative English Skill I         | 5           | None                        |
|               |                 |                                |             | FLEn1012    | Communicative English Skill II        | 5           | None                        |
| 1             | General         | Social sciences and humanities |             | Psch 1011   | General Psychology                    | 5           | None                        |
|               |                 |                                |             | LoCT 1011   | Critical Thinking                     | 5           | None                        |
|               |                 |                                |             | GeES1011    | Geography of Ethiopia & the Horn      | 5           | None                        |
|               |                 |                                |             | Anth 1012   | Social Anthropology                   | 4           | None                        |
| 1             | General         |                                |             | Math 1011   | Math for Natural Sciences             | 5           | None                        |
| 1             | General         |                                |             | Phyc-1011   | General Physics                       | 5           | None                        |
| 1             | General         |                                |             | SpSc 1011   | Physical Fitness                      | 0           | None                        |
| 1             | General         |                                |             | MCiE 1012   | Moral and Civic Education             | 4           | None                        |
| 1             | General         |                                |             | SNIE 1012   | Inclusiveness                         | 4           | None                        |
| 1             | General         |                                |             | MGMT 1012   | Entrepreneurship                      | 5           | None                        |
| 1             | General         |                                |             | GlTr 1012   | Global Trends                         | 4           | None                        |
| 2             | General         |                                |             | EmTe 1012   | Introduction to Emerging Technologies | 5           | None                        |
|               | General         |                                |             | Econ 1011   | Economics                             | 5           | None                        |

## 10. Module Sequencing by Semester and Year

### Year I Semester I

| Module code           | Module title | Course Code | Course Title                     | Course ECTS     | Mode of delivery | Duration (hrs) | Lecture+ Tutorial+ Lab per week (hrs) |
|-----------------------|--------------|-------------|----------------------------------|-----------------|------------------|----------------|---------------------------------------|
|                       |              | FLEn1011    | Communicative English Skill I    | 5               | Parallel         | 135            | 3+0+0                                 |
|                       |              | Psyc 1011   | General Psychology               | 5               | Parallel         | 135            | 2+1+0                                 |
|                       |              | LoCT1011    | Critical Thinking                | 5               | Parallel         | 135            | 3+0+0                                 |
|                       |              | GeES1011    | Geography of Ethiopia & the Horn | 5               | Parallel         | 135            | 3+0+0                                 |
|                       |              | Math1011    | Math for Natural Sciences        | 5               | Parallel         | 135            | 2+1+0                                 |
|                       |              | Phyc 1011   | General Physics                  | 5               | Parallel         | 135            | 2+1+0                                 |
|                       |              | SpSc 1011   | Physical Fitness                 | NC*=2ECTS       | Parallel         | 81             | *2 contact hours                      |
| <b>Semester Total</b> |              |             |                                  | <b>30 ECTS*</b> |                  | 891            |                                       |

**\* There will be a total ECTS of 33 if the NC Physical Fitness course is added**

**Year I Semester II\***

| <b>Module code</b>    | <b>Module title</b>   | <b>Course Code</b> | <b>Course Title</b>                   | <b>Course ECTS</b> | <b>Mode of delivery</b> | <b>Duration</b> | <b>Lecture+ Tutorial+ Lab per week (hrs)</b> |
|-----------------------|-----------------------|--------------------|---------------------------------------|--------------------|-------------------------|-----------------|----------------------------------------------|
|                       |                       | FLEn1012           | Communicative English Skill II        | 5                  | Parallel                | 135             | 3+0+0                                        |
|                       |                       | EmTe 1012          | Introduction to Emerging Technologies | 5                  | Parallel                | 135             | 12+4+0                                       |
|                       |                       | Anth 1012          | Social Anthropology                   | 4                  | Parallel                | 108             | 3+0+0                                        |
|                       |                       | MCiE 1012          | Moral and Civic Education             | 4                  | Parallel                | 108             | 4+0+0                                        |
| Biom-M1033            | Biomedical sciences-I | Anat1031           | Human Anatomy and Histology           | 7                  | Parallel                | 189             | 4+1+2                                        |
| Biom-M1033            | Biomedical sciences-I | Phyl1032           | Human Physiology-I                    | 5                  | Parallel                | 135             | 3+1+0                                        |
| Chem-M1043            | Chemistry             | Chem1041           | Organic chemistry                     | 5                  | Parallel                | 135             | 12+4+0                                       |
| Chem-M1043            | Chemistry             | Chem1042           | Organic chemistry laboratory          | 2                  | Parallel                | 54              | 0+0+9                                        |
| <b>Semester Total</b> |                       |                    |                                       | <b>37 ECTS</b>     |                         | 999             |                                              |

**\* 20 Weeks semester**

### Year II Semester I\*

| Module code           | Module title                             | Course Code | Course Title                              | Course ECTS    | Mode of delivery | Duration | Lecture+ Tutorial+ Lab per week (hrs) |
|-----------------------|------------------------------------------|-------------|-------------------------------------------|----------------|------------------|----------|---------------------------------------|
| Biom-M2053            | Biomedical sciences-II                   | Path2054    | Pathology                                 | 5              | Parallel         | 135      | 5+3+0                                 |
| Phar-M2081            | Pharmacognosy and Alternative Medicine I | Phar2081    | Chemistry of Natural products             | 5              | Parallel         | 135      | 3+0+0                                 |
| Biom-M2053            | Biomedical sciences- II                  | Phyl2055    | Human Physiology-II                       | 5              | Parallel         | 135      | 2.5+1+0                               |
| Biom-M2053            | Biomedical sciences- II                  | Bioc2051    | Biochemistry-I                            | 5              | Parallel         | 135      | 5+3+0                                 |
| Biom-M2053            | Biomedical sciences- II                  | Bioc2052    | Biochemistry-II                           | 5              | Parallel         | 135      | 5+3+0                                 |
| Biom-M2053            | Biomedical sciences-II                   | Mbio2053    | Microbiology, Immunology and Parasitology | 7              | Parallel         | 189      | 8+3+6                                 |
| Phar-M2071            | Introductory Pharmacy Module             | Phar 2071   | Introduction to pharmacy                  | 2              | Parallel         | 54       | 1+0+2                                 |
| Phar-M2071            | Introductory Pharmacy Module             | Phar2072    | Pharmaceutical calculations               | 2              | Parallel         | 54       | 1+3+0                                 |
|                       |                                          | Econ 1011   | Economics                                 | 5              | Parallel         | 135      | 3+0+0                                 |
| <b>Semester Total</b> |                                          |             |                                           | <b>41 ECTS</b> |                  | 1107     |                                       |

### \* 20 Weeks semester

The following were the considerations for amendment:

- Universities, which have **not** offered the Organic Chemistry Lab course in the second semester of the first year, should arrange to offer the course in the first week of Year II-Semester I and retrospectively register the students for Year I-Semester II.
- Universities, which already completed McIe-1012, can replace it with a similar module of 3 – 5 ECTS.
- In case of universities, which already offered any one of the listed courses, it is important that they replace the course/s with an equivalent course, in order to avoid credit spillover to the later years/semesters.

### Year II Semester II\*

| Module code           | Module title                             | Course Code | Course Title                                                 | Course ECTS    | Mode of delivery | Duration | Lecture+ Tutorial+ Lab per week (hrs) |
|-----------------------|------------------------------------------|-------------|--------------------------------------------------------------|----------------|------------------|----------|---------------------------------------|
| Phar-M2081            | Pharmacognosy and Alternative Medicine I | Phar2082    | Pharmacognosy                                                | 7              | Parallel         | 189      | 4 +1+3                                |
| Phar-M2101            | Pharmacology module I                    | Phar2101    | Pharmacology-I                                               | 7              | Parallel         | 189      | 4+1+3                                 |
| Phar-M2111            | Medicinal Chemistry module I             | Phar2111    | Medicinal chemistry-I                                        | 7              | Parallel         | 189      | 4+1+3                                 |
| Phar-M2091            | Dosage form Sciences I                   | Phar2091    | Integrated physical pharmacy and pharmaceuticals-I           | 7              | Parallel         | 189      | 4+1+0                                 |
| Phar-M2091            | Dosage form Sciences I                   | Phar2092    | Practical Integrated physical pharmacy and pharmaceuticals-I | 2              | Parallel         | 54       | 0+0+3                                 |
| PUBH2063              | Biostatistics and Epidemiology           | ComH2061    | Biostatistics**                                              | 3              | Parallel         | 81       | 2+0+0                                 |
| PUBH2063              | Biostatistics and Epidemiology           | ComH2062    | Epidemiology**                                               | 3              | Parallel         | 81       | 2+0+0                                 |
|                       |                                          | SNIE 1012   | Inclusiveness                                                | 4              | Parallel         | 81       | 2+0+0                                 |
| <b>Semester Total</b> |                                          |             |                                                              | <b>40 ECTS</b> |                  | 1053     |                                       |

\* 20 Weeks semester

\*\* In the event that the contents of the course History of Ethiopia & the Horn (Hist.1012) are completed before students start this semester, the course can be offered in place of Biostatistics and Epidemiology. Then Epidemiology can be offered in Year III Semester I or Year III Semester II and Biostatistics can be offered in Year III Semester II or Year IV Semester I.

**Year III Semester I\***

| <b>Module code</b>       | <b>Module title</b>           | <b>Course Code</b> | <b>Course Title</b>                                         | <b>Course ECTS</b> | <b>Mode of delivery</b> | <b>Duration</b> | <b>Lecture+ Tutorial+ Lab per week (hrs)</b> |
|--------------------------|-------------------------------|--------------------|-------------------------------------------------------------|--------------------|-------------------------|-----------------|----------------------------------------------|
| Phar-M3121               | Dosage form Sciences II       | Phar3121           | Integrated physical pharmacy and pharmaceutics-II           | 7                  | Parallel                | 189             | 4+1+0                                        |
| Phar-M3121               | Dosage form Sciences II       | Phar3122           | Practical Integrated physical pharmacy and pharmaceutics-II | 2                  | Parallel                | 54              | 0+0+3                                        |
| Phar-M3131               | Pharmacology module II        | Phar3131           | Pharmacology II                                             | 7                  | Parallel                | 189             | 4+1+3                                        |
| Phar-M3141               | Medicinal Chemistry module II | Phar3141           | Medicinal chemistry-II                                      | 5                  | Parallel                | 135             | 3+1+0                                        |
| Phar-M3151               | Pharmaceutical Analysis       | Phar3151           | Pharmaceutical analysis-I                                   | 7                  | Parallel                | 189             | 4+1+3                                        |
| Phar-M3161               | Pharmacotherapeutics Module I | Phar3162           | Integrated therapeutics-I                                   | 7                  | Parallel                | 189             | 4+1+1                                        |
| Phar-M3161               | Pharmacotherapeutics Module I | Intm 3161          | Physical assessment                                         | 2**                | Parallel                | 54              | 1+0+3                                        |
| <b>Semester total***</b> |                               |                    |                                                             | <b>37 ECTS</b>     |                         | 999             |                                              |

**\* 20 Weeks semester**

**\*\* The physical assessment course is co-requisite to Integrated therapeutics-I**



**Year III Semester II \***

| <b>Module code</b>      | <b>Module title</b>                            | <b>Course Code</b> | <b>Course Title</b>                            | <b>Course ECTS</b> | <b>Mode of delivery</b> | <b>Duration</b> | <b>Lecture+ Tutorial+ Lab per week (hrs)</b> |
|-------------------------|------------------------------------------------|--------------------|------------------------------------------------|--------------------|-------------------------|-----------------|----------------------------------------------|
| Phar-M3191              | Biopharmaceutics and Clinical Pharmacokinetics | Phar3191           | Biopharmaceutics and Clinical Pharmacokinetics | 7                  | Parallel                | 189             | 4+1+0                                        |
| Phar-M3171              | Pharmaceutical Technology I                    | Phar3171           | Immunological and biological products          | 3                  | Parallel                | 81              | 4 +2+0                                       |
| Phar-M3131              | Pharmacology module II                         | Phar 3132          | Clinical toxicology                            | 3                  | Parallel                | 81              | 2 +2+0                                       |
| Phar-M3181              | Social and administrative pharmacy module I    | Comh3181           | Health service management and policies         | 5                  | Parallel                | 135             | 4 +0+0                                       |
| Phar-M3161              | Pharmacotherapeutics Module I                  | Phar3163           | Integrated therapeutics-II                     | 7                  | Parallel                | 189             | 4+1+2                                        |
|                         |                                                | MGMT 1012          | Entrepreneurship                               | 5                  | Parallel                | 135             | 3 + 1+ 0                                     |
| Phar-M3151              | Pharmaceutical Analysis                        | Phar3152           | Pharmaceutical analysis-II                     | 7                  | Parallel                | 189             | 4+1+3                                        |
| <b>Semester total**</b> |                                                |                    |                                                | <b>37</b>          |                         | 999             |                                              |

**\* 20 Weeks semester**

**Year IV Semester I\***

| <b>Module code</b>    | <b>Module title</b>                          | <b>Course Code</b> | <b>Course Title</b>                      | <b>Course ECTS</b> | <b>Mode of delivery</b> | <b>Duration</b> | <b>Lecture+ Tutorial+ Lab per week (hrs)</b> |
|-----------------------|----------------------------------------------|--------------------|------------------------------------------|--------------------|-------------------------|-----------------|----------------------------------------------|
| Phar-M4201            | Pharmaceutical Technology II                 | Phar4201           | Industrial pharmacy                      | 7                  | Parallel                | 189             | 4+0.5+3                                      |
| Phar-M4211            | Social and administrative pharmacy module II | Phar4211           | Introduction to Pharmacoeconomics        | 3                  | Parallel                | 81              | 2+2+0                                        |
| Phar-M4211            | Social and administrative pharmacy module II | Phar4212           | Pharmaceutical Supply Chain management   | 7                  | Parallel                | 189             | 4+1+0                                        |
| Phar-M4221            | Pharmacotherapeutics Module II               | Phar4221           | Integrated therapeutics-III              | 7                  | Parallel                | 189             | 4+1+2                                        |
| Phar-M4231            | Pharmacognosy and Alternative Medicine II    | Phar4231           | Complementary and alternative medicine** | 3                  | Parallel                | 81              | 2+1+1**                                      |
| Phar-M4241            | Pharmacy practice module                     | Phar4241           | Drug informatics                         | 3                  | Parallel                | 81              | 4+2+2                                        |
| Phar-M4241            | Pharmacy practice module                     | Phar4242           | Communication skills for pharmacists     | 3                  | Parallel                | 81              | 4+2+2                                        |
| Phar-M4241            | Pharmacy practice module                     | Phar4243           | Pharmacy law and ethics                  | 3                  | Parallel                | 81              | 4 +1+0                                       |
| <b>Semester total</b> |                                              |                    |                                          | <b>36 ECTS</b>     |                         | 972             |                                              |

**\* 20 Weeks semester**

**\*\* Consider visits to botanical gardens/registered traditional healers or their settings/Health bureau regulating the traditional healers**

**Year IV Semester II \***

| <b>Module code</b>  | <b>Module title</b>                          | <b>Course Code</b> | <b>Course Title</b>                      | <b>Course ECTS</b> | <b>Mode of delivery</b> | <b>Duration</b> | <b>Lecture+ Tutorial+ Lab per week (hrs)</b> |
|---------------------|----------------------------------------------|--------------------|------------------------------------------|--------------------|-------------------------|-----------------|----------------------------------------------|
|                     |                                              | GlTr 1012          | Global Trends                            | 4                  | Parallel                | 108             | 4+0+0                                        |
| Phar-M4211          | Social and administrative pharmacy module II | Phar4213           | Medical supplies, equipment and reagents | 3                  | Parallel                | 81              | 4+1+0                                        |
| Phar-M4211          | Social and administrative pharmacy module II | Phar4214           | Pharmaceutical Marketing and promotion   | 3                  | Parallel                | 81              | 4+2+2                                        |
| Phar-M4221          | Pharmacotherapeutics Module II               | Phar4222           | Integrated therapeutics-IV               | 7                  | Parallel                | 189             | 4+1+2                                        |
| Phar-M4241          | Pharmacy practice module                     | Phar4244           | Pharmacy practice***                     | 7                  | Parallel                | 189             | 4 +0+6                                       |
| Phar-M4241          | Pharmacy practice module                     | Nurs4245           | First aid                                | 3                  | Parallel                | 81              | 4 +2+10                                      |
| Phar-M4241          | Pharmacy practice module                     | PUBH4246           | Nutrition                                | 3                  | Parallel                | 81              | 4 +2+0                                       |
| Phar-M4252          | Professional elective module                 | *                  | Professional elective course             | 5                  | Parallel                | 135             | 4+1+1                                        |
| Phar-M4261          | Pharmaceutical Research I                    | Phar4261           | Research Methods                         | 3                  | Parallel                | 81              | 4+1+0                                        |
| Semester total***** |                                              |                    |                                          | <b>38</b>          |                         | 1026            |                                              |

**\* 20 Weeks semester**

**\*\*\*The course shall be owned by all departments/units in the school/department of pharmacy and the theory and practice should give enough time for learning and practicing on the various areas of pharmacy practice such as pharmaceutical manufacturing, regulatory affairs, quality control, supply chain management, etc.**

## Year V Semester I and Year V Semester II

**Final year attachment clerkship will be year-based. Student assessment will be made at the end of the year**

| Module code           | Module title                    | Course Code | Course Title                                         | Course ECTS*   | Mode of delivery | Duration | Number of weeks         |
|-----------------------|---------------------------------|-------------|------------------------------------------------------|----------------|------------------|----------|-------------------------|
| Phar-M5271            | Pharmacy clerkship I            | Phar5271    | Ambulatory care clerkship                            | 5              | Parallel         | 135      | 3                       |
| Phar-M5271            | Pharmacy clerkship I            | Phar5272    | Drug information service clerkship                   | 3              | Parallel         | 81       | 2                       |
| Phar-M5271            | Pharmacy clerkship I            | Phar5273    | Internal medicine clerkship                          | 7              | Parallel         | 189      | 4                       |
| Phar-M5271            | Pharmacy clerkship I            | Phar5274    | Hospital pharmacy clerkship                          | 7              | Parallel         | 189      | 4                       |
| Phar-M5281            | Pharmacy clerkship II           | Phar5281    | Pediatric clerkship                                  | 7              | Parallel         | 189      | 4                       |
| Phar-M5281            | Pharmacy clerkship II           | Phar5282    | Gynecology, obstetrics and family planning clerkship | 3              | Parallel         | 81       | 2                       |
| Phar-M5281            | Pharmacy clerkship II           | Phar5283    | Pharmaceutical Manufacturing clerkship               | 5              | Parallel         | 135      | 3                       |
| Phar-M5281            | Pharmacy clerkship II           | Phar5284    | Community pharmacy clerkship                         | 5              | Parallel         | 135      | 3                       |
| Phar-M5292            | Professional elective clerkship | ***         | Elective attachment                                  | 5              | Parallel         | 135      | 3                       |
| Phar-M5301            | Pharmaceutical Research II      | Phar5301    | Directed study                                       | 5              | Parallel         | 135      | NA                      |
| PUBHM5311             | Team Training Program           | ComH5311    | Team training program                                | 7              | Parallel         | 189      | 6                       |
| Phar-M5321            | Comprehensive exit exam         | Phar5321    | Comprehensive exit exam                              | Non credit     | Parallel         | NA       | 3 weeks for preparation |
| <b>Semester total</b> |                                 |             |                                                      | <b>59 ECTS</b> |                  | 1593     | 37**                    |

\* To be determined based on a student's preference

\*\* A one-week off after major pharmacy clerkships is recommended and students are encouraged to plan their time so that they can **find extra time for data collection and/or experiments to fulfil the directed study requirements.**

## **11. Teaching Methodology**

### **Mode of delivery**

The program use a Parallel mode of delivery.

### **Method of teaching**

The following methods will be used as strategies to teach the courses within the different modules in the program:

- Lectures
- Laboratory practices
- Demonstrations
- Group works
- Home study
- Seminars
- Tutorials
- Home take assignments
- Skill lab
- Hospital attachment
- Health center attachment
- Pharmaceutical industries attachments
- Community based learning
- Field supervision
- Team training program
- Research project
- Role play
- Symposium
- Talk show
- Drama etc.

## 12. Assessment

Evaluation will be in the form of formative and summative assessment

- Formative assessment consists of,
  - Continuous assessment (test, quiz, case presentation, assignment /individual – group exercise)
  - Bed side, round
  - Supervision
  - Log book
  - Portfolio
  - Assessment of practical skills
- Summative assessment consists of:
  - Final written exam
  - Objective Structured Clinical Examination (OSCE)
  - Oral exam
  - Exit exam

N.B:

- Grading for exit exam will be P/F, and the student should score greater than or equal to 50% to pass.

### Grading System

The program uses criteria referenced method of evaluation to value the performance of students at the end of each semester and/or year. The instructor uses the scale fixed and grade based on the pre-settled criteria, which are driven from the learning objectives of the topic/course. Thus, we do not use norm-referenced method.

Accordingly, students will be graded on the letter grade as well as on percentage grading system. The grading subsystem has to be adjusted to calculate the GPA based on the fixed scale range shown in the following table. The grading system that has been agreed upon by all public Schools of Pharmacy and subsequently approved by the then MOSHE shall be observed. However, the following table provides guidance on the raw marks and their corresponding fixed number and letter grades.

The grading of the comprehensive exit exam and the physical fitness courses will be on a “Pass” or “Fail” basis.

| Raw Mark Interval<br>[100%] | Corresponding fixed<br>number Grade | Corresponding<br>Letter Grade |
|-----------------------------|-------------------------------------|-------------------------------|
| [90, 100]                   | 4.00                                | A+                            |
| [85, 90)                    | 4.00                                | A                             |
| [80, 85)                    | 3.75                                | A-                            |
| [75, 80)                    | 3.50                                | B+                            |
| [70, 75)                    | 3.00                                | B                             |
| [65, 70)                    | 2.75                                | B-                            |
| [60, 65)                    | 2.50                                | C+                            |
| [50, 60)                    | 2.00                                | C                             |
| [45, 50)                    | 1.75                                | C-                            |
| [40, 45)                    | 1.00                                | D                             |
| < 40                        | 0.00                                | F                             |

### **Promotion criteria**

- A student who scored “F” grade on a pre-requisite course shall not be allowed to take the next course.
- A student who scored “C-” and/or “D” in one or two major professional courses can be allowed to take remedial exam once. The student shall repeat the course/s if he/she could not score “C” or above on the remedial exam.
- A student who scored “C-” or below in three or more major professional courses shall not be promoted to the next semester, irrespective of his/her cGPA. However, a student who scored “C-” in only three major professional courses can be allowed to take remedial exam

for one of the courses and can be promoted if he/she scored a minimum grade of “C” in the remedial exam.

- Irrespective of his/her cGPA, a student is expected to score a minimum of “C” grade on all major professional courses to be eligible for the final year clerkship attachment.

### 13. Quality Assurance

The school/department quality assurance committee is responsible for the management and monitoring of the program. Both formative and summative evaluations will be done to assess the realization of the curriculum’s objectives. The following mechanisms will be employed to evaluate whether or not the courses offered in the program meet the standards:

- A course syllabus according to the course content indicated in this curriculum should be prepared for each course with time frame.
- The respective School/departments will evaluate the agreement between examination contents and the course syllabi.
- Recruitment of qualified staff
- Examination and continuous assessment (formative and summative)
- Periodic acquisition of up – to – date references, laboratory equipment and reagents
- Supervised practices in the training health facilities
- Periodic evaluation of the curriculum and the program in general
- Periodic evaluation and revision of the curriculum based on the feedback from stakeholders/employers, graduates and students.

### 14 Resource Profile

#### 14.1 Names and ranks of active teaching and technical staff

| No. | Full Name        | Academic Rank       | Department                         |
|-----|------------------|---------------------|------------------------------------|
| 1   | Abebe Ejigu      | Assistant Professor | Pharmacology and Clinical Pharmacy |
| 2   | Abebech Adane    | Technical assistant | Pharmacology and Clinical Pharmacy |
| 3   | Abel Degu        | Lecturer            | Pharmacology and Clinical Pharmacy |
| 4   | Akeberegn Gorems | Lecturer            | Pharmacology and Clinical Pharmacy |
| 5   | Alemseged Beyene | Assistant Professor | Pharmacology and Clinical Pharmacy |
| 6   | Alemu Belayneh   | Assistant lecturer  | Pharmacology and Clinical Pharmacy |
| 7   | Alfoalem Araba   | Assistant lecturer  | Pharmacology and Clinical Pharmacy |



|    |                         |                     |                                    |
|----|-------------------------|---------------------|------------------------------------|
| 8  | Assefa Mulu             | Assistant Professor | Pharmacology and Clinical Pharmacy |
| 9  | Atalay Mulu             | Lecturer            | Pharmacology and Clinical Pharmacy |
| 10 | Beshir Bedru            | Lecturer            | Pharmacology and Clinical Pharmacy |
| 11 | Chalelgn Kassaw         | Lecturer            | Pharmacology and Clinical Pharmacy |
| 12 | Dagmawit Awko           | Lecturer            | Pharmacology and Clinical Pharmacy |
| 13 | Dr. Legese Chelkeba     | Associate Professor | Pharmacology and Clinical Pharmacy |
| 14 | Dr. Teshome Nedi        | Associate Professor | Pharmacology and Clinical Pharmacy |
| 15 | Dr. Zelealem Petros     | Assistant Professor | Pharmacology and Clinical Pharmacy |
| 16 | Dr. Anteneh Kassahun    | Lecturer            | Pharmacology and Clinical Pharmacy |
| 17 | Dr. Shemsu Umer         | Assistant Professor | Pharmacology and Clinical Pharmacy |
| 18 | Dr. Solomon Mequanente  | Associate Professor | Pharmacology and Clinical Pharmacy |
| 19 | Dr. Workineh Shibeshi   | Associate Professor | Pharmacology and Clinical Pharmacy |
| 20 | Eskinder Ayalew         | Lecturer            | Pharmacology and Clinical Pharmacy |
| 21 | Fantu Assefa            | Technical assistant | Pharmacology and Clinical Pharmacy |
| 22 | Fikreselam Habte        | Lecturer            | Pharmacology and Clinical Pharmacy |
| 23 | Firehiwot Amare         | Assistant Professor | Pharmacology and Clinical Pharmacy |
| 24 | Getachew Alemkere       | Lecturer            | Pharmacology and Clinical Pharmacy |
| 25 | Girma Tekle             | Lecturer            | Pharmacology and Clinical Pharmacy |
| 26 | Gobezie Temesgen        | Lecturer            | Pharmacology and Clinical Pharmacy |
| 27 | Hailemeskel Mekonnen    | Lecturer            | Pharmacology and Clinical Pharmacy |
| 28 | Hannan Muzzeyin         | Assistant lecturer  | Pharmacology and Clinical Pharmacy |
| 29 | Hayat Hamid             | Technical assistant | Pharmacology and Clinical Pharmacy |
| 30 | Mammo Feyissa           | Lecturer            | Pharmacology and Clinical Pharmacy |
| 31 | Melak Gedamu            | Lecturer            | Pharmacology and Clinical Pharmacy |
| 32 | Melaku Tileku           | Lecturer            | Pharmacology and Clinical Pharmacy |
| 33 | Merema Hussien          | Technical assistant | Pharmacology and Clinical Pharmacy |
| 34 | Minyahil Alebachew      | Assistant Professor | Pharmacology and Clinical Pharmacy |
| 35 | Mohammed Gebre          | Assistant Professor | Pharmacology and Clinical Pharmacy |
| 36 | Oumer Sada              | Assistant Professor | Pharmacology and Clinical Pharmacy |
| 37 | Prof. Ephrem Engdawerk  | Professor           | Pharmacology and Clinical Pharmacy |
| 38 | Prof. Eyasu Mekonnen    | Professor           | Pharmacology and Clinical Pharmacy |
| 39 | Prof. Tefera Abula      | Professor           | Pharmacology and Clinical Pharmacy |
| 41 | Samuel Worku            | Lecturer            | Pharmacology and Clinical Pharmacy |
| 42 | Solomon Assefa          | Assistant Professor | Pharmacology and Clinical Pharmacy |
| 43 | Tamrat Assefa           | Assistant Professor | Pharmacology and Clinical Pharmacy |
| 44 | Tenawork Fikre          | Technical assistant | Pharmacology and Clinical Pharmacy |
| 45 | Teshale Ayele           | Assistant Professor | Pharmacology and Clinical Pharmacy |
| 46 | Tigistu Alemu           | Lecturer            | Pharmacology and Clinical Pharmacy |
| 47 | Wondmagegn Tamiru       | Assistant Professor | Pharmacology and Clinical Pharmacy |
| 48 | Yewubdar Belew          | Lecturer            | Pharmacology and Clinical Pharmacy |
| 49 | Yohannes Tsegayie       | Lecturer            | Pharmacology and Clinical Pharmacy |
| 50 | Zelalem Tilahun Tesfaye | Assistant Professor | Pharmacology and Clinical Pharmacy |
| 51 | Zenebe Negash           | Lecturer            | Pharmacology and Clinical Pharmacy |
| 52 | Abrham Temesgen Mengstu | Lecturer            | Pharmaceutics and Social Pharmacy  |

|    |                                 |                     |                                     |
|----|---------------------------------|---------------------|-------------------------------------|
| 53 | Alemayehu Adinew Teshale        | Lecturer            | Pharmaceutics and Social Pharmacy   |
| 54 | Dr. Anteneh Belete              | Assoc. Prof.        | Pharmaceutics and Social Pharmacy   |
| 55 | Arebu Issa Bilal                | Assist.Prof         | Pharmaceutics and Social Pharmacy   |
| 56 | Asrat Hordofa Janko             | Chief Tech. Assist  | Pharmaceutics and Social Pharmacy   |
| 57 | Bezawit Negash Demissie         | Assist. Lecturer    | Pharmaceutics and Social Pharmacy   |
| 58 | Birhanu Nigusse                 | Lecturer            | Pharmaceutics and Social Pharmacy   |
| 59 | Dr. Bruck Messele Habte         | Assist. Prof.       | Pharmaceutics and Social Pharmacy   |
| 60 | Dawit Teshome                   | Lecturer            | Pharmaceutics and Social Pharmacy   |
| 61 | Esayas Ayele Tilahun            | Assist. Lecturer    | Pharmaceutics and Social Pharmacy   |
| 62 | Dr. Eskinder Eshetu Ali         | Assist. Prof.       | Pharmaceutics and Social Pharmacy   |
| 63 | Eyerusalem Berhanemeskel        | Lecturer            | Pharmaceutics and Social Pharmacy   |
| 64 | Fantahun Mola Kasa              | Assist. Prof.       | Pharmaceutics and Social Pharmacy   |
| 65 | Fekade Tefera Deneke            | Chief Tech. Assist  | Pharmaceutics and Social Pharmacy   |
| 66 | Dr. Gebremariam Birhanu         | Assist. Prof.       | Pharmaceutics and Social Pharmacy   |
| 67 | Jemal Dilebo Jabir              | Lecturer            | Pharmaceutics and Social Pharmacy   |
| 68 | Kebede Wondu Tilayeneh          | Lecturer            | Pharmaceutics and Social Pharmacy   |
| 69 | Menbere Getaneh                 | Assist. Lecturer    | Pharmaceutics and Social Pharmacy   |
| 70 | Muluken Nigatu Selam            | Assist. Prof.       | Pharmaceutics and Social Pharmacy   |
| 71 | Dr. Mesfin Haile Kahissay       | Assist. Prof.       | Pharmaceutics and Social Pharmacy   |
| 72 | Muluwork Sahile                 | Lecturer            | Pharmaceutics and Social Pharmacy   |
| 73 | Nathan Seyoum H/Selassie        | Assist. Lecturer    | Pharmaceutics and Social Pharmacy   |
| 74 | Dr. Nisha Mary Joseph           | Assoc. Prof.        | Pharmaceutics and Social Pharmacy   |
| 75 | Samuel Abera                    | Lecturer            | Pharmaceutics and Social Pharmacy   |
| 76 | Semayawit Bahiru Bezabih        | Assist. Lecturer    | Pharmaceutics and Social Pharmacy   |
| 77 | Sine Gebreegziabhere Abay       | Assist. Lecturer    | Pharmaceutics and Social Pharmacy   |
| 78 | Sisay Endale Biru               | Lecturer            | Pharmaceutics and Social Pharmacy   |
| 79 | Sofia Seid Yimam                | Lecturer            | Pharmaceutics and Social Pharmacy   |
| 80 | Prof. Teferi Gedif Fenta        | Professor           | Pharmaceutics and Social Pharmacy   |
| 81 | Tesfa Marew Wallelign           | Lecturer            | Pharmaceutics and Social Pharmacy   |
| 82 | Tesfaye Gabriel Dalalo          | Lecturer            | Pharmaceutics and Social Pharmacy   |
| 83 | Prof. Tsige GebreMariam         | Professor           | Pharmaceutics and Social Pharmacy   |
| 84 | Yohannes Mulualem Tsehai        | Lecturer            | Pharmaceutics and Social Pharmacy   |
| 85 | Yonas Brhane Hadera             | Lecturer            | Pharmaceutics and Social Pharmacy   |
| 86 | Zelalem Tilahun Mekonnen        | Assist. Prof        | Pharmaceutics and Social Pharmacy   |
| 87 | Alemu Tadesse Feroche           | Lecturer            | Pharmaceutical Chem & Pharmacognosy |
| 88 | Prof. Ariaya Hymete Sahledingle | Professor           | Pharmaceutical Chem & Pharmacognosy |
| 89 | Awrraris Anbessie Telile        | Lecturer            | Pharmaceutical Chem & Pharmacognosy |
| 90 | Ayenew Ashenif Alemu            | Assistant Professor | Pharmaceutical Chem & Pharmacognosy |
| 91 | Ayinalem Redae Asegedom         | Assistant Lecturer  | Pharmaceutical Chem & Pharmacognosy |
| 92 | Besufikad Abebe Woldeyes        | Lecturer            | Pharmaceutical Chem & Pharmacognosy |
| 93 | Biniam Paulos Bura              | Assistant Professor | Pharmaceutical Chem & Pharmacognosy |
| 94 | Bonsa Amsalu Haile              | Lab Technician      | Pharmaceutical Chem & Pharmacognosy |
| 95 | Daniel Bisrat Mekuria           | Associate Professor | Pharmaceutical Chem & Pharmacognosy |
| 96 | Eshetu Assefa Abera             | Lab Technician      | Pharmaceutical Chem & Pharmacognosy |
| 97 | Eyael Tewolde Habtemariam       | Lecturer            | Pharmaceutical Chem & Pharmacognosy |

|     |                               |                     |                                     |
|-----|-------------------------------|---------------------|-------------------------------------|
| 98  | Feruza Ahmed Mengesha         | Assistant Lecturer  | Pharmaceutical Chem & Pharmacognosy |
| 99  | Frehiwot Beyene               | Assistant Lecturer  | Pharmaceutical Chem & Pharmacognosy |
| 100 | Hiwot Ayalew Demisse          | Lecturer            | Pharmaceutical Chem & Pharmacognosy |
| 101 | Prof. Kaleab Asres Abay       | Professor           | Pharmaceutical Chem & Pharmacognosy |
| 102 | Dr. Mariamawit Yonthan Yishak | Associate Professor | Pharmaceutical Chem & Pharmacognosy |
| 103 | Dr. Solomon Tadesse Zeleke    | Assistant Professor | Pharmaceutical Chem & Pharmacognosy |
| 104 | Tadele Ethicha Yadeta         | Associate Professor | Pharmaceutical Chem & Pharmacognosy |
| 105 | Yitagesu Tewabe Cherkos       | Lecturer            | Pharmaceutical Chem & Pharmacognosy |
| 106 | Yonathan Alebachew Wolde      | Lecturer            | Pharmaceutical Chem & Pharmacognosy |
| 107 | Zewude Tagesse Sugamo         | Lab Technician      | Pharmaceutical Chem & Pharmacognosy |

## 14.2 List of laboratories

There are seven laboratories that are used for the B. Pharm program as listed below.

| No. | Laboratory                         | Size              |
|-----|------------------------------------|-------------------|
| 1.  | Dispensing Laboratory              | 20 in one section |
| 2   | Physical Pharmacy Laboratory       | 20 in one section |
| 3   | Industrial Pharmacy Laboratory     | 20 in one section |
| 4   | Clinical Skill Laboratory          | 20 in one section |
| 5   | Pharmacology Laboratory            | 20 in one section |
| 6   | Pharmacognosy Laboratory           | 20 in one section |
| 7   | Pharmaceutical Analysis Laboratory | 20 in one section |

## 14.3 Library and Classrooms

The B. Pharm program shares the library and classroom facilities of the College of Health Sciences.

**15. Module/Course Equivalency** (in case module/course title or ECTS is changed; or module/course substituted equivalency should be established)

| Module/Course Equivalency                    |             |                                                               |             |                                    |             |                                                     |             |
|----------------------------------------------|-------------|---------------------------------------------------------------|-------------|------------------------------------|-------------|-----------------------------------------------------|-------------|
| NEW                                          |             |                                                               |             | OLD                                |             |                                                     |             |
| Module Title                                 | Module ECTS | Course title                                                  | Course ECTS | Module Title                       | Module ECTS | Course title                                        | Course ECTS |
| Pharmacology module II                       | 10          | Clinical toxicology                                           | 3           | Pharmacology module                | 17          | Applied toxicology                                  | 3           |
| Social and administrative pharmacy module II | 18          | Pharmaceutical Supply Chain Management                        | 7           | Social and administrative pharmacy | 21          | Drug supply management                              | 5           |
| Pharmacy practice module                     | 22          | Pharmacy practice                                             | 7           | Pharmacy practice                  | 25          | Pharmacy practice II                                | 5           |
|                                              |             |                                                               |             | Pharmacy practice                  | 25          | Pharmacy practice I                                 | 5           |
| Dosage form Sciences I                       | 9           | Integrated physical pharmacy and pharmaceuticals-I            | 7           | Dosage form Sciences               | 18          | Integrated physical pharmacy and pharmaceuticals-I  | 9           |
|                                              |             | Practical Integrated physical pharmacy and pharmaceuticals-I  | 2           |                                    |             |                                                     |             |
| Dosage form Sciences II                      | 9           | Integrated physical pharmacy and pharmaceuticals-II           | 7           | Dosage form Sciences               | 18          | Integrated physical pharmacy and pharmaceuticals-II | 9           |
|                                              |             | Practical Integrated physical pharmacy and pharmaceuticals-II | 2           |                                    |             |                                                     |             |

## 16. Module Description

Module 1 [Various General Education modules have been designated as module one as the courses are being offered by various departments and schools/faculties]

**Module name:** English language skill module

**Module category:** General

**Module code:** Enla-M1014

**Module Number:** 01

**Module weight in ECTS:** 10 ECTS

**Courses:**

| Course Name                       | Course Code | ECTS |
|-----------------------------------|-------------|------|
| Communication English language I  | Enla1011    | 5    |
| Communication English language II | Enla1012    | 5    |

### Module Description

The module focuses on enhancing students' language competencies. The first course comprehensively presents students with the opportunity to develop their language skills (Listening, Speaking, Reading, Writing, Vocabulary and Grammar). The course focuses on the development of communication skills of the students both in academic and non-academic contexts. As a result it has a big contribution to the success of students in their other university courses. The second course entirely focuses developing the students' writing skill in both academic and non-academic contexts. Both courses must be given on semester basis as the development of the skills that the courses provide is enhanced with the extension of the period at least to the extent that they can associate them with other courses.

**Module objective:** At the end of this module, students will be able to:

- Involve in various communicative contexts
- Read and understand texts with ease
- Differentiate oral and written discourses
- Listen to conversations (communications) in English and decode message easily
- Write reports (paragraphs, essays) in academic contexts

**Module competency:** Develop writing and communication skills which facilitates university studies

**Mode of delivery:** Parallel

## English Language skill Module Course Syllabi

**Course title:** Communicative English Skills I  
**Course code:** EnLa1011  
**Module name:** English Language skill  
**Module code:** Enla-M1014

**Course ECTS:** 5 ECTS (135 hrs)  
Lecture 48 hrs  
Tutorial 7 hrs  
Group Work 10 hrs  
Assessment 20 hrs  
Home Study 50 hrs  
Total 135 hrs

**Year/Semester Course is offered:** Year I, Semester I

**Contact hours/ week:** 135

**Pre-requisite:** None

### **Course description:**

This course is intended to develop and improve students' language competence. This course is aimed at developing trainees' communicative abilities in English which will help students to develop their communicative skills and overall language competence in English. Generally, this course will cover the specific language aspects described below. Developing basic functions of English language skills: reading (scanning, skimming, reading for details, summarizing, understanding the structure of a text); listening (listening for the gist, listening for details, recognizing discourse markers, noticing the structure of a lecture, understanding speaker intentions, recognizing signposting, attending and following skills); writing (summarizing a text, synthesizing choppy sentences, writing argumentative texts, writing research report, writing a project report); speaking (introducing oneself and others, interviewing, discussions, stating and supporting propositions, stating one's opinions, organizing and taking part in a debate, making a persuasive speech, questioning); vocabulary (working out meanings from context, synonyms, antonyms, collocations, definitions); grammar (relative clauses, modals, voice, conditionals, tense, reported speech).

### **Course Objectives:**

Upon completing the course, students will be able to:

- ✓ Express their ideas in various communicative contexts (in group/ pair discussion, in public speaking settings)
- ✓ Present oral reports
- ✓ Write short reports
- ✓ Read various materials and make their own notes
- ✓ Identify the structure of oral and written discourses
- ✓ Attend their academic work at ease and with clarity

**Course mode of delivery:** Parallel

### **Course learning and teaching methods**

Classroom contact/Lecture, group work, interactive tutorial sessions (group and pair work/discussions and individual work (independent learning)

### **Assessment techniques:**

Students will be assessed out of 100% in this course. Of which 60% will be allotted for the Continuous Assessment (CA) that will be done throughout the semester. The remaining 40 % will be for the final examination. The CA includes varied types of activities that will allow the students to express themselves like real speaker or communicator. Thus, Students will be assessed continuously at least once in each of the six components. A final exam is administered to assess students'. Break down of the assessment can be seen bellow:

#### **Continuous Assessment**

|                                              |           |
|----------------------------------------------|-----------|
| ➤ Debates                                    | 10%       |
| ➤ Speech Delivery (2) (Impromptu & Prepared) | (5×2) 10% |
| ➤ Group Assignment                           | 10%       |
| ➤ Report (Oral & Written)                    | 10%       |
| ➤ Summary & Review                           | 10%       |
| ➤ Listening                                  | 10%       |
| ➤ Final Examination                          | 40%       |

### **References:**

- Dean, M.1988. Write it; Writing Skills for intermediate learners of English. Cambridge University Press
- DEFL, 1996.College English: volume I and II.AAU.AAU Printing Press
- Gregory.1999.Public speaking for college and career (Fifth Ed).New York: McGraw Hill College
- Hewings, M. 1999.Advanced Grammar in use: self-study Reference Practice Book for Advanced Learners of English. Cambridge: CUP.
- MOE, 2005.Improve Your English: A Course for Ethiopian Teachers (Grade 1-4)-Face to Face Learner's Books 1&2.Addis Ababa: EMPDE
- Strong, W.1991. Writer's Choice: Grammar and Composition. Illinois: McGraw Hall

### Course Schedule:

| Weeks | Study Hours                                                  | Main Topic/Sub topic/s/ Chapter                                                                                                                                                                       | Reading material /assignments                                                                    | Student Activities                                                                        |
|-------|--------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| 1st   | Lecture Hours= 2hrs<br>Home Study= 2hrs<br>Discussion = 4hrs | 1.Introductory Unit<br>1.1. Listening and Speaking: Finding out about other people<br>1.2. Vocabulary: Learning to learn vocabulary<br>1.3. Grammar: Learning to use grammar for facilitating meaning | <i>College English VL.I PP 4-10</i><br><br><i>English Communicative Grammar pp 34-48</i>         | - Introduce themselves to their partners<br><br>-find out information about others        |
| 2nd   | LH= 2hrs<br>DH= 2hrs<br>PH= 2hrs<br>HS=2hrs                  | 1.4. Reading: What is involved in understanding text?<br>1.5. Speaking: Introducing oneself and others<br>1.6. Writing: A short Personal description or story                                         | - College English VL.I<br><br>- Communicative English Skills II-unpublished<br>- Writer's Choice | -Participate in group discussions<br>introduce themselves<br>write a personal description |



|                         |                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                       |
|-------------------------|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3rd<br>&4 <sup>th</sup> | LH= 4hrs<br>HS=6hrs<br>DH= 6hrs           | <p>2.AIDS</p> <p>2.1. Listening and Speaking:</p> <p>2.1.1. Understanding markers of addition and relating</p> <p>2.1.2. Listening for gist</p> <p>2.1.3. Responding to the speaker's purpose</p> <p>2.1.4. Writing a brief summary of a talk</p> <p>2.2. Vocabulary</p> <p>2.2.1. Using component parts of a word as clues to meaning</p> <p>2.2.2. Using topic relationships in order to learn words</p> <p>2.2.3. Being aware of how words collocate with each other</p> <p>2.2.4. Working out word meanings from context</p> <p>2.3. Grammar</p> <p>2.3.1. Using relative clauses</p> <p>2.3.2. Expressing warning and advice</p> | <p>College English-Teacher's Guide</p><br><p>College Reading + McCarthy</p><br><p>Advanced Grammar in Use + Grammar for English Language Teachers 350-79</p><br><p>College English VL.I</p> <p>College English VL.I</p> | <p>Listen to texts and identify markers of addition and relating,</p><br><p>identify the gist of the talk,</p> <p>write summary of the talk</p> <p>-guess the meaning of words depending on clues, topic relationship and collocation</p><br><p>-</p> |
| 5th<br>&6 <sup>th</sup> | LH=4hrs<br>DH=4hrs<br>PH=4hrs<br>HS= 4hrs | <p>2.4 Reading</p> <p>2.4.1. Identifying the intended audience of a text and other critical reading skills</p> <p>2.4.2. Relating a diagram to a text</p><br><p>2.5 Speaking</p> <p>2.4.3. Brain storming</p> <p>2.4.4. Public speaking</p>                                                                                                                                                                                                                                                                                                                                                                                           | <p>College English VL.I +</p><br><p>Public Speaking for College and Career</p>                                                                                                                                          | <p>-read passages and work on comprehension questions</p><br><p>-practice and present public speeches</p>                                                                                                                                             |

|                                       |                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                |                                                                                                                                                                                      |
|---------------------------------------|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7 <sup>th</sup> & 8 <sup>th</sup>     | SH=6hrs<br>LH=4hrs<br>DH=6hrs   | 3.Culture and Values<br>3.1. Listening and Speaking<br>3.1.1. Identifying the structure of a talk<br>3.1.2. Completing a note framework<br><br>3.2. Vocabulary<br>3.2.1. Using topic relationships to learn new words<br>3.2.2. Words of Greek and Latin origin<br>3.2.3. Using a vocabulary network to learn words<br><br>3.3. Grammar<br>3.3.1. Using active and passive constructions for descriptive writing<br>3.3.2. using time clauses for descriptive writing | College English-Teacher's Guide<br><br>College English VL.I<br><br>Grammar for English Language Teachers p.287 | Listen to texts and identify structure of the talk<br><br>Guess meaning of words based on their origin and topic relationship<br><br>Practice using active and passive constructions |
| 9 <sup>th</sup> and 10 <sup>th</sup>  | SH= 4<br>DH= 5<br>LH= 4<br>PH=3 | 3.4. Reading<br>3.4.1. Critical reading<br>3.4.2. Reading for main ideas<br>3.4.3. Reading for detail<br>3.5. Speaking<br>3.5.1. Understanding reference<br>3.5.2. Brainstorming<br>3.5.3. Organizing and taking part in a debate                                                                                                                                                                                                                                     | College English VL.I<br><br>Public Speaking for College and Career                                             | -read passage and identify main idea and specific details<br><br>-participate in debating organized in the classroom<br>-write summary and                                           |
| 11 <sup>th</sup> and 12 <sup>th</sup> | LH=5hrs<br>HS=5hrs<br>DH= 6hrs  | 4.Improving Study Practices<br>4.1. Listening and speaking<br>4.1.1. Thinking about what you do when you listen to a lecture and take notes<br>4.1.2. Understanding listening and                                                                                                                                                                                                                                                                                     | College English-Teacher's Guide                                                                                | listen to lectures and take notes identify main sections of a lecture                                                                                                                |

|                                        |                                |                                                                                                                                                                                                                                                                                                                                                        |                                                                          |                                                                                                         |
|----------------------------------------|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| 13 <sup>th</sup> &<br>14 <sup>th</sup> | LH=5hrs<br>HS=5hrs<br>DH= 6hrs | 4.4. Reading<br>4.4.1. Skimming for gist<br>4.4.2. Critical reading and evaluating<br>4.4.3. Using reference/textual markers<br>4.5. Speaking<br>4.5.1. Brainstorming and discussing on what makes a good learner<br>4.6 Writing<br>4.6.1. Summarizing a talk<br>4.6.2. Summarizing an academic article<br>4.6.3. Writing an essay on learning English | College English VL.I<br><br>College English VL. I<br><br>Writers' Choice | read passage and identify references and textual markers<br><br><br>practice writing summary and essays |
|----------------------------------------|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|

## **Communicative English Language Skills II**

**Course title:** Communicative English Skills II

**Course code:** EnLa1012

**Module name:** English Language skill

**Module code:** Enla-M1014

**Course ECTS:** 5 ECTS (135 hrs)

|            |         |
|------------|---------|
| Lecture    | 48 hrs  |
| Tutorial   | 7 hrs   |
| Group Work | 10 hrs  |
| Assessment | 20 hrs  |
| Home Study | 50 hrs  |
| Total      | 135 hrs |

**Year/Semester Course is offered:** Year I, Semester II

**Contact hours/ week:**

**Pre-requisite:** None

**Course description:**

Communicative English Language Skills II Module is a continuation of Communicative English I Module, and it mainly aims to provide first year University students proficiency with reading, speaking and writing skills. It also aims to help students learn vocabularies that are assumed unfamiliar to them. In the grammar part, with the intention of providing explanations, brief notes are given in each unit.

The module consists of five units with three supplementary reading at the end of the Module. The supplementary readings are included to support ideas included in the reading passages in units 1-3.

Students are advised to read the references put in the box for further learn the grammar points included in the Module.

**Course Objectives:**

**Course mode of delivery:** Parallel

**Course learning and teaching methods:**

**Assessment techniques:**

**Teachers and Students Role**

**References:**

Azar, B. S. (2003). Fundamentals of English grammar. Longman.

Eggenchwiler, J., & Biggs, E.D. (2001). Writing: Grammar, Usage, and Style. New York. Hungry Minds. Inc

Lucy, J. A., & Lucy, L. A. (Eds.). (1993). Reflexive Language: Reported Speech and Meta pragmatics. Cambridge University Press.

Murphy, R. (2012). English Grammar in Use. Ernst Klett Sprachen.  
Naylor, H., & Murphy, R. (2007). Essential Grammar in Use. Supplementary Exercises. With Answers. Ernst Klett Sprachen

### **Course Contents**

#### **Unit I : Life Skills**

**Part I** Reading passage: The concept of life skills

**Part II** Grammar: Active and passive voices

**Part III** Speaking

**Part IV** Writing

#### **Unit II: Speculations about the future of Science**

**Part I** Reading passage: Grassroots attack in bilharzia

**Part II** Grammar: Future Tense

**Part III** Speaking

**Part IV** Writing

#### **Unit III: Environmental protection**

**Part I** Reading: Environmental Challenges: A river run through it

**Part II** Grammar: Modal verbs

**Part III** Speaking

**Part IV** Writing

#### **Unit IV: Indigenous Knowledge**

**Part I** Reading: A local Pathway to Global Development

**Part II** Grammar: Reported Speech

**Part III** Speaking

**Part IV** Writing

#### **Unit V: Cultural Heritage**

**Part I** Reading: Cultural Heritage What is it? Why is it important

**Part II** Grammar: Relative Clauses

**Part III** Speaking

**Part IV** Writing

### **Supplementary Readings**

A. Environmental Problems

B. The Origin of Humans: The Record From the Afar of Ethiopia

C. Tourism Can be Used to Preserve Ethiopia's Cultural and Historic Wealth

## Module 1: Social Sciences and humanities

**Module name:** Social sciences and humanities module

**Module category:** General

**Module code:** Sshm-M1024

**Module weight in ECTS:** 20 ECTS

**Courses:**

| Course Name                        | Course Code | ECTS |
|------------------------------------|-------------|------|
| General Psychology                 | Psyc 1021   | 5    |
| Critical Thinking                  | LoCT 1011   | 5    |
| Geography of Ethiopia and the Horn | GeES1011    | 5    |
| Social Anthropology                | Anth 1012   | 5    |

**Module description:** the module will try familiarizing students with social, psychological and ethical issues of the society and human being. The module covers key concepts of psychology & civic and ethics.

**Module objective:** to equip students with the psychological ethical approaches to live up ethically with the complex human social life.

**Module competency:**

- Develop skills to enhance students' ability to analyze critically the dynamics of society and current social issues.
- Develop critical thinking and problem solving skills
- Enhance students' Social research skills
- Improve both your communication and group interaction skills
- Gain knowledge about the theoretical discourses and practices of state/government, society and citizenship and their mutual interplay especially in the context of Ethiopia;
- 

**Mode of delivery:** Parallel

**Mode of Assessment:**

### Assessment Criteria

#### A. Assessment Criteria

The assessment criteria are based on continuous assessment of class activities, individual and group assignment, field visit and report writing, test and final exams. This in turn can be broken down in to;

- Group assignments ..... 30%
- Tests/quizzes ..... 30%
- Final Exam. .... 40%
- Total. .... 100%

### Learning activities and teaching methods

#### A. Learning Activities

Students are expected to undertake among others the following major learning activities:

- Engage in learning by doing (independent study, group assignments, presentation, report writing, and etc ..)
- Participation and note takings during class lectures and debates and discussions;
- Analysis, summarization and presentations of chapter/article, motions on selected issues;

#### B. Teaching Methods

- The course facilitator is expected to lecture for introducing concepts and topics, and give
- references, facilitate discussions, ask questions, give and correct assessments,
- Arrange and facilitate seminar sessions by inviting professionals for public lectures or debates on subject related issues.

## **Social sciences and humanities module Course syllabi**

**Course Title:** General Psychology

**Course Code:** Psyc 1021

**Module name:** Social sciences and humanities module

**Module code:** Sshm-M1024

**Course ECTS:** 5 ECTS

**Year/Semester Course is offered:** Year I Semester I

**Contact hours/ week:** 135

|                              |         |
|------------------------------|---------|
| Lecture                      | 48 hrs  |
| Tutorial and Problem Solving | 10 hrs  |
| Group Discussion             | 20 hrs  |
| Assessment                   | 8 hrs   |
| Presentation                 | 14 hrs  |
| Home Study                   | 35 hrs  |
| Total                        | 135 hrs |

**Prerequisite course:** None

### **Course Description:**

- This introductory course will provide students with an overview of the current body of knowledge and methods of the science of psychology. It is a general overview course focusing on the scientific study of both the behavioral and mental processes of human beings and animals. More specifically, topics will be covering: historical foundations of psychology, scientific thoughts in psychology, research methodology, biological basis of behavior, human development, sensation and perception, learning, memory and forgetting, motivation and emotion, personality, psychological disorders and psychotherapy.

### **Course Objectives:**

An overriding course goal is to introduce students about the basic concepts of psychology and to provide access about the ways psychologists apply psychological knowledge, principles, and theories to understand their lives and the lives of others. Toward this goal, upon completion of this course, students will be able to:

- Clearly describe psychological concepts
- Compare and contrast the major perspectives in Psychology
- Explain the various research methods in Psychology
- Recognize the link between human biology and behavior
- Discuss different aspects of human development



- Comprehend how people sense and give meaning to their environment
- Explain the process of learning a new behavior from different theoretical basis
- Elucidate about memory and forgetting processes
- Describe motivational and emotional processes
- Discuss personality theories
- Describe the characteristics of major psychological disorders
- Appreciate the practical value of psychology

## **Course learning and teaching methods**

### **Assessment Method:**

Dominantly, there will be formative continuous assessment (quizzes, individual and group work, discussion, class activity, assignments) just at the end of each week. Moreover, summative assessments such as mid semester and final examination will be carried out.

#### **Assessment Arrangement**

|                                                 |     |
|-------------------------------------------------|-----|
| Quizzes/Tests                                   | 20% |
| Group/ individual Assignments with presentation | 30% |
| Class participation and attendance              | 10% |
| Final Examination                               | 40% |

### **Teachers and Students Role**

#### **References:**

- Feldman, R.S. (1999). Essentials of Understanding psychology: McGraw Hill college
- Lahey, B. (2004) Psychology: An Introduction (8<sup>th</sup>ed.) Boston: McGraw Hill Book Company.
- Morgan C. (2003). Introduction to Psychology. 6<sup>th</sup> ed McGraw Hill Book Company
- McMahan, J, McMahan, F, and Ramano, T. (1995). Psychology & you (2<sup>nd</sup>ed.) New York: McGraw Hill Book Company.
- Miles H., Frank D. and Jonathan F. (2005). Psychology. Alden Press, Oxford, UK.
- Note: Students are also recommended to read other possible sources like the research articles, newsletters, magazines, etc

**Course Schedule:** By Time, Contents and Reading Materials

| Days | Contact Hrs | Topic/Subtopics/ Chapters                                                                                                                                                                                 | Reference                                                                                                                                                                                                          | Remark |
|------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 1    | 3:12 hrs    | Unit 1: Introduction to Psychology<br>1.1. Definition of psychology<br>1.2. The Goals of Psychology<br>1.3. The subject Matter of psychology<br>1.1. Historical development of psychology                 | <ul style="list-style-type: none"> <li>Robert S. Feldman (1996). Essentials of Understanding psychology. Pp.4-238</li> <li>Morgan C. (2003). Introduction to Psychology. 6<sup>th</sup> ed. Pp1-24</li> </ul>      |        |
| 2    | 3:12 hrs    | 1.2. Perspectives in psychology<br>1.3. Major Areas in psychology<br>1.4. Research Methods in Psychology                                                                                                  |                                                                                                                                                                                                                    |        |
| 3    | 3:12 hrs    | Unit 2: Biological Basis of Behaviors<br>2.1. Heredity (gene) Vs Behavior<br>2.2. Nerve system Vs Behavior<br>2.3. Endocrine system Vs Behavior                                                           | <ul style="list-style-type: none"> <li>Robert S. Feldman (1996). Essentials of Understanding psychology. Pp.39-79</li> <li>Morgan C. (2003). Introduction to Psychology. 6<sup>th</sup> ed. Pp 25-79</li> </ul>    |        |
| 4    | 3:12 hrs    | Unit 3: Human development<br>3.1. The nature of human development<br>2.2. Issues or controversies in development                                                                                          | <ul style="list-style-type: none"> <li>Robert S. Feldman (1996). Essentials of Understanding psychology. Pp.331-376</li> <li>Morgan C. (2003). Introduction to Psychology. 6<sup>th</sup> ed. Pp461-126</li> </ul> |        |
| 5    | 3:12 hrs    | 3.3. Theories of human development <ul style="list-style-type: none"> <li>cognitive development</li> <li>psychosexual development</li> <li>psychosocial development</li> <li>moral development</li> </ul> |                                                                                                                                                                                                                    |        |
| 6    | 3:12 hrs    | Unit 4: Sensations and Perception<br>4.1 Definition: Sensation and Perception<br>4.2 Sensing the environment                                                                                              | <ul style="list-style-type: none"> <li>Robert S. Feldman (1996). Essentials of Understanding psychology. Pp.81-126</li> <li>Morgan C. (2003). Introduction to Psychology. 6<sup>th</sup>ed. Pp 80-136</li> </ul>   |        |

|    |          |                                                                                                                                                                                                                                |                                                                                                                                                                                                                            |  |
|----|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|    |          | 4.3 Perceptual processes<br>4.3.1 Attention<br>4.3.2 Organization<br>4.3.3 Interpretation                                                                                                                                      |                                                                                                                                                                                                                            |  |
| 7  | 3:12 hrs | Unit 5: Learning<br>5.1 Definition and characteristics of learning<br>5.2 Theories of learning<br>5.2.1. Pavlov's classical conditioning<br>5.2.2. Operant conditioning                                                        | <ul style="list-style-type: none"> <li>• Robert S. Feldman (1996). Essentials of Understanding psychology. Pp.163-201</li> <li>• Morgan C. (2003).). Introduction to Psychology. 6<sup>th</sup> ed , Pp 137-180</li> </ul> |  |
| 8  | 3:12 hrs | 5.2.3. Social learning theory<br>5.2.4. Cognitive view of learning                                                                                                                                                             |                                                                                                                                                                                                                            |  |
| 9  | 3:12 hrs | Unit 6: Memory and forgetting<br>6.1 Processes of memory<br>6.2 Sensory memory<br>6.3 Short term memory<br>6.4 Long term memory<br>6.5 Theories of forgetting                                                                  | <ul style="list-style-type: none"> <li>• Robert S. Feldman (1996). Essentials of Understanding psychology. Pp.203-234</li> <li>• Morgan C. (2003).). Introduction to Psychology. 6<sup>th</sup> ed. Pp 181-224</li> </ul>  |  |
| 10 | 3:12 hrs | Unit 7: Motivation and Emotion<br>7.1. The nature of motivation<br>7.2. Theories of motivation<br>7.3. Conflict motives & frustration<br>7.4. Definition of emotions<br>7.5. Components of emotion<br>7.6. Theories of emotion | <ul style="list-style-type: none"> <li>• Robert S. Feldman (1996). Essentials of Understanding psychology. Pp.293-330</li> <li>• Morgan C. (1999). Introduction to Psychology. 6<sup>th</sup>ed. Pp 265-306</li> </ul>     |  |
| 11 | 3:12 hrs | Unit 8: Stress and Coping mechanisms<br>8.1. The nature of stress<br>8.2. Sources of stress (stressors)<br>8.3. Coping mechanisms of stress                                                                                    | <ul style="list-style-type: none"> <li>• Robert S. Feldman (1996). Essentials of Understanding psychology. Pp.411-451</li> <li>• Morgan C. (2003). Introduction to Psychology. 6<sup>th</sup> ed. Pp307-338</li> </ul>     |  |

|    |          |                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                         |  |
|----|----------|----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| 12 | 3:12 hrs | Unit 9: Personality<br>9.1. The nature of personality<br>9.2. Theories of personality                                            | <ul style="list-style-type: none"> <li>• Robert S. Feldman (1996). Essentials of Understanding psychology. Pp379-409</li> <li>• Morgan C. (2003).). Introduction to Psychology. 6<sup>th</sup> ed. Pp 563-611</li> <li>• Robert S. Feldman (1996). Essentials of Understanding psychology. Pp.411-451</li> <li>• Morgan C. (2003). Introduction to Psychology. 6<sup>th</sup> ed. Pp-612-724</li> </ul> |  |
| 13 | 3:12 hrs | 9.3. The measurement of personality<br>Unit 10: Abnormal Behaviors and psychotherapy<br>10.1. Criteria/approaches of abnormality |                                                                                                                                                                                                                                                                                                                                                                                                         |  |
| 14 | 3:12 hrs | 10.2. Classifications of abnormal behaviors<br>10.3. Treatment of psychological disorders                                        |                                                                                                                                                                                                                                                                                                                                                                                                         |  |
| 15 | 3:12 hrs | Exam Preparation week                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                         |  |

**Social sciences and humanities module Course syllabi**

**Course Title:** Critical Thinking

**Course Code:** LoCT 1011

**Module name:** Social sciences and humanities module

**Module code:** Sshm-M1024

**Course ECTS:** 5 ECTS

**Year/Semester Course is offered:** Year I Semester I

**Contact hours/ week:** 135

|                              |         |
|------------------------------|---------|
| Lecture                      | 48 hrs  |
| Tutorial and Problem Solving | 10 hrs  |
| Group Discussion             | 20 hrs  |
| Assessment                   | 8 hrs   |
| Presentation                 | 14 hrs  |
| Home Study                   | 35 hrs  |
| Total                        | 135 hrs |

**Details of the course syllabus to be communicated by MoSHE**

**Social sciences and humanities module Course syllabi**

**Course Title:** Geography of Ethiopia & the Horn

**Course Code:** GeES1011

**Module name:** Social sciences and humanities module

**Module code:** Sshm-M1024

**Course ECTS:** 5 ECTS

**Year/Semester Course is offered:** Year I Semester I

**Contact hours/ week:** 135

|                              |         |
|------------------------------|---------|
| Lecture                      | 48 hrs  |
| Tutorial and Problem Solving | 10 hrs  |
| Group Discussion             | 20 hrs  |
| Assessment                   | 8 hrs   |
| Presentation                 | 14 hrs  |
| Home Study                   | 35 hrs  |
| Total                        | 135 hrs |

**Details of the course syllabus to be communicated by MoSHE**

## **Social sciences and humanities module Course syllabi**

**Course Title:** Social Anthropology

**Course Code:** Anth 1012

**Module name:** Social sciences and humanities module

**Module code:** Sshm-M1024

**Course ECTS:** 5 ECTS

**Year/Semester Course is offered:** Year I Semester II

**Contact hours/ week:** 135

|                              |         |
|------------------------------|---------|
| Lecture                      | 48 hrs  |
| Tutorial and Problem Solving | 10 hrs  |
| Group Discussion             | 20 hrs  |
| Assessment                   | 8 hrs   |
| Presentation                 | 14 hrs  |
| Home Study                   | 35 hrs  |
| Total                        | 135 hrs |

**Prerequisite course:** None

### **Course Description:**

Dear Learner! Welcome to this course, Anth101. The course is expected to acquaint you with essential concept of anthropology covering a wide array of questions revolving around our very existence. It cover issues such as what makes human beings similar to each other? How do we differ one another? What do anthropologist mean when they talk about diversity, multiculturalism, marginalization, inclusion and exclusion?

The course will enable learners grasp the different ways of being human by dealing with themes such as culture, kinship, marriage, cultural relativism, ethnocentrism, humanity, human origins, cosmologies, race, ethnicity, ethnic relations, ethnic boundaries, marginalization, minorities, local systems of governance, legal pluralism, indigenous knowledge systems, and indigenous practices and development.

### **Course Objectives:**

p on the successful completion of the course, students will be able to:

- ✓ Develop an understanding of the nature of anthropology and its broader scope in making sense of humanity in a global perspective;
- ✓ Understand the cultural and biological diversity of humanity and unity in diversity across the world and in Ethiopia;
  - ✓ Analyze the problems of ethnocentrism against the backdrop of cultural relativism;
- ✓ Realize the socially constructed nature of identities & social categories such as gender, ethnicity, race and sexuality;

- ✓ Explore the various peoples and cultures of Ethiopia;
- ✓ Understand the social, cultural, political, religious & economic life of different ethno-linguistic & cultural groups of Ethiopia;
  - ✓ Understand different forms marginalization and develop skills inclusiveness;
  - ✓ Appreciate the customary systems of governance and conflict resolution institutions of the various peoples of Ethiopia;
  - ✓ Know about values, norms and cultural practices that maintain society together;
  - ✓ Recognize the culture area of peoples of Ethiopia and the forms of interaction developed over time among themselves; and

Develop broader views and skills to deal with people from a wide variety of socio-economic and cultural backgrounds

### **Course learning and teaching methods**

The teacher or course facilitator who is assigned to deliver is recommended to make use of different active learning methods including: brainstorming, question and answer, group discussion, buzz-group, cross-over, home-works, reading assignments, peer teaching, and seldom active lecturing.

### **Assessment Method:**

To assess the progress of student, the instructor/ the course facilitator is expected to employ a continuous assessment technique in the form of quizzes, group and individual assignments, take-home exam, final exam, term paper. The purpose of using various assessment techniques is to improve the process of students' learning



**Course Title:** Moral and Civic Education

**Course Code:** CvEd 1022

**Module name:** Social sciences and humanities module

**Module code:** Sshm-M1024

**Course ECTS:** 3 (81 hrs)

**Year/Semester Course is offered:** Year I Semester II

**Contact hours/ week:** 2

**Prerequisite course:** None

**Course Description:**

As the Ethiopian Education Development Roadmap (2018-30) stated that, since one of the challenges for quality education is missing the proper moral and civic education, the education policy objectives should be revisited and formulated to reflect the creation of holistic development in all citizens, confident and competent citizens, critical thinkers, competent professionals who satisfy the requirements of the global market; entrepreneurs and innovative, strong ethical and moral values, stand for justice; peace, and unity in diversity.

The benchmarking moral, ethical and citizenship education are part of the curriculum of the educational system to address diversity and national unity. The education system should promote these realities and be able to produce adequate and capable graduates to satisfy both the domestic and global markets.

Given this, the Ethiopian government has designed and implemented moral and civic education curricula to aim at educating students about democratic culture, ethical values and principles, supremacy of constitution, and the rule of law and so on. These elements are imperative in the process of producing self-confident citizens and a generation who has the capability to shoulder responsibility. Accordingly, this module is basically aspires to equip the learners with relevant knowledge, respect for the worth and human dignity of every individual, right attitudes and requisite skills to enable them perform their roles as a credible members of their society. Through the module, learners will also acquire nature of Ethiopian federalism and parliamentary system of government, ways of making responsible decisions, solve problems, care about others, contribute to society, and be tolerant and respectful of diversity.

This module is organized into five chapters. The first chapter deals with the definition of concepts and terms, differences between civics and ethics, goals of civics and ethics as well as competences of a good citizen. The second chapter presents the major rival theories and perspectives on ethics

and morality. The third chapter dwells with ethical decision making and the justification behind the moral judgments, while chapter four contains about the concepts of citizenship, state and government particularly the state structures and theories of state, systems of government, theories of citizenship, ways of acquiring and losing citizenship and the interplay between citizens, state and government and final fifth chapter deals with constitution, human rights and democracy.

### **Course Objectives:**

After the successful completion of this module students will be able to:

- Conceptualize what morality, ethics and civics mean.
- Comprehend the goals of civics and ethics as well as the competences of a good citizen.
- Discuss the relations between society, state and government.
- Differentiate federal state structure from unitary and discuss the advantages and disadvantages of the state structures.
- Discuss the processes of modern Ethiopian state formation and nation building.
- Comprehend the features of Ethiopian federalism.
- Conceptualize constitution, its classification and unique features.
- Define the term human rights, the unique features and its classifications.
- Differentiate the teleological, deontological and virtue theories.

### **Course learning and teaching methods:**

The mode of the delivery of the course includes lecture, tutorials, home study, group discussions, intensive readings, role play and class debates, independent assignments. Based on these methodologies of teaching, the course should have highly participatory that helps students to develop habits of critical thinking, inquisitive, critical, analytic, integrative and morally balanced student, and exhibits higher ethical standards like open-mindedness, rational thinking, evidence-oriented personality and problem solving skills.

### **Assessment Method:**

Evaluation will be made based on continuous assessment (50%) and final exam (50%) results. The continuous assessment comprises group and individual assignments, presentation, class attendance and participation, and quizzes. In all kinds of assessments students are expected to clearly demonstrate their horizon of thinking, rational reasoning proper use of language by ensuring clear, effective and meaningful communication.

### **Continuous assessment**

Group assignment with presentation ..... 20%

|            |              |     |
|------------|--------------|-----|
| Quiz I     | Day 2 .....  | 10% |
| Quiz II    | Day 5 .....  | 10% |
| Quiz III   | Day 10 ..... | 10% |
| Quiz IV    | Day 12 ..... | 10% |
| Final exam | Day 18.....  | 40% |

### **Competence to be measured:**

(Knowledge, skill and attitude)

- Describing key concepts like civics, ethics, democracy, profession and so on; institutions, policies, strategies and legal provisions of the country;
- Explaining their rights and duties as well as governments' rights over them and duties towards them.
- Demonstrating their understanding of government institutions, policies, strategies and legal provisions of their country

### **Teachers and Students Role**

#### **References:**

1. Assefa Fiseha (2005) *Federalism and the Accommodation of Diversity in Ethiopia: A Comparative Study*, Netherlands, Wolf Legal Publishers.
2. Bayles, Michael (1989). *Professional Ethics*. 2d ed. Belmont, Calif : Wadsworth.
3. Bahru Zewde, (1991), *A History of Modern Ethiopia: 1855-1974*. Addis Ababa: AUU Press.
4. Clapham, C., *Haile-Sellase's Government*, (London: Longman, 1969).
5. Fasil Nahum (1997), *Constitution for a Nation of Nations: The Ethiopian Prospect*. Asmara: The Red Sea press.
6. Johari, J.C (1987) *Contemporary Political Theory: New Dimensions, Basic Concepts and major Trends*. New Delhi: Sterling publishers Put. Ltd.
7. Kassahun Berhanu (1998) 'Democracy, State-Building and Nations in Ethiopia: 1974-1995.' In Gros, Jean- Germain (ed.) *Democratization in Late Twentieth- Century Africa coping with Uncertainty*.
8. Merera Gudina, (2003) *Ethiopia: Competing Ethnic Nationalities and the Quest for Democracy, 1960-2000*. Chamber printing house: Addis Ababa

9. Tesfaye Molla (2010) *Civics and Ethics Distance Learning Material*, Hawassa University, Department of Governance and Development Studies.
10. Tsegaye Regassa, (2001). *Ethnic Federalism and The Right to Self-Determination As A Constitutional Legal Solution to the Problem of Multi-Ethnic Societies: The Case of Ethiopia* (LLM Thesis, Ethiopian Civil Service College, Law Library, Unpublished) Policy/legal Documents
11. The Federal Democratic Republic of Ethiopia Constitution of 1995 Proclamation No. 1/1995, 21<sup>st</sup> August, 1995, adopted on 8<sup>th</sup> of December

**Course Schedule:**

| Days     | Contact Hrs          | Topic to be discussed                                                                                                                                                                                                                                                                                                                                                          | Reading assignment                                                                                   | Guided study questions                                                                                                                                                                                                                                                           |
|----------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Day<br>1 | 4 hrs<br>(morning)   | First class meeting General introduction about the course, and setting ground rules<br><br>Chapter One: Civics and Ethics for Professionals<br>Conceptualizing Citizenship and Morality;<br>Civics and Ethics: meaning; Definition; Why Civics and Ethics?; Historical Development of Civics and Ethics in Ethiopia, Sources of civics and ethics, goals of civics and ethics. | Bayles (1989), pp1-6<br><i>Civics and Ethics Teaching Material</i> , Module , Chapter One, pp 1-9    | After introducing students the objectives of learning civics and ethics they attempt to address the questions: What do we mean by <i>Civics and Ethics</i> ? What about by <i>morality</i> ??<br><br>Why you take this course?                                                   |
|          | 2 hrs<br>(afternoon) | Profession and Professional Ethics in Ethiopia; What is profession; and who are professionals?<br><br>Distinguishing Features of profession                                                                                                                                                                                                                                    | Bayles (1989), pp 6-18<br><i>Civics and Ethics Teaching Material</i> , Module , Chapter One, pp 9-29 | Students reflect on the meaning of profession, and elements that are to be considered in defining profession<br><br>Identifying the distinguishing features of profession; and enumerating the attributes of ethical and moral principles of profession will be made by students |
| Day<br>2 | ½ hrs<br>(morning)   | Quiz-I                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                      |                                                                                                                                                                                                                                                                                  |

|          |                      |                                                                                                                                                       |                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                     |
|----------|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | 2 hrs<br>(morning)   | CHAPTER TWO<br>Perspectives on Society,<br>state and government,<br>Definition and essential<br>elements of state, theories<br>on the origin of state | <i>Teaching Material</i> , Module<br>Johari, J.C (1987), pp 1-20<br>The 1933 Montevideo<br>Convention on the Rights<br>and Duties of States<br><i>Civics and Ethics Teaching<br/>Material</i> , Module , Chapter<br>Two, pp 30-34 | Students reflect on the meaning of, Society, state and government and the triangular<br>relationship among them. Students describe the elements of modern state and reflect views<br>on theories on the origin of state.                            |
|          | 4 hrs<br>(afternoon) | Structures of State, Forms<br>of Government                                                                                                           | <i>Teaching Material</i> , Module<br>, Chapter<br>Two, pp 34-44                                                                                                                                                                   | Identify the two structures of state. Compare and contrast the structures of state in Ethiopian<br>context.<br><br>Reflect view on the different forms of government, emphasis on parliamentary and<br>presidential systems as well as sovereignty. |
| Day<br>3 | 3 hrs                | Tutorial-I                                                                                                                                            |                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                     |
| Day<br>4 | 4hrs<br>(morning)    | Society, State and<br>Government in Ethiopia.                                                                                                         | <i>Civics and Ethics Teaching<br/>Material</i> , Module , Chapter<br>Two, pp 44-73<br>Johari, J.C (1987), pp 1-20<br>The 1933 Montevideo<br>Convention on the Rights<br>and Duties of States                                      | Reflect view on the form and structure of the successive Governments of Ethiopia<br>Discuss the state-society relations of successive governments of Ethiopia<br>Students reflect their views on rationales and dynamics of federalism in Ethiopia. |
|          |                      | Chapter Three:<br>Citizenship, Patriotism and<br>civic Participation.                                                                                 | <i>Civics and Ethics Teaching<br/>Material</i> , Module , Chapter<br>Three, pp 74-89                                                                                                                                              | Students understand the concepts of citizen and citizenship;<br>Discuss on the historical survey of citizenship;<br>Explain the aspects of citizenship;<br>Know ways of acquiring and losing citizenship;                                           |

|           |                            |                                                                                                                                                                |                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                         |
|-----------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|           |                            | Definition of Citizenship, Historical survey of Citizenship, aspects of Citizenship, qualifications for citizenship: ways of acquiring and loosing citizenship |                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                         |
|           | 4hrs<br>(afternoon)        | The rights and Duties of Citizens, Citizenship and Patriotism in the Ethiopian Context, Civic Participation.                                                   | <i>Civics and Ethics Teaching Material</i> , Module , Chapter Three, pp 89-112<br>FDRE constitution (Articles 13 -44),<br>Protections of Nationality under the 2003 Nationality Proclamation Articles 14-17)<br>Criminal Code of Ethiopia (Arts 561-600 | To assess students’ ability to know their rights and duties as well as governments’ rights over them and duties towards; identify their responsibilities towards their community.<br><br>Explain the concept of patriotism and its linkage with citizenship<br><br>Describe the forms and forums of civic participation |
| Day 5     | ½ hr<br>(afternoon)        | Quiz-II                                                                                                                                                        |                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                         |
| Day 6 & 7 | 1 <sup>st</sup> Week Break |                                                                                                                                                                |                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                         |
| Day 8     | 3 hrs                      | Tutorial-II                                                                                                                                                    |                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                         |

|           |                      |                                                                                                                                               |                                                                                                                                                            |                                                                                                                                                                                                                                                     |
|-----------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Day<br>9  | 4hrs<br>(morning)    | CHAPTER FOUR:<br>Democracy and Good Governance in Ethiopia<br>Definition and the Historical Survey of Democracy, Forms and Types of democracy | <i>Civics and Ethics Teaching Material</i> , Module , Chapter Four, pp 113-116                                                                             | Students reflect their views on the concept and meaning of democracy; Describe the forms and types of democracy                                                                                                                                     |
|           |                      | Fundamental principles and values of democracy, Actors in the democratization process                                                         | <i>Civics and Ethics Teaching Material</i> , Module Chapter Four, pp 116-133<br>FDRE constitutions , basic principles of the constitution, (Articles 8-12) | List down the basic fundamental values and principles of democracy; Evaluate the role of different actors in the democratization process                                                                                                            |
|           | 4hrs<br>(afternoon)  | Democracy and good governance in the context of Africa and Ethiopia.                                                                          | <i>Civics and Ethics Teaching Material</i> , Module , Chapter Four, pp 133-137<br>Johari, J.C (1987 ) pp95-122                                             | After introducing the foundations of democracy in general, students reflect their views on foundations of democracy in Africa and Ethiopia; attributes of good governance, the conditions required for a political system to qualify as a democracy |
| Day<br>10 | ½ hrs<br>(afternoon) | Quiz-III                                                                                                                                      |                                                                                                                                                            |                                                                                                                                                                                                                                                     |
| Day<br>11 | 3hrs<br>(morning)    | Tutorial-III                                                                                                                                  |                                                                                                                                                            |                                                                                                                                                                                                                                                     |



|        |                            |                                                                                                                                                                                                          |                                                                                                                                              |                                                                                                                                                                                                                                                                                               |
|--------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        | 4hrs<br>(afternoon)        | Chapter five: Constitution and constitutionalism<br><br>Definition of constitution, definition of constitutionalism, purposes and classification of constitution, Contents and validity of constitution, | <i>Civics and Ethics Teaching Material</i> , Module Chapter five, pp 138-146                                                                 | Students identify the basic features of constitution and constitutionalism; list the major purposes and function of constitution; appreciate why countries needs to have constitution ;distinguish modern classification of constitutions; describe the contents and validity of constitution |
|        |                            | The Constitutional Experience of Ethiopia, the pre-1931 traditional constitutional experience, The 1931 Constitution, The 1955 Revised Constitution.                                                     | <i>Civics and Ethics Teaching Material</i> , Module , Chapter Five, pp 146-153<br>The 1931 constitution, the Revised 1955 Constitution       | The students evaluate the major achievements of traditional constitution of Ethiopia; explain the motives and progressive political elements of the 1931 and the 1955 Ethiopian constitution;<br><br>compare the 1931 and the 1955 revised constitution.                                      |
| Day 12 | ½ hr<br>(afternoon)        | Quiz IV                                                                                                                                                                                                  |                                                                                                                                              |                                                                                                                                                                                                                                                                                               |
| 13-14  | 2 <sup>nd</sup> Week Break |                                                                                                                                                                                                          |                                                                                                                                              |                                                                                                                                                                                                                                                                                               |
| Day 15 | 3hrs<br>(morning)          | Tutorial-IV                                                                                                                                                                                              |                                                                                                                                              |                                                                                                                                                                                                                                                                                               |
|        | 4hrs<br>(afternoon)        | The 1987 PDRE Constitution, the Transitional Charter of 1991, The 1995 FDRE Constitution.                                                                                                                | <i>Civics and Ethics Teaching Material</i> , Module , Chapter Five, pp 153-163<br>The 1987 PDRE constitution and The 1995 FDRE constitution. | The students reflect their views on the silent features of the 1987 PDRE constitution; 1991 Transitional Charter and 1995 FDRE constitution; Compare and contrast the ideological basis of Ethiopian constitutions under the three successive regimes.<br><br>;                               |

|           |                      |                                                                                                                                                                          |                                                                                                                                                                                                                              |                                                                                                                                                                               |
|-----------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Day<br>16 | 4hrs<br>(morning)    | CHAPTER SIX :<br>Globalization<br>Introduction, Globalization<br>and the changing world,<br>dimensions and values of<br>globalization, major actors<br>of globalization. | <i>Civics and Ethics Teaching<br/>Material</i> , Module , Chapter<br>Six, pp 164-180<br>Douglas, S., and Y. Wind<br>(1987)<br><i>The Myth of Globalization</i> .<br>NY: www.ingentaconnect.<br>com/content/mcb/036/2001<br>. | Identify the major value cracks which are believed to supply the ever worsening value<br>crises over work as a spiritual and material source of ethical problems in Ethiopia. |
|           | 4 hrs<br>(afternoon) | Challenges of globalization<br>in developing countries,<br>Ethiopia; a state in a<br>globalized world, Global<br>citizenship                                             | <i>Civics and Ethics Teaching<br/>Material</i> , Module, Chapter<br>Six, pp 180-187<br>Swann, D. (1999) <i>The<br/>Economics of the Common<br/>Market</i> , 6th ed., London:<br>Penguin Books                                | Reflect views on the meaning by globalization and its drivers                                                                                                                 |
| Day<br>17 |                      | One day break for final exam preparation                                                                                                                                 |                                                                                                                                                                                                                              |                                                                                                                                                                               |

**Course Title:** Mathematics for Natural Sciences

**Course Code:** Math 1011

**Module name:**

**Module code:**

**Course ECTS:** 5

**Year/Semester Course is offered:** Year I Semester I

**Contact hours/ week:** 135

## **Content Page**

### **Chapter 1:** Propositional Logic and set Theory

#### 1.1. Propositional Logic

##### 1.1.1 Definition and examples of Propositions

##### 1.1.2 Logical connectives

##### 1.1.3 Compound (or complex) proposition

##### 1.1.4 Tautology and contradiction

#### 1.2. Open propositions and quantifiers

#### 1.3. Arguments and Validity

#### 1.4. Set Theory

##### 1.4.1 The Concept of a set

##### 1.4.2 Description of sets

##### 1.4.3 Set operations and Venn diagrams

### **Chapter 2:** The Real and Complex Number Systems

#### 2.1 The real number system

##### 2.1.1 The natural numbers, principle of mathematical induction and the well ordering axiom

##### 2.1.2 The set of integers

##### 2.1.3 The set of rational numbers

##### 2.1.4 The set of real numbers, upper bound and lower bound, least Upper bound and greatest lower bound; completeness property of real numbers

#### 2.2 The set of complex numbers

##### 2.2.1 Plotting complex numbers

##### 2.2.2 Operations on complex numbers

##### 2.2.3 Conjugate of a complex number

##### 2.2.4 Modulus (Norm) of a complex number

##### 2.2.5 Additive and multiplicative inverse

##### 2.2.6 Argument of a complex number

##### 2.2.7 Polar form of a complex numbers

##### 2.2.8 Extraction of roots

### **Chapter 3:** Functions

#### 3.1 Review of relations and functions

#### 3.2 Real valued functions and their properties

- 3.3 Types of functions and inverse of a function
- 3.4 Polynomials, zeros of polynomials, rational functions and their graphs
- 3.5 Definition and basic properties of logarithmic, exponential, trigonometric and hyperbolic functions, and their graphs

#### **Chapter 4: Analytic Geometry**

- 4.1 Distance Formula and Equation of Lines
  - 4.1.1 Distance between two points and division of segments
  - 4.1.2 Equations of lines
  - 4.1.3 Distance between a point and a line
- 4.2 Circles
  - 4.2.1 Definition of a circle
  - 4.2.2 Equation of a circle
  - 4.2.3 Intersection of a circle with a line and tangent line to a circle
- 4.3 Parabolas
  - 4.3.1 Definition of parabola
  - 4.3.2 Equation of parabolas
- 4.4 Ellipse
  - 4.4.1 Definition of ellipse
  - 4.4.2 Equation of ellipse
- 4.5 Hyperbola
  - 4.5.1 Definition of a hyperbola
  - 4.5.2 Equation of a hyperbola
- 4.6 The general second degree equation
  - 4.6.1 Rotation of coordinate axes
  - 4.6.2 Analysis of the general second degree equations

#### **References**

1. Alemayehu Haile and Yismaw Alemu, *Mathematics an Introductory Course*, Department of Mathematics, Addis Ababa University
2. Demisu Gemedu and Seid Mohammed, *Fundamental Concepts of Algebra*, AAU
3. Semu Mitiku Kassa, Berhanu Guta Wordofa and Tilahun Abebaw Kebede, *Engineering Mathematics I*, Galaxy University Books Series, , 2017.
4. Edwin J. Purcell, Dale Varberg, *Calculus with Analytic Geometry*
5. G. Chartrand, A. D. Polimeni and P. Zihang, *Mathematical proofs: a transition to advanced mathematics* 3rd edition, Pearson Education. Inc.
6. Goodman Hirsch, *Precalculus-Understanding functions*, 2000
7. James Ward Brown and Ruel V. Churchill: *Complex Numbers and Applications*, 7th edition
8. Michael D. Alder: *An Introduction to Complex Analysis to Engineers*, 1997

**Course Title:** General Physics

**Course Code:** Phyc-1011

**Module name:**

**Module code:**

**Course ECTS:** 5 ECTS

**Year/Semester Course is offered:** Year I Semester I

**Contact hours/ week:** 135

**Details of the course syllabus to be communicated by MoSHE**

**Course Title:** Physical Fitness

**Course Code:** SpSc-1011

**Module name:**

**Module code:**

**Course ECTS:** NC

**Year/Semester Course is offered:** Year I Semester I

**Contact hours/ week:** 2

**Details of the course syllabus to be communicated by MoSHE**

**Course Title:** Inclusiveness

**Course Code:** Incl-1011

**Module name:**

**Module code:**

**Course ECTS:** 3

**Year/Semester Course is offered:** Year II Semester II

**Contact hours/ week:**

**Details of the course syllabus to be communicated by MoSHE**

**Course Title:** Entrepreneurship

**Course Code:** MGMT-1012

**Module name:**

**Module code:**

**Course ECTS:** 5

**Year/Semester Course is offered:** Year III Semester II

**Contact hours/ week:**

**Details of the course syllabus to be communicated by MoSHE**



**Course Title:** Global Trends

**Course Code:** GTr-1012

**Module name:**

**Module code:**

**Course ECTS:** 3

**Year/Semester Course is offered:** Year IV Semester II

**Contact hours/ week:**

**Details of the course syllabus to be communicated by MoSHE**

**Course Title:** Introduction to Emerging Technologies

**Course Code:** EmTe-1012

**Module name:**

**Module code:**

**Course ECTS:** 5

**Year/Semester Course is offered:** Year I Semester II

**Contact hours/ week:**

**Details of the course syllabus to be communicated by MoSHE**

**Module 3: Biomedical Sciences-I****Module name:** Biomedical science I**Module category:** Basic**Module code:** Biom-M1033**Module number:** 03**Module weight in ECTS:** 12**Courses:**

| Course Name                 | Course Code | ECTS |
|-----------------------------|-------------|------|
| Human anatomy and histology | Anat 1031   | 7    |
| Human physiology I          | Physl 1032  | 5    |

**Module description**

Biomedical science I module emphasizes on structural organization of the human body at the gross (macroscopic) and histological (microscopic) level and also it addresses basic understanding of the function and regulation of the systems and organs of the human body. It will be delivered over a period of 2 semesters.

**Module objective:** The module is designed to provide students with a basic understanding of the structure, function and regulation of the human body

**Module competency:** Apply the normal anatomic and physiologic conditions of the human body to understand drugs effect

**Module Mode of delivery:** Parallel

**Module learning teaching methods**

Active learning methods (brain storming, buzz group, discussion, etc), Lecture, group & individual presentation, assignment, project work, and laboratory work.

**Module Assessment techniques:**

Quizzes

Mid exam

Final Exam

Seminar

Laboratory

Assignment

## **BIOMEDICAL MODULE I COURSES SYLLABI**

**Course title:** Human Anatomy and Histology

**Course code:** Anat 1031

**Module name:** Biomedical Module I

**Module code:** Biom-M1033

**Course EtCTS:** 7 EtCTS (189 hrs)

- |                       |        |
|-----------------------|--------|
| ▪ Lecture             | 64 hrs |
| ▪ Tutorial            | 12 hrs |
| ▪ Demonstration       | 36 hrs |
| ▪ Independent Study   | 63 hrs |
| ▪ Seminar, Assignment | 8 hrs  |
| ▪ Assessment          | 6 hrs  |

**Year/Semester Course is offered:** Year I Semester II

**Contact hours/ week:** 3

**Pre-requisite:** None

**Course description:** This course covers the facts and concepts of human anatomy, intended for application to Pharmacy practice. The major goal is to enable students to increase knowledge and build upon their professional skills through understanding the relationships of the human body structure & their clinical relevance, in general

### **Course Objectives:**

To provide the basic anatomical & histological knowledge of the human body structures, their relationship and functions

1. Assess the basic knowledge of human body development.
2. Analyze the basic structure, location and functions of body tissues
3. Describe the knowledge of systemic Anatomy and their relationships
4. Apply the knowledge they acquired during the course to the profession

**Course mode of delivery:** Block/Parallel

### **Course learning and teaching methods**

- Active learning methods (brain storming, buzz group, discussion, etc), Lecture, group and individual presentation, assignment, project work, and laboratory work.

### **Assessment techniques:**

- Continuous assessments ..... 50%
  1. First test
  2. Second test
  3. Assignment, Group work and Oral Presentations
  4. 3d assessment

5. Final Examination-----50 %

**Teachers and Students Role**

**References:**

1. Van de Graaf Kent 4<sup>th</sup> ed. and above. Human Anatomy
2. Tortora, G.J. & Bryan D. 11<sup>th</sup> edition and above. Principles of Anatomy & Physiology
3. Anthony L Mescher. Junqueira's Basic Histology 11th ed. or above. McGraw-Hill Medical
4. T. W. Sadler. Langman's Medical Embryology, 10th ed. or above. Lippincott Williams & Wilkins
5. Netter 2008. Netter's Atlas Human Anatomy 5<sup>th</sup> ed. or latest Lippincott Williams & Wilkins

**Course Schedule:**

| Week | Contact Hrs | Topic/sub-topic/Chapter                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Reading Material |
|------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
|      | 2hrs        | Unit-1 Introduction to human Anatomy<br>1.1. History, Definition and Divisions of Anatomy <ul style="list-style-type: none"> <li>• Divisions of Anatomy               <ul style="list-style-type: none"> <li>a. Gross anatomy(Macroscopic anatomy)                   <ul style="list-style-type: none"> <li>○ Systemic anatomy</li> <li>○ Regional anatomy</li> </ul> </li> <li>b. Microscopic anatomy</li> </ul> </li> <li>• Other subdivisions of anatomy               <ul style="list-style-type: none"> <li>• Applied (Clinical) anatomy</li> <li>• Surgical anatomy</li> <li>• Surface anatomy</li> <li>• Radiological anatomy</li> </ul> </li> </ul> 1.2. Anatomical terminologies<br>1.3. Body Parts, Views, Planes and Body Movement<br>1.3.1 Anatomical positions<br>1.3.2 Anatomical planes and sections <ul style="list-style-type: none"> <li>• Anatomical planes               <ul style="list-style-type: none"> <li>♣ Frontal (coronal) plane</li> <li>♣ Sagittal planes</li> <li>♣ Median (Midsagittal) plane</li> <li>♣ Transverse Plane</li> </ul> </li> <li>• Anatomical sections</li> </ul> |                  |

|                 |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |
|-----------------|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| 1 <sup>st</sup> |     | ♠ Coronal, median, horizontal, longitudinal, oblique, and cross-sections.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |
|                 | 1hr | 1.3.3 Directional terms in Anatomy<br>*Terms of relationship (position)<br>*Terms of movement<br>1.4. Body regions and regional names. <ul style="list-style-type: none"> <li>• Body regions <ul style="list-style-type: none"> <li>• abdominopelvic regions</li> <li>• abdominopelvic quadrants</li> </ul> </li> <li>• Regional names <ul style="list-style-type: none"> <li>○ The head</li> <li>○ The Neck</li> <li>○ The trunk</li> <li>○ The upper limb</li> <li>○ The lower limb</li> </ul> </li> </ul> 1.5. Body cavities and membranes <ul style="list-style-type: none"> <li>• Body cavities <ul style="list-style-type: none"> <li>♣ Dorsal /posterior <ul style="list-style-type: none"> <li>• cranial cavities</li> <li>• Vertebral cavity</li> </ul> </li> <li>♣ Ventral/anterior body cavities <ul style="list-style-type: none"> <li>• Thoracic cavity</li> <li>• The Abdominopelvic cavity</li> </ul> </li> </ul> </li> </ul> |  |
| 1 <sup>st</sup> | 1hr | 1.6 Levels of structural organization <ul style="list-style-type: none"> <li>• Chemical level of organization</li> <li>• The cellular levels of structural organization</li> <li>• The tissue levels of organization</li> <li>• The organ levels of organization</li> <li>• The system levels</li> <li>• The organismal level</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |

|                 |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                               |
|-----------------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2 <sup>nd</sup> | 1hr  | Unit – 2: Cellular Organization of the Body<br>2.1. Introduction<br>2.2. The cytoplasm, cytoplasmic organelles and cytoplasmic inclusions<br>2.3. The plasma membrane<br>2.4. Cell nucleus<br>2.5. Cell cycle<br>2.6 Cell extensions and connection<br>✓ Cilia, Flagella and cytoskeleton<br>• Applied Anatomy                                                                                                                                                                                                                   | 1. Van de Graaf Kent 4 <sup>th</sup> ed. and above. Human Anatomy<br>Anthony L Mescher.<br>Junqueira’s Basic Histology 11th ed or above. McGraw-Hill Medical. |
|                 | 2hrs | UNIT -3: HUMAN BODY TISSUES<br>Introduction<br>3.1. Tissue types<br>• The primary tissue classes<br>• Embryonic tissues<br>3.2. Epithelial tissues<br>• Covering epithelium and epithelial membranes<br>• Glandular epithelium<br>3.3. Connective tissues<br>• Classification<br>• Embryonic connective tissue<br>• Connective tissue proper<br>• Connective tissue supportive: bone and cartilage<br>• Cartilage<br>• Hyaline Cartilage<br>• Elastic Cartilage<br>• Fibrocartilage<br>• Bone<br>• Compact bone<br>• Spongy bone |                                                                                                                                                               |
|                 | 3hr  | • Connective tissue with special properties:<br>Blood<br>♣ Red Blood Cells (erythrocytes)<br>♣ White Blood Cells (leukocytes)<br>♣ Platelets<br>3.4. Muscle tissue<br>• Skeletal muscle<br>• Cardiac muscle                                                                                                                                                                                                                                                                                                                      | Anthony L Mescher.<br>Junqueira’s Basic Histology 11th ed or above. McGraw-Hill Medical.                                                                      |

|                 |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                              |
|-----------------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
|                 |      | <ul style="list-style-type: none"> <li>• Smooth muscle</li> </ul> 3.5. Nerve tissues <ul style="list-style-type: none"> <li>♣ Neurons</li> <li>♣ Neuroglia</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                              |
|                 | 1hr  | Applied anatomy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                              |
| 4 <sup>th</sup> | 2hrs | Unit – 4. GENERAL EMBRYOLOGY<br>4.1. Introduction<br>4.2. Gametogenesis <ul style="list-style-type: none"> <li>♣ Formation of male gamete</li> <li>♣ Formation female gamete</li> </ul> 4.3. Fertilization<br>4.4.1 <sup>st</sup> week of embryonic development <ul style="list-style-type: none"> <li>• Fertilization</li> <li>• Cleavage of the zygote</li> <li>• Morula</li> <li>• Blastocyst formation</li> <li>• Implantation</li> </ul> 4.5. 2 <sup>nd</sup> week of embryonic development <ul style="list-style-type: none"> <li>♣ Formation of syncytiotrophoblast</li> <li>♣ Formation of cytotrophoblast</li> </ul> |                                                                                              |
| 4 <sup>th</sup> | 1hr  | 4.6.3 <sup>rd</sup> week of embryonic development <ul style="list-style-type: none"> <li>○ Differentiation of germ layers <ul style="list-style-type: none"> <li>• Ectoderm</li> <li>• Mesoderm</li> <li>• Endoderm</li> </ul> </li> </ul> 4.7. Placenta <ul style="list-style-type: none"> <li>• function of placenta</li> <li>• structures of placenta</li> </ul> 4.8 Twins <ul style="list-style-type: none"> <li>• Fraternal twins</li> <li>• Identical twins</li> </ul>                                                                                                                                                  | T. W. Sadler. Langman's Medical Embryology, 10th ed or above. Lippincott Williams & Wilkins. |
| 5 <sup>th</sup> | 1hr  | UNIT-5: INTEGUMENTARY SYSTEM<br>5.1. Structure of the skin ,functions of skin<br>5.2. Epidermis <ul style="list-style-type: none"> <li>♣ Four types of cells <ul style="list-style-type: none"> <li>• Keratinocytes</li> <li>• Melanocytes</li> <li>• Merkel cells</li> <li>• Langerhans cells</li> </ul> </li> </ul> Layers (from deep to superficial) <ul style="list-style-type: none"> <li>• Stratum basale or germinativum Stratum spinosum**</li> <li>• Stratum granulosum**</li> <li>• Stratum lucidum **</li> <li>• Stratum corneum **</li> </ul> (**Thick skin only)                                                 |                                                                                              |



|                 |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                           |
|-----------------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5 <sup>th</sup> |     | <ul style="list-style-type: none"> <li>• Stratum basale or germinativum*</li> <li>• Stratum spinosum*</li> <li>• Stratum granulosum*</li> <li>• Stratum corneum*</li> </ul> (*Thin skin only)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                           |
|                 | 1hr | 5.3. Dermis<br><br>Two layers: <ul style="list-style-type: none"> <li>• Papillary</li> <li>• Reticular</li> </ul> Fiber types: <ul style="list-style-type: none"> <li>♣ collagen</li> <li>♣ elastic</li> <li>♣ Reticular</li> </ul> <ul style="list-style-type: none"> <li>• Hypodermis(superficial fascia)</li> </ul> 5.4. Appendages of the skin           5.4.1. Subcutaneous glands <ul style="list-style-type: none"> <li>• Sweat glands</li> </ul> Types of sweat glands: <ul style="list-style-type: none"> <li>♣ Eccrine or merocrine</li> <li>♣ Apocrine</li> </ul> <ul style="list-style-type: none"> <li>• Sebaceous glands</li> <li>• Ceraminous glands*</li> <li>• Mammary glands*</li> </ul> *Modified apocrine glands |                                                                                                                                                           |
|                 | 1hr | 5.4.2. Hair <ul style="list-style-type: none"> <li>• Parts:               <ul style="list-style-type: none"> <li>♣ Root imbedded in skin</li> <li>♣ Shaft projecting above skin surface</li> </ul> </li> <li>• Three concentric layers:               <ul style="list-style-type: none"> <li>• Medulla (core)</li> <li>• Cortex (surrounds medulla)</li> <li>• Cuticl(single layers, overlapping)</li> </ul> </li> <li>• Types of hair:               <ul style="list-style-type: none"> <li>• Vellus: fine, short hairs</li> <li>• Intermediate hairs</li> <li>• Terminal: longer, courser hair</li> </ul> </li> </ul>                                                                                                              |                                                                                                                                                           |
|                 | 1hr | 5.4.3. Nails<br><br>5.5. Skin color<br><br>Three skin pigments: <ul style="list-style-type: none"> <li>• Melanin</li> <li>• Carotene</li> <li>• Hemoglobin</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <ul style="list-style-type: none"> <li>• Applied Anatomy Anthony L Mescher. Junqueira's Basic Histology 11th ed or above. McGraw-Hill Medical.</li> </ul> |
|                 |     | UNIT-6: THE SKELETAL SYSTEM                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                           |

|                 |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                         |
|-----------------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| 6 <sup>th</sup> | 2hrs | 6.1. Surface making and their functions<br>6.2. Type of bones and their histology<br>The Structure of a Typical Bone: <ul style="list-style-type: none"> <li>• Compact bone<br/>             The Histological Features of compact bone:             <ul style="list-style-type: none"> <li>♣ Osteon (Haversian System)</li> <li>♣ Central (Haversian) canal</li> <li>♣ Perforating (Volkmann's) canal</li> </ul> </li> <li>• Spongy bone<br/>             The Histological Features of Spongy Bone:             <ul style="list-style-type: none"> <li>• Lamellae</li> <li>• <i>Trabeculae</i></li> </ul> </li> </ul> 6.3 Types of Bone Cells <ul style="list-style-type: none"> <li>• Osteoblasts</li> <li>• Osteocytes</li> <li>• Osteoclasts</li> </ul> Classification of Bones: <ul style="list-style-type: none"> <li>• Long bones</li> <li>• Short bones</li> <li>• Flat bones</li> <li>• Irregular bones</li> <li>• Sesamoid bones</li> <li>• Accessory bones</li> </ul> 6.4. Division, Location, and Functional anatomy of the bone of the human body<br>6.4.1. Axial skeleton <ul style="list-style-type: none"> <li>• The Skull             <ul style="list-style-type: none"> <li>♣ 8 cranial bones</li> <li>♣ 14 facial bones</li> </ul> </li> <li>• The hyoid bone(1)</li> <li>• The Auditory ossicles(3pairs)             <ul style="list-style-type: none"> <li>○ Malleus</li> <li>○ incus</li> <li>○ stapes</li> </ul> </li> <li>• Vertebral column- vertebrae             <ul style="list-style-type: none"> <li>• 7cervical vertebrae</li> <li>• 12 thoracic</li> <li>• 5 lumbar</li> <li>• 1 sacrum (5 fused )</li> <li>• 1 coccyx (4 fused</li> </ul> </li> <li>• Thoracic cages             <ul style="list-style-type: none"> <li>▪ Sternum</li> <li>▪ Ribs</li> <li>▪ Thoracic vertebrae</li> <li>▪ costal cartilages</li> </ul> </li> </ul> |                                                                         |
| 6 <sup>th</sup> | 1hr  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                         |
| 6 <sup>th</sup> |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                         |
| 7 <sup>th</sup> | 1hr  | 6.4.2. Appendicular skeleton <ul style="list-style-type: none"> <li>• Bones of Pectoral girdle</li> <li>• Clavicle (collarbone)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <ul style="list-style-type: none"> <li>• The Applied Anatomy</li> </ul> |

|                 |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                |
|-----------------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7 <sup>th</sup> | 2hrs | <ul style="list-style-type: none"> <li>• Scapula (shoulder blade)</li> <li>• The bones of upper limbs <ul style="list-style-type: none"> <li>○ humerus</li> <li>○ ulna</li> <li>○ radius</li> <li>○ Carpal</li> <li>○ metacarpal</li> <li>○ phalanges</li> </ul> </li> <li>• Bones of the pelvic girdle <p>Formed by:</p> <ul style="list-style-type: none"> <li>♠ hipbones(Ilium, Ischium and Pubis)</li> <li>♠ sacrum of the</li> <li>♠ coccyx</li> </ul> </li> <li>• Difference b/n male &amp; female pelvis</li> <li>• Types of pelvis <ul style="list-style-type: none"> <li>-True pelvis(lesser)</li> <li>-False pelvis(greater)</li> </ul> </li> <li>• The bones of lower limbs <ul style="list-style-type: none"> <li>♣ curural (Leg) bones-tibia &amp; fibula</li> <li>♣ The foot (Pes)</li> <li>♣ Tarsus – ankle</li> <li>♣ Metatarsals – sole</li> <li>♣ Phalanges – toes</li> </ul> </li> <li>• Bone Fractures <ul style="list-style-type: none"> <li>• Blood and nerve supply to bones</li> </ul> </li> </ul> <p>6.5. Joints /Articulations</p> <p>1. Structurally are of three types (i.e., based on presence or absence of joint cavity).</p> <ul style="list-style-type: none"> <li>• Fibrous- Immoveable</li> <li>• Cartilaginous- slightly movable</li> <li>• Synovial- freely movable <ul style="list-style-type: none"> <li>♣ Hinge</li> <li>♣ Ball &amp; socket</li> <li>♣ Gliding</li> <li>♣ Saddle</li> </ul> </li> </ul> <p>2. Functionally are three types of joints (i.e. based on the degree of movement)</p> <ul style="list-style-type: none"> <li>♣ Synarthroses – immovable joints</li> <li>♣ Amphiarthroses – slightly movable</li> <li>♣ Diarthroses – freely movable</li> </ul> | <p>Anatomy 5<sup>th</sup> edn or latest<br/>Lippincott Williams &amp; Wilkins</p> <p>2. Tortora, G.J. &amp; Bryan D. 11<sup>th</sup> edition and above.<br/>Principles of Anatomy &amp; Physiology</p> <p>3. Van de Graaf Kent 4<sup>th</sup> ed. and above.<br/>Human Anatomy</p> <p>Netter 2008. Netter's Atlas of Human</p> |
|                 |      | <p>UNIT-7: THE MUSCULAR SYSTEM</p> <p>7.1. The Skeletal muscle tissue</p> <p>7.2. The Connective tissue components</p> <ul style="list-style-type: none"> <li>• endomysium</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                |

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| 7 <sup>th</sup>  | 1hr  | <ul style="list-style-type: none"> <li>• perimysium</li> <li>• epimysium</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 8 <sup>th</sup>  | 1hr  | <p>7.3. Criteria for naming the skeletal system</p> <ul style="list-style-type: none"> <li>○ Named on the basis of: <ul style="list-style-type: none"> <li>♣ shape</li> <li>♣ location,</li> <li>♣ attachment</li> <li>♣ orientation of fibers</li> <li>♣ relative position, or function</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                  | 2hrs | <p>7.4. The principal skeletal muscles in the body</p> <ul style="list-style-type: none"> <li>• Name</li> <li>• Origin and Insertion</li> <li>• Nerve Supply</li> <li>• Blood supply</li> <li>• Function ( action) <ul style="list-style-type: none"> <li>♠ Maintainance of body posture</li> <li>♠ Movement</li> </ul> </li> </ul>                                                                                                                                                                                                                                                         | <ul style="list-style-type: none"> <li>• Applied Anatomy</li> </ul> <ol style="list-style-type: none"> <li>1. Tortora, G.J. &amp; Bryan D. 11<sup>th</sup> edition and above. Principles of Anatomy &amp; Physiology</li> <li>2. Van de Graaf Kent 4<sup>th</sup> ed. and above. Human Anatomy Netter 2008. Netter's Atlas oHuman Anatomy 5<sup>th</sup> edn or latest Lippincott Williams &amp;Wilkins.</li> </ol> |
|                  | 1hr  | <p>UNIT – 8: THE NERVOUS SYSTEM</p> <p>8.1. Protection &amp; coverings</p> <p>8.1.1. The meninges</p> <ul style="list-style-type: none"> <li>• Dura mater</li> <li>• Arachnoid membrane</li> <li>• Pia mater</li> <li>• Sub arachnoid space</li> </ul>                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                  | 2hrs | <p>8.2.2. The cerebrospinal fluid</p> <p>8.2.1. The brain ventricles</p> <ul style="list-style-type: none"> <li>• The lateral ventricles</li> <li>• The 3<sup>rd</sup> ventricle</li> <li>• The 4<sup>th</sup> ventricle</li> </ul>                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 10 <sup>th</sup> | 3hrs | <p>8.2.2. The formation, circulation and absorption of CSF</p> <p>8.2. Division, organization and functional anatomy of nervous system</p> <p>8.2.1. The Central Nervous System (CNS)</p> <ul style="list-style-type: none"> <li>• The Brain <ul style="list-style-type: none"> <li>- Principal parts</li> <li>- External structure</li> <li>- Internal structure</li> <li>- Function</li> <li>- Cranial nerves</li> </ul> </li> <li>• The spinal cord <ul style="list-style-type: none"> <li>- External structure</li> <li>- Internal structure</li> <li>- Function</li> </ul> </li> </ul> | <ul style="list-style-type: none"> <li>• Applied Anatomy</li> </ul> <ol style="list-style-type: none"> <li>1. Tortora, G.J. &amp; Bryan D. 11<sup>th</sup> edition and above. Principles of Anatomy &amp; Physiology</li> <li>2. Van de Graaf Kent 4<sup>th</sup> ed. and above. Human Anatomy Netter 2008. Netter's Atlas oHuman Anatomy 5<sup>th</sup> edn or latest Lippincott Williams &amp;Wilkins.</li> </ol> |

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|                  |      | <ul style="list-style-type: none"> <li>- Spinal nerves</li> </ul> <p>8.2.2. The Peripheral Nervous System (PNS)</p> <ul style="list-style-type: none"> <li>• Functional classification of nerves</li> <li>• Somatic nervous system</li> <li>• Autonomic nervous system <ul style="list-style-type: none"> <li>- Sympathetic division</li> <li>- Parasympathetic division</li> </ul> </li> </ul> <p>8.3. Sensory Organs</p> <ul style="list-style-type: none"> <li>• Special senses</li> </ul> <p>8.4. General senses</p>                                            |                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                  | 1hr  | <p>UNIT-9: THE ENDOCRINE SYSTEM</p> <p>9.1. Types and locations of the Endocrine glands in the body</p> <ul style="list-style-type: none"> <li>• The pituitary gland</li> <li>• The thyroid gland</li> </ul>                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 11 <sup>th</sup> | 2hrs | <ul style="list-style-type: none"> <li>• The Parathyroid gland</li> <li>• The Adrenal glands</li> <li>• The Gonads</li> <li>• Pancreas</li> <li>• Thymus and Pineal glands</li> <li>♠ Blood and nerve supply to Endocrine glands</li> </ul>                                                                                                                                                                                                                                                                                                                         | <ul style="list-style-type: none"> <li>• Applied Anatomy</li> </ul> <ol style="list-style-type: none"> <li>1. Tortora, G.J. &amp; Bryan D. 11<sup>th</sup> edition and above. Principles of Anatomy &amp; Physiology</li> <li>2. Van de Graaf Kent 4<sup>th</sup> ed. and above. Human Anatomy Netter 2008. Netter's Atlas of Human Anatomy 5<sup>th</sup> edn or latest Lippincott Williams &amp; Wilkins.</li> </ol> |
|                  | 2hrs | <p>UNIT-10: THE CARDIOVASCULAR SYSTEM</p> <p>10.1. General consideration</p> <p>10.2. The Heart</p> <ul style="list-style-type: none"> <li>• Location and coverings</li> <li>• The structure of its wall</li> <li>• Heart chambers <ul style="list-style-type: none"> <li>- Right atrium</li> <li>- Left atrium</li> <li>- Right ventricle</li> <li>- Left ventricle</li> </ul> </li> <li>• Valves of the heart</li> <li>• Conducting system of the heart</li> <li>• Great vessels connected to the heart</li> <li>• Blood and nerve supply to the heart</li> </ul> |                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 12 <sup>th</sup> | 2hrs | <p>10.3. The blood vessels</p> <ul style="list-style-type: none"> <li>• Types</li> <li>• General structure of blood vessels</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                        |

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|                  |      | <ul style="list-style-type: none"> <li>Distribution of blood vessels in the body <ul style="list-style-type: none"> <li>Arterial distribution</li> <li>Venous drainage</li> </ul> </li> </ul> <p>10.4. Circulatory routes</p> <ul style="list-style-type: none"> <li>Systemic circulation</li> <li>Pulmonary circulation</li> <li>Hepatic circulation</li> <li>Cerebral circulation</li> </ul> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                  | 2hrs | <p>10.5. The Lymphatic System</p> <ul style="list-style-type: none"> <li>The Lymph vessel</li> <li>The Lymph nodes</li> <li>The Lymph Circulation</li> <li>The Lymph organs</li> </ul>                                                                                                                                                                                                         | <ul style="list-style-type: none"> <li>Applied Anatomy <ol style="list-style-type: none"> <li>Tortora, G.J. &amp; Bryan D. 11<sup>th</sup> edition and above. Principles of Anatomy &amp; Physiology</li> <li>Van de Graaf Kent 4<sup>th</sup> ed. and above. Human Anatomy</li> </ol> </li> </ul> <p>Netter 2008. Netter's Atlas of Human Anatomy 5<sup>th</sup> edn or latest Lippincott Williams &amp; Wilkins.</p>                                             |
| 13 <sup>th</sup> | 3hrs | <p>UNIT-11: THE RESPIRATORY SYSTEM</p> <p>11.1. The pleurae</p> <p>11.2. The Lungs</p> <p>11.3. Function and structure</p> <p>11.4. The Respiratory pathways</p> <p>Nose, Larynx, Trachea, Bronchi and Alveoli</p>                                                                                                                                                                             | <ul style="list-style-type: none"> <li>Applied AnatomPharynx</li> </ul> <p><b>Recommended References</b></p> <ol style="list-style-type: none"> <li>Tortora, G.J. &amp; Bryan D. 11<sup>th</sup> edition and above. Principles of Anatomy &amp; Physiology</li> <li>Van de Graaf Kent 4<sup>th</sup> ed. and above. Human Anatomy</li> <li>Netter 2008. Netter's Atlas of Human Anatomy 5<sup>th</sup> edn or latest Lippincott Williams &amp; Wilkins.</li> </ol> |
|                  | 1hr  | <p>UNIT-12: THE DIGESTIVE SYSTEM</p> <p>12.1. The Peritoneum</p> <p>12.2. General organizations</p>                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 14 <sup>th</sup> | 2hrs | <p>12.3. Structure and functional anatomy of the digestive system</p> <ul style="list-style-type: none"> <li>The Oral cavity <ul style="list-style-type: none"> <li>Tongue</li> <li>Salivary glands</li> <li>Teeth</li> </ul> </li> </ul>                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                  | 2hrs | <ul style="list-style-type: none"> <li>The pharynx</li> <li>The esophagus</li> </ul>                                                                                                                                                                                                                                                                                                           | <ul style="list-style-type: none"> <li>Applied Anatomy</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                  |

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|                  |       | <ul style="list-style-type: none"> <li>• The stomach</li> <li>• The intestines</li> <li>• The accessory organs               <ul style="list-style-type: none"> <li>- Liver</li> <li>- Gall bladder</li> <li>- Pancreases</li> </ul> </li> </ul> | <b>Recommended References</b> <ol style="list-style-type: none"> <li>1. Tortora, G.J. &amp; Bryan D. 11<sup>th</sup> edition and above. Principles of Anatomy &amp; Physiology</li> <li>2. Van de Graaf Kent 4<sup>th</sup> ed. and above. Human Anatomy</li> </ol> Netter 2008. Netter's Atlas of Human Anatomy 5 <sup>th</sup> edn or latest Lippincott Williams & Wilkins.                                                                     |
| 15 <sup>th</sup> | 2hrs  | UNIT-13: THE URINARY SYSTEM<br>13.1. The kidneys <ul style="list-style-type: none"> <li>- External structure</li> <li>- Internal structure</li> </ul> 13.2. The Ureter<br>13.3. The Urinary bladder<br>13.4. The Urethra                         | <ul style="list-style-type: none"> <li>• Applied Anatomy</li> </ul> <b>Recommended References</b> <ol style="list-style-type: none"> <li>1. Tortora, G.J. &amp; Bryan D. 11<sup>th</sup> edition and above. Principles of Anatomy &amp; Physiology</li> <li>2. Van de Graaf Kent 4<sup>th</sup> ed. and above. Human Anatomy</li> </ol> Netter 2008. Netter's Atlas of Human Anatomy 5 <sup>th</sup> edn or latest Lippincott Williams & Wilkins. |
|                  | 2hrs  | UNIT-14: REPRODUCTIVE SYSTEM<br>14.1. The male reproductive system <ul style="list-style-type: none"> <li>14.1.1. The spermatic cord</li> <li>14.1.2. The testes</li> <li>14.1.3. The Epididymis</li> <li>14.1.4. The duct system</li> </ul>     |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 16 <sup>th</sup> | 2hrs  | 14.2. The female reproductive system <ul style="list-style-type: none"> <li>14.2.1. The ovaries</li> <li>14.2.2. The fallopian tubes</li> <li>14.2.3. The uterus</li> <li>14.2.4. Endocrine relation</li> </ul>                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                  | 2 hrs | 14.3.5. The vagina and vulva<br>14.3.6. The Breast (mammary glands)                                                                                                                                                                              | <ul style="list-style-type: none"> <li>• Applied Anatomy</li> </ul> <b>Recommended References</b> <ul style="list-style-type: none"> <li>• Tortora, G.J. &amp; Bryan D. 11<sup>th</sup> edition and above. Principles of Anatomy &amp; Physiology</li> </ul>                                                                                                                                                                                      |

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|  |  |  | <ul style="list-style-type: none"> <li>• Van de Graaf Kent<br/>4<sup>th</sup> ed. and above.<br/>Human Anatomy<br/>Netter 2008. Netter's Atlas<br/>oHuman Anatomy 5<sup>th</sup> edn or<br/>latest Lippincott Williams &amp;<br/>Wilkins.</li> </ul> |
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**Course Title: Human Physiology I**

**Course Code:** Phyl1032

**Module name: Biomedical Module I**

**Module code:** Biom-M1033

**Course ECTS:** 5

- Lecture: 48 hours
- Tutorial:16 hours
- Presentations: 15 hours
- Assignment:7.5 hours
- Assessment (continuous and final) 8 hours
- Independent study (alone or in groups) 40.5 hours

**Mode of delivery:** Parallel

**Pre-requisite if any:** None

**Course description:**

This module will give an overview of a range of physiological systems, including the homeostasis, the cell and cell membrane transport, composition of the body Fluid, physiology of blood, physiology of the nerve, physiology of the muscle, autonomic NS, cardiovascular physiology, respiratory physiology, renal physiology, gastro intestinal system, energy metabolism, endocrine system, male and female reproductive system, central nervous system and the special senses.

**Course objectives:**

At the end of the module the student should be able to:

- Explain the composition of and levels of organization of human body.
- Describe the basic physiological principle of the “internal environment” of the body.
- Explain basic principles of homeostasis and homeostatic regulatory mechanisms.
- Describe functional importance of different organ systems of human body and their integrated role in the maintenance of homeostasis.
- Appreciate the various physiological regulatory mechanisms of the body in the maintenance of homeostasis.
- Describe the various structures of the GIT, Secretory functions of GIT, Digestive and Absorptive functions and Pathophysiology of the GIT.
- Explain about Energy and Metabolism, Body Temperature Regulation, Feeding regulation and its abnormalities.

- Describe in detail the various endocrine glands of the body and various hormones secreted, the pituitary gland function and malfunction, the thyroid gland function and malfunction, the adrenal gland hormonal abnormalities on the body functions, Calcium homeostasis and Glucose homeostasis.
- Describe the various structures of the male and female reproductive system, Pregnancy and contraception, Parturition and lactation.
- Explain the nervous mechanisms, which govern the regulation and homeostasis of the principal physiological systems.
- Describe in detail the general organization of the NS, Sensory and Motor functions of the NS, Higher motor centers - Functions and lesions.
- Describe the various special senses of the human body

### Course learning and teaching methods

- Active learning methods (brain storming, buzz group, discussion, etc), Lecture, group and individual presentation, assignment, project work, and laboratory work.

### Assessment techniques:

- Assignments and Seminars (20%)
- Quizzes (30%)
- Final-exam (50%)

### Teachers and Students Role

### References:

1. Guyton A. C 1995-2006. Textbook of Medical physiology. 9<sup>th</sup> -11<sup>th</sup> editions
2. Ganong WF 1993-2006. Review of Medical physiology, 18<sup>th</sup> -22<sup>nd</sup> editions
3. Tortora G. J 1993. Principles of Anatomy and physiology. 7<sup>th</sup> edition.
4. Salah Abu-Sitta. Handouts containing different chapters (eight separate handouts.)

**Course Schedule:** By contact time, contents/topics & reading/reference materials for each topic

| Week            | Contact hrs | Topic/sub-topic/chapter                                                                                                                                                                                                                                                                                           | Reading Materials                                     |
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| 1 <sup>st</sup> | 3           | <u>General Introduction and Cell Physiology (7Hrs)</u> <ul style="list-style-type: none"> <li>- Definition</li> <li>- Regulatory mechanisms in Physiology</li> <li>- Composition of human body</li> <li>- Cell and its functions</li> <li>- Cell membrane, functional structure</li> <li>- Homeostasis</li> </ul> | Guyton (page 3-9)<br>Ganong (p 1-28)<br>Ganong (p 48) |

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|                 |   | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                |
| 2 <sup>nd</sup> | 3 | <ul style="list-style-type: none"> <li>- Cell organelles and their functions</li> <li>- Intracellular connections and Communications</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Guyton(p 9-17)<br>Ganong(p36-48)                                                               |
| 3 <sup>rd</sup> | 3 | <ul style="list-style-type: none"> <li>- Diffusion and osmosis</li> <li>- Endocytosis and exocytosis</li> <li>- Transports across cell membrane</li> <li>- Body fluid and electrolytes</li> </ul> <u>Physiology of Nerve (4 Hrs)</u> <ul style="list-style-type: none"> <li>- Functional structure of neurons</li> <li>- Classification of neurons and neuroglia cells</li> <li>- Membrane potential (resting membrane potential)</li> </ul>                                                                                                                                                                                                                                                                                     | Guyton(p 45-55)<br>Ganong(28-36)<br>Guyton(p 57-70)<br>Ganong(p 60-63)                         |
| 4 <sup>th</sup> | 3 | <ul style="list-style-type: none"> <li>- Electrical signals and excitable cells</li> <li>- Action potential (nerve impulse)</li> <li>- Better to add Graded potential</li> </ul> Propagation of nerve impulse <ul style="list-style-type: none"> <li>- Synapses</li> <li>- Synaptic transmission at neuronal synapses</li> </ul> <u>Physiology of muscles (6 Hrs)</u> <ul style="list-style-type: none"> <li>- Classification of muscles</li> <li>- Muscle structure [Functional structure of skeletal muscle, Molecular basis of contraction, Muscle excitation (skeletal muscle &amp; smooth muscle), Excitation contracting coupling]</li> <li>- Mechanism of muscle contraction</li> <li>- Neuromuscular junction</li> </ul> | Guyton(p 57-70)<br>Ganong(p 51-58)<br>Guyton(p 559-570)<br>Guyton(p 72-83)<br>Ganong(p 85-119) |
| 5 <sup>th</sup> | 3 | <ul style="list-style-type: none"> <li>- Excitation-Contraction coupling mechanism (skeletal &amp; smooth muscle, Molecular basis of contraction)</li> <li>- Tetanus &amp; clonus</li> <li>- Myasthenia gravis</li> <li>- Rigor mortis</li> <li>- General aspects of cardiac &amp; smooth muscles               <ul style="list-style-type: none"> <li>- Sub divisions of smooth muscle</li> <li>- Control of smooth muscle contraction</li> <li>- Membrane potential and action potential</li> </ul> </li> </ul>                                                                                                                                                                                                                | Guyton(p 85-99)<br>Ganong(78-84)                                                               |
| 6 <sup>th</sup> | 3 | <u>The autonomic nervous system (4 Hrs)</u> <ul style="list-style-type: none"> <li>- General organization of the NS</li> <li>- Basic difference between Somatic NS and ANS</li> <li>- Autonomic ganglia, Autonomic reflex</li> <li>- Divisions of ANA: Basic features of Sympathetic NS</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                               | Guyton(p 748-757)<br>Ganong(223-226)<br>Guyton(p 757)                                          |

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|                  |   | <ul style="list-style-type: none"> <li>- Effect of ANA in various organs of the body</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                     |
| 7 <sup>th</sup>  | 3 | <ul style="list-style-type: none"> <li>- Autonomic transmitters and receptors, their mechanism of action</li> <li>- Introduction to the Pharmacology of the ANS</li> </ul> <p><b><u>Physiology of Blood(6 Hrs)</u></b></p> <ul style="list-style-type: none"> <li>- General Introduction of blood               <ul style="list-style-type: none"> <li>o Blood volume &amp; constituents</li> <li>o Plasma constituents</li> <li>o Erythrocytes &amp; blood grouping, anemia &amp; polycythemia</li> <li>o Leukocytes &amp; immune responses</li> <li>o Platelets &amp; Homeostasis, coagulation disorders</li> </ul> </li> <li>- Plasma and plasma proteins</li> </ul>           | Guyton (p 759)<br>Ganong(p 223-226)<br>Guyton (p 419-428)<br>Ganong(p 515-516)                      |
| 8 <sup>th</sup>  | 3 | <ul style="list-style-type: none"> <li>- RBCs; anemia's and polycythemia</li> <li>- WBCs &amp; Immunology</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Guyton(p 429-449)<br>Ganong(p 516- 540)                                                             |
| 9 <sup>th</sup>  |   | <ul style="list-style-type: none"> <li>- Homeostasis, coagulation and disorders</li> <li>- Blood groups and blood transfusion</li> </ul> <p><b><u>Physiology of Cardiovascular System (9 Hrs)</u></b></p> <ul style="list-style-type: none"> <li>- Physiological anatomy of the heart</li> <li>- Basic properties of the cardiac muscle</li> <li>- Electro physiology of the cardiac muscle</li> <li>- Excitation contraction coupling of the myocardium</li> <li>- Electro cardio gram (ECG)</li> <li>- Cardiac cycle</li> <li>- The heart sounds</li> <li>- General Introduction; functional structure of the heart</li> <li>- Electrophysiology of the heart muscle</li> </ul> | Guyton(p 451-467)<br>Ganong(p 540-546)<br><br>Guyton (p 103-110)<br>Ganong (p 547-561)              |
| 10 <sup>th</sup> | 3 | <ul style="list-style-type: none"> <li>- The cardiac cycle</li> <li>- The E.C.G</li> <li>- The heart sounds</li> <li>- Ischemic heart disease and heart failure</li> <li>- The heart rate and its regulation</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Guyton(p 116-121)<br>Guyton(p 123-129)<br>Ganong(p 565-570)<br>Ganong(p 597-602)                    |
| 11 <sup>th</sup> | 3 | <ul style="list-style-type: none"> <li>- The cardiac output in normal and in failing heart</li> <li>- The arterial blood pressure and its regulation</li> <li>- Tissue fluid formation and drainage, Edema</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Guyton(p232-244)                                                                                    |
| 12 <sup>th</sup> | 3 | <ul style="list-style-type: none"> <li>- The coronary circulation</li> <li>- Hypertension: causes, types, complications, Rx</li> <li>- Hypotension (Shock): stages and types</li> </ul> <p><b><u>Physiology of the Respiratory System (6 Hrs)</u></b></p> <ul style="list-style-type: none"> <li>- Introduction to RS</li> <li>- Mechanism of breathing</li> </ul>                                                                                                                                                                                                                                                                                                                | Guyton(p 161-179)<br>Ganong(620-643)<br>Guyton(p 278-287)<br>Guyton(p 471-476)<br>Ganong(p 647-666) |

|                  |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                |
|------------------|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
|                  |   | <ul style="list-style-type: none"> <li>- Diffusion and gas transport (O<sub>2</sub> and CO<sub>2</sub>)</li> <li>- Functional anatomy of respiratory system</li> <li>- Mechanics of Pulmonary ventilation</li> <li>- Pleural &amp; Alveolar pressure</li> <li>- Pulmonary volumes &amp; Capacities</li> <li>- Ventilation</li> <li>- Exchange of gasses</li> <li>- Transport of gasses</li> <li>- Non respiratory function of respiratory system</li> <li>- Regulation of respiration</li> <li>- Patho-Physiological aspects of respiratory system</li> </ul>                                                                                                                                |                                                                                |
| 13 <sup>th</sup> | 3 | <ul style="list-style-type: none"> <li>- Regulation of breathing</li> <li>- Hypoxia, cyanosis</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Guyton(p514-522)<br>Ganong(p 671-678)                                          |
| 14 <sup>th</sup> | 3 | <ul style="list-style-type: none"> <li>- Pathophysiology of respiratory system</li> </ul> <p><b><u>Renal Physiology (8 Hrs)</u></b></p> <ul style="list-style-type: none"> <li>- Physiology of the body fluids</li> <li>- Body fluids &amp; their subdivision</li> <li>- Body fluids &amp; their compartments</li> <li>- Disturbance of volume concentration of body fluid</li> <li>- General renal function</li> <li>- Functional anatomy of the kidney</li> <li>- Renal blood flow &amp; its control</li> <li>- Function of the kidneys</li> <li>- Structural function of kidneys, nephrons</li> <li>- Urine formation, GFR, tubular load, T<sub>m</sub> &amp; Plasma clearance</li> </ul> | Guyton(p524-532)<br>Ganong(p 683-695)<br>Guyton(p 291-325)<br>Ganong(p 699-70) |
| 15 <sup>th</sup> | 3 | <ul style="list-style-type: none"> <li>- Concentration and dilution of urine</li> <li>- Micturation and its abnormalities in some diseases</li> <li>- Pathophysiology of the renal system</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Guyton(p 402-414)<br>Ganong(p 723-726)                                         |
| 16 <sup>th</sup> | 3 | <ul style="list-style-type: none"> <li>- Acid-base balance</li> <li>- Chemical and physiological regulation</li> <li>- Acid-base imbalance and disturbances</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Guyton(p 383-400)<br>Ganong(p 724-726)                                         |

**Module 4: Chemistry Module****Module name:** Chemistry module**Module category:** Basic**Module code:** Chem-M1043**Module Number:** 04**Module weight in ECTS:** 7 ECTS**Courses:**

| Course Name                  | Course Code | ECTS |
|------------------------------|-------------|------|
| Organic Chemistry            | Chem 1041   | 5    |
| Organic Chemistry Laboratory | Chem1042    | 2    |

**Module description**

Organic chemistry is a chemistry based discipline that deals with carbon related compounds. The module builds on the students' knowledge and understanding of functional groups, (structure, nomenclature, stereochemistry), reaction mechanisms, biological molecules and biotransformation. Students will acquire knowledge of some reactions of organic chemistry, particularly carbonyl reactions, substitution, addition and elimination reactions, which are, examined in terms of reaction type, mechanism and stereochemical implications. They will then apply this knowledge to devise organic synthetic pathways. The student will also examine the synthesis and reactivity of aromatic and heteroaromatic compounds with an emphasis on the named reactions.

**Module objective:** Students will demonstrate knowledge and understanding of the fundamental theories and practices of organic chemistry and apply the concepts and principles to solve problems related to Pharmacy. The course emphasize about classification, naming, identification, reaction, mechanism and synthesis or preparation of organic compounds and organic biological molecules. Differentiate organic chemicals in terms of their usefulness, hazards, and cautions to be taken in the manufacture, storage and use.

**Module competencies:**

- Discuss the chemical bonding theories and influence of bonding types on properties of compounds
- Determine the stereochemistry of organic molecules
- Explain mechanisms in organic reactions and describe the factors affecting reaction rates
- Explain the physical and chemical behaviors of organic compounds based on their functional groups
- Explain the properties, preparation and reactions of organic compounds
- Describe different classes of Biological molecules and apply the knowledge and principles to Medicinal chemistry
- Predict the existence of the kinds of stereo isomers, represent and designate their structures
- Review of the classes of organic compounds and give systematic name to different organic compounds

- Develop practical laboratory skills in chemical and analytical procedures and realize the importance of chemistry in everyday life

**Mode of delivery:** Block

Total time: 189hrs

Lecture: 54 hrs

Tutorials: 6hrs

Practical lab: 42 hrs

Independent study hour: 65 hrs

Seminar, Assignments and assessment: 12 hrs

Assessment: 10 hrs

**Mode of Assessment:**

Laboratory Reports

Laboratory Presentation

Laboratory written Exam

Assignments and Seminars

Practical exam

Quizzes

Final-exam

**Chemistry Module syllabi**

**Course title:** Organic Chemistry

**Course code:** Chem 1041

**Module name:** Chemistry module

**Module code:** Chem-M1043

**Course EtCTS:** 5 EtCTS (135 hrs)

Lecture: 48hrs

Tutorials: 16hrs

Independent study hour: 57hrs

Seminar, Assignments: 8hrs

Assessment: 6 hrs

**Year/Semester Course is offered:** Year I Semester II

**Contact hours/ week:**

**Pre-requisite:** None

**Course description:**

This course is intended to provide the students basic understanding of different classes of organic functional groups ( alkanes, , alkenes, and alkynes, alkyl halides , alcohols, ethers, aromatic rings, ketones, aldehydes, carboxylic acids and their derivatives, amines biomolecules(carbohydrates, ,proteins, lipids and nucleic acid) with a special focus on their physical properties (boiling point, melting point, solubility,) and chemical properties(reactivity and reaction mechanism) and basics of stereochemistry and basics of drug activities in relation to stereochemistry and Chemical bonding. In addition it deals with Chemistry of Aromatic Compounds; Carbonyl Reactions; Introduction to biological molecules

**Course Objectives:**

Upon completion of this course the students would be able to: discuss the chemical bonding theories and influence of bonding types on properties of compounds predict the existence of the kinds of stereo isomers, represent and designate their structures determine the stereochemistry of organic molecules describe the factors affecting reaction rates and explain mechanisms in organic reactions give systematic name to different organic compounds review of the classes of organic compounds explain the physical and chemical behaviors of organic compounds based on their functional groups explain the properties, preparation and reactions of aromatic compounds discuss different types of reactions of carbonyl compounds describe different classes of Biological molecules.

**Supporting Objectives:**

- Familiar with types of bonding and principle of formation and hybridization



- Explain the structures of organic compounds.
- Familiar with the most important classes of organic functional groups and their physical and chemical properties.
- Use the rules of nomenclature to give correct names for organic compounds
- Draw correct structures that correspond to a name, and correctly use and recognize common names.
- Use principles of stereochemistry to locate stereocenters and label stereoisomers,
- Identify chiral compounds, give stereochemical relationships between molecules,
- Use Fischer projections, solve optical activity problems, and identify stereochemical results of a reaction.
- Be able to identify typical chemical reactions on the basis of their structure and properties
- Give starting materials, reagents, and products for reactions of organic compounds.
- Classify various biological molecules such as carbohydrates, lipids, amino acids and proteins, and their important chemical properties.

### **Course mode of delivery: Block/Parallel**

### **Learning activities and teaching methods**

Interactive lectures, case studies, computer assisted learning, formative problem-solving exercises, self-directed learning through virtual learning environment and technologies, practical activities and assignments.

### **Assessment techniques:**

Assignments and Seminars; 15%

Quizzes: 35%

Final-exam: 50%

### **Teachers and Students Role**

#### **Teacher's role**

Course instructors are expected to:

- Provide lecture and guide students
- Providing assignments to be done and feedback for students
- Prepare lecture note, Assignment topics and title for group discussions
- Select seminar title and advice students in preparation and presentations
- Prepare assessing questions and examine students
- Organize laboratory sessions and demonstrate some laboratory activities
- Prepare laboratory manuals

#### **Student's role**

Students are expected to:

- Read text books, lecture handouts and reference books
- Be an active participant in class discussion (ask questions and answering questions)
- Prepare paper under the title given for seminars, assignments and present it
- Analyze and evaluate different literatures, reference books and journal articles
- Perform laboratory activities
- Write laboratory reports
- Taking exams

**References:**

- Organic reaction Mechanisms 1993. A.C. Knope and w.E. Watts, University of Ulster Northern Ireland.
- Organic reaction Mechanisms, Ronald Brewlow, Columbia University, second edition.
- Organic Chemistry. Graham solomons, 6th ed, Univ of South Florida.
- A textbook of Organic Chemistry, K.S. Tewair, S.N. Mehrotra, N.K. Vishnoi

**Course Schedule:**

| Course Content                                          |  |
|---------------------------------------------------------|--|
| 1. Structure                                            |  |
| 1.1. Energy levels and Atomic orbital                   |  |
| 1.2. Covalent bonds                                     |  |
| 1.3. Molecular orbital theory                           |  |
| 1.4. Orbital hybridization                              |  |
| 2. Nomenclature                                         |  |
| 2.1. Alkanes                                            |  |
| 2.2. Alkenes and Alkynes                                |  |
| 2.3. Alcohols                                           |  |
| 2.4. Aldenydes                                          |  |
| 2.5. Ketones                                            |  |
| 2.6. Amines                                             |  |
| 2.7. Ethers                                             |  |
| 2.8. Aromatics                                          |  |
| 3. Stereochemistry                                      |  |
| 3.1. Symmetry and dissymmetry                           |  |
| 3.2. The asymmetric carbon                              |  |
| 3.3. Optical isomerism                                  |  |
| 3.4. Fischer projections                                |  |
| 3.5. Multiple asymmetric centers                        |  |
| 3.6. Configuration                                      |  |
| 4. Substitution reactions                               |  |
| 4.1. SN1 and SN2 mechanisms                             |  |
| 4.2. Applications of substitution Reactions             |  |
| 4.2.1. Alcohols                                         |  |
| 4.2.2. Ethers                                           |  |
| 4.2.3. Carboxylic acids                                 |  |
| 4.2.4. Alkanes, Alkenes, and Alkynes                    |  |
| 4.2.5. Amines                                           |  |
| 4.2.6. Epoxide Ring opening                             |  |
| 4.2.7. Reactions of malonic ester and acetoacetic ester |  |
| 5. Elimination reactions                                |  |
| 5.1. Mechanisms                                         |  |

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| <ul style="list-style-type: none"> <li>5.2. Evidences for mechanisms of elimination reaction</li> <li>5.3. E1 versus E2</li> <li>5.4. Elimination versus substitution</li> <li>5.5. Applications of elimination reactions <ul style="list-style-type: none"> <li>5.5.1. Dehydration of Alcohols</li> <li>5.5.2. Dehydrohalogenation of alkylhalides</li> <li>5.5.3. Vicinal Dihalides</li> <li>5.5.4. Hofmann Elimination</li> <li>5.5.5. Acetate pyrolysis</li> <li>5.5.6. Cope reaction</li> </ul> </li> <li>6. Addition Reactions <ul style="list-style-type: none"> <li>6.1. Mechanism</li> <li>6.2. Reactivity</li> <li>6.3. Rules of addition reactions <ul style="list-style-type: none"> <li>6.3.1. Markovnikov Rule</li> <li>6.3.2. Michael Addition</li> <li>6.3.3. Radical addition</li> </ul> </li> <li>6.4. Applications of Addition Reactions <ul style="list-style-type: none"> <li>6.4.1. Addition of halogen</li> <li>6.4.2. Addition of hydrogen halide</li> <li>6.4.3. Addition of hypohalous acids</li> <li>6.4.4. Hydration of alkenes</li> <li>6.4.5. Hydroboration</li> <li>6.4.6. Diels-Alder addition <ul style="list-style-type: none"> <li>6.4.6.1. Kinetic vs Thermodynamic control of the Diels-Alder reaction</li> <li>6.4.6.2. Stereochemistry of the Diels-Alder reaction</li> </ul> </li> <li>6.4.7. Catalytic hydrogenation</li> <li>6.4.8. Ozonization</li> <li>6.4.9. Peracid oxidation</li> <li>6.4.10. Glycol formation</li> </ul> </li> </ul> </li> <li>7. Aromatic substitution reactions <ul style="list-style-type: none"> <li>7.1. Introduction</li> <li>7.2. Aromaticity</li> <li>7.3. Aromatic substitution</li> <li>7.4. Directing effects</li> <li>7.5. Application of electrophilic substitutions <ul style="list-style-type: none"> <li>7.5.1. Halogenation</li> <li>7.5.2. Sulfonation</li> <li>7.5.3. Nitration</li> <li>7.5.4. Friedel-Crafts Alkylation</li> <li>7.5.5. Friedel-Crafts Acylation</li> <li>7.5.6. Diazotization of Amines</li> </ul> </li> </ul> </li> </ul> |  |
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|--------------------------------------------------------|--|
| 7.5.7. Reactions of aromatic side chains               |  |
| 8. Carbonyl reactions                                  |  |
| 8.1. Carbonyl addition                                 |  |
| 8.2. Addition Elimination                              |  |
| 8.3. Enolization Ketonization                          |  |
| 8.4. Application of Addition reactions                 |  |
| 8.4.1. Hydrate formation                               |  |
| 8.4.2. Hemiacetals and Hemiketals                      |  |
| 8.4.3. Cyanohydrins                                    |  |
| 8.4.4. Carbinolamins                                   |  |
| 8.4.5. addition of Grignard reagents                   |  |
| 8.4.6. Addition of hydrogen                            |  |
| 8.4.7. Lithiumaluminiumhydride and sodiumborohydride   |  |
| 8.5. Application of addition-elimination reactions     |  |
| 8.5.1. Imines and related compounds                    |  |
| 8.5.2. Wittig reaction                                 |  |
| 8.5.3. Acetal and ketal formation in acid media        |  |
| 8.5.4. Acids and their derivatives                     |  |
| 8.5.5. Ester hydrolysis and formation in acid media    |  |
| 8.5.6. Acid chlorides                                  |  |
| 8.5.7. Acid anhydrides                                 |  |
| 8.5.8. Reduction of acid derivatives                   |  |
| 8.6. Application of enolization-ketonization reactions |  |
| 8.6.1. Halogenation                                    |  |
| 8.6.2. Alkylation                                      |  |
| 8.6.3. aldol condensation                              |  |
| 8.6.4. Claisen-Schmidt condensation                    |  |
| 8.6.5. Mannich condensation                            |  |
| 8.6.6. Perkin condensation                             |  |
| 8.6.7. Claisen condensation                            |  |
| 9. Rearrangement reactions                             |  |
| 9.1. Rearrangement to an electro-deficient carbon      |  |
| 9.2. Rearrangement to an electro-deficient oxygen      |  |
| 9.3. Rearrangement to an electro-deficient nitrogen    |  |
| 10. Oxidation-reduction reaction                       |  |
| 10.1. Introduction                                     |  |
| 10.2. Oxidation reaction                               |  |
| 10.2.1. Alcohols                                       |  |
| 10.2.2. Aldehydes                                      |  |
| 10.3. Reduction Reactions                              |  |
| 10.3.1. Catalytic hydrogenation                        |  |
| 10.3.2. Chemical Reduction                             |  |
| 10.3.3. Dissolving metal reductions                    |  |

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| 10.3.4. Acyloin condensation<br>11. Electrocyclic Reactions<br>11.1. Molecular Orbitals<br>11.2. Electrocyclic reactions<br>11.3. Stereospecificity of cyclic reactions<br>12. Biological molecules<br>12.1. Glucose: An introduction to carbohydrate chemistry<br>12.2. Disaccharides and polysaccharides<br>12.3. Amino-acids, peptides, and proteins<br>12.3.1. The structure and properties of alpha-amino acids<br>12.3.2. Analysis of alpha-amino acids<br>12.3.3. Synthesis of alpha-amino acids<br>12.4. Peptides and proteins<br>12.5. Peptide synthesis<br>13. Biological transformation<br>13.1. Glycolysis<br>13.2. Thiamine<br>13.3. Tricarboxylic acid cycle<br>13.4. Vitamin B6-Transamination<br>13.5. Mechanism of chymotrypsin action storage of metabolic energy<br>13.6. Generation and storage of metabolic energy<br>13.7. Biosynthetic pathways |           |
| <b>Total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>48</b> |

**Chemistry Module syllabi**

**Course title:** Organic Chemistry Laboratory

**Course code:** Chem 1042

**Module name:** Chemistry module

**Module code:** Chem-M1043

**Year/Semester Course is offered:** Year I Semester II

**To be included as soon as it is ready (received from the owner department)**

## **Module 5: Biomedical Sciences-II**

**Module name:** Biomedical science II

**Module category:** Basic

**Module code:** Biom-M2053

**Module Number: 05**

**Module weight in ECTS: 27**

**Courses:**

| Course Name                               | Course Code | ECTS |
|-------------------------------------------|-------------|------|
| Biochemistry I                            | Bioc 1051   | 5    |
| Biochemistry II                           | Bioc 1052   | 5    |
| Microbiology, immunology and parasitology | Mbio1053    | 7    |
| Pathology                                 | Path2054    | 5    |
| Human Physiology-II                       | Phyl2055    | 5    |

### **Module description**

The module provides students with basic knowledge and understanding of the normal chemical and metabolic processes of the body and how this is affected by certain disorders; knowledge of the molecular basis of certain types of diseases, and the biochemical basis of drug actions and effects of toxins; help the student to understand and appropriately identify biochemical disorders and recommend specific treatment regimen. This module is also designed to enable the students understand the background to the development of microbiology; classification, nomenclature, microscopic characterization and morphology, reproduction and growth of microorganisms and parasitic agents. It also explains the cultivation techniques and nutritional requirements of microorganisms. The immunology part helps the student learn about the basis of immunity, preparation and characterization of antibodies, the mechanism of antibody-antigen reactions and diagnostic applications of immunological principles. The practical sessions include identification of bacteria, staining techniques, media and culture preparation, sterilization, antibiotic testing, assay and sensitivity test

**Module objective:** This module provides students with basic understanding of; the normal chemical and metabolic processes of the body and how this is affected by certain disorders; identification and characterization of different type of microbial and parasites and also the disease they cause; basics of immunological principle

**Module competency:**

- Apply the concept of Biochemistry to drug therapy

- Apply knowledge of common disease causing organism to drug therapy

**Module mode of delivery:** Blocked for biochemistry courses and parallel for Microbiology and immunology course

**Module mode of Assessment:**

Quizzes

Mid exam

Final Exam

Seminar

Laboratory

Assignment

**Module learning teaching methods**

Active learning methods (brain storming, buzz group, discussion, etc), Lecture, group & individual presentation, assignment, project work, and laboratory work.



**Course title:** Biochemistry I

**Course code:** Bioc. 2051

**Module name:** Biomedical II module

**Module code:** Biom-M2053

**Course ECTS:** 5 (135 hrs)

- Lecture: 48 hours
- Tutorial: 9 hours
- Home study: 60 hours
- Assignment and presentation: 18 hours

**Year/Semester Course is offered:** Year II Semester I

**Contact hours/ week:** 135-87= 48 hours/ 16 weeks= 3 hours

**Pre-requisite if any:** Organic Chemistry

**Course description:**

This Biochemistry course is designed to prepare B.Sc. graduate anesthesia students who are competent providers of anesthesia services. Students will be able to explain the biochemical aspects of human life & describe the biochemistry of carbohydrates, proteins, lipids and nucleic acids. It contains only the theoretical part that the trainee should pass through to acquire the basic competence in accomplishing organizational activities relating to its coverage.

**Course objectives:**

- After completing this course, the student will be able to explain the biochemical aspects of human life; describe the chemistry and metabolism of biomolecules (carbohydrates, lipids, proteins and nucleic acids); explain central metabolism; outline the transmission & expression of genetic information and correlate the biochemical processes with health & disease.

**Specific Objectives:**

1. Describe the structure and classification of carbohydrates
2. Outline the metabolic pathways for anaerobic glycolysis, pentose shunt, and gluconeogenesis, including substrates, unique enzymes and regulatory mechanisms.
3. Outline the metabolic pathways for synthesis and degradation of glycogen.
4. Differentiate the structure and composition of lipids.
5. Describe the composition and functions of different lipoproteins present in plasma.

6. Outline the sequence of reactions involved in oxidation of fatty acids in the mitochondrion.
7. Explain the rationale for the pathway of ketogenesis and identify the major intermediates and products of this pathway.
8. Describe the synthesis of fatty acids and triglycerides.
9. Outline the sequence of reactions in the tri carboxylic acid cycle and explain the purpose of the cycle.
10. Outline the mitochondrial electron transport system and define membrane potential and explain its role in ATP synthesis and thermogenesis.
11. Describe the structure and classification of amino acids & proteins.
12. Describe the mechanism of oxygen binding to myoglobin and hemoglobin.

**Course mode of delivery: Block/Parallel**

**Course learning and teaching methods:**

- Active learning methods (brain storming, discussion, etc), Lecture, group and individual presentation, assignment.

**Assessment techniques:**

Continuous assessment & summative assessment

- Class attendance (%)
- Quiz (%)
- Assignments (%)
- Final Exam (50)

**Teachers and Students Role**

**References:**

■ **Textbook:**

□ Pamela C.C, and Richard A.H., Lippincott's Illustrated Reviews: Biochemistry, 3<sup>rd</sup> edition, J.B.Lippincott Company Philadelphia, 2004.

**Recommended Reading**

1. Stryer L. et al. Biochemistry, 5<sup>th</sup> edition, W.H. Freeman and Company & Sumanas, Inc., 2004.
2. Murray R.K et al. Harper's Illustrated Biochemistry 27<sup>th</sup> edition, The McGraw-Hill Companies, Inc., 2006.
3. Lehninger A.L. Principles of Biochemistry, CBS publishers and distributors, 2006.

4. Lieberman M. et al. Marks' Essential Medical Biochemistry, 2<sup>nd</sup> Edition, Lippincott Williams & Wilkins, 2007.
5. Zubay P. et al. Principles of Biochemistry, W.M.C. Brown Publishers USA, 1995.
6. Smith E.L. et al., Principles of Biochemistry, McGraw-Hill-International, 1993 or recent edition.

**Course Schedule:** By contact time, contents/topics & reading/reference materials for each topic

| Week | Contact hrs | Topic/sub-topic/chapter                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Reading materials                                                                                           |
|------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| 1    | 2           | 1. INTRODUCTION TO BIOCHEMISTRY <ul style="list-style-type: none"> <li>– Introduction: <ul style="list-style-type: none"> <li>• Definitions</li> <li>• Role of biochemistry</li> </ul> </li> <li>– Cellular components</li> <li>– Brief introduction to metabolism &amp; Enzymes</li> </ul>                                                                                                                                                                                                        | Lehninger Principles of Biochemistry (Page No. 3-12)                                                        |
| 1-2  |             | 2. WATER & pH <ul style="list-style-type: none"> <li>– Role of water in biological system</li> <li>– Acid base theories <ul style="list-style-type: none"> <li>• Definition of pH, pKa and pKb</li> </ul> </li> <li>– Buffers &amp; Acid-base balance</li> </ul>                                                                                                                                                                                                                                   | Harper's Biochemistry (Page No. 5-13)                                                                       |
| 3-7  | 15          | 3. CARBOHYDRATES <ul style="list-style-type: none"> <li>– Structure &amp; classification of carbohydrates</li> <li>– Digestion &amp; absorption of carbohydrates</li> <li>– Metabolism of carbohydrates: <ul style="list-style-type: none"> <li>• Glycolysis</li> <li>• Oxidation of pyruvate</li> <li>• Pentose phosphate path way</li> <li>• Glycogen metabolism</li> <li>• Gluconeogenesis</li> <li>• Metabolism of major non-glucose sugars</li> </ul> </li> </ul> Regulation of blood glucose | Lehninger Principles of Biochemistry (page No. 239-255)<br>Harper's Biochemistry Page No. 130-136, 136-163) |
| 8-12 | 15          | 4. LIPIDS <ul style="list-style-type: none"> <li>– Structure and classification of lipids</li> <li>– Biological membranes</li> <li>– Digestion &amp; absorption of lipids</li> <li>– Metabolism of fatty acids:</li> </ul>                                                                                                                                                                                                                                                                         | Harper's Biochemistry (Page No. 111-122)<br>Pamela C.C. (Page No. 163-205)                                  |

|       |   |                                                                                                                                                                                                                                                                                                                                                                    |                                                                                       |
|-------|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|       |   | <ul style="list-style-type: none"> <li>• Biosynthesis and storage of fatty acids</li> <li>• Oxidation of Fatty Acids</li> <li>• Ketogenesis &amp; Ketolysis</li> </ul> <p>– Lipid transport and storage:</p> <ul style="list-style-type: none"> <li>• Structure and function of Lipoproteins</li> </ul> <p>– Cholesterol synthesis, transport, &amp; excretion</p> |                                                                                       |
| 13-14 | 6 | <p>5. CENTRAL METABOLISM</p> <p>– Tricarboxylic acid (Krebs') cycle</p> <p>– Bioenergetics (thermodynamics):</p> <ul style="list-style-type: none"> <li>• Related to nutrition and obesity</li> <li>• The Electron transport system</li> <li>• Oxidative phosphorylation</li> </ul>                                                                                | <p>Pamela C.C. (Page No. 105-109)</p> <p>Harper' Biochemistry (Page No. 92-102)</p>   |
| 15-16 | 6 | <p>6. AMINO ACIDS AND PROTEINS</p> <p>– Structure and classification of amino acids</p> <ul style="list-style-type: none"> <li>• Physico-chemical properties of amino acids</li> </ul> <p>– Structure &amp; functions of proteins</p> <ul style="list-style-type: none"> <li>• Mechanism of oxygen binding to myoglobin and hemoglobin.</li> </ul>                 | <p>Pamela C.C. (Page No. 229-267)</p> <p>Harper's Biochemistry (Page No. 249-264)</p> |

**Course title:** Biochemistry II

**Course code:** Bioc 2052

**Module name:** Biomedical II module

**Module code:** Biom-M2053

**Course ECTS:** 5 (135 hrs)

- Lecture: 48 hours
- Tutorial: 9 hours
- Home study: 60 hours
- Assignment and presentation: 18 hours

**Contact hours/ week:** 135-87= 48 hours/ 16 weeks= 3 hours

**Pre-requisite if any:** Organic Chemistry

**Course description:**

- This Biochemistry course is designed to prepare B.Sc. graduate anesthesia students who are competent providers of anesthesia services. Students will be able to explain the biochemical aspects of human life & describe the biochemistry of carbohydrates, proteins, lipids and nucleic acids. It contains only the theoretical part that the trainee should pass through to acquire the basic competence in accomplishing organizational activities relating to its coverage.

**Course Objectives:**

- After completing this course, the student will be able to explain the biochemical aspects of human life; describe the chemistry and metabolism of biomolecules (carbohydrates, lipids, proteins and nucleic acids); explain central metabolism; outline the transmission & expression of genetic information and correlate the biochemical processes with health & disease.

**Specific Objectives:**

1. Describe the structure and classification of amino acids & proteins.
2. Describe the mechanism of oxygen binding to myoglobin and hemoglobin.
3. Describe the mechanisms used by humans for removal of the nitrogen from amino acids prior to the metabolism of their carbon skeletons.
4. Discuss the structure and composition of enzymes, including cofactors, and conditions that affect enzymatic reactions.
5. Compare and contrast the structure and biosynthesis of purines and pyrimidines, highlighting the differences between de novo and salvage pathways.

6. Describe the compositions and structures of DNA and RNA.
7. Explain how replication of DNA is achieved with high fidelity in a bidirectional manner and in a semi-conservative fashion.
8. Describe the major steps in transcription of an RNA molecule.
9. Describe how the different RNAs involved in protein synthesis interact to produce a polypeptide.

**Delivery mode/methodology: Block**

**Course learning and teaching methods**

- Active learning methods (brain storming, discussion, etc), Lecture, group and individual presentation, assignment

**Assessment mechanisms:**

Continuous assessment & summative assessment

- Class attendance (%)
- Quiz (%)
- Assignments (%)
- Final Exam (50%)

**Teachers and Students Role**

**References:**

■ **Textbook:**

- Pamela C.C, and Richard A.H., Lippincott's Illustrated Reviews: Biochemistry, 3<sup>rd</sup> edition, J.B.Lippincott Company Philadelphia, 2004.

■ **Recommended Reading:**

1. Stryer L.et al.Biochemistry,5<sup>th</sup> edition, W.H. Freeman and Company &Sumanas, Inc., 2004.
2. Murray R.K et al. Harper's Illustrated Biochemistry 27<sup>th</sup> edition,The McGraw-Hill Companies, Inc., 2006.
3. Lehninger A.L. Principles of Biochemistry, CBS publishers and distributors, 2006.
4. Lieberman M. et al.Marks' Essential Medical Biochemistry, 2<sup>nd</sup> Edition,Lippincott Williams & Wilkins, 2007.
5. Zubay P. et al.Principles of Biochemistry, WM.C. Brown Publishers USA, 1995.
6. Smith E.L et. al., Principles of Biochemistry, McGraw-Hill-International, 1993 or recent edition.

**Course Schedule:** By contact time, contents/topics & reading/reference materials for each topic

|      |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                  |
|------|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| 1-7  | 20 | <p>1. AMINO ACIDS AND PROTEINS</p> <ul style="list-style-type: none"> <li>• Digestion &amp; absorption of proteins</li> <li>• Metabolism of proteins &amp; amino acids: <ul style="list-style-type: none"> <li>○ Biosynthesis of nonessential amino acids</li> <li>○ Catabolism of Proteins &amp; of Amino Acid Nitrogen- Urea cycle</li> <li>○ Catabolism of the carbon skeletons of amino acids</li> <li>○ Conversion of amino acids to specialized products</li> <li>○ Metabolism of Haem:</li> <li>○ Porphyrins &amp; Bile Pigments</li> </ul> </li> </ul> | <p>Pamela C.C.<br/>(Page No. 229-267)<br/>Harper's Biochemistry<br/>(Page No. 249-264)<br/><br/>Harper's Biochemistry<br/>(Page No- 270-286)</p> |
| 7-9  | 6  | <p>8. VITAMINS</p> <ul style="list-style-type: none"> <li>– Classification of vitamins</li> <li>– Structure and function of: <ul style="list-style-type: none"> <li>• Water &amp; Fat soluble vitamins</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                  | <p>Harper's Biochemistry<br/>(Page No. 49-72)<br/>Pamela C.C.<br/>(Page No. 47-58)</p>                                                           |
| 9-16 | 18 | <p>9. MOLECULAR BIOLOGY</p> <ul style="list-style-type: none"> <li>– Nucleotide structure</li> <li>– Biosynthesis &amp; Degradation of nucleotides</li> <li>– DNA structure and Replication</li> <li>– RNA structures and Transcription</li> <li>– Protein Synthesis: <ul style="list-style-type: none"> <li>• The Genetic Code</li> </ul> </li> </ul>                                                                                                                                                                                                         | <p>Pamela C.C.<br/>(Page No. 319-330)<br/>Harper's Biochemistry<br/>(Page No. 481-497)</p>                                                       |

|  |  |                                                                                                                                                                                                                                      |  |
|--|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  |  | <ul style="list-style-type: none"> <li>• Translation</li> <li>– Mutation</li> <li>– Regulation of gene expression in               <ul style="list-style-type: none"> <li>• Prokaryotes</li> <li>• Eukaryotes</li> </ul> </li> </ul> |  |
|--|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|



**Course title: Microbiology, Parasitology & immunology**

**Course code:** Mbio2053

**Module name: Biomedical II module**

**Module code:** Biom-M2053

**Course ECTS:** 7 (189 hrs)

Lecture: 48 hrs

Tutorial: 20 hrs

Lab practice: 48 hrs

Home study individual work: 45 hrs

Total: 189 hrs

**Year/Semester Course is offered: Year II Semester I**

**Contact hours/ week:** 135-87= 48 hours/ 16 weeks= 3 hours

**Pre-requisite if any:** Medical Physiology, Medical Biochemistry and Human Anatomy

**Course description:**

The course includes Introduction to microbiology, theories on origin of microorganisms, classification of microorganisms, morphology and cytology of bacteria, host parasite relationship, common pathogenic bacteria, introduction to immunology, introduction to mycology and virology.

**Course objectives:**

- This course helps the students:
- To understand brief history of Medical Microbiology and important events, discoveries and inventions significantly contributed to its development as a science.
- To appreciate the relationship between microbes, the immune system, and disease outcomes.
- To understand how the immune system functions in a specific and non-specific way, to defend the host against infections by bacteria, fungi and viruses.
- To recognize the structural components of microbes (bacteria, fungi, parasites and viruses) and how these impact the pathogenesis of disease.
- To explain the methods of microorganisms control ( chemotherapy & vaccines, disinfection and sterilization )
- To know the common microorganisms associated with specific clinical diseases and what factors are involved in pathogenesis.
- To appreciate the role of immune system in allergic diseases.
- To appreciate the role of the clinical laboratory in diagnosis and management of infectious diseases.

- To develop the ability to correlate the clinical picture with laboratory information to establish a diagnosis and select for appropriate treatment options.
- To understand mechanism of action of anti-microbial agents
- To appreciate mechanisms how microbes resist anti-microbial agents
- Discuss the concepts of parasitism, the relationships between parasites and host, between parasites and environment
- Recognize the general epidemiological aspects of parasites that affect human
- Illustrate the life cycle of specific parasites and identify the important parasitic agent affecting human health
- Describe some important arthropods responsible for the transmission of disease causing parasites
- Describe commonly used methods for microscopic examination of parasites
- Describe the transmission and pathogenesis of helminthes infections
- Describe the basic concepts and principles how to control protozoan infections.

**Course mode of delivery: Block/Parallel**

**Course learning and teaching methods:**

- Classroom contact/Lecture
- Presentation and group discussion
- Computer assisted instruction

**Assessment techniques:**

- Four individual assignments 10%
- Two exams (15%)
  - Exam 1     15%       week 7
  - Exam 2     15 %      week 10
- Mid Exam    30%.
- Final exam     40    % Week 16

**Teachers and Students Role**

**References:**

- Markell, Voge, Jhon. Medical Parasitology. 6<sup>th</sup> ed. 1986. W.b. Saunders Company.
- Paul Chester Beaver, Rodney Clifton jung, Eddie Wayne Cupp. Clinical Parasitology. 9<sup>th</sup> ed. 1984. K.M. Varghese company

- Herbert M. Gilles. Protozoal Diseases. 1999. Arnold
- David L. Belding. Text book of Parasitology. 3<sup>rd</sup> ed. 1965.
- Monica Cheesbrough. District Laboratory Practice in Tropical countries. Part I -2<sup>nd</sup>ed updated. 1998. Tropical Health Technology. Cambridge
- Judith S. Heelan, Frances W. Ingersoll. Essentials of Human Parasitology. 2002. Delmar Thomson Learning
- VigarZaman. Atlas of Medical Parasitology. 1979
- Harold W. Brown, Franklin A. Neva. Basic Clinical Parasitology. 5<sup>th</sup> ed. 1983
- Mohammed AwolAdem and WaqtolaCheneke. Medical Parasitology: Lecture note for medical laboratory technology students: upgraded lecture note serious. 2006
- Modern Parasitology A text book of Parasitology ( Cox 2<sup>nd</sup>edn)
- Clinical parasitology (Beaver et. al 9<sup>th</sup>ed.)
- Atlas of Medical Helminthology and Protozoology (Jaffeey and Leach 2<sup>nd</sup> edition)
- District laboratory practice in tropical counties (Monica CheesbroughVol I)
- Essentials of Parasitology (Murray D. Dailey 6<sup>th</sup> ed. 1996)
- Essentials of parasitology (Gerald D. Schmidt 4<sup>th</sup> ed. 1994)
- Parasitology for medical Laboratory Technology students – Lecture note series (GirmaM. and Mohammed A. 2003)
- Craig ad Faust's clinical parasitology (Ernest C. Faust 8<sup>th</sup> ed. 1977)
- Web materials - DPDx

### Course Schedule:

| Week              | Lecture Topics of Parasitology                                                                                                                                                                                                                                                                                                                                                                                                                                               | Contact Hours | Reading Materials                                                                                                                                                                                                                                        |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 <sup>st</sup> ^ | 1. Chapter I- Introduction to Medical Parasitology <ul style="list-style-type: none"> <li>• Features of parasites</li> <li>• Source of infection</li> <li>• Mode of transmission</li> <li>• Direct mode of transmission</li> <li>• Indirect mode of transmission</li> <li>• Routes of transmission</li> <li>• General life cycle of parasites               <ul style="list-style-type: none"> <li>• Direct life cycle</li> <li>• Indirect life cycle</li> </ul> </li> </ul> | 1             | Judith S. Heelan, Frances W. Ingersoll. Essentials of Human Parasitology. 2002. Delmar Thomson Learning<br>Markell, Voge, Jhon. Medical Parasitology. 6th ed. 1986. W.b. Saunders company.<br>David L. Belding. Text book of Parasitology. 3rd ed. 1965. |

|                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2 <sup>nd</sup> & 3 <sup>rd</sup> | <p>Chapter II - Nemathelminthes /Round worms/</p> <ul style="list-style-type: none"> <li>○ General characteristics</li> <li>○ Classification (Intestinal &amp; tissue) <ul style="list-style-type: none"> <li>▪ Intestinal round worms <ul style="list-style-type: none"> <li>• <i>Ascaris lumbricoides</i></li> <li>• <i>Trichuris trichura</i></li> <li>• <i>Enterobius vermicularis</i></li> <li>• <i>Ancylostoma duodenale</i></li> <li>• <i>Necator americanus</i></li> <li>• <i>Strongyloides stercoralis</i></li> </ul> </li> </ul> </li> </ul> <p>Assignment I</p> <ul style="list-style-type: none"> <li>▪ Tissue round worms <ul style="list-style-type: none"> <li>• <i>Wuchereria bancrofti</i></li> <li>• <i>Onchocerca volvulus</i></li> <li>• <i>Trichinella spiralis</i></li> </ul> </li> </ul> | 2 | <p>Monica Cheesbrough. District Laboratory Practice in Tropical countries. Part I -2nd updated. 1998. Tropical Health Technology. Cambridge</p> <p>Judith S. Heelan, Frances W. Ingersoll. Essentials of Human Parasitology. 2002. Delmar Thomson Learning</p> <p>Mohammed AwolAdem and WaqtolaCheneke. Medical Parasitology: Lecture note for medical laboratory technology students: upgraded lecture note series. 2006</p>                                                                                                                                                        |
| 4 <sup>th</sup> & 5 <sup>th</sup> | <p>Chapter III –Platyhelminthes</p> <p>Cestodes /The tape worms/</p> <ul style="list-style-type: none"> <li>○ General characteristics <ul style="list-style-type: none"> <li>▪ <i>Taenia saginata</i></li> <li>▪ <i>Taenia solium</i></li> <li>▪ <i>Hymenolepis nana</i></li> <li>▪ <i>Echinococcus granulosus</i></li> </ul> </li> </ul> <p>Tematodes /The flukes/</p> <ul style="list-style-type: none"> <li>○ General characteristics</li> <li>○ Classification (blood, liver &amp; intestinal flukes) <ul style="list-style-type: none"> <li>▪ Blood flukes <ul style="list-style-type: none"> <li>• <i>Schistosoma mansoni</i></li> <li>• <i>Schistosoma haematobium</i></li> </ul> </li> </ul> </li> </ul> <p>Mid Exam</p>                                                                                | 2 | <p>Monica Cheesbrough. District Laboratory Practice in Tropical countries. Part I -2nd updated. 1998. Tropical Health Technology. Cambridge</p> <p>Judith S. Heelan, Frances W. Ingersoll. Essentials of Human Parasitology. 2002. Delmar Thomson Learning</p> <p>Mohammed AwolAdem and WaqtolaCheneke. Medical Parasitology: Lecture note for medical laboratory technology students: upgraded lecture note series. 2006</p> <p>Markell, Voge, Jhon. Medical Parasitology. 6th ed. 1986. W.b. Saunders company</p> <p>David L. Belding. Text book of Parasitology. 3rd ed. 1965</p> |
| 6 <sup>th</sup>                   | <p>Chapter IV – Introduction</p> <ul style="list-style-type: none"> <li>▪ Protozoa</li> <li>▪ General Morphology</li> <li>▪ Life Cycle</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 1 | <p>Monica Cheesbrough. District Laboratory Practice in Tropical countries. Part I -2nd updated.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

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|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                 | <ul style="list-style-type: none"> <li>▪ Classification</li> </ul>                                                                                                                                                                                                                                                                                                                                                                         |   | <p>1998. Tropical Health Technology. Cambridge</p> <p>Judith S. Heelan, Frances W. Ingersoll. Essentials of Human Parasitology. 2002. Delmar Thomson Learning</p> <p>Mohammed AwolAdem and WaqtolaCheneke. Medical Parasitology: Lecture note for medical laboratory technology students: upgraded lecture note serious. 2006</p> <p>Markell, Voge, Jhon. Medical Parasitology. 6th ed. 1986. W.b. Saunders company</p> <p>David L. Belding. Text book of Parasitology. 3rd ed. 1965</p>                                                                                                |
| 7 <sup>th</sup> | <p>Chapter V - The Amoeba:</p> <p>Alimentary canal</p> <ul style="list-style-type: none"> <li>○ General <i>characteristics</i> <ul style="list-style-type: none"> <li>▪ <i>Entamoeba histolytica/dispar</i></li> </ul> </li> </ul> <p><i>Reading assignments</i></p> <ul style="list-style-type: none"> <li>- <i>Mechanism of action of anti-protozoan drugs</i></li> <li>- <i>Mechanism of action of anti-helminthic drugs</i></li> </ul> | 1 | <p>Monica Cheesbrough. District Laboratory Practice in Tropical countries. Part I -2nded updated. 1998. Tropical Health Technology. Cambridge</p> <p>Judith S. Heelan, Frances W. Ingersoll. Essentials of Human Parasitology. 2002. Delmar Thomson Learning</p> <p>Mohammed AwolAdem and WaqtolaCheneke. Medical Parasitology: Lecture note for medical laboratory technology students: upgraded lecture note serious. 2006</p> <p>Markell, Voge, Jhon. Medical Parasitology. 6th ed. 1986. W.b. Saunders company</p> <p>David L. Belding. Text book of Parasitology. 3rd ed. 1965</p> |

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|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8 <sup>th</sup> & 10 <sup>th</sup> | <p>Chapter VI - Flagellate Protozoa:</p> <p>Digestive and urogenital tract</p> <ul style="list-style-type: none"> <li>○ General <i>characteristics</i> <ul style="list-style-type: none"> <li>▪ <i>Giardia lamblia</i></li> <li>▪ <i>Trichomonas vaginalis</i></li> </ul> </li> </ul> <p>Hemoflagellates / Blood and Tissue/</p> <ul style="list-style-type: none"> <li>i. The Leishmania <ul style="list-style-type: none"> <li>a. <i>Leishmaniatropica complex</i></li> <li>b. <i>L. tropica minor</i></li> <li>c. <i>L. tropica major</i></li> <li>d. <i>L. aethiopica</i></li> </ul> </li> <li>ii. Leishmaniamexicana complex <ul style="list-style-type: none"> <li>a. <i>L. m. mexicana</i></li> <li>b. <i>L. m. Amazonensis</i></li> <li>c. <i>L. M. pifanoi</i></li> </ul> </li> <li>iii. Leishmanibraziliensis complex <ul style="list-style-type: none"> <li>a. <i>L. B. braziliensis</i></li> <li>b. <i>L. B. guyanensis</i></li> <li>c. <i>L. B. panamensis</i></li> <li>d. <i>L. peruviana</i></li> </ul> </li> <li>iv. Leishmaniadonovani complex <ul style="list-style-type: none"> <li>a. <i>L. d. donovani</i></li> <li>b. <i>L. d. chagasi</i></li> <li>c. <i>L. infantum</i></li> </ul> </li> <li>v. The Trypanosome <ul style="list-style-type: none"> <li>a. Trypanosoma brucei complex <ul style="list-style-type: none"> <li>i. <i>T. b. Gambiense</i></li> <li>ii. <i>T. b. Rhodesiense</i></li> </ul> </li> </ul> </li> </ul> | 3 | <p>Monica Cheesbrough. District Laboratory Practice in Tropical countries. Part I -2nded updated. 1998. Tropical Health Technology. Cambridge</p> <p>Judith S. Heelan, Frances W. Ingersoll. Essentials of Human Parasitology. 2002. Delmar Thomson Learning</p> <p>Mohammed AwolAdem and WaqtolaCheneke. Medical Parasitology: Lecture note for medical laboratory technology students: upgraded lecture note serious. 2006</p> <p>Markell, Voge, Jhon. Medical Parasitology. 6th ed. 1986. W.b. Saunders company</p> <p>David L. Belding. Text book of Parasitology. 3rd ed. 1965</p> |
| 11 <sup>th</sup> -15 <sup>th</sup> | <p>Chapter VII – Apicomplexa</p> <p>The Plasmodium Species</p> <ul style="list-style-type: none"> <li>• General characteristics <ul style="list-style-type: none"> <li>▪ <i>Plasmodium falciparum</i></li> <li>▪ <i>Plasmodium vivax</i></li> <li>▪ <i>Plasmodium malariae</i></li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 4 | <p>Monica Cheesbrough. District Laboratory Practice in Tropical countries. Part I -2nded updated. 1998. Tropical Health Technology. Cambridge</p> <p>Judith S. Heelan, Frances W. Ingersoll. Essentials of Human Parasitology. 2002. Delmar Thomson Learning</p>                                                                                                                                                                                                                                                                                                                        |

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|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                  | <ul style="list-style-type: none"> <li>▪ <i>Plasmodium ovale</i></li> <li>• Drug resistance in malaria</li> </ul> <p>The Coccidia and related Protozoa</p> <ul style="list-style-type: none"> <li>• General characteristics <ul style="list-style-type: none"> <li>▪ Genus <i>Cryptosporidium</i></li> <li>▪ Genus <i>Isospora</i></li> <li>▪ Genus <i>Sarcocystis</i></li> <li>▪ Genus <i>Toxoplasma</i></li> <li>▪ Genus <i>Pneumocystis</i></li> </ul> </li> </ul> |  | <p>Mohammed AwolAdem and WaqtolaCheneke. Medical Parasitology: Lecture note for medical laboratory technology students: upgraded lecture note serious. 2006</p> <p>Markell, Voge, Jhon. Medical Parasitology. 6th ed. 1986. W.b. Saunders company</p> <p>David L. Belding. Text book of Parasitology. 3rd ed. 1965</p> |
| 16 <sup>th</sup> | Final Exam                                                                                                                                                                                                                                                                                                                                                                                                                                                            |  |                                                                                                                                                                                                                                                                                                                        |

**Course title: Pathology**

**Course code: Path2054**

**Course ECTS: 5 (135 hrs)**

- □ Lecture: 48 hours
- □ Tutorial: 9 hours
- □ Home study: 60 hours
- □ Assignment and presentation: 18 hours

Contact hours/ week:  $135-87=48$  hours/ 16 weeks= 3 hours

**Year/Semester Course is offered: Year II Semester I**

**Pre-requisite/co-requisite if any:** Physiology, and Biochemistry

**Course description:** This General Pathology course is designed to help B.Pharm students to better understand pharmacotherapy of both communicable and non communicable diseases which will prepare them to be competent providers of pharmaceutical service. Students will be able to explain the concepts of pathology and the pathophysiology of cellular injury, wound healing, hemodynamics of cardiovascular diseases, immunopathology, neoplasia, endocrine disorders, common infectious diseases and CNS disorders.

**Course objectives:**

- □ After completing this course, the student will be able to explain the pathophysiologic mechanisms of common disorders.

| Week | Lecture # | Topic                                                                                   | References |
|------|-----------|-----------------------------------------------------------------------------------------|------------|
| 1    | 1         | Chapter 1: Pathophysiology basics (2 hrs)                                               | 1 – 8      |
|      |           | • Definition of pathology, the structure of cells and how cells reproduce, age, and die |            |
|      |           | • Homeostasis                                                                           |            |
|      |           | • Diagnostic techniques in pathology                                                    |            |
|      |           | • Categories of the causes of diseases                                                  |            |
|      |           | • Causes, outcome, consequences of diseases                                             |            |
|      | 2         | Chapter 2: Cellular Reaction to Injury (3 hrs)                                          | 1 – 8      |
|      |           | • Definition of hyperplasia, hypertrophy, atrophy, & Metaplasia                         |            |
|      |           | • Reversible & irreversible forms of cell injury                                        |            |



|                 |    |                                                                                                                                                        |       |
|-----------------|----|--------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
|                 |    | <ul style="list-style-type: none"> <li>• Mechanisms of necrosis</li> </ul>                                                                             |       |
| 2               | 3  | <ul style="list-style-type: none"> <li>• Types of necrosis and their causes</li> </ul>                                                                 |       |
|                 |    | <b>Chapter 3: Inflammation (4 hrs)</b>                                                                                                                 | 1 – 8 |
|                 |    | <ul style="list-style-type: none"> <li>• Causes and processes of inflammation</li> </ul>                                                               |       |
|                 | 4  | <ul style="list-style-type: none"> <li>• Etiopathogeneses of granulomatous inflammations</li> </ul>                                                    |       |
| 3               | 5  | <ul style="list-style-type: none"> <li>• Acute and chronic inflammations</li> </ul>                                                                    |       |
|                 |    | <b>Chapter 4: Healing (3 hrs)</b>                                                                                                                      | 1 – 8 |
|                 |    | <ul style="list-style-type: none"> <li>• Processes and patterns of healing</li> </ul>                                                                  |       |
|                 |    | <ul style="list-style-type: none"> <li>• Factors that influence wound healing</li> </ul>                                                               |       |
|                 | 6  | <ul style="list-style-type: none"> <li>• Complications of wound healing</li> </ul>                                                                     |       |
|                 |    | <ul style="list-style-type: none"> <li>• Fracture healing</li> </ul>                                                                                   |       |
| 4               |    | <b>Chapter 5: Hemodynamic Disorders (5 hrs)</b>                                                                                                        | 1 – 8 |
|                 | 7  | <ul style="list-style-type: none"> <li>• Fluid balance</li> </ul>                                                                                      |       |
|                 |    | <ul style="list-style-type: none"> <li>• Cause and pathogenesis of clinical conditions like ischemia, infarction, thrombosis, embolism, DIC</li> </ul> |       |
|                 | 8  | <ul style="list-style-type: none"> <li>• Pathogenesis of edema of congestive heart failure, nephrotic syndrome, cirrhosis</li> </ul>                   |       |
| 5               | 9  | <ul style="list-style-type: none"> <li>• Types of shock and their pathogenesis, manifestations, and complications</li> </ul>                           |       |
|                 |    | <b>Chapter 6: Genetics Disease (4 hrs)</b>                                                                                                             | 1 – 8 |
|                 |    | <ul style="list-style-type: none"> <li>• Basis of genetic diseases</li> </ul>                                                                          |       |
|                 | 10 | <ul style="list-style-type: none"> <li>• Categories of genetic diseases</li> </ul>                                                                     |       |
|                 |    | <ul style="list-style-type: none"> <li>• Types of chromosomal disorders</li> </ul>                                                                     |       |
| 6               | 11 | <ul style="list-style-type: none"> <li>• Multifactorial disorders</li> </ul>                                                                           |       |
| <b>MID EXAM</b> |    |                                                                                                                                                        |       |
|                 |    | <b>Chapter 7: Immunoajj (6 hrs)</b>                                                                                                                    | 1 – 8 |
|                 | 12 | <ul style="list-style-type: none"> <li>• Hypersensitivity reaction</li> </ul>                                                                          |       |
| 7               | 13 | <ul style="list-style-type: none"> <li>• Etiologic factors in autoimmune disease</li> </ul>                                                            |       |
|                 | 14 | Immunodeficiency states                                                                                                                                |       |

|    |    |                                                                                                                                                                                                   |              |
|----|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
|    |    | <b>Chapter 8: Selected Infectious Diseases (6 hrs)</b>                                                                                                                                            | <b>1 – 8</b> |
|    |    | Etiology, pathogenesis and clinical features of typhoid fever, leishmaniasis, schistosomiasis and malaria, osteomyelitis                                                                          |              |
| 8  | 15 | Etiology, pathogenesis and clinical features of pneumonia, tuberculosis and leprosy                                                                                                               |              |
|    | 16 | Etiology, pathogenesis and clinical features of syphilis, bacterial meningitis and HIV/AIDS,                                                                                                      |              |
| 9  | 17 | <b>Chapter 9: Metabolic Diseases (4 hrs)</b>                                                                                                                                                      | <b>1 – 8</b> |
|    |    | <ul style="list-style-type: none"> <li>Thyroid diseases and adrenal disorders (definition, classification, diagnosis criteria, pathogenesis, clinical manifestation and complications)</li> </ul> |              |
|    | 18 | Diabetes mellitus (definition, classification, diagnosis criteria, pathogenesis, clinical manifestation and complications)                                                                        |              |
| 10 | 19 | <b>Chapter 10: Neoplasm (7hrs)</b>                                                                                                                                                                | <b>1 – 8</b> |
|    |    | <ul style="list-style-type: none"> <li>Causes of abnormal cell growth</li> </ul>                                                                                                                  |              |
|    |    | <ul style="list-style-type: none"> <li>Difference between neoplastic lesions and non-neoplastic ones</li> </ul>                                                                                   |              |
|    | 20 | <ul style="list-style-type: none"> <li>Benign versus malignant tumours</li> </ul>                                                                                                                 |              |
|    |    | <ul style="list-style-type: none"> <li>The warning signs of cancer</li> </ul>                                                                                                                     |              |
| 11 | 21 | <ul style="list-style-type: none"> <li>Methods and mechanisms of metastasis</li> </ul>                                                                                                            |              |
|    |    | <ul style="list-style-type: none"> <li>Classifications of cancer</li> </ul>                                                                                                                       |              |
|    | 22 | <ul style="list-style-type: none"> <li>Etiologic factors of carcinogenesis</li> </ul>                                                                                                             |              |
|    |    | <ul style="list-style-type: none"> <li>Clinical effects of neoplasms</li> </ul>                                                                                                                   |              |
| 12 | 23 | <b>Chapter 11: Diseases of CNS (4hrs)</b>                                                                                                                                                         | <b>1 – 8</b> |
|    |    | <ul style="list-style-type: none"> <li>Pathophysiology of degenerative disorders (Parkinsonism, Alzheimer, MS, ALS)</li> </ul>                                                                    |              |
|    |    | <ul style="list-style-type: none"> <li>Classification and pathogenesis of epilepsy</li> </ul>                                                                                                     |              |

|            |    |                                                                                                                                               |  |
|------------|----|-----------------------------------------------------------------------------------------------------------------------------------------------|--|
|            | 24 | <ul style="list-style-type: none"> <li>• Pathophysiology of psychiatric disorders (anxiety, depression, mania, OCD, schizophrenia)</li> </ul> |  |
|            |    | <ul style="list-style-type: none"> <li>• Acute alcohol intoxication and chronic alcoholism</li> </ul>                                         |  |
| FINAL EXAM |    |                                                                                                                                               |  |

### **Delivery mode/methodology:**

- □ Active learning methods (brain storming, discussion, etc), Lecture, group and individual presentation, assignment.

**Assessment mechanisms:** Continuous assessment & summative assessment

- Class attendance (%)
- Quiz (%)
- Assignments (%)
- Final Exam (40%)

### **Learning Materials**

- Handouts
- Audiovisual aids: Computer & LCD
- Textbooks

### **References:**

1. Cotran RS, Kumar V, Collins T. Robbins pathologic basis of diseases. Philadelphia, J.B. Saunders Company. Latest edition
2. Emanuel Rubin, and John L. Farber, Essential Pathology, Philadelphia, Latest edition
3. William Boyd; Textbook of Pathology, structure and Function in disease, Philadelphia, Latest edition
4. James E. Pointer; Alan B. Fletcher; Basic life support, California, Latest edition
5. F.B. Walter and M.S Israel; General Pathology, Churchill Livingstone Edinburgh and London, Latest edition
6. Macfarlane, Reid, callander, Illustrated Pathology, Churchill Livingstone, Latest edition
7. Muir's Textbook of Pathology Latest edition
8. Lecture note on General Pathology for Health Science Students, 2004

**Course Title: Human Physiology II**

**Course Code:** Phyl2055

**Module name: Biomedical Module I**

**Module code:** Biom-M1033

**Course ECTS:** 5

- Lecture: 48 hours
- Tutorial:16 hours
- Presentations: 15 hours
- Assignment:7.5 hours
- Assessment (continuous and final) 8 hours
- Independent study (alone or in groups) 40.5 hours

**Mode of delivery:** Parallel

**Pre-requisite if any:** Human Physiology I

**Course description:**

This module will give an overview of a range of physiological systems, including the homeostasis, the cell and cell membrane transport, composition of the body Fluid, physiology of blood, physiology of the nerve, physiology of the muscle, autonomic NS, cardiovascular physiology, respiratory physiology, renal physiology, gastro intestinal system, energy metabolism, endocrine system, male and female reproductive system, central nervous system and the special senses.

**Course objectives:**

At the end of the module the student should be able to:

- Explain the composition of and levels of organization of human body.
- Describe the basic physiological principle of the “internal environment” of the body.
- Explain basic principles of homeostasis and homeostatic regulatory mechanisms.
- Describe functional importance of different organ systems of human body and their integrated role in the maintenance of homeostasis.
- Appreciate the various physiological regulatory mechanisms of the body in the maintenance of homeostasis.
- Describe the various structures of the GIT, Secretory functions of GIT, Digestive and Absorptive functions and Pathophysiology of the GIT.
- Explain about Energy and Metabolism, Body Temperature Regulation, Feeding regulation and its abnormalities.

- Describe in detail the various endocrine glands of the body and various hormones secreted, the pituitary gland function and malfunction, the thyroid gland function and malfunction, the adrenal gland hormonal abnormalities on the body functions, Calcium homeostasis and Glucose homeostasis.
- Describe the various structures of the male and female reproductive system, Pregnancy and contraception, Parturition and lactation.
- Explain the nervous mechanisms, which govern the regulation and homeostasis of the principal physiological systems.
- Describe in detail the general organization of the NS, Sensory and Motor functions of the NS, Higher motor centers - Functions and lesions.
- Describe the various special senses of the human body

### Course learning and teaching methods

- Active learning methods (brain storming, buzz group, discussion, etc), Lecture, group and individual presentation, assignment, project work, and laboratory work.

### Assessment techniques:

- Assignments and Seminars (20%)
- Quizzes (30%)
- Final-exam (50%)

### Teachers and Students Role

### References:

5. Guyton A. C 1995-2006. Textbook of Medical physiology. 9<sup>th</sup> -11<sup>th</sup> editions
6. Ganong WF 1993-2006. Review of Medical physiology, 18<sup>th</sup> -22<sup>nd</sup> editions
7. Tortora G. J 1993. Principles of Anatomy and physiology. 7<sup>th</sup> edition.
8. Salah Abu-Sitta. Handouts containing different chapters (eight separate handouts.)

**Course Schedule:** By contact time, contents/topics & reading/reference materials for each topic (physiology II)

| Week            | Contact hrs | Topic/sub-topic/chapter                                                                                                                                                                                                                            | Reading materials                      |
|-----------------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| 1 <sup>st</sup> | 3           | <b><u>Physiology of Digestive System(8 Hrs)</u></b> <ul style="list-style-type: none"> <li>- General overview of the GIT</li> <li>- Functional Structures of the GIT</li> <li>- Secretary function of GIT</li> <li>- Salivary Secretion</li> </ul> | Guyton(p 791-799)<br>Ganong(p 467-477) |

|                 |   |                                                                                                                                                                                                                                                                                                     |                                                                                                       |
|-----------------|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
|                 |   | - Gastric secretion                                                                                                                                                                                                                                                                                 |                                                                                                       |
| 2 <sup>nd</sup> | 3 | <ul style="list-style-type: none"> <li>- Pancreatic secretion</li> <li>- Intestinal secretion</li> <li>- Bile secretion, jaundice</li> </ul>                                                                                                                                                        | Guyton(p 799-800)<br>Ganong(p 479-508)<br>Guyton(p 806)                                               |
| 3 <sup>rd</sup> | 3 | <ul style="list-style-type: none"> <li>- Digestive &amp; absorptive function of GIT</li> <li>- Pathophysiology of GIT</li> </ul> <b><u>Energy and Metabolism(4 Hrs)</u></b> <ul style="list-style-type: none"> <li>- Introduction to Energy and Metabolism</li> <li>- The metabolic rate</li> </ul> | Guyton(p 800-817)<br>Ganong(p 479-508)<br>Guyton(p 819-824)<br>Guyton(p 830-839)<br>Ganong(p 279-285) |
| 4 <sup>th</sup> | 3 | <ul style="list-style-type: none"> <li>- Energy balance</li> <li>- Feeding and its regulation</li> <li>- Body temperature regulation</li> <li>- FEVER</li> <li>- Obesity and the balanced diet</li> </ul>                                                                                           | Guyton(p 865-878)<br>Ganong(p 279-285)                                                                |
| 5 <sup>th</sup> | 3 | <b><u>Physiology of Endocrine Glands(12 Hrs)</u></b> <ul style="list-style-type: none"> <li>- Introduction to Endocrine physiology</li> <li>- Mechanism of action of hormones</li> <li>- Hypothalamus-pituitary relationship</li> </ul>                                                             | Guyton(p 905-916)<br>Ganong(p 918-921)                                                                |
| 6 <sup>th</sup> | 3 | <ul style="list-style-type: none"> <li>- The pituitary gland function and malfunction</li> <li>- The thyroid gland function and malfunction</li> </ul>                                                                                                                                              | Guyton(p 922-942)<br>Ganong(p 317-328)<br>Ganong(p 396-409)                                           |
| 7 <sup>th</sup> | 3 | <ul style="list-style-type: none"> <li>- Adrenal medulla: catecholamine function and malfunction</li> <li>- Adrenal cortex: function and malfunction</li> </ul>                                                                                                                                     | Guyton(p 944-959)<br>Ganong(p 356-380)                                                                |
| 8 <sup>th</sup> | 3 | Calcium homeostasis <ul style="list-style-type: none"> <li>- Bone formation and growth</li> <li>- Vitamin D<sub>3</sub>: synthesis, function and regulation</li> <li>- Parathyroid gland function and dysfunction</li> </ul> Glucose homeostasis                                                    | Guyton(p 978-992)<br>Ganong(p 382-395)<br>Guyton(p 961-976)<br>Ganong(p 333-353)                      |

|                 |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                    |
|-----------------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
|                 |  | <ul style="list-style-type: none"> <li>- Pancreas: endocrine and exocrine function of pancreas</li> <li>- Insulin: synthesis, function, mechanism of action</li> <li>- Glucagon, function, mechanism of action</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                    |
| 9 <sup>th</sup> |  | <p><b><u>Reproductive System(8 Hrs)</u></b></p> <ul style="list-style-type: none"> <li>- Introduction to Male Reproductive system</li> <li>- Functional anatomy of male reproductive system</li> <li>- Primary &amp; secondary sexual characteristics</li> <li>- The male sexual act</li> <li>- Abnormalities of male sexual function</li> <li>- Functional anatomy of female reproductive system</li> <li>- Menstruation cycle</li> <li>- Female sexual act</li> <li>- Fertility &amp; birth control</li> <li>- Abnormalities &amp; secretion by ovaries</li> <li>- Fertilization</li> <li>- Implantation</li> <li>- Physiological changes during pregnancy</li> <li>- Parturition</li> <li>- Lactation</li> <li>- Fetal physiology</li> <li>- Physiological adjustments at birth</li> <li>- Nutrition of the neonate</li> <li>- Reproductive and hormonal function of the male testis</li> <li>- Spermatogenesis</li> </ul> | <p>Guyton(p 996-1001)</p> <p>Ganong(p 441-451)</p> |

|                  |   |                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                |
|------------------|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| 10 <sup>th</sup> | 3 | <ul style="list-style-type: none"> <li>- Function of seminal vesicles</li> <li>- Function of the prostate gland</li> <li>- The male sexual act</li> <li>- Androgens</li> <li>- Abnormalities of male sex</li> <li>- Reproductive function of female</li> </ul>                                                                                                                                              | Guyton(p 996-1011)<br>Ganong(p 441-433)                                        |
| 11 <sup>th</sup> | 3 | <ul style="list-style-type: none"> <li>- The menstrual cycle</li> <li>- Pregnancy and contraception</li> <li>- Parturition and lactation</li> </ul> <p><b><u>Physiology of Nervous System</u></b><br/> <b><u>(Neurophysiology)(12 Hrs)</u></b></p> <ul style="list-style-type: none"> <li>- General overview and General organization of the NS</li> <li>- General tissue; neurons and neuralgia</li> </ul> | Guyton(p 1011-1034)<br>Ganong(p 433-451)<br>Guyton(p 555-559)<br>Ganong(p 129) |
| 12 <sup>th</sup> | 3 | <ul style="list-style-type: none"> <li>- Generation and transmission of nerve impulse</li> <li>- Neurotransmitters</li> <li>- Sensory function of the NS</li> <li>- Sensory receptors, classification</li> <li>- Somatic sensation and their pathways</li> </ul>                                                                                                                                            | Guyton(p 559-583)<br>Ganong(p 138-148)                                         |
| 13 <sup>th</sup> | 3 | <ul style="list-style-type: none"> <li>- Motor function of the NS</li> <li>- Reflexes; reflex arc, examples</li> <li>- Higher motor centers</li> <li>- Cerebral cortex</li> <li>- Basal ganglia function and disease</li> </ul>                                                                                                                                                                             | Guyton(p 673-684)<br>Ganong(p 129-137)<br>Guyton(p 685-697)                    |
| 14 <sup>th</sup> | 3 | <ul style="list-style-type: none"> <li>- Hypothalamus function and disease</li> <li>- Thalamus function and disease</li> <li>- Cerebellum function and disease</li> <li>- The brain stem; reticular formation</li> <li>- Pyramidal and extra pyramidal tracts, lesion</li> </ul>                                                                                                                            | Guytonp 698-713)<br>Ganong(p 202-217)                                          |
| 15 <sup>th</sup> | 3 | <ul style="list-style-type: none"> <li>- Limbic system</li> <li>- Reticular activating system</li> </ul>                                                                                                                                                                                                                                                                                                    | Guyton(p 728-738)<br>Ganong(p 192-196)                                         |



|                  |   |                                                                                                                                                                                                                                                                                                                                                     |                                                   |
|------------------|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
|                  |   | <ul style="list-style-type: none"> <li>- Brain electrical activity: EEG and sleep</li> <li>- Cerebral cortex, memory, speech, and aphasia</li> </ul> <p><b><u>Physiology of the Special senses (4 Hrs)</u></b></p> <ul style="list-style-type: none"> <li>- Introduction to Physiology of the Special senses</li> <li>- Visual sensation</li> </ul> | <p>Guyton(p 613-649)</p> <p>Ganong(p 148-168)</p> |
| 16 <sup>th</sup> | 3 | <ul style="list-style-type: none"> <li>- Auditory sensation</li> <li>- Olfactory sensation</li> <li>- Gustatory sensation</li> <li>- Physiology of equilibrium</li> </ul>                                                                                                                                                                           | <p>Guyton(p 651-668)</p> <p>Ganong(p 171-188)</p> |

## **Module 06: Biostatistics and Epidemiology**

**Module name:** Biostatistics and Epidemiology module

**Module category:** Basic

**Module code:** PUBH2063

**Module Number:** 06

**Module weight in ECTS:** 6 ECTS

**Courses:**

| <b>Course Name</b> | <b>Course Code</b> | <b>ECTS</b> |
|--------------------|--------------------|-------------|
| Biostatistics      | PUBH2061           | 3           |
| Epidemiology       | PUBH2062           | 3           |

**Module description:** This module will introduce students the principles and concept of Biostatistics and epidemiology.

**Module competency:** Involve in research and public health promotion

**Mode of delivery:** Block

**Module objectives:** At the end of this module students will be able to:

- Analyze data using various statistical techniques and soft wares
- Understand the principles of Epidemiology and biostatistics

### **Module learning teaching methods**

#### **Learning activities and teaching methods**

##### **A. Learning Activities**

Students are expected to undertake among others the following major learning activities:

- Engage in learning by doing (independent study, group assignments, presentation, report writing, and etc...)
- Participation and note takings during class lectures and debates and discussions;
- Analysis, summarization and presentations of chapter/article, motions on selected issues;

##### **B. Teaching Methods**

- The course facilitator is expected to lecture for introducing concepts and topics, and give
- references, facilitate discussions, ask questions, give and correct assessments,
- Arrange and facilitate seminar sessions by inviting professionals for public lectures or debates on subject related issues.

**Course title: Biostatistics**

**Course Code:** PUBH2061

**Module name: Biostatistics and Epidemiology module**

**Module code:** PUBH2063

**Course ECTS:** 3 ECTS (81 hrs)

- Lecture:- 32 hours
- Tutorial:- 12 hours
- Home study : 30 hours
- Assessment : 7 hours

**Year/Semester Course is offered: Year II Semester II**

**Contact hours/ week:**

**Pre-requisite:**

**Course Description:**

**Course Objectives:**

At the end of this course students should be able to:

- Describe the different types and methods of data collection and identify advantages and limitations of the different methods
- Outline the steps in designing a questionnaire and identify the different interviewing techniques
- Describe the different methods of data organization and summarization and identify the advantage and disadvantages of the different methods
- Describe the different measures of mortality and fertility
- Compute probability of an event and composite events
- Identify type of events
- Describe commonly used probability distributions of discrete and continuous random variables
- Identify the different sampling methods
- Identify the different estimation techniques in one and two samples situation
- Estimate sample size for cross-sectional study
- Do test of hypothesis on means and proportions in one and two sample situations

**Course mode of delivery:** Parallel

**Course learning and teaching methods**

**Assessment techniques:**

**Teachers and Students Role**

**References:**

1. Daniel, W.W., 1991. Biostatistics: a foundation for analysis in health Sciences, 5<sup>th</sup> ed. John Willy & Sons, New York
2. Getu Degu and Fasil Tessema, 2003. Biostatistics for Health Science Students. Lecture Note Series. The Carter Center
3. Douglas G. Altman, 1991. Practical Statistics for Medical Research. Chapman & Hall
4. Bernard Rosner, 1995. Fundamentals of Biostatistics. 4<sup>th</sup> ed. Duxbury Press
5. Theodore Colton, 1974. Statistics in Medicine. Little, Brown and Company
6. Betty R. Kirkwood, 1988. Essentials of Medical Statistics. Blackwell Science ltd
7. Richard D. Remington, M. Anthony Schork, 1985. Statistics with Applications to Biological and Health Sciences, 2<sup>nd</sup> ed. Prentice Hall, New Jersey, USA
8. Abramson J. H., 1990. Survey Methods in Community Medicine. Epidemiological Studies Programme Evaluation Clinical Trials. 4<sup>th</sup> ed. Churchill Livingstone.
9. William G. Cochran, 1977. Sampling Techniques. 3<sup>rd</sup> ed. John Willy & Sons Inc.
10. Lwanga, S.K. and Lemeshow, S., 1991. Sample size determination in health studies. A practical Manual. World Health Organization, Geneva

**Contents:**

1. Introduction
2. Methods of data collection
  - 2.1. Data types and measurement scales
  - 2.2. Data collection methods
  - 2.3. Questionnaire design and interviewing techniques
3. Methods of data processing, organization, presentation and summarization
  - 3.1. Tables and diagrams
  - 3.2. Measures of central tendency
  - 3.3. Measures of variation
4. Demographic statistics
5. Introduction to probability
  - 5.1 Rules of probability and types of events
  - 5.2 Probability distributions : binomial and normal distributions
6. Introduction to Sampling
  - 6.1 Non-probability sampling techniques
  - 6.2 Probability sampling techniques

7. Estimation techniques – point and interval estimation on one and two sample situation of means and proportions and sample size estimation for cross sectional study
8. Test of hypothesis: Type I and Type II errors, Power of the test, Critical and P-value methods, test on means and proportions in one and two sample situation

## **Epidemiology course syllabus**

**Course title: Epidemiology**

**Course Code:** PUBH2062

**Module name: Biostatistics and Epidemiology module**

**Module code:** PUBH2063

**Course ECTS:** 3 ECTS (81 hrs)

- Lecture:- 32 hours
- Tutorial:- 12 hours
- Home study : 30 hours
- Assessment : 7 hours

**Year/Semester Course is offered: Year II Semester II**

**Contact hours/ week:**

**Pre-requisite:**

**Course Description:**

This course is designed to equip students with the basic concepts of epidemiology (definition of epidemiology), communicable disease epidemiology, measures of disease occurrence, establishment of disease causation, epidemiological study designs, outbreak investigation and management, screening in disease control and epidemiological surveillance.

**Course Objectives:**

At the end of the course the student will be able to:

- Understand the principles of Epidemiology
- Describe concepts of disease causation
- Calculate the measures of disease and death
- Understand types of study design
- Investigate and control outbreaks and epidemics
- Describe the purpose and types of surveillance
- Understand the factors that affect validity of studies

**Course mode of delivery: Block**

**Course learning and teaching methods**

Brain storming, buzz group, discussion, Lecture, group and individual presentation, assignment

**Assessment techniques:**

- Class participation, quizzes, assignment, [50%], and
- Final exam (50%)

## Teachers and Students Role

### References:

- Charles H. Hennekens, DrPH. Julie E. Buring, Sc D. Epidemiology in medicine. Lippincott Williams and Wilkins, USA.
- Kifle Wolde Michael, Yigzaw Kebede, Kidist Lulu. Epidemiology for health science students: Lecture Note Series.
- Lilienfield, MA. Lilienfield ED. *Foundations of epidemiology*, 1980, Oxford University Press, New York.
- Zein Ahmed Zein and H. Kloos. *The Ecology of Health and Disease in Ethiopia*, 1993.

**Course Schedule:** By contact time, contents/topics and reading/reference materials for each topic

| Week | Contact hrs/week | Topic/subtopic chapter                                                                                                                                                                                                                                                                                                                                            | Reading materials                                                                                                                                                                                                                                                                                                                                                                   |
|------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | 3                | Introduction to Epidemiology <ul style="list-style-type: none"><li>• Definition</li><li>• History of Epidemiology</li><li>• Use/applications of Epidemiology</li><li>• Scope of epidemiology</li><li>• Basic assumptions of epidemiology</li><li>• Theories disease causation</li><li>• Levels of disease occurrence</li><li>• Branches of epidemiology</li></ul> | Charles H. Hennekens, DrPH. Julie E. Buring, Sc D. Epidemiology in medicine. In: definition and background and design strategies in epidemiologic research. Lippincott Williams and Wilkins, USA. 1987;p(1-16)<br><br>Kifle Wolde Michael, Yigzaw Kebede, Kidist Lulu. Epidemiology for health science students: Lecture Note Series. In: chapter 2 and 3. Ethiopia. 2003; p(10-28) |
| 2    | 3                | Natural history of disease and levels of prevention <ul style="list-style-type: none"><li>• natural history of disease</li></ul>                                                                                                                                                                                                                                  | Kifle Wolde Michael, Yigzaw Kebede, Kidist Lulu. Epidemiology for health science students: Lecture Note Series. In: chapter 4. Ethiopia. 2003; p(29-38)                                                                                                                                                                                                                             |

|         |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                         |
|---------|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
|         |   | <ul style="list-style-type: none"> <li>• stages in the natural history disease</li> <li>• levels of disease prevention</li> <li>• applications to common diseases</li> </ul>                                                                                                                                                                                                                                                                                                   |                                                                                                                                                         |
| 3       | 3 | <p>The infectious disease cycle</p> <ul style="list-style-type: none"> <li>• agent</li> <li>• reservoir</li> <li>• portal of exit</li> <li>• modes of transmission</li> <li>• portal of entry</li> <li>• host</li> <li>• spread of disease through person to person transmission</li> <li>• infection vs. disease</li> <li>• time course of an infectious disease</li> <li>• carries and their role in disease transmission</li> <li>• individual and herd immunity</li> </ul> | Kifle Wolde Michael, Yigzaw Kebede, Kidist Lulu. Epidemiology for health science students: Lecture Note Series. In: chapter 5. Ethiopia. 2003; p(39-48) |
| 4 and 5 | 6 | <p>Basic measurement in epidemiology</p> <ul style="list-style-type: none"> <li>• Number , ratio, proportion , and rate</li> <li>• Measures of morbidity -incidence and prevalence</li> </ul>                                                                                                                                                                                                                                                                                  | Kifle Wolde Michael, Yigzaw Kebede, Kidist Lulu. Epidemiology for health science students: Lecture Note Series. In: chapter 7. Ethiopia. 2003; p(57-77) |



|           |   |                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                            |
|-----------|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|           |   | <ul style="list-style-type: none"> <li>Measures of mortality               <ul style="list-style-type: none"> <li>-crude vs. specific rates</li> <li>- Standardization of rates</li> </ul> </li> </ul>                   |                                                                                                                                                                                                                                                                                                                                                            |
| 6         | 3 | Source of epidemiologic data <ul style="list-style-type: none"> <li>Census</li> <li>Vital records</li> <li>Data from health institutions</li> <li>Data from morbidity surveys</li> <li>Other sources</li> </ul>          | Kifle Wolde Michael, Yigzaw Kebede, Kidist Lulu. Epidemiology for health science students: Lecture Note Series. In: chapter 6. Ethiopia. 2003; p(49-56)                                                                                                                                                                                                    |
| 7         | 3 | Public health surveillance <ul style="list-style-type: none"> <li>Definition</li> <li>Purpose of surveillance</li> <li>Types of surveillance</li> <li>Activities in surveillance</li> <li>Modifiable diseases</li> </ul> | Kifle Wolde Michael, Yigzaw Kebede, Kidist Lulu. Epidemiology for health science students: Lecture Note Series. In: chapter 12. Ethiopia. 2003; p(153-169)                                                                                                                                                                                                 |
| 8 and 9   | 6 | Descriptive study designs <ul style="list-style-type: none"> <li>Purpose of descriptive studies</li> <li>Types of descriptive study designs</li> </ul>                                                                   | Kifle Wolde Michael, Yigzaw Kebede, Kidist Lulu. Epidemiology for health science students: Lecture Note Series. In: chapter 8. Ethiopia. 2003; p(78-90)<br><br>Charles H. Hennekens, DrPH. Julie E. Buring, Sc D. Epidemiology in medicine. In: types of epidemiologic studies: descriptive studies. Lippincott Williams and Wilkins, USA. 1987;p(101-132) |
| 10 and 11 | 6 | Analytical epidemiology <ul style="list-style-type: none"> <li>Purpose of analytical epidemiology</li> </ul>                                                                                                             | Kifle Wolde Michael, Yigzaw Kebede, Kidist Lulu. Epidemiology for health science students: Lecture Note Series. In: chapter 9. Ethiopia. 2003; p(91-106)                                                                                                                                                                                                   |

|           |   |                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                              |
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|           |   | <ul style="list-style-type: none"> <li>• Observational analytic study designs vs. experimental analytical studies</li> <li>• Case control studies</li> <li>• Cohort studies</li> <li>• Cross sectional studies</li> <li>• Intervention studies</li> <li>• Types of intervention studies</li> <li>• Analysis and interpretation</li> </ul> | <p>Charles H. Hennekens, DrPH. Julie E. Buring, Sc D. Epidemiology in medicine. In: types of epidemiologic studies: case control, cohort and interventional studies. Lippincott Williams and Wilkins, USA. 1987;p(133- 215)</p>                                                                                                                              |
| 12 and 13 | 6 | Measures of strength of association                                                                                                                                                                                                                                                                                                       | <p>Kifle Wolde Michael, Yigzaw Kebede, Kidist Lulu. Epidemiology for health science students: Lecture Note Series. In: chapter 9. Ethiopia. 2003; p(107-118)</p> <p>Charles H. Hennekens, DrPH. Julie E. Buring, Sc D. Epidemiology in medicine. In: measures of disease frequency and association. Lippincott Williams and Wilkins, USA. 1987;p(54-100)</p> |
| 14        | 3 | <p>Analysis of cause effect relationship</p> <ul style="list-style-type: none"> <li>• Validity of studies</li> <li>• Role of chance</li> <li>• Role of bias</li> <li>• Role of confounding factors</li> <li>• Evaluation of overall evidence for a cause-effect relationship</li> </ul>                                                   | <p>Kifle Wolde Michael, Yigzaw Kebede, Kidist Lulu. Epidemiology for health science students: Lecture Note Series. In: chapter 10. Ethiopia. 2003; p(119-133)</p> <p>Charles H. Hennekens, DrPH. Julie E. Buring, Sc D. Epidemiology in medicine. In: analysis of cause effect relationship. Lippincott Williams and Wilkins, USA. 1987;p(30-53)</p>         |

|    |   |                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                            |
|----|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15 | 3 | <p>Screening in disease control</p> <ul style="list-style-type: none"> <li>• Definition</li> <li>• Diseases appropriate for screening program</li> <li>• Criteria for establishing screening program</li> <li>• Validity and reliability of tests</li> <li>• Sensitivity and specificity</li> <li>• Predictive value of a test</li> </ul>           | <p>Kifle Wolde Michael, Yigzaw Kebede, Kidist Lulu. Epidemiology for health science students: Lecture Note Series. In: chapter 13. Ethiopia. 2003; p(170-179)</p> <p>Charles H. Hennekens, DrPH. Julie E. Buring, Sc D. Epidemiology in medicine. In: screening. Lippincott Williams and Wilkins, USA. 1987;p(327-350)</p> |
| 16 | 3 | <p>Investigation of an epidemic</p> <ul style="list-style-type: none"> <li>• Definition of terms (endemic, hypo-endemic, hyper-endemic, holo-endemic, cluster of cases, outbreak, epidemic, pandemic)</li> <li>• Types of epidemics</li> <li>• Steps in epidemic investigation</li> <li>• Prevention and control strategies of epidemics</li> </ul> | <p>Kifle Wolde Michael, Yigzaw Kebede, Kidist Lulu. Epidemiology for health science students: Lecture Note Series. In: chapter 11. Ethiopia. 2003; p(134-152)</p>                                                                                                                                                          |

**Module Name:** Introductory Pharmacy Module

**Module Category:** Core

**Module Code:** Phar-M2071

**Module Number:** 07

**Module Weight:** 4 ECTS

**Courses:**

| Course name                 | Course code | ECTS |
|-----------------------------|-------------|------|
| Introduction to pharmacy    | Phar2071    | 2    |
| Pharmaceutical calculations | Phar2072    | 2    |

**Module description:**

The module covers evolution and scope of pharmacy; pharmaceutical terminologies; and pharmacist role in the health care delivery. The module also introduces students with some fundamentals of measurement and calculations, calculation of doses and formulas, dilution and concentration and isotonic, buffer and electrolyte solutions. The field study will also provide an opportunity to the student to have practical exposure to the various pharmacy settings.

**Module objective:**

The aim of the module is to introduce the students with the pharmacy profession and its evolutionary development in the context of both the local and the global sense and to familiarize students with basic calculations related with pharmacy practices.

**Module competencies:**

Upon a successful completion of this module, students will be able to calculate doses for different groups of patients, quantify the ingredients required in dosage forms preparation & dispensing, prepare different strength of solutions and interpret prescriptions.

**Mode of delivery (Parallel/Block):** Parallel

**Module teaching/learning method:**

- Active participation during class lectures and excursions
- Engage in learning by doing
- The course instructor is expected to introduce concepts and topics and give references, facilitate discussions, ask questions, correct assignments
- Arrange and facilitate excursions

**Module mode of assessment:**

- Quizzes
- Report writing
- Assignments
- Seminar presentation
- Final exam

**Course Name:** Introduction to pharmacy  
**Course code:** Phar2071  
**Module Name:** Introductory Pharmacy Module  
**Module Code:** Phar-M2071  
**Course ECTS:** 2 ECTS  
**Year/Semester Course is offered:** Year II Semester I

**Course prerequisite/s:** None

**Course description:**

The theoretical aspect of the course covers overview about Ethiopia's health care system; history and evolving scope of pharmacy practice; education pathway in pharmacy; pharmaceutical terminologies; and practice areas in pharmacy. The field study provides an opportunity to the student to have practical exposure to the various pharmacy settings.

**Course objective:**

After completion of this course students will be able to:

- Describe the structure of the Ethiopian health care system
- Discuss the history of pharmacy
- Read and interpret commonly used pharmaceutical and medical terminologies
- Identify the different practice areas in the profession of pharmacy

**Course mode of delivery:** Parallel

**Course learning and teaching methods**

- Illustrated lecture
- Individual and group exercises and assignments presentation
- Visits to various pharmacy settings (15 hrs: 3 hrs x 5 weeks)
- Alumni pharmacists' experience sharing in a class

**Assessment techniques:**

- Tests: 15
- Quizzes: 10%
- Presentation: 10%
- Assignments: 10%
- Report writing: 15%
- Final exam: 40%

**Teachers' and students' role**

**Roles of Instructors**

The instructor will be expected to:

- Be a facilitator (introduce the subject; give guidance, moderate discussions, etc.)
- Read and provide feedback to students' assignment submissions on time;
- Prepare his/her lessons and deliver lectures;

- Provide available and necessary reference materials;
- Encourage active participation of students in the teaching learning process;
- Assist students with learning difficulties;

### **Roles of Students**

Students are expected to:

- Have a minimum of 85% class attendance
- Read all reading assignments in advance
- Submit all group and/or individual assignments on due date
- Take all continuous assessments as scheduled.

### **References:**

#### **Required readings (Text)**

1. Remington's Pharmaceutical Science, 21<sup>st</sup> ed., Lippincott Williams & Wilkins, Pennsylvania, 2006.

#### **Recommended readings**

2. Stone, P. and Curtis, S. J., Pharmacy Practice, 2<sup>nd</sup> ed., Farrand Press, London, 1995.
3. Sonnedeker, G., Kremer and Urdag's History of Pharmacy, Lippincott, Philadelphia, 1976.
4. Winfield, A. J. and Richards, R. M. E. (edrs.), Pharmaceutical Practice, 2nd ed., Churchill Livingstone, London, 1998.
5. Michael L. P., Pharmacy, An Introduction to the Profession, The American Pharmacists Association, Washington DC, 2003.
6. Whalley, B. J., Fletcher K.E., Weston S.E., Howard R.L. and Rawlinson C.F., Foundation in Pharmacy Practice, Pharmaceutical Press, London, 2008.
7. Sneader W. Drug discovery: a history. John Wiley & Sons; 2005 Jun 23.
8. International Pharmaceutical Federation – FIP. Transforming Our Workforce. The Hague, The Netherlands: International Pharmaceutical Federation, 2016
9. Gall, D., Bates, I. and Bruno, A., 2012. FIP Global Pharmacy Workforce Report 2012.
10. MoH. Ethiopian Health Sector Transformation Plan (2015-2020)
11. Gosselin MC, Robbins J, Cupolo J. Inside pharmacy: The anatomy of a profession. CRC Press; 1998 Oct 2.
12. Pfizer Inc. Full preparation: the Pfizer guide to careers in pharmacy. Pfizer Pharmaceuticals Group; 2001.
13. Kelly WN. Pharmacy: what it is and how it works. CRC press; 2011 Jul 26.
14. Ethiopian Pharmaceutical Association (EPA) 1974 – 2015. 40th Anniversary Special Publication

### **Course schedule\***

| <b>Week</b> | <b>Contact Hours</b> | <b>Topic/sub-topic/chapter/Assessments/Assignments</b>                                                       |
|-------------|----------------------|--------------------------------------------------------------------------------------------------------------|
| 1           | 1                    | 1. Introduction to the Ethiopian Healthcare System<br>1.1. Health status<br>1.2. Health policies, strategies |

|    |   |                                                                                                                                                                                                                   |
|----|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |   | 1.3. Organization and governance<br>1.4. Financing<br>1.5. Physical and Human resources<br>1.6. Provision of services<br>1.7. MoH, EFDA, EPHI, EPSA                                                               |
| 2  | 1 | 2. The history of Pharmacy and its evolving scope of practice<br>2.1. A Brief History of Pharmacy (pre-historic pharmacy, antiquity, the middle ages, renaissance, discoveries and background of modern pharmacy) |
| 3  | 1 | 2.2. History of drugs and dosage forms<br>2.3. Pharmacy in Ethiopia: development, types of pharmacy settings                                                                                                      |
| 4  | 1 | 3. Education pathways in pharmacy<br>3.1. The Early Years<br>3.2. The Five-Year Versus the Six-Year Pharmacy Entry-Level Degree<br>3.3. Pharmacy education capacity and training institution distribution         |
| 5  | 1 | 3.4. Accreditation Standards<br>3.5. Continuing Education Programs<br>3.6. Advanced study opportunities                                                                                                           |
| 6  | 1 | 4. Commonly used pharmaceutical and medical terminologies<br>4.1. Latin terms and abbreviations, types of dispensed pharmaceutical preparations                                                                   |
| 7  | 1 | 5. Practice Areas in Pharmacy<br>5.1. Academic Pharmacist<br>5.2. Community Pharmacist                                                                                                                            |
| 8  | 1 | 5.3. Compounding Pharmacist<br>5.4. Critical Care Pharmacist                                                                                                                                                      |
| 9  | 1 | 5.5. Drug Information Specialist<br>5.6. Home Care Pharmacist                                                                                                                                                     |
| 10 | 1 | 5.7. Hospital Staff Pharmacist<br>5.8. Long-term Care Pharmacist<br>5.9. Managed Care Pharmacist                                                                                                                  |
| 11 | 1 | 5.10. Military Pharmacist<br>5.11. Nutrition Support Pharmacist<br>5.12. Operating Room Pharmacist<br>5.13. Infectious Disease Pharmacist                                                                         |
| 12 | 1 | 5.14. Pediatric Pharmacist<br>5.15. Industry-Based Pharmacist<br>5.16. Nuclear Pharmacist<br>5.17. Oncology Pharmacist                                                                                            |
| 13 | 1 | 6. Pharmacists in Non-traditional Settings<br>6.1. Pharmacy Benefit Manager                                                                                                                                       |
| 14 | 1 | 6.2. Poison Control Pharmacist<br>6.3. Primary Care Pharmacist<br>6.4. Regulatory Pharmacist<br>6.5. Veterinary Pharmacist                                                                                        |
| 16 |   | <b>FINAL EXAM</b>                                                                                                                                                                                                 |

**Course Name:** Pharmaceutical Calculations

**Course code:** Phar2072  
**Module Name:** Introductory Pharmacy Module  
**Module Code:** 07  
**Course ECTS:** 2

**Totally required hours for the course: 54hrs**

**Lecture hours:** 13

**Study hours:** 20

**Group work:** 14

**Project work:** 0

**Presentation(s):** 0

**Tutorial:** 7

**Assessment:**

**Year/Semester Course is offered:** Year II Semester I

**Course prerequisite/s:** None

**Course Description:**

This course is designed to familiarize students with the basic calculations related with pharmacy practices. The course introduces students with some fundamentals of measurement and calculations, calculation of doses and formulas, dilution and concentration and isotonic, buffer and electrolyte solutions, percentage calculations, calculations involving parenteral admixtures and radiopharmaceuticals.

**Course Objectives:**

After completion of this course students will be able to:

- Understand the basic concepts about balance sensitivity, significant figures, accuracy and percentage of errors, measurement of volume and weight, aliquot method of weighing and measuring, density, specific gravity, specific volume which are important in pharmacy practices.
- Understand the basic calculations in percentage preparations, dosage conversions, formula reduction and enlargement
- Understand the basic calculations of dilution and concentration
- Understand the basic calculations in isotonic, buffer and electrolyte solutions
- Understand the basic calculations involving parenteral admixtures and radiopharmaceuticals

**Course mode of delivery:** Parallel

**Course learning and teaching methods:**



- Active participation during class lectures and excursions
- Engage in learning by doing
- The course instructor is expected to introduce concepts and topics and give references, facilitate discussions, ask questions, correct assignments
- Arrange and facilitate excursions

#### **Assessment techniques:**

- Assignments: 15%
- Tests: 30%
- Quizzes: 15%
- Final exam: 40%

#### **Teachers' and students' role**

##### **Roles of Instructors**

##### **The instructor will be expected to:**

- Be a facilitator (introduce the subject; give guidance, moderate discussions, etc.)
- Read and comment assignments & exercises of students on time;
- Prepare his/her lessons and deliver lectures;
- Provide available and necessary reference materials;
- Encourage active participation of students in the teaching learning process;
- Assist students with learning difficulties

##### **Roles of Students**

##### **Students are expected to:**

- Engage in learning by doing (independent study, group works/exercises, etc.)
- Be active learners (participate effectively in group assignments, class activities/exercises, etc.);
- Attend classes regularly

#### **References:**

##### **Required readings (Text)**

1. Ansel, H. C. Pharmaceutical calculations, 13th ed., Lippincott Williams & Wilkins, 2010.

##### **Recommended readings**

2. Joel L. Zatz and M. G. Teixeira. Pharmaceutical calculations, 5<sup>th</sup> ed., John Wiley & Sons, Inc., New Jersey, 2017.

3. Mansoor A Kahn, and Indra K Reddy. Pharmaceutical and Clinical calculation 2<sup>nd</sup> edition. CRC Press, New York, 2000.

### Course Schedule

| Week | Contact Hours | Topic/sub-topic/chapter/Assessments/Assignments                                                                                                                                                                                                                                                                                                                                                                                            | Reading Materials                      |
|------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| 1    | 2             | <ul style="list-style-type: none"> <li>Some fundamentals of measurement and calculations               <ul style="list-style-type: none"> <li>Balance sensitivity, significant figures, accuracy and percentage of errors</li> <li>Measurement of volume and weight</li> <li>Aliquot method of weighing and measuring</li> <li>Density, specific gravity, specific volume</li> </ul> </li> </ul>                                           | Ref. 1&2, instructor provided handouts |
| 2    | 2             | <ul style="list-style-type: none"> <li>Percentage calculations               <ul style="list-style-type: none"> <li>Percentage, ratio and proportion</li> <li>Percentage preparations (percentage W/V, V/V, W/W)</li> <li>Conversions of concentration to mg/mL, Parts per million (ppm)</li> </ul> </li> </ul>                                                                                                                            | Ref. 1&2, instructor provided handouts |
| 3    | 2             | <ul style="list-style-type: none"> <li>Group &amp; Individual exercises</li> <li>Quiz</li> </ul>                                                                                                                                                                                                                                                                                                                                           | Ref. 1, instructor provided worksheet  |
| 4    | 2             | <ul style="list-style-type: none"> <li>Calculation of doses and formulas               <ul style="list-style-type: none"> <li>Calibration of Droppers</li> <li>Calculations of doses (dose size, number of doses, amount dispensed, quantity of ingredient)</li> <li>Drug Dosage based on Age (pediatrics and geriatrics)</li> <li>Drug Dosage based on Body Weight</li> <li>Drug Dosage based on Body Surface Area</li> </ul> </li> </ul> | Ref. 1&2, instructor provided handouts |

|    |   |                                                                                                                                                                                                                                                                                                                                         |                                        |
|----|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
|    |   | <ul style="list-style-type: none"> <li>○ Enlarging and reducing formulas</li> </ul>                                                                                                                                                                                                                                                     |                                        |
| 5  | 2 | <ul style="list-style-type: none"> <li>• Dilution and concentration               <ul style="list-style-type: none"> <li>○ Strength and total quantity</li> <li>○ Dilution and Concentration of liquids</li> <li>○ Dilution and Concentration of solids</li> <li>○</li> </ul> </li> </ul>                                               | Ref. 1&2, instructor provided handouts |
| 6  | 2 | <ul style="list-style-type: none"> <li>• Group &amp; Individual exercises</li> <li>• Quiz</li> </ul>                                                                                                                                                                                                                                    | Ref. 1, instructor provided worksheet  |
| 7  | 2 | <ul style="list-style-type: none"> <li>• Dilution and concentration (Continued...)               <ul style="list-style-type: none"> <li>○ Triturations</li> <li>○ Alligation medial and Alligation Alternate</li> <li>○ Specific gravity of Mixture</li> </ul> </li> </ul>                                                              | Ref. 1&2, instructor provided handouts |
| 8  |   | <ul style="list-style-type: none"> <li>• Test</li> </ul>                                                                                                                                                                                                                                                                                |                                        |
| 9  | 2 | <ul style="list-style-type: none"> <li>• Isotonic, buffer and electrolyte solutions               <ul style="list-style-type: none"> <li>○ Calculation for Isotonic Solution preparation</li> <li>○ Sodium Chloride equivalent of a substance</li> <li>○ Isotonicity, osmolarity, milliequivalents, milliosmoles</li> </ul> </li> </ul> | Ref. 1&2, instructor provided handouts |
| 10 | 1 | <ul style="list-style-type: none"> <li>• Isotonic, buffer and electrolyte solutions               <ul style="list-style-type: none"> <li>○ Buffers and buffer solutions</li> <li>○ Buffer equation</li> <li>○ Isotonic buffer solutions</li> </ul> </li> </ul>                                                                          | Ref. 1&2, instructor provided handouts |
| 11 | 2 | <ul style="list-style-type: none"> <li>• Some calculations involving parenteral admixtures               <ul style="list-style-type: none"> <li>○ Dry powders for reconstitution, parenteral admixtures, additives, hyperalimentation solutions, rate of flow of IV fluids</li> </ul> </li> </ul>                                       | Ref. 1&2, instructor provided handouts |
| 12 | 1 | <ul style="list-style-type: none"> <li>• Some calculations involving radiopharmaceuticals</li> </ul>                                                                                                                                                                                                                                    |                                        |

|    |   |                                                                                                                                    |                                        |
|----|---|------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
|    |   | ○ Radioisotopes, radioactivity, units of radioactivity                                                                             | Ref. 1&2, instructor provided handouts |
| 13 | 2 | <ul style="list-style-type: none"> <li>Some calculations involving Biological products (vaccines and immunizing agents)</li> </ul> | Ref. 1&2, instructor provided handouts |
| 14 | 2 | <ul style="list-style-type: none"> <li>Group &amp; Individual exercises</li> <li>Quiz</li> </ul>                                   | Ref. 1, instructor provided worksheet  |
| 15 |   | <ul style="list-style-type: none"> <li>Assignment and Exercises</li> </ul>                                                         | Selected exercises from each chapter   |
| 16 |   | <b>FINAL EXAM</b>                                                                                                                  |                                        |

## **Pharmacognosy and Alternative Medicine I**

**Module Name: Pharmacognosy and Alternative Medicine I**

**Module Category: Core**

**Module Code: Phar-M2081**

**Module Number: 08**

**Module Weight: 12 ECTS**

**Courses:**

| <b>Course name</b>            | <b>Course code</b> | <b>ECTS</b> |
|-------------------------------|--------------------|-------------|
| Chemistry of Natural Products | (Phar1081)         | (5 ECTS)    |
| Pharmacognosy                 | (Phar1082)         | (7 ECTS)    |

**Module description:** The module studies about medicines derived from natural source; and natural substances and their chemistry. It is designed in such a way that the trainee gets well acquainted with the study of the physical, chemical, biochemical and biological properties of drugs, drug substances, or potential drugs or drug substances of natural origin as well as the search for new drugs from natural sources, various alternative and complementary medicine practices including the Ethiopian traditional medicine. It also deals with chemical structure, chemical reactions and synthesis of natural products.

**Module objective:** At the end of this module students will understand and demonstrate the source, chemical & biological features of drugs and drug substances of natural origin & explain different forms of complementary & alternative medicines.

**Module competencies:**

- Define and describe natural sources of drugs and drug substances
- Explain the physical, chemical, biochemical and biological properties of drugs or drug substances of natural origin
- Associate pharmaceutical application of natural products and related services
- Devise research protocols on drug discovery from natural products
- Demonstrate procedures of obtaining drug/drug substance from natural sources
- Perform physical, chemical & biological characterization of natural products
- Describe & compare the role of various forms of complementary & alternative medicines in primary health care service
- Display & perform regulatory & quality control activities on natural products
- Display rational usage of natural products (as drugs, foods, alternative medicines)

- Comply laboratory safety precautions and standards
- Assist research activities related to drug discoveries from natural products
- Follow scientific protocols to perform and report experimental works
- Ready to provide service that ensure rational usage of natural products

**Mode of delivery (Parallel/Block):**

- Total study hour: 324 hours

**Module teaching/learning method:**

Learning Activities:

- Attend lectures and demonstrations, take notes, and ask questions
- Engage in learning by doing (independent study, group assignments, presentation, report writing, and etc.)
- Participation and discussions
- Practical laboratory works including sample preparation, extraction, interpretation and report writing.

Teaching Methods

- The course facilitator is expected to lecture for introducing concepts and topics, and give references, facilitate discussions, ask questions, give and correct assessments, guide practical sessions, correct and give feedbacks of reports of practical sessions.
- Arrange and facilitate seminar sessions, discussions and give comments and feedbacks.
- Field visit and visiting traditional practitioners

**Module mode of assessment:**

- Seminars and assignments
- Quizzes
- Practical exam
- Laboratory report
- Laboratory written exam
- Final exam

**Course Name:** Chemistry of Natural Products

**Course code:** Phar2081

**Module Name:** Pharmacognosy and Alternative Medicine I

**Module Code:** Phar-M2081

**Course ECTS:** 5 (135hr)

**Totally required hours for the module:**

- Lecture: 40 hours
- Laboratory 8 h
- Tutorial: 8 hours
- Individual presentation, group discussions, audiovisuals, seminars, and assignments: 10 hours
- Assessment (continuous & final): 8 hours
- Independent study (alone or in groups): 61 hours

**Year/Semester Course is offered:** Year II Semester I

**Course prerequisite/s:** Organic Chemistry, Organic Chemistry Laboratory

**Course description:** The course covers some selected topics in natural products chemistry. The goal is to acquaint students to the basic evidences, which are results of observations carried out over generations that are in use to chemically characterize natural products of primary and secondary metabolism. It also realizes that the same reactions organic chemists know so well are apparently only mimics of what occurs naturally.

**Course objective:**

After completion of this course students will be able to:

- Students will be able to describe the chemical structure, chemical reactions, some chemical classes and related properties of natural products as well as their applicability.

**Course mode of delivery:** Parallel

**Course learning and teaching methods**

**Assessment techniques:**

- Seminars and assignments... ..25%
- Quizzes... ..25%
- Final exam. ....50%

**Teachers' and students' role****Roles of Instructors**

The instructor will be expected to:

- Be a facilitator (introduce the subject; give guidance, moderate discussions, etc.)
- Read and comment assignments of students on time;
- Prepare his/her lessons and deliver lectures;
- Provide available and necessary reference materials;
- Encourage active participation of students in the teaching learning process;
- Assist students with learning difficulties and
- Arrange and follow up practical sessions

**Roles of Students**

Students are expected to:

- Engage in learning by doing (independent study, project work; group work, etc.)
- Be active learners (participate effectively in laboratory activities, in group assignments, make presentations, write reports, etc.);

**References:****Required readings (Text)**

1. Sarker SD, Nahar L, *Chemistry for Pharmacy Students. General, Organic and Natural Product Chemistry. John Willey and Sons Ltd, UK, 2007*
2. Dewick PM. *Medicinal Natural Products: A biosynthetic Approach, 3rd edition. Jhon Wiley and Sons, LTD, England 2009.*
3. David R. Klein, *Organic Chemistry as a Second Language: First Semester Topics, Wiley; 5 edition (September 11, 2019)*



### Course schedule\*

| Week   | Contact Hours               | Topic/sub-topic/chapter/Assessments/Assignments                                                                                                                                                                                                                                | Reading Materials                                                                                         |
|--------|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| 1      | 4hrs                        | Stereochemistry <ul style="list-style-type: none"> <li>Isomerism, conformation and configuration, chirality, enantiomers, optical activity, diastereomers, racemic mixtures, meso-compounds</li> </ul>                                                                         | Sarker SD, Nahar L, Chemistry for Pharmacy Students. G<br>Argawell, Organic and Natural Product Chemistry |
| 2, 3&4 | 9hrs                        | Carbohydrates <ul style="list-style-type: none"> <li>Introduction, classification, reactions and configuration of carbohydrates, cyclic structures of monosaccharides, conformation of monosaccharides, chemistry of disaccharides and polysaccharides</li> </ul>              | Sarker SD, Nahar L, Chemistry for Pharmacy Students. G                                                    |
| 5      | 4hrs                        | Lipids <ul style="list-style-type: none"> <li>Definition, occurrence and composition of fats, oils and waxes; Reactions of fats and oils; Determination of analytical values for fats and oils</li> </ul>                                                                      |                                                                                                           |
| 6&7    | 6hrs                        | Amino acids and proteins <ul style="list-style-type: none"> <li>Structure, nomenclature, physical and chemical properties of amino acids, structure and nomenclature of peptides, classification and properties of proteins, synthesis of peptides</li> </ul>                  |                                                                                                           |
| 8&9    | 5hrs                        | Terpenes <ul style="list-style-type: none"> <li>Introduction, properties and isolation of terpenoids, overviIsoprene rule, Classification of terpenoids, Significance of terpenoids in pharmacy</li> <li></li> </ul>                                                           |                                                                                                           |
| 10&11  | 4hrs<br>(after terpenoids ) | Steroids <ul style="list-style-type: none"> <li>Introduction, sources for steroids, significance of steroids in pharmacy, chemistry and nomenclature of steroids, sterols</li> <li>Basics of the biosynthetic concept of steroids must be included.</li> </ul>                 |                                                                                                           |
| 12     | 3hrs                        | Purines and nucleic acids <ul style="list-style-type: none"> <li>Introduction, Uric acid, Purine derivatives (adenine, xanthine, hypoxanthine, and guanine), Xanthine bases (theophylline, theobromine and caffeine), Introduction to and structure of nucleic acid</li> </ul> |                                                                                                           |
| 13&14  |                             | Heterocyclic compounds                                                                                                                                                                                                                                                         |                                                                                                           |

|  |                             |                                                                                                                                                                                                                                                                                                            |  |
|--|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | (before<br>terpene)<br>5hrs | <ul style="list-style-type: none"> <li>• Introduction, classification and nomenclature, physical and chemical properties, significance of some heterocyclic compounds in pharmacy (furan, pyrrole, thiophene, pyrazole, imidazole, pyridine, pyrimidine, oxazole, isoxazole, and phenothiazine)</li> </ul> |  |
|--|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

**Course Name:** Pharmacognosy

**Course code:** Phar 2082

**Module Name:** Pharmacognosy and Alternative Medicine I

**Module Code:** 08

**Course ECTS:** 7

**Totally required hours for the module:** 189 hours

**Assessment:**

- Seminar/Assignments/ quizzes: 35%
- Laboratory report writing and written exam: 15%
- Practical exam: 10%
- Final Exam: 40%

**Year/Semester Course is offered:** Year II Semester II

**Course prerequisite/s:** Chemistry of Natural Products

**Course description:** The course is designed in such a way that the trainee gets well acquainted with the study of the physical, chemical, biochemical and biological properties of drugs, drug substances, or potential drugs or drug substances of natural origin as well as the search for new drugs from natural sources. The course familiarizes trainees with the basic scientific knowledge and skill needed to obtain and characterize active substances from natural sources. It also helps trainees to understand and realize the fact that nature provides the origin and continuous supply of drugs or drug substances, and think about the proper management and utilization of such natural products.

**Course objective:**

After completion of this course students will be able to:

- To familiarize themselves to general aspects of crude drugs, extraction and isolation methods and the distribution, properties and uses of various primary and secondary metabolites of plant, animal and mineral origin.

**Course mode of delivery:** Parallel

### **Course learning and teaching methods**

- Illustrated Lecture: 48 hours
- Practical sessions: 48 hours
- Tutorial: 12 hours
- Seminars, assignments and presentation: 16 hours
- Assessment (continuous & final): 10 hours
- Independent study (alone or in groups): 45 h hours
- Field visits (10 h)

### **Assessment techniques:**

- Seminar/Assignments: 10%
- Quizzes/Continuous assessment: 25%
- Laboratory written exam: 15%
- Practical exam: 10%
- Final Exam: 40%

### **Teachers' and students' role**

#### **Roles of Instructors**

- The instructor will be expected to:
- Be a facilitator (introduce the subject; give guidance, moderate discussions, etc.)
- Read and comment assignments of students on time;
- Prepare his/her lessons and deliver lectures;
- Provide available and necessary reference materials;
- Encourage active participation of students in the teaching learning process;
- Assist students with learning difficulties and
- Arrange and follow up practical sessions
- Roles of Students

#### **Students are expected to:**

- Engage in learning by doing (independent study, project work; group work, etc.)
- Be active learners (participate effectively in laboratory activities, in group assignments, make presentations, write reports, etc.);

**References:****Required readings (Text)**

- William C. Evans, George E. Trease, and Daphne Evans, Trease and Evans' Pharmacognosy (16<sup>th</sup> ed.), Elsevier (2009).

**Recommended readings**

- Dewick PM. Medicinal Natural Products: A biosynthetic Approach, 3<sup>rd</sup> edition. Jhon Wiley and Sons, LTD, England 2009.
- Sarker D, Latif Z, Gray A. Methods in Biothechnology Natural Products Isolation, 2<sup>nd</sup> edition, Human Press, Totowa, New Jersey, 2006
- Pulok K. Mukherejee. Quality control of herbal drugs; an approach to evaluation of botanicals. Business Horizons pharmaceutical publishers 2002

**Course schedule\***

| Week | Contact Hours | Topic/sub-topic/chapter/Assessments/Assignments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1&2  |               | <ol style="list-style-type: none"><li>General Introduction<ol style="list-style-type: none"><li>Definition, History and scope of Pharmacognosy (1 hr)</li><li>Crude drugs (6 h)<ol style="list-style-type: none"><li>Pharmaceutical botany</li><li>Definition and Nomenclature of crude drugs</li><li>Classification of crude drugs</li><li>Evaluation of crude drugs</li><li>Types of preparations from plants</li><li>Schemes for pharmacognostic studies of crude drugs.</li><li>Official and Unofficial drugs</li><li>Factors affecting crude drug quality</li></ol></li><li>Steps in the scientific analysis of drugs from natural resources<ol style="list-style-type: none"><li>Selection of plant material</li><li>Taxonomic identification of the plant</li><li>Literature survey on the identified plant</li><li>Design of appropriate extraction and separation methods</li><li>Checking extracts/ fractions for pharmacological activity</li><li>Identification of classes of compounds found in the plant (phytochemical screening)</li><li>Isolation of active compounds or fractions responsible for the pharmacological activity of the plant</li></ol></li></ol></li></ol> |
| 3&4  | 8hrs          | <ol style="list-style-type: none"><li>General methods in studying constituents of crude drugs</li></ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

|                            |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                            |        | <p><b>2.1. Extraction (3 hrs)</b></p> <p>2.1.1. Definition and the need for extraction</p> <p>2.1.2. Preparation of plant material for solvent extraction</p> <p>2.1.3. Choice of suitable solvents</p> <p>2.1.4. Methods of extraction</p> <p><b>2.2. Isolation and purification of active constituents (5hrs)</b></p> <p>2.2.1. Classical methods of separation</p> <p>2.2.2. Modern methods/chromatographic methods</p> <p>2.2.2.1.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 5,6,7,8,<br>9,10,<br>11&12 | 29 hrs | <p><b>3. Major plant constituents and their botanical sources</b></p> <p><b>3.1. Primary and secondary plant metabolites (1 h)</b></p> <p><b>3.2. Carbohydrates (2 h)</b></p> <p>3.2.1. Sugars and sugar containing drugs</p> <p>3.2.2. Compounds related to sugars</p> <p>3.2.3. Polysaccharides</p> <p>3.2.4. Gums and mucillages</p> <p><b>3.3. Glycosides (6 h)</b></p> <p>3.3.1. General properties of glycosides</p> <p>3.3.2. Classification of glycosides</p> <p>3.3.3. Classes of glycosides: Anthraquinones, Saponins, Cardiac glycosides, Simplephenolic glycosides, Flavonoid glycosides, Isothiocyanate glycosides, Cyanogenetic glycosides, Coumarin glycosides, lignans</p> <p><b>3.4. Tannins (2 h)</b></p> <p>3.4.1. General properties and Chemistry</p> <p>3.4.2. Classification: Hydrolysable, Nonhydrolysable (condensed), Pseudotannins</p> <p>3.4.3. Significance of tannins</p> <p><b>3.5. Lipids and waxes (2 h)</b></p> <p>3.5.1. Lipids: Physical and chemical properties, Extraction methods</p> <p>3.5.2. Official fixed oils/fats and their composition</p> <p>Arachis oil, castor oil, almond oil, sesame oil, theobroma oil, codeliver oil etc</p> <p>3.5.3. Waxes: Definition and general properties, Animal waxes, Vegetable waxes</p> <p><b>3.6. Volatile oils (5 h)</b></p> <p>3.6.1. Distribution and occurrence, Uses, Methods of preparation (Distillation, Expression, Extraction with solvent, enzymatic hydrolysis), Physical properties, Chemistry, Biosynthesis</p> <p>3.6.2. Constituents of volatile oils: Hydrocarbons, Alcohols, Aldehydes, Ketones, Esters, Phenols and phenolic ethers Oxides Peroxides Sulfur containing compounds Nitrogen containing compounds</p> <p><b>3.7. Resins and resin combinations (3 h)</b></p> <p>3.7.1. General properties and chemistry</p> |

|       |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       |              | <p>3.7.2. Examples of drugs containing resins: Rosin, Podophyllin, Jalap, Mastic, Cannabis (Preparation, Constituents, Factors affecting the narcotic activity, Legal aspects, Analysis), Oleoresins Oleo-gum resins, Balsams</p> <p><b>3.8. Alkaloids (8 h)</b></p> <p>3.8.1. Definition, Nomenclature, Occurrence, Physical and chemical properties, Detection, Extraction and isolation</p> <p>3.8.2. Classification</p> <p>3.8.3. Classes of alkaloids</p> <p>3.8.3.1. Ornithine derived alkaloids: Tropane alkaloids Solanaceae alkaloids, Coca alkaloids, Pyrolizidine alkaloids</p> <p>3.8.3.2. Lysine derived alkaloids: Lobelia alkaloids, Lupine alkaloids</p> <p>3.8.3.3. Nicotinic acid derived alkaloids: Ricinine, Areca alkaloids, Tobacco alkaloids</p> <p>3.8.3.4. Tyrosine derived alkaloids: (1) Simple phenylethylamines and tetrahydroisoquinolines: Mescaline, Ephedrine, Alkaloids of 'Khat'; (2) Colchicine; (3) Papaver alkaloids: Phenanthrene group, Benzyloisoquinoline group; (4) Emetine and related alkaloids; (5) Tubocurarine</p> <p>3.8.3.5. Tryptophan derived alkaloids: Simple indoles, Tricyclic alkaloids: Pegamum alkaloids, Physostigma alkaloids, Ergot alkaloids (Historical background, Life cycle of ergot, Commercial production, Chemistry and occurrence, Biosynthesis and use, LSD), Rauwolfia alkaloids, Strychnos alkaloids, Cinchona alkaloids, Periwinkle alkaloids</p> <p>3.8.3.6. Histidine derived alkaloids: Pilocarpine</p> <p>3.8.3.7. Polyacetate derived alkaloids: Hemlock alkaloids</p> <p>3.8.3.8. Psedoalkaloids: Steroidal alkaloids: Veratrum alkaloids, Solanum alkaloids, Holarrhena alkaloids, Buxus alkaloids, Purine alkaloids</p> |
| 13&14 | <b>(4 h)</b> | <p>4. Natural compounds and cancer</p> <ul style="list-style-type: none"> <li>• Anticancer drugs of natural origin</li> <li>• Natural compounds as carcinogens</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

**Module Name: Dosage Form Sciences I****Module Category:** Core**Module Code:** Phar-M2091**Module Number:** 09**Module Weight:** 9 ECTS**Courses:**

| S/N | Course name                                                | Course code | ECTS |
|-----|------------------------------------------------------------|-------------|------|
| 1   | Integrated Physical Pharmacy and Pharmaceutics I           | Phar2091    | 7    |
| 3   | Practical Integrated Physical Pharmacy and Pharmaceutics I | Phar2092    | 2    |

■

**Module Description**

The module deals with the science and arts of converting drugs into medicines rendering students with practical insight into drug formulation principles at the very outset. It gives students the basic sciences in physical pharmacy that play a role during large scale production and extemporaneous compounding of liquid dosage forms (solutions, emulsions and suspensions). Moreover, this module will give students about critical issue to be considered in formulation development including micromeritics, rheology, drug release from dosage forms (diffusion and dissolution), issue of components incompatibilities, drug stability and methods to determine shelf life of different products.

**Module Objective**

This module aims at providing the student with a broad understanding of physicochemical principles that govern the behavior of drugs, excipients and dosage forms. It also enables the student to prepare extemporaneous preparations based on the basic science knowledge she/he to gain.

**Module Competency**

This module enables student develop the knowledge, skill and attitude required in preparing extemporaneous preparations and play a role in formulation development.

**Mode of deliver**

- Parallel

**Module teaching/learning methods****Learning activities**

Students are expected to undertake among others the following major learning activities:

- Engage in learning by doing (independent study, group assignments, presentation, report writing, and etc...)
- Participation and note takings during class lectures and debates and discussions;
- Analysis, summarization and presentations of chapter/article, scientific papers (and be able present or submit in a concise and shorten form)
- Search validated formula from standard books, journals, scientific papers; elaborate the purpose of each component of the formula; understanding the compounding procedure; prepare extemporaneous preparations and then write label to it

**Teaching Methods**



- The course facilitator is expected to lecture for introducing concepts and topics, and give references, facilitate discussions, ask questions, give and correct assessments,
- Arrange and facilitate seminar sessions by inviting professionals from other units/departments in the department/school and other interested staffs as well.

**Module mode of Assessment:**

- Group assignments
- Presentations
- Laboratory reports
- Practical exam
- Tests/quizzes
- Final Exam

**Course Name:** Integrated Physical Pharmacy and Pharmaceutics I  
**Course code:** Phar2091  
**Module Name:** Dosage Form Sciences Module  
**Module Code:** Phar-M2091  
**Course ECTS:** 7

**Totally required hours for the course:** 189hrs

|                         |     |
|-------------------------|-----|
| <b>Lecture hours:</b>   | 64  |
| <b>Study hours:</b>     | 100 |
| <b>Group work:</b>      | 0   |
| <b>Project work:</b>    | 0   |
| <b>Presentation(s):</b> | 10  |
| <b>Tutorial:</b>        | 15  |

**Year/Semester Course is offered:** Year II Semester II

**Course prerequisite/s:** Pharmaceutical Calculations

**Course Description:**

The design of the course is based on the integration of the study of physico-chemical principles of pharmacy with the formulation and preparation of pharmaceutical dosage forms. The integration is done within each main class of pharmaceutical dosage forms. The study of the physico-chemical principles of pharmacy serves as a prologue to the materials covered in each section. The main focus of this course is the application of the knowledge of the physico-chemical principles of pharmacy to the rational formulation, compounding, quality control, packaging and storage of pharmaceutical dosage forms.

**Course Objectives:**

After completion of this course students will be able to:

- Understand the different types of dosage forms and routes of administration
- Understand the types of intermolecular interaction forces, the phase rule and phase equilibria of one, two and three component systems
- Understand the concepts surface and interfacial tensions, adsorption at liquid and solid interfaces
- Define solubility, understand different solubility expressions, the basic concepts behind gas/liquid and liquid/liquid solutions and solubility of different types of solids in liquids
- Understand the distribution law, and its applications

- Understand different types of containers, packaging materials, storage conditions and labeling requirements of pharmaceutical dosage forms
- Understand the different types of solution dosage forms and develop skills to compound them
- Define and differentiate Newtonian and Non-newtonian systems, understand thixotropic property of fluids, understand the methods of determination of viscosity and its pharmaceutical applications
- Understand sedimentation in suspensions, interfacial property of suspended particles, the electric double layer and DLVO theory
- Differentiate between flocculated and deflocculated suspensions and understand rheologic property of suspensions, preparation methods, and labeling & storage conditions
- Understand types of emulsions and methods to identify emulsion type, theories of emulsification and physical instabilities in emulsions, preservation of emulsions & rheology of emulsions
- Understand methods of extemporaneous compounding of emulsions, labeling and storage conditions

**Course mode of delivery:** Parallel

**Course learning and teaching methods:**

- Active participation during class lectures and excursions
- Engage in learning by doing
- The course instructor is expected to introduce concepts and topics and give references, facilitate discussions, ask questions, correct assignments
- Arrange and facilitate excursions

**Assessment techniques:**

- Assignments: 15%
- Tests: 30%
- Quizzes: 15%
- Final exam: 40%

**Teachers' and students' role**

**Roles of Instructors**

**The instructor will be expected to:**

- Be a facilitator (introduce the subject; give guidance, moderate discussions, etc.)
- Read and comment on the assignments & exercises of students on time;

- Prepare his/her lessons and deliver lectures;
- Provide available and necessary reference materials;
- Encourage active participation of students in the teaching learning process;
- Assist students with learning difficulties

### **Roles of Students**

#### **Students are expected to:**

- Engage in learning by doing (independent study, group works/exercises, etc.)
- Be active learners (participate effectively in group assignments, class activities/exercises, etc.);
- Attend classes regularly

### **References:**

#### **Required readings (Text)**

1. P. J. Sinko, Martin's Physical Pharmacy and Pharmaceutical Sciences, 7th Edition, Lippincott Williams & Wilkins, Philadelphia, 2016.
2. M. E. Aulton, Pharmaceutics: the science of dosage form design, 7th ed., Churchill Livingstone, Edinburgh.

#### **Recommended readings**

3. M. J. Wilson, Pharmaceutical Compounding and Dispensing, 2nd Edition, Pharmaceutical press, 2010.
4. L. V. Allen, N. G Popovich, H. C Ansel, Ansel's Pharmaceutical Dosage Forms and Drug Delivery Systems, 11th edition, Lippincott Williams & Wilkins, 2017.
5. J. E. Thompson and L. Davidow, A Practical Guide to Contemporary Pharmacy Practice, 3rd edition, Lippincott Williams & Wilkins.
6. A. Martin, J. Swarbrick and A. Cammarata, Physical Pharmacy, 3rd Edition, Lea & Febiger, Philadelphia, 1983.
7. A. T. Florence and D. Attwood, Physico-chemical Principles of Pharmacy, Macmillan Publishers Ltd., London, 1981
8. S. C. Wallwork and D. J. W. Grant, Physical Chemistry for Students of Pharmacy and Biology, 3rd Edition, Longman Group Ltd., London, 1977.

### **Course Schedule**

| <b>Week</b> | <b>Contact Hours</b> | <b>Topic/sub-topic/chapter/Assessments/Assignments</b> |
|-------------|----------------------|--------------------------------------------------------|
|             |                      |                                                        |

|   |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | 4 | <ul style="list-style-type: none"> <li>• Introduction to dosage forms and routes of drug administration <ul style="list-style-type: none"> <li>○ Definition, the need for dosage forms, classification, overview of dosage form design</li> <li>○ Introduction to pharmaceutical ingredients (definition, importance)</li> <li>○ Routes of administration</li> </ul> </li> </ul>                                                                                                                                                  |
| 2 | 3 | <ul style="list-style-type: none"> <li>• Phase Equilibria <ul style="list-style-type: none"> <li>○ Introduction to intermolecular force of interaction</li> <li>○ The phase rule</li> <li>○ Phase equilibria of single, two and three component systems (principles and applications)</li> </ul> </li> </ul>                                                                                                                                                                                                                      |
| 3 | 4 | <ul style="list-style-type: none"> <li>• Interfacial Phenomena <ul style="list-style-type: none"> <li>○ Liquid interface (surface/interfacial tension, measurement of surface/interfacial tension, surface free energy, spreading &amp; spreading coefficient)</li> </ul> </li> <li>• Quiz</li> </ul>                                                                                                                                                                                                                             |
| 4 | 4 | <ul style="list-style-type: none"> <li>• Interfacial Phenomena (Continued...) <ul style="list-style-type: none"> <li>○ Adsorption at liquid interfaces (surfactants: basic concepts, the HLB system and applications)</li> <li>○ Adsorption at solid interfaces (Solid/gas interface, solid/liquid interface, Adsorption isotherms)</li> </ul> </li> </ul>                                                                                                                                                                        |
| 5 | 4 | <ul style="list-style-type: none"> <li>• Solubility and Distribution Phenomena <ul style="list-style-type: none"> <li>○ Terminologies (solute, solvent, solution, solubility)</li> <li>○ Solute-solvent interactions (polar, nonpolar and semi polar solvents)</li> <li>○ Solubility expressions</li> <li>○ Solubility of gases in liquids <ul style="list-style-type: none"> <li>▪ Factors affecting solubility of gases, Solubility calculations</li> </ul> </li> <li>○ Solubility of liquids in liquids</li> </ul> </li> </ul> |

|   |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |   | <ul style="list-style-type: none"> <li>▪ Ideal and real solutions, complete and partial miscibility, factors affecting solubility of liquids</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 6 | 4 | <ul style="list-style-type: none"> <li>• Solubility and Distribution Phenomena (Continued...)               <ul style="list-style-type: none"> <li>○ Solubility of solids in liquids                   <ul style="list-style-type: none"> <li>▪ Ideal and non-ideal solubility</li> <li>▪ Solubility and the heat of solution</li> <li>▪ Solubility of strong and slightly soluble electrolytes</li> <li>▪ Solubility of weak electrolytes (effect of pH)</li> <li>▪ The influence of solvents on the solubility of drugs</li> <li>▪ Influence of surfactants</li> <li>▪ Complexation as solubility enhancing mechanism</li> <li>▪ Influence of solid state (polymorphs, amorphous, solvates)</li> </ul> </li> <li>○ Distribution phenomena                   <ul style="list-style-type: none"> <li>▪ Distribution law</li> <li>▪ Effect of molecular association and ionic dissociation</li> <li>▪ Applications</li> </ul> </li> </ul> </li> </ul> |
| 7 |   | <ul style="list-style-type: none"> <li>• Test I</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 7 | 4 | <ul style="list-style-type: none"> <li>• Packaging and storage of pharmaceuticals               <ul style="list-style-type: none"> <li>○ Introduction (definitions and terminologies)</li> <li>○ Packaging materials</li> <li>○ Closures</li> <li>○ Labeling pharmaceutical dosage forms</li> <li>○ Storage, stability of pharmaceuticals and beyond use date</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 8 | 1 | <ul style="list-style-type: none"> <li>• Packaging and storage of pharmaceuticals (Continued...)               <ul style="list-style-type: none"> <li>○ Labeling pharmaceutical dosage forms</li> <li>○ Storage, stability of pharmaceuticals and beyond use date</li> </ul> </li> <li>• Quiz</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 9 | 4 | <ul style="list-style-type: none"> <li>• Pharmaceutical Solutions               <ul style="list-style-type: none"> <li>○ Introduction</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

|    |   |                                                                                                                                                                                                                                                                                                                                                                 |
|----|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |   | <ul style="list-style-type: none"> <li>○ Formulation of solutions (API and Excipients)</li> <li>○ General methods of preparation</li> <li>○ Solutions taken orally</li> </ul>                                                                                                                                                                                   |
| 10 | 4 | <ul style="list-style-type: none"> <li>● Pharmaceutical Solutions (Continued...) <ul style="list-style-type: none"> <li>○ Solutions used in the mouth and throat</li> <li>○ Solutions instilled into body cavities</li> <li>○ Topical solutions</li> <li>○ Injectables (sterile products)</li> </ul> </li> </ul>                                                |
| 11 | 4 | <ul style="list-style-type: none"> <li>● Rheology <ul style="list-style-type: none"> <li>○ Introduction</li> <li>○ Newtonian and Non-Newtonian systems</li> </ul> </li> </ul>                                                                                                                                                                                   |
| 12 | 3 | <ul style="list-style-type: none"> <li>● Rheology (Continued...) <ul style="list-style-type: none"> <li>○ Thixotropy</li> <li>○ Determination of viscosity</li> <li>○ Pharmaceutical applications of rheology</li> </ul> </li> </ul>                                                                                                                            |
| 13 |   | <ul style="list-style-type: none"> <li>● Test II</li> </ul>                                                                                                                                                                                                                                                                                                     |
| 13 | 4 | <ul style="list-style-type: none"> <li>● Colloids <ul style="list-style-type: none"> <li>○ Introduction (definition, classification and applications)</li> <li>○ Optical properties of colloids</li> <li>○ Kinetic properties of colloids</li> <li>○ Electrical properties of colloids</li> </ul> </li> </ul>                                                   |
| 14 | 3 | <ul style="list-style-type: none"> <li>● Pharmaceutical Suspensions <ul style="list-style-type: none"> <li>○ Introduction</li> <li>○ Desirable properties</li> <li>○ Sedimentation in suspensions</li> <li>○ Interfacial properties of suspended particles</li> <li>○ Electrical properties of suspended particles (EDL and DLVO theory)</li> </ul> </li> </ul> |
| 15 | 4 | <ul style="list-style-type: none"> <li>● Pharmaceutical Suspensions (Continued...)</li> </ul>                                                                                                                                                                                                                                                                   |

|    |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |   | <ul style="list-style-type: none"> <li>○ Flocculated/deflocculated suspensions (properties and evaluations)</li> <li>○ Formulation approaches (structure vehicle, controlled flocculation and combination)</li> <li>○ Rheology of suspension</li> <li>○ Preparation of suspensions (diffusible, indiffusible, poorly wettable solids)</li> <li>○ Label and storage</li> </ul>                                                                                                                                                         |
| 16 | 4 | <ul style="list-style-type: none"> <li>● Pharmaceutical Emulsions <ul style="list-style-type: none"> <li>○ Introduction</li> <li>○ Determination of emulsion type</li> <li>○ Theories of emulsification (surface free energy, mechanisms of stabilization by emulsifying agents)</li> <li>○ Physical instabilities (creaming, flocculation, cracking, phase inversion)</li> <li>○ Preservation of emulsion</li> <li>○ Rheology of emulsion</li> <li>○ Preparation of emulsion</li> <li>○ Labelling and storage</li> </ul> </li> </ul> |
|    |   | <b>FINAL EXAM</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |



**Course Name:** Practical Integrated Physical Pharmacy and Pharmaceutics I

**Course code:** Phar2092

**Module Name:** Dosage form Sciences Module

**Module Code:** Phar-M2091

**Course ECTS:** 2

**Totally required hours for the course:** 54hrs

**Illustrated lecture:** 8

**Practical lab:** 32

**Assignments and assessment:** 4

**Home Study:** 10

**Year/Semester Course is offered:** Year II Semester II

**Course prerequisite/s:** Pharmaceutical Calculations & Integrated Physical Pharmacy and Pharmaceutics I

### **Course Description:**

The course is designed to give basic understanding and concepts of practical Physical Pharmacy and Pharmaceutics. In this course students will be introduced to pharmaceutical measurements and the basic weight and volume measuring techniques. Furthermore, they will learn practically the effect of concentration and temperature on miscibility of partially miscible liquids, determination of equilateral diagram of determining three component systems, determination of the adsorption isotherms, the effect of temperature on solubility of slightly soluble drugs, techniques of determination of the solubility product, determination of dissociation constant, determination of distribution ratio, determination of the solubilizing action of Tweens, and formulation of liquid dosage forms.

### **Course Objectives:**

After completion of this course students will be able to:

- Describe Qualitative and Quantitative Accuracy
- Identify the different equipment used for measurement of weight and volume
- Describe the different cares that should be taken in weighing and volume measurement
- Identify the weight and volume measuring techniques
- Investigate the effect of concentration and temperature on miscibility of partially miscible liquids, and on solubility of drugs
- Describe three component systems
- Determine adsorption isotherms

- Understand the basic techniques of determination of the solubility product, dissociation constant, and distribution ratio
- To determine the solubilizing action of surface active agents
- Formulate different types liquid formulations

**Course mode of delivery:** Parallel

**Course learning and teaching methods:**

- Illustrated lectures and discussions, student presentations, individual and group laboratory practicals & demonstrations.

**Assessment techniques:**

- Practical skill and competency based exams: 50%
- Written exam: 25%
- Laboratory reports: 15%
- Presentation: 10%

**Teachers' and students' role**

**Roles of Instructors**

**The instructor will be expected to:**

- Be a facilitator (introduce the subject; give guidance, moderate discussions, etc.)
- Read and comment lab reports, assignments & presentations of students on time;
- Prepare his/her lessons and deliver lectures;
- Provide available and necessary reference materials;
- Encourage active participation of students in the teaching learning process;
- Assist in laboratory practice & demonstrations,
- Assist students with learning difficulties

**Roles of Students**

**Students are expected to:**

- Engage in learning by doing (independent study, group works/exercises, etc.)
- Be active learners (participate effectively in group assignments, lab activities/presentations, etc.);
- Attend lab sessions regularly

**References:**

**Required readings (Text)**

1. Practical Manual

**Recommended readings**

2. P. J. Sinko, Martin's Physical Pharmacy and Pharmaceutical Sciences, 7th Edition, Lippincott Williams & Wilkins, Philadelphia, 2016.
3. M. E. Aulton, Pharmaceutics: the science of dosage form design, 7th ed., Churchill Livingstone, Edinburgh.

### Course Schedule

| Week | Contact Hours | Topic/sub-topic/chapter/Assessments/Assignments |
|------|---------------|-------------------------------------------------|
| 1    | 3             | • Experiment No 1:                              |
| 2    | 3             | • Experiment No 2:                              |
| 3    | 3             | • Experiment No 3:                              |
| 4    | 3             | • Experiment No 4:                              |
| 5    | 3             | • Experiment No 5:                              |
| 6    | 3             | • Experiment No 6:                              |
| 7    | 3             | • Experiment No 7:                              |
| 8    | 3             | • Experiment No 8:                              |
| 9    | 3             | • Experiment No 9:                              |
| 10   | 3             | • Experiment No 10:                             |
| 11   | 3             | • Experiment No 11:                             |
| 12   | 3             | • Experiment No 12:                             |
| 13   | 3             | • Experiment No 13:                             |
| 14   | 3             | • Experiment No 14:                             |
| 15   | 3             | • Experiment No 15:                             |
| 16   | 3             | • Experiment No 16:                             |

|    |  |                                                                    |
|----|--|--------------------------------------------------------------------|
| 17 |  | <ul style="list-style-type: none"> <li>• Practical exam</li> </ul> |
| 20 |  | <b>FINAL WRITTEN EXAM</b>                                          |

## **Module 10: Pharmacology Module I**

**Module name:** Pharmacology module I

**Module category:** Core

**Module code:** Phar-M2101

**Module number:** 10

**Module weight in ECTS:** 7

**Courses:**

| Course name    | Course Code | ECTS |
|----------------|-------------|------|
| Pharmacology I | Phar 3101   | 7    |

### **Module description**

The pharmacology module will familiarizes the pharmacy students about the drugs, their pharmacokinetics, pharmacodynamic, clinical indication, contraindication, drug interaction and adverse effect of the therapeutically used drugs. In addition to that the module introduces the students about poisons, and management of poisoning agents. By incorporating what they learn in the theoretical aspect in to the laboratory attachment they will become well organized and well oriented professional.

**Module objective:** The objective of the module is to impart fundamental knowledge, skill on the pharmacokinetics, pharmacodynamic, therapeutic use and toxic effects drugs of both therapeutically benefit or toxic/poisoning agents.

**Module competency:**

- Apply the knowledge and skill of Pharmacology and toxicology in drug therapy decision

**Mode of delivery:** Parallel

- Total time: 189 hrs
- Lecture: 64 hrs
- Practical/lab session: 30 hrs
- Tutorial: 32 hrs
- Independent study hour: 40 hrs
- Seminar/Presentation: 16 hrs
- Assessment: 7 hrs

**Module learning teaching methods**

Active learning methods (brain storming, buzz group, discussion, etc), Lecture, group & individual presentation, assignment, project work, and laboratory work.

**Mode of Assessment:**

- Quizzes
- Mid exam
- Final Exam
- Practical exam
- Seminar
- Assignment

## **PHARMACOLOGY MODULE COURSES SYLLABI**

**Course title:** Pharmacology I

**Course code:** Phar 2101

**Module name:** Pharmacology module I

**Module code:** Phar-M2101

**Course ECTS:** 7 ECTS (189 hrs)

- Lecture: 64 hours
- Laboratory: 30 hours
- Tutorial: 32 hours
- Home study: 40 hours
- Assignment and presentation: 16 hours
- Assessment : 7 hours

**Year/Semester Course is offered:**

**Contact hours/ week:** 189- 40= 149 hours/ 19 weeks= 8 hours

**Pre-requisite:**

- Biochemistry I and II
- Physiology I and II
- Human Anatomy

**Course description:**

- This course is designed to enable graduate Pharmacists comprehensively provide the student with the fundamental concepts of Pharmacology and provides students with knowledge about drugs used for treatment, diagnosis and prevention of various diseases. The course starts with basic pharmacologic principles including pharmacokinetic (absorption, distribution metabolism and elimination) and pharmacodynamics (mechanisms of action, drug-receptor interactions, receptor-response coupling and effect of drugs ) and synaptic transmission .The course also describes the Pharmacology of autonomic nervous system, central nervous system, respiratory system and gastro intestinal system. Moreover, the course also covers autacoids and drugs affecting inflammation.

**Course Objectives:**

- At the end of this course, students will be able to describe drugs acting on the nervous system, respiratory system, gastrointestinal system and explain autacoids and drug therapy of inflammation.

## **Learning Objectives**

- Up on completion of this course, students will be able to
  - Understand the general principles of pharmacology.
  - Explain the pharmacokinetics of drugs affecting the nervous system, respiratory system, gastrointestinal system and autacoids.
  - Explain the pharmacological actions, mechanism of actions, and therapeutic uses and pharmacokinetics of drugs affecting the nervous system, respiratory system, gastrointestinal system and autacoids.
  - Explain side effects of drugs acting on autonomic nervous system, central nervous system, respiratory, GI, and drugs acting on inflammation.
  - Apply concept and principles of pharmacology to ensure and proper use of drugs.
  - To work in Pharmacology Laboratory and will be able to practice selected basic experimental demonstration

## **Course mode of delivery: Block/Parallel**

## **Course learning and teaching methods**

- Active learning methods (brain storming, buzz group, discussion, etc), Lecture, group and individual presentation, assignment, project work, and laboratory work.

## **Assessment techniques:**

Continuous assessment & summative assessment

- Quiz (10%)
- Assignments with Presentation (10%)
- Tests (15%)
- Laboratory (25 %)
- Final Exam (40%)

## **Teachers and Students Role**

### **Role of Instructor**

The instructor will be expected to:

- Facilitates students' individual and group activities
- Organize students' hospital visit, laboratory work, workshop practices (if any) and project work presentation(s) and discussion sessions
- Assess students' performances (written and oral presentations)
- Provide timely feedback orally and in writing
- Make follow-up on developments made



- Plan and implement students' consultation program

## Role of Students

Students are expected to:

- Attend sessions
- Carry out individual and group tasks
- Active participant
- Reflect on feedbacks and take actions
- Carry out reading assignment

## References:

Required reading (text)

A. Katzung B.G.: Basic and Clinical Pharmacology: 14<sup>th</sup> or later editions.

## Recommended reading:

B. Goodman and Gilman's: The Pharmacological Basis of Therapeutics; 13<sup>th</sup> or later editions.

C. Rang H.P. and Dale M.M. : Pharmacology; 8<sup>th</sup> edition or later editions.

D. Mycek M.J. Harvey R.A. Lipincott's Illustrated Reviews: Pharmacology; 8<sup>nd</sup> or later editions.

E. Richard A. LEHNE. Pharmacology for Nursing care. 5<sup>th</sup> or later editions.

**Course Schedule:** contact time, contents/topics & reading/reference materials for each topic

| Week | Contact Hours | Topic/sub-topic/chapter/Assessment/Assignments                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Reading Materials |
|------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| 1    | 4             | <p><b>1. General pharmacology</b></p> <ul style="list-style-type: none"> <li>• Introduction: Definition, Scope and Branches of Pharmacology<br/>History of Pharmacology, Drug: Definition, Sources and Nomenclature</li> <li>• Pharmacokinetics: <ul style="list-style-type: none"> <li>➤ Passage of drugs across a bio – membrane (passive, specialized or carrier mediated transport mechanisms)</li> <li>➤ Routes of drug administration (<b>Individual Reading</b>)</li> </ul> </li> </ul> | A, B              |

|   |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |         |
|---|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
|   |   | ➤ Drug absorption (factors modifying absorption, first – pass effects, bio availability, drug formulations, special drug delivery systems)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |         |
| 2 | 4 | <ul style="list-style-type: none"> <li>➤ Drug distribution (volumes, plasma protein bindings, distribution spaces – fat, BBB, placenta)</li> <li>➤ Drug biotransformation (phase I &amp; II reactions, consequences of enzyme induction &amp; inhibition, formation of toxic metabolites, factors influencing metabolism)</li> <li>➤ Excretion of drugs (renal – filtration, passive reabsorption, active secretion; biliary excretion &amp; entero – hepatic – circulation; other routes of excretion)</li> <li>➤ Pharmacokinetic variables (Vd, half – life, clearance, steady state, maintenance dose, loading dose, dosing intervals)</li> </ul> |         |
| 3 | 4 | <ul style="list-style-type: none"> <li>• Pharmacodynamics: <ul style="list-style-type: none"> <li>➤ Introduction, Receptors and General Mechanisms of Drug Action</li> <li>➤ Drug Receptor Interaction: Drug Receptor Theories,</li> <li>➤ Dose-Response Relationships (concepts: affinity, intrinsic activity, agonist, partial agonist/antagonist, agonist – antagonist interactions)</li> <li>➤ Receptor – effector coupling (signaling mechanisms, second messengers)</li> </ul> </li> <li>➤ <b>Quiz 1</b></li> </ul>                                                                                                                            | A, B    |
| 3 | 9 | <ul style="list-style-type: none"> <li>• Practical Laboratory Sessions; e.g. <ul style="list-style-type: none"> <li>➤ Lab animal handling techniques and routes of administration.</li> <li>➤ Introduction to Lab instruments &amp; route of drug administration</li> <li>➤ Effect of route of administration on onset and duration of action of drugs</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                        |         |
| 4 | 4 | <ul style="list-style-type: none"> <li>• Drug Interactions (Classification and Mechanisms)</li> <li>• Adverse Drug Reactions, Describing Drug Toxicity (Types)</li> <li>• Therapeutic Index (LD50 and LD50 determination)</li> <li>• Gene Therapy</li> <li>• Clinical Pharmacology (Clinical drug development, fundamental concepts, application)</li> <li>• Pharmacogenetics</li> </ul>                                                                                                                                                                                                                                                             | A, B, D |

|    |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |         |
|----|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| 5  | 4 | <b>2. Drug affecting the autonomic nervous system</b> <ul style="list-style-type: none"> <li>• Introduction to Autonomic Neurotransmission</li> <li>• Pharmacology of Autonomic Drugs <ul style="list-style-type: none"> <li>• Cholinoreceptor agonists and cholinesterase inhibiting drugs</li> <li>• Cholinoreceptor Blocking Drugs: Antimuscarinic Drugs</li> </ul> </li> </ul>                                                                                | B       |
| 6  | 2 | <ul style="list-style-type: none"> <li>• Nicotinic Pharmacology</li> <li>• Ganglionic Blocking Drugs, Neuromuscular Blocking Drugs</li> </ul>                                                                                                                                                                                                                                                                                                                     | B       |
| 6  | 2 | <ul style="list-style-type: none"> <li>• Sympathomimetic Drugs: Direct Acting Sympathomimetics, Indirect Acting Sympathomimetics</li> </ul>                                                                                                                                                                                                                                                                                                                       |         |
| 7  | 4 | <ul style="list-style-type: none"> <li>• Adrenergic Receptor Blocking Drugs: Adrenergic Receptor Antagonists, <math>\beta</math>-Adrenergic Receptor Antagonists, Combined, Adrenergic Receptor Blocking Agents</li> <li>• <b>TEST</b></li> </ul>                                                                                                                                                                                                                 | B       |
| 7  | 6 | <ul style="list-style-type: none"> <li>• Practical Laboratory Sessions; e.g. <ul style="list-style-type: none"> <li>➤ Effect of pilocarpine and atropine on the eye</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                        |         |
| 8  | 4 | <b>3. Drugs acting on the kidney</b> <ul style="list-style-type: none"> <li>• Introduction: Urine Formation, Renal Tubular Transport Processes, Principles of Diuretic Action (1hr)</li> <li>• Diuretics: Carbonic Anhydrase Inhibitors, Loop Diuretics, Thiazide Diuretics, Potassium Sparing Diuretics and Osmotic Diuretics <ul style="list-style-type: none"> <li>• Vasopressin and Other Agents Affecting Renal Conservation of Water</li> </ul> </li> </ul> | A, B    |
| 8  | 9 | Practical Laboratory Session; e.g. <ul style="list-style-type: none"> <li>• Effect of diuretic drugs on urine volume</li> </ul>                                                                                                                                                                                                                                                                                                                                   |         |
| 9  | 4 | <b>4. Cardiovascular Drugs</b> <ul style="list-style-type: none"> <li>• Drugs used for the Treatment of Hypertension/ Antihypertensive Agents</li> <li>• Drugs Used for the Treatment of Angina</li> </ul>                                                                                                                                                                                                                                                        | A, B, D |
| 9  | 4 | <ul style="list-style-type: none"> <li>• Drugs Used for the Treatment of Heart Failure</li> <li>• Drugs for the Treatment of Cardiac Dysrhythmias</li> </ul>                                                                                                                                                                                                                                                                                                      | A, B, D |
| 10 | 4 | <ul style="list-style-type: none"> <li>• Lipid Regulating Drugs</li> <li>• Drugs for Hypotensive States (IV fluids, correction of electrolyte and acid base balance)</li> </ul>                                                                                                                                                                                                                                                                                   | A, B, D |

|    |   |                                                                                                                                                                                                                                                                                                                       |         |
|----|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| 11 | 4 | <b>5. Drugs acting on blood and blood forming organs</b> <ul style="list-style-type: none"> <li>Hematopoiesis; Anemias and Anti-anemic Agents <ul style="list-style-type: none"> <li>Coagulants and Anticoagulants</li> <li>Thrombolytics and Antiplatelets</li> </ul> </li> </ul>                                    | A, B, D |
| 12 | 4 | <b>6. Autacoids and drug therapy of inflammation</b> <ul style="list-style-type: none"> <li>Histamine and Its Antagonists</li> <li>5-Hydroxytryptamine and Its Antagonists; Bradykinin and Its Antagonists</li> <li>Lipid Derived Autocoids/Ecosanoids/ and Platelet Activating Factor/PAF/ , Nitric Oxide</li> </ul> | B, D    |
| 13 | 4 | <ul style="list-style-type: none"> <li>Analgesic and Antipyretics: Non-Steroidal</li> <li>Anti-inflammatory Drugs, Treatment of Rheumatoid Arthritis and Gout</li> </ul>                                                                                                                                              | B, D    |
| 13 | 9 | <b>7. Drugs Acting on the Respiratory System</b> <ul style="list-style-type: none"> <li>Drugs for the Treatment of Asthma <b>(Presentation)</b></li> <li>Antitussives Expectorants and Nasal Decongestants <b>(Presentation)</b></li> </ul>                                                                           | C       |
| 14 | 4 | <b>8. Drugs Acting on the Gastrointestinal System</b> <ul style="list-style-type: none"> <li>Drugs for the treatment of Peptic Ulcer Disease</li> </ul>                                                                                                                                                               | B       |
| 15 | 4 | <ul style="list-style-type: none"> <li>Drugs for the Treatment of Constipation: Laxatives and Cathartics; <b>(Presentation)</b></li> <li>Drugs for the Treatment of Diarrhea: Antidiarrheals <b>(Presentation)</b></li> <li>Emetics and Antiemetics, Digestants <b>(Presentation)</b></li> </ul>                      |         |
| 15 | 9 | <ul style="list-style-type: none"> <li>Practical Laboratory Sessions; e.g. <ul style="list-style-type: none"> <li>➤ Antispasmodic effect of Atropine</li> <li>➤ Anti-diarrheal effect of Loperamide</li> </ul> </li> </ul>                                                                                            |         |
|    |   | <b>Final Exam</b>                                                                                                                                                                                                                                                                                                     |         |

**Module 11: Medicinal Chemistry Module I**  
**Module Name: Medicinal Chemistry Module I**

**Module Category:** Core

**Module Code:** Phar-M2111

**Module Number:** 11

**Module weight in ECTS:** 7 ECTS

**Courses:**

**Medicinal Chemistry I (Phar2111) (7 ECTS)**

**Module description**

Medicinal chemistry is a chemistry-based discipline involving aspects of biological, medical and pharmaceutical sciences. It is the application of chemistry in the context of human medicine. The general purpose of this Module is to train highly qualified pharmacists who are competent in the invention, discovery, design, identification and preparation of biologically active compounds, the study of their metabolism, the interpretation of their mode of action at the molecular level and the construction of structure-activity relationships. Besides, this module helps the student in their future carrier especially in pharmaceutical industry drug research and development sections, in research institutions and in universities. Students will gain an appreciation for the drug development process, together with brief introduction to the drug discovery and designing methods, and also deals with the chemistry of various classes of drugs that act on different systems and organs of human body, and reviews the general principles of drug action and the pharmacological activities of various classes of drugs. The major focus is on the molecular mechanisms of drug action, with a detailed discussion of one or more prototypes of each drug class, which includes drugs acting on; autonomic nervous system, central nervous system, Histamine and histamine antagonists, non-narcotic analgesics, drugs used in gout; Antidiabetics, cardiovascular drugs; vitamins; pesticides; diagnostic agents; expectorants and antitussives; non-steroidal and steroidal hormones, local and general anesthetics, chemotherapeutic and products of biotechnology

**Module objective:** Upon completion of the module; students have concept of drug at molecular level to which they understand the effect of structure on the pharmacokinetics and Pharmacodynamics. Students are able to apply the knowledge in drug design, discovery and development.

**Module competencies:**

- Understand and demonstrate principles and practice of medicinal chemistry
- Discuss and Practice on different methods employed in drug design that helps to drug discovery and development
- Ability to follow and critically interpret the latest advances in the theory and practice of medicinal Chemistry
- Describe, identify and classify drugs based on their chemical structure, pharmacological action and site of drug action
- Relate the relevance of structure to pharmacological action
- Explain the principles of drug action and the role of bonding in drug-target interactions
- Discuss and Analyze the structural activities relationships of different compounds
- Develop skill to identify and synthesize biologically active compounds using standard methods of synthesis
- Identify and Practice on naming of pharmaceutical products

- Understand the basic biotransformation of organic compounds
- Suggest chemistry based application of biologically active compounds in advance; evaluate the probable side effect and adverse reactions
- Participate in problem solving drug development strategies
- Describe the physicochemical properties of biologically active compounds and currently available drugs
- Apply the knowledge and skill in drug therapy decision making process
- Transfer knowledge obtained from medicinal chemistry

**Mode of Delivery:** Parallel

**Mode of Assessment:**

Quizzes (10%)

Laboratory (20%)

Tests (20%)

Assignments (10%)

Final Exam (40%)

### **Learning activities and teaching methods**

Interactive lectures, case studies, computer assisted learning, formative problem-solving exercises, self-directed learning through virtual learning environment and technologies.

### **Teachers' and students' role**

#### **Teacher's role**

Course instructors are expected to:

- Organize group discussions
- Provide lecture and guide students
- Providing assignments and feedbacks for students (reading, working)
- Prepare lecture note, Assignment topics and title for group discussions
- Select seminar title and advice students in preparation and presentations
- Prepare assessing questions and examine students
- Prepare cases

#### **Student's role**

Students are expected to:

- Attend each lecture classes and Be an active participant in class discussion (ask questions and answering questions)
- Read text books, lecture handouts and reference books
- Prepare and present seminar papers
- Analyze and evaluate different literatures, reference books and journal articles
- Present case studies
- Take exams

### Medicinal Chemistry Module Syllabi

|                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Course Title</b>               | <b>Medicinal Chemistry I</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Course Code</b>                | Phar2111                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Course EtCTS (Course hour)</b> | 7 (189 hrs)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Pre-requisite</b>              | Organic Chemistry                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Co-requisite</b>               | Pharmacology I                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Course Description</b>         | Medicinal chemistry is the application of chemistry in the context of human medicine. In this course students will gain an appreciation for the drug development process, together with brief introduction to the drug discovery and designing methods, and also deals with the chemistry of various class of drugs that act on different systems and organs of human body, which includes drugs acting on Autonomic nervous system, Central nervous system, Respiratory system, Antioxidants and Autoxidation, Diagnostic agents, Antihistaminic agents, Non-narcotic analgesics and related drugs, expectorants, Pesticides and Antitussives, Gastrointestinal and related drugs.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Course Objectives</b>          | After completion of this course, students will be able to understand the drug discovery and designing methods, and also deals with the chemistry of various class of drugs that act on different systems and organs of human body                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Supporting objectives</b>      | <ul style="list-style-type: none"> <li>• Describe the basic concepts in medicinal chemistry,</li> <li>• Describe the basic concepts of drug design (the drug discovery and development process), and the strategies to achieve it.</li> <li>• To describe the chemical basis of drug absorption, distribution, metabolism and Elimination,</li> <li>• To recognize the important functional groups that act as weak acids and bases and to recognize the molecular and environmental factors that influence their precise ionization profiles.</li> <li>• To explore the fundamentals of drug metabolism (both biotransformation and conjugation pathways) through an identification of drug mechanisms that include the activation of some pro-drugs.</li> <li>• To describe the chemical basis of drug-target interactions.</li> <li>• To explore the structure-activity concepts related to the presence of specific functional groups in agonist and antagonist drug structures.</li> <li>• To understand the links with specific therapeutic applications for agonists and antagonists acting at specific drug receptors</li> <li>• To develop the ability to recognize superior therapeutic drug mechanisms and properties of drugs</li> </ul> |
| <b>Course Content</b>             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |





[illegible]

|                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Text Book</b>       | <i>Lemke, T.L. and Williams, D.A., Roche, V.F., Zito, W.S. Foye's Principles of Medicinal Chemistry, 6th. ed. Lippincott Williams &amp; Wilkins, 2008.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Reference Books</b> | <ol style="list-style-type: none"> <li>1. <i>Wilson-Gisvold-Doerge; Text book of organic medicinal chemistry and pharmaceutical chemistry. 12<sup>th</sup> edn.; Lippincott (USA), 2011.</i></li> <li>2. <i>Gareth Thomas, Medicinal Chemistry An introduction, 2<sup>nd</sup> edition, 2007</i></li> <li>3. <i>Rama Rao Nadendla; Principles of Organic Medicinal Chemistry, 2005.</i></li> <li>4. <i>Burger's Medicinal Chemistry and Drug discovery, Sixth edition, Volume 1: Drug Discovery; 2003</i></li> <li>5. <i>Burger's Medicinal Chemistry and Drug discovery and drug development, Sixth edition, Volume 2: Drug Discovery; 2003</i></li> <li>6. <i>Burger's Medicinal Chemistry and Drug discovery, Sixth edition, Volume 6: Nervous System Agents; 2003</i></li> <li>7. <i>Donald J. Abraham (Ed.). Burger's's medicinal Chemistry and Drug Discovery, 2003, 6<sup>th</sup> edn., vol1-6, wiley-interscience (USA)</i></li> <li>8. <i>Thomas, L.Lemeke and David, A. Williams. Principle of Medicinal Chemistry, 2002, 5<sup>th</sup> edn. A Lea and Febiger book, Williams and Wilkins</i></li> </ol> |

**Module Name: Dosage Form Sciences II****Module Category:** Core**Module Code:** Phar-M3121**Module Number:** 12**Module Weight:** 9 ECTS**Courses:**

| S/N | Course name                                                 | Course code | ECTS |
|-----|-------------------------------------------------------------|-------------|------|
| 1   | Integrated Physical Pharmacy and Pharmaceutics II           | Phar3121    | 7    |
| 2   | Practical Integrated Physical Pharmacy and Pharmaceutics II | Phar3122    | 2    |

**Module Description**

The module deals with the science and arts of converting drugs into medicines rendering students with practical insight into drug formulation principles at the very outset. It gives students the basic sciences in physical pharmacy that play a role during large scale production and extemporaneous compounding of semisolid dosage forms (ointment, cream, paste and jelly). Moreover, this module will give students about critical issue to be considered in formulation development including micromeritics, rheology, drug release from dosage forms (diffusion and dissolution), issue of components incompatibilities, drug stability and methods to determine shelf life of different products.

**Module Objective**

This module aims at providing the student with a broad understanding of physicochemical principles that govern the behavior of drugs, excipients and dosage forms. It also enables the student to prepare extemporaneous preparations based on the basic science knowledge she/he to gain.

**Module Competency**

This module enables student develop the knowledge, skill and attitude required in preparing extemporaneous preparations and play a role in formulation development.

**Mode of deliver**

- Parallel

**Module teaching/learning methods****Learning activities**

Students are expected to undertake among others the following major learning activities:

- Engage in learning by doing (independent study, group assignments, presentation, report writing, and etc...)
- Participation and note takings during class lectures and debates and discussions;
- Analysis, summarization and presentations of chapter/article, scientific papers (and be able present or submit in a concise and shorten form)
- Search validated formula from standard books, journals, scientific papers; elaborate the purpose of each component of the formula; understanding the compounding procedure; prepare extemporaneous preparations and then write label to it

**Teaching Methods**

- The course facilitator is expected to lecture for introducing concepts and topics, and give references, facilitate discussions, ask questions, give and correct assessments,
- Arrange and facilitate seminar sessions by inviting professionals from other units/departments in the department/school and other interested staffs as well.

**Module mode of Assessment:**

- Group assignments
- Presentations
- Laboratory reports
- Practical exam
- Tests/quizzes
- Final Exam

**Course Name:** Integrated Physical Pharmacy and Pharmaceutics II  
**Course code:** Phar3121  
**Module Name:** Dosage Form Sciences Module  
**Module Code:** 07  
**Course ECTS:** 7

**Totally required hours for the course:** 189hrs

|                         |    |
|-------------------------|----|
| <b>Lecture hours:</b>   | 64 |
| <b>Study hours:</b>     | 80 |
| <b>Group work:</b>      | 0  |
| <b>Project work:</b>    | 0  |
| <b>Presentation(s):</b> | 14 |
| <b>Tutorial:</b>        | 15 |
| <b>Assessment:</b>      | 16 |

**Year/Semester Course is offered:** Year III Semester I

**Course prerequisite/s:** Integrated Physical Pharmacy and Pharmaceutics I

**Course Description:**

The design of the course is based on the integration of the study of physico-chemical principles of pharmacy with the formulation, preparation and stabilization of semisolid and solid pharmaceutical dosage forms. In line with this, the course deals with scientific principles related to diffusion and dissolution theories associated with drug release kinetics from dosage forms. The module is also designed to enable students apply these theories and principles for the formulation and production of semisolid and solid dosage forms in the practical compounding sessions and thereafter during their professional career. The principles of drug degradation mechanisms, rate and kinetic theories of chemical reactions are also included for their application in the determination of product shelf-life and stability studies. The applications of pharmaceutics in cosmetics, overview of radiopharmaceuticals and veterinary dosage forms are also included.

**Course Objectives:**

After completion of this course students will be able to:

- Apply the physicochemical, electrical and thermodynamic properties of colloidal particles for the formulation of acceptable disperse systems
- Develop the knowledge and skills of formulation, compounding and dispensing of semisolid and solid dosage forms
- Select and characterize appropriate formulation excipients and packaging materials for pharmaceutical dosage forms and labeling thereof

- Describe the theories of diffusion and dissolution and apply for the determination of drug release kinetics from a dosage form
- Describe the different approaches of product stability studies and determine the shelf-life and expiry date of pharmaceutical products
- Identify the different types of formulation incompatibilities and their effect on the physicochemical and therapeutic performance of products
- Describe the different types of radiopharmaceuticals and their application, handling and storage precautions
- Apply the knowledge and skills of pharmaceutics in cosmetics
- Identify the important considerations while dealing with veterinary dosage forms

**Course mode of delivery:** Parallel

**Course learning and teaching methods:**

- Active participation during class lectures and excursions
- Engage in learning by doing
- The course instructor is expected to introduce concepts and topics and give references, facilitate discussions, ask questions, correct assignments
- Arrange and facilitate excursions

**Assessment techniques:**

- Assignments: 15%
- Tests: 30%
- Quizzes: 15%
- Final exam: 40%

**Teachers' and students' role**

**Roles of Instructors**

**The instructor will be expected to:**

- Be a facilitator (introduce the subject; give guidance, moderate discussions, etc.)
- Read and comment assignments & exercises of students on time;
- Prepare his/her lessons and deliver lectures;
- Provide available and necessary reference materials;
- Encourage active participation of students in the teaching learning process;
- Assist students with learning difficulties

**Roles of Students**

**Students are expected to:**

- Engage in learning by doing (independent study, group works/exercises, etc.)
- Be active learners (participate effectively in group assignments, class activities/exercises, etc.);
- Attend classes regularly

## References:

### Required readings (Text)

1. P. J. Sinko, Martin's Physical Pharmacy and Pharmaceutical Sciences, 7th Edition, Lippincott Williams & Wilkins, Philadelphia, 2016.
2. M. E. Aulton, Pharmaceutics: the science of dosage form design, 7th ed., Churchill Livingstone, Edinburgh.

### Recommended readings

3. M. J. Wilson, Pharmaceutical Compounding and Dispensing, 2nd Edition, Pharmaceutical press, 2010.
4. L. V. Allen, N. G Popovich, H. C Ansel, Ansel's Pharmaceutical Dosage Forms and Drug Delivery Systems, 11th edition, Lippincott Williams & Wilkins, 2017.
5. J. E. Thompson and L. Davidow, A Practical Guide to Contemporary Pharmacy Practice, 3rd edition, Lippincott Williams & Wilkins.
6. A. Martin, J. Swarbrick and A. Cammarata, Physical Pharmacy, 3rd Edition, Lea & Febiger, Philadelphia, 1983.
7. A. T. Florence and D. Attwood, Physico-chemical Principles of Pharmacy, Macmillan Publishers Ltd., London, 1981
8. S. C. Wallwork and D. J. W. Grant, Physical Chemistry for Students of Pharmacy and Biology, 3rd Edition, Longman Group Ltd., London, 1977.
9. Hardee, G. E. and Baggot, J. D., Development and Formulation of Veterinary Dosage Forms, 2nd ed. Marcel Dekker, Inc. New York, 1998.
10. Breuer, M. M., Cosmetic Science, Academic Press, London, 1978.

## Course Schedule

| Week | Contact Hours | Topic/sub-topic/chapter/Assessments/Assignments                                                                                                                                                                                                                                       |
|------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | 4             | <ul style="list-style-type: none"> <li>• Semisolid dosage forms               <ul style="list-style-type: none"> <li>○ Introduction</li> <li>○ Classification of semisolids</li> <li>○ Desired properties of semisolids</li> <li>○ <b>Ointments and Pastes</b></li> </ul> </li> </ul> |
|      |               |                                                                                                                                                                                                                                                                                       |

|                   |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                   |   | <ul style="list-style-type: none"> <li>▪ Introduction (Definitions, Properties and applications)</li> <li>▪ Formulation (ideal properties of bases, types of bases)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 2                 | 4 | <ul style="list-style-type: none"> <li>• Semisolid dosage forms (Continued...) <ul style="list-style-type: none"> <li>○ <b>Ointments and Pastes</b> (Continued...) <ul style="list-style-type: none"> <li>▪ Preparation of ointment and pastes</li> <li>▪ Packaging, labeling and storage</li> </ul> </li> <li>○ <b>Creams</b> <ul style="list-style-type: none"> <li>▪ Introduction (Definition, types and properties of creams)</li> <li>▪ Formulation</li> <li>▪ Methods of preparation</li> <li>▪ Packaging, labeling and storage</li> </ul> </li> <li>○ <b>Gels</b> <ul style="list-style-type: none"> <li>▪ Introduction (Definition and applications)</li> <li>▪ Classification</li> <li>▪ Formulation (Gelling agents, factors affecting gelation)</li> <li>▪ Syneresis and swelling of gels</li> <li>▪ Preparation of gels</li> <li>▪ Packaging, labeling and storage</li> </ul> </li> </ul> </li> </ul> |
| 3                 | 4 | <ul style="list-style-type: none"> <li>• Suppositories and Pessaries <ul style="list-style-type: none"> <li>○ Introduction (Definition and applications)</li> <li>○ Formulations (desirable properties of bases, classification of bases, other excipients)</li> </ul> </li> <li>• Quiz</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 4                 | 3 | <ul style="list-style-type: none"> <li>• Suppositories and Pessaries (Continued...) <ul style="list-style-type: none"> <li>○ Preparation of suppositories <ul style="list-style-type: none"> <li>▪ Calibration of moulds, determination of displacement value</li> <li>▪ Methods of preparation: fusion and compression</li> </ul> </li> <li>○ Packaging, labeling and storage</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 5                 | 4 | <ul style="list-style-type: none"> <li>• Micromeritics <ul style="list-style-type: none"> <li>○ Particle size and size distribution</li> <li>○ Methods for determination of particle size and size distribution</li> <li>○ Particle shape and surface area</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 6                 |   | <ul style="list-style-type: none"> <li>• Test I</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 6 [day other than | 4 | <ul style="list-style-type: none"> <li>• Micromeritics (Continued...) <ul style="list-style-type: none"> <li>○ Methods for determination of surface area</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |



|               |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the test day] |   | <ul style="list-style-type: none"> <li>Derived properties of powders (densities, porosity, packing arrangement, flowability)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                       |
| 7             | 4 | <ul style="list-style-type: none"> <li>Powders and Granules               <ul style="list-style-type: none"> <li>Powders as a dosage forms</li> <li>Introduction (definition, classification and applications)</li> <li>Preparation (size reduction, mixing and packing)</li> <li>Challenges of powder dosage forms; eg, eutectic mixtures</li> <li>Granules as dosage forms</li> <li>Granulated preparations</li> <li>Effervescent granules and methods of preparations</li> </ul> </li> <li>Quiz</li> </ul> |
| 8             | 4 | <ul style="list-style-type: none"> <li>Diffusion and Dissolution               <ul style="list-style-type: none"> <li>Introduction (osmosis, dialysis, diffusion)</li> <li>Fick's law of diffusion (steady state, diffusion through a membrane)</li> <li>Applications of diffusion</li> <li>Dissolution of particles (Noyes-whetney equation, factors affecting dissolution)</li> </ul> </li> </ul>                                                                                                           |
| 9             | 2 | <ul style="list-style-type: none"> <li>Diffusion and Dissolution (Continued...)               <ul style="list-style-type: none"> <li>Intrinsic dissolution rate</li> <li>Sink conditions, Lag time and burst effects</li> <li>Hixon-crowel equation</li> </ul> </li> </ul>                                                                                                                                                                                                                                    |
| 10            | 2 | <ul style="list-style-type: none"> <li>Kinetics and Drug Stability               <ul style="list-style-type: none"> <li>Introduction</li> <li>Rates and orders of reactions</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                            |
| 11            | 4 | <ul style="list-style-type: none"> <li>Kinetics and Drug Stability (Continued...)               <ul style="list-style-type: none"> <li>Physical degradation</li> <li>Chemical degradation (mechanisms and stabilization approaches)</li> <li>Factors affecting stability of drugs</li> <li>Influence of temperature on reaction rates (Arrhenius equation)</li> <li>Stability study (real time and accelerated stability study)</li> <li>Prediction of shelf life</li> </ul> </li> </ul>                      |
| 12            |   | <ul style="list-style-type: none"> <li>Test II</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 12 [day other | 4 | <ul style="list-style-type: none"> <li>Introduction to radiopharmaceuticals               <ul style="list-style-type: none"> <li>Formulation aspects, stability and handling of radiopharmaceuticals</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                   |

|                          |   |                                                                                                                                                                                                                                                                          |
|--------------------------|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| than<br>the test<br>day] |   | <ul style="list-style-type: none"> <li>• Incompatibilities in formulation</li> </ul>                                                                                                                                                                                     |
| 13                       | 4 | <ul style="list-style-type: none"> <li>• Cosmetics <ul style="list-style-type: none"> <li>○ Introduction</li> <li>○ Hair cosmetics</li> <li>○ Skin cosmetics</li> </ul> </li> </ul>                                                                                      |
| 14                       | 4 | <ul style="list-style-type: none"> <li>• Cosmetics (Continued...) <ul style="list-style-type: none"> <li>○ Deodorants and antiperspirants</li> <li>○ Oral care products</li> <li>○ Nail products</li> <li>○ Eye cosmetics</li> <li>○ Lip products</li> </ul> </li> </ul> |
| 15                       | 4 | <ul style="list-style-type: none"> <li>• Veterinary dosage forms</li> </ul>                                                                                                                                                                                              |
| 16                       |   | <ul style="list-style-type: none"> <li>• Test III</li> </ul>                                                                                                                                                                                                             |
|                          |   | <b>FINAL EXAM</b>                                                                                                                                                                                                                                                        |

**Course Name:** Practical Integrated Physical Pharmacy and Pharmaceutics II  
**Course code:** Phar3122  
**Module Name:** Dosage form Sciences Module  
**Module Code:** Phar-M3121  
**Course ECTS:** 2

|                                               |       |
|-----------------------------------------------|-------|
| <b>Totally required hours for the course:</b> | 54hrs |
| <b>Illustrated lecture:</b>                   | 8     |
| <b>Practical lab:</b>                         | 32    |
| <b>Assignments and assessment:</b>            | 7     |
| <b>Home Study:</b>                            | 7     |

**Year/Semester Course is offered:**

**Course prerequisite/s:** Integrated Physical Pharmacy and Pharmaceutics II

**Course Description:**

The course is designed to give basic understanding and concepts of practical Physical Pharmacy and Pharmaceutics, including: determination of particle size by different techniques, determination of some derived properties of powders and granules, preparation of different extemporaneous semisolid preparations, preparation of suppositories and pessaries, preparation of powder and granule dosage forms, and determination of order and rate constant.

**Course Objectives:**

After completion of this course students will be able to:

- Understand methods and techniques used in determination of particle size using microscopy and by sieving,
- Determination of some derived properties of powders and granules, including: flowability, density and porosity
- Prepare different types of semisolid dosage forms, including: ointments, pastes, creams, and gels.
- Prepare suppositories and pessaries
- Prepare Powder and Granule dosage forms
- Understand the techniques of how to determine the specific rate constants of drugs
- Understand the techniques of how to determine the order of reaction
- Appreciate techniques to detected changes in the physical, chemical, or therapeutic qualities due to incompatibilities.

**Course mode of delivery:** Parallel

**Course learning and teaching methods:**

- Illustrated lectures and discussions, student presentations, individual and group laboratory practicals & demonstrations.

#### **Assessment techniques:**

- Practical skill and competency based exams: 50%
- Written exam: 25%
- Laboratory reports: 15%
- Presentation: 10%

#### **Teachers' and students' role**

##### **Roles of Instructors**

##### **The instructor will be expected to:**

- Be a facilitator (introduce the subject; give guidance, moderate discussions, etc.)
- Read and comment lab reports, assignments & presentations of students on time;
- Prepare his/her lessons and deliver lectures;
- Provide available and necessary reference materials;
- Encourage active participation of students in the teaching learning process;
- Assist in laboratory practice & demonstrations,
- Assist students with learning difficulties

##### **Roles of Students**

##### **Students are expected to:**

- Engage in learning by doing (independent study, group works/exercises, etc.)
- Be active learners (participate effectively in group assignments, lab activities/presentations, etc.);
- Attend lab sessions regularly

#### **References:**

##### **Required readings (Text)**

1.

##### **Recommended readings**

2.

#### **Course Schedule**

| <b>Week</b> | <b>Contact Hours</b> | <b>Topic/sub-topic/chapter/Assessments/Assignments</b> |
|-------------|----------------------|--------------------------------------------------------|
|             |                      |                                                        |

|    |   |                           |
|----|---|---------------------------|
| 1  | 3 | • Experiment No 1:        |
| 2  | 3 | • Experiment No 2:        |
| 3  | 3 | • Experiment No 3:        |
| 4  | 3 | • Experiment No 4:        |
| 5  | 3 | • Experiment No 5:        |
| 6  | 3 | • Experiment No 6:        |
| 7  | 3 | • Experiment No 7:        |
| 8  | 3 | • Experiment No 8:        |
| 9  | 3 | • Experiment No 9:        |
| 10 | 3 | • Experiment No 10:       |
| 11 | 3 | • Experiment No 11:       |
| 12 | 3 | • Experiment No 12:       |
| 13 | 3 | • Experiment No 13:       |
| 14 | 3 | • Experiment No 14:       |
| 15 | 3 | • Experiment No 15:       |
| 16 | 3 | • Experiment No 16:       |
|    |   | • Practical exam          |
|    |   | <b>FINAL WRITTEN EXAM</b> |

## **Module 13: Pharmacology Module II**

**Module name:** Pharmacology module II

**Module category:** Core

**Module code:** Phar-M3131

**Module number:** 13

**Module weight in ECTS:** 10

**Courses:**

| <b>Course name</b>  | <b>Course Code</b> | <b>ECTS</b> |
|---------------------|--------------------|-------------|
| Pharmacology II     | Phar 3131          | 7           |
| Clinical Toxicology | Phar 3132          | 3           |

### **Module description**

The pharmacology module will familiarizes the pharmacy students about the drugs, their pharmacokinetics, pharmacodynamic, clinical indication, contraindication, drug interaction and adverse effect of the therapeutically used drugs. In addition to that the module introduces the students about poisons, and management of poisoning agents. By incorporating what they learn in the theoretical aspect in to the laboratory attachment they will become well organized and well oriented professional.

**Module objective:** The objective of the module is to impart fundamental knowledge, skill on the pharmacokinetics, pharmacodynamic, therapeutic use and toxic effects drugs of both therapeutically benefit or toxic/poisoning agents.

### **Module competency:**

- Apply the knowledge and skill of Pharmacology and toxicology in drug therapy decision

### **Mode of delivery:** Parallel

- Total time: 270 hrs
  - Lecture: 96 hrs
  - Practical/lab session: 30 hrs
  - Tutorial: 48 hrs
  - Independent study hour: 60 hrs
  - Seminar/Presentation: 24 hrs
  - Assessment: 12 hrs

### **Module learning teaching methods**

Active learning methods (brain storming, buzz group, discussion, etc), Lecture, group & individual presentation, assignment, project work, and laboratory work.

**Mode of Assessment:**

Quizzes

Mid exam

Final Exam

Practical exam

Seminar

Assignment

**Course title: Pharmacology II**

**Course code:** Phar 3131

**Module name: Pharmacology module II**

**Module code: Phar-M3131**

**Course ECTS:** 7 ECTS (189 hrs)

- Lecture: 64 hours
- Laboratory: 30 hours
- Tutorial: 32 hours
- Home study: 40 hours
- Assignment and presentation: 16 hours
- Assessment : 7 hours

**Year/Semester Course is offered: Year III, Semester I**

**Contact hours/ week:** 189- 40= 149 hours/ 16 weeks= 9 hours

**Pre-requisite:**

- Pharmacology I

**Course description:**

This course is a continuation of Pharmacology I. It is designed to enable graduate Pharmacists describe drugs that are not addressed in Pharmacology I and are used for treatment, diagnosis and prevention of diseases. The course starts with renal and cardiovascular pharmacology and tries to cover diuretics and drugs used for the treatment of hypertension, angina, congestive heart failure, arrhythmia, and hyperlipidemia. Then it describes drugs affecting blood and blood forming organs, Chemotherapeutic drugs which will include antibacterial, antiseptics, antifungal drugs, antiviral, antiprotozoal, antiparasitic and anticancer agents. Moreover this course also describes drugs acting on the endocrine system and finally dermatological agents used for various skin disorders.

**Course Objectives:**

At the end of this course, students will be able to describe drugs acting on various organs and systems and drugs used for the treatment of Infectious Diseases and Neoplastic Diseases.

**Learning Objectives**

- Up on completion of this course, students will be able to
  - Explain the pharmacokinetics of drugs acting on the renal, cardiovascular, blood and blood forming organs, and chemotherapeutic drugs



- Explain the pharmacological actions, mechanism of actions, and therapeutic uses of drugs renal, cardiovascular, blood and blood forming organs, and chemotherapeutic drugs
- Describe side effects of drugs acting on renal, cardiovascular, blood and blood forming organs, and chemotherapeutic drugs.
- Apply concept and principles of pharmacology to ensure and proper use of drugs.
- To work in Pharmacology Laboratory and will be able to practice selected basic experimental demonstration

**Course mode of delivery: Block/Parallel**

**Course learning and teaching methods**

Active learning methods (brain storming, buzz group, discussion, etc), Lecture, group and individual presentation, assignment, project work, case studies and laboratory work.

**Assessment techniques:**

Continuous assessment & summative assessment

- Quiz (10%)
- Tests (15%)
- Assignments with Presentation (10 %)
- Laboratory (25 %)
- Final Exam (40%)

**Teachers and Students Role**

**Role of Instructor**

The instructor will be expected to:

- Facilitates students' individual and group activities
- Organize students' hospital visit, laboratory work, workshop practices (if any) and project work presentation(s) and discussion sessions
- Assess students' performances (written and oral presentations)
- Provide timely feedback orally and in writing
- Make follow-up on developments made
- Plan and implement students' consultation program

**Role of Students**

Students are expected to:

- Attend sessions
- Carry out individual and group tasks
- Active participant
- Reflect on feedbacks and take actions
- Carry out reading assignment

### References:

Required reading (text)

A. Katzung B.G.: Basic and Clinical Pharmacology: 14<sup>th</sup> or later editions.

### Recommended reading:

B. Goodman and Gilman's: The Pharmacological Basis of Therapeutics; 13<sup>th</sup> or later editions.

C. Rang H.P. and Dale M.M. : Pharmacology; 8<sup>th</sup> edition or later editions.

D. Mycek M.J. Harvey R.A. Lipincott's Illustrated Reviews: Pharmacology; 8<sup>th</sup> or later editions.

E. Richard A. LEHNE. Pharmacology for Nursing care. 5<sup>th</sup> or later editions.

**Course Schedule:** contact time, contents/topics & reading/reference materials for each topic

| Week | Contact hrs | Topic/sub-topic/chapter                                                                                                                                                                                                             | Reading materials |
|------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| 1    | 4           | <b>1. Drugs acting on the central nervous system</b> <ul style="list-style-type: none"> <li>• Central Neurotransmitters, Classification of CNS Drugs</li> <li>• General Anesthetics</li> <li>• Local Anesthetics</li> </ul>         | A, E              |
| 1    | 9           | <ul style="list-style-type: none"> <li>• Practical Laboratory Sessions; e.g. <ul style="list-style-type: none"> <li>➤ Anesthetic potency of ether and halothane</li> <li>➤ Solubility of general anesthetics</li> </ul> </li> </ul> |                   |
| 2    | 4           | <ul style="list-style-type: none"> <li>• Sedatives and Hypnotics: Benzodiazepines, Non-Benzodiazepines, Barbiturates, Newer Sedative Hypnotics</li> </ul>                                                                           | A, E              |

|   |   |                                                                                                                                                                                                                                                                                                                                                                                               |      |
|---|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
|   |   | <ul style="list-style-type: none"> <li>Alcohols</li> <li>Centrally Acting Muscle Relaxants</li> </ul>                                                                                                                                                                                                                                                                                         |      |
| 2 | 6 | <ul style="list-style-type: none"> <li>Practical Laboratory Sessions; e.g. <ul style="list-style-type: none"> <li>➤ The sedative-hypnotic effect of diazepam</li> <li>➤ Effect of diazepam in induced skeletal muscle relaxation in mice by using inclined plane</li> </ul> </li> </ul>                                                                                                       |      |
| 3 | 4 | <ul style="list-style-type: none"> <li>Opioid Analgesics and Antagonists</li> <li>Analeptics/ CNS Stimulants/ and Psychotomimetics</li> <li>General Aspects of Psychopharmacology</li> </ul>                                                                                                                                                                                                  | A, E |
| 3 | 9 | <ul style="list-style-type: none"> <li>Practical Laboratory Sessions; e.g <ul style="list-style-type: none"> <li>➤ Analgesic effect of morphine in mice using tail immersion method;</li> <li>➤ The analgesic effect of morphine after a pain is induced by chemical stimuli</li> <li>➤ Chemical induced comparative analgesic effect opioid and non-opioid analgesics</li> </ul> </li> </ul> |      |
| 4 | 4 | <ul style="list-style-type: none"> <li>Antipsychotics</li> <li>Antidepressants</li> </ul>                                                                                                                                                                                                                                                                                                     | A, E |
| 4 | 4 | <ul style="list-style-type: none"> <li>Drugs for the treatment of Anxiety and Mood Disorders</li> <li>Drugs for the treatment of Neurodegenerative Disorders</li> </ul>                                                                                                                                                                                                                       |      |
| 4 |   | <ul style="list-style-type: none"> <li>Practical Laboratory Sessions; e.g <ul style="list-style-type: none"> <li>➤ Atropine like Anti-Parkinson's drug</li> </ul> </li> </ul>                                                                                                                                                                                                                 |      |
| 5 | 4 | <ul style="list-style-type: none"> <li>Antiepileptic Drugs</li> <li>Social Pharmacology: Drug Dependence</li> </ul>                                                                                                                                                                                                                                                                           | A, E |
| 5 | 4 | <p><b>2.Anti-infective drugs</b></p> <ul style="list-style-type: none"> <li>Principles of Antimicrobial Therapy</li> </ul>                                                                                                                                                                                                                                                                    | A, B |

|    |   |                                                                                                                                                                                                                                                                                                                                                                  |         |
|----|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
|    |   | <ul style="list-style-type: none"> <li>Antiseptics and Disinfectants (<b>Presentation</b>)</li> </ul>                                                                                                                                                                                                                                                            |         |
| 6  | 4 | <ul style="list-style-type: none"> <li>Cell Wall Synthesis Inhibitors: <math>\beta</math>-lactam Antibiotics and Other Cell Wall Synthesis Inhibitors</li> </ul>                                                                                                                                                                                                 | A, B    |
| 7  | 4 | <ul style="list-style-type: none"> <li>Protein Synthesis Inhibitors: Aminoglycosides, Tetracyclines, Chloramphenicol, Macrolides, Streptogramins, Oxazolidinone, Others</li> </ul>                                                                                                                                                                               | A, D    |
| 8  | 4 | <ul style="list-style-type: none"> <li>Antimetabolites: Sulfonamides, Trimetoprim, Trimetoprim-Sulfamethoxazole</li> <li>Fluoroquinolones</li> </ul>                                                                                                                                                                                                             | A, D    |
| 9  | 4 | <ul style="list-style-type: none"> <li>Antimycobacterial Drugs: Drugs for the Treatment of Tuberculosis</li> <li>Antileprosy drugs</li> </ul>                                                                                                                                                                                                                    | A, B    |
| 10 | 4 | <ul style="list-style-type: none"> <li>Antifungal Drugs: <ul style="list-style-type: none"> <li>➤ Systemic Antifungal Agents; Amphotericin B, Flucytosine, The Azoles, Echinocandins, Allylamines</li> <li>➤ Topical Antifungal Agents; Nystatin, Topical Azoles, Topical Allylamines</li> <li>➤ Other antifungal drugs like Griseofulvin</li> </ul> </li> </ul> | A, D    |
| 11 | 4 | <ul style="list-style-type: none"> <li>Antiviral Agents: <ul style="list-style-type: none"> <li>Introduction,</li> <li>Agents to Treat Herpes Simplex Virus (HSV) &amp; Varicella Zoster Virus (VZV) Infections,</li> <li>Agents to Treat Cytomegalovirus (CMV) Infections, Hepatitis Virus</li> </ul> </li> </ul>                                               | A, C    |
| 12 | 4 | <ul style="list-style-type: none"> <li>Drugs for Influenza Virus Infection ,</li> <li>Drugs for Respiratory Syncytial Virus Infection</li> <li>Antiretroviral Drugs</li> </ul>                                                                                                                                                                                   | A, B, D |

|    |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |      |
|----|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 12 | 4 | <ul style="list-style-type: none"> <li>• Drugs Used for the Treatment of Parasitic Infections (<b>Presentation</b>)</li> <li>• Drugs Used for the Treatment of Malaria</li> <li>• Drugs Used for the Treatment of Amebiasis, Giardiasis, Trichomoniasis, Leishmaniasis, and Trypanosomiasis (<b>Presentation</b>)</li> <li>• Anthelmintics (<b>Presentation</b>)</li> </ul>                                                                               | B, D |
| 13 | 4 | <b>3. Anti-cancer Drugs</b> (Alkylating agents, Antimetabolites, anticancer antibiotics, Plant derived anticancer drugs,...and others anticancer drugs                                                                                                                                                                                                                                                                                                    | A    |
| 13 | 4 | <b>4. Immunomodulators</b>                                                                                                                                                                                                                                                                                                                                                                                                                                | B    |
| 14 | 4 | <b>5. Vitamins, hormones &amp; hormone antagonists</b> <ul style="list-style-type: none"> <li>• Vitamins: Water Soluble Vitamins and Fat Soluble Vitamins</li> <li>• Introduction; Anterior Pituitary Hormones: Growth Hormone and Its Antagonists, Prolactin, Gonadotropins Thyroid and Antithyroid Drugs</li> </ul>                                                                                                                                     | B    |
| 15 | 4 | <ul style="list-style-type: none"> <li>• Pancreatic Hormones: Drugs for the Treatment of Diabetes Mellitus- Insulin and Oral Hypoglycemic Agents</li> <li>• Corticosteroids</li> <li>• Sex Hormones: Estrogens, Anti-Estrogens and Estrogen Receptor Modulators Progestins, Anti-Progestins and Progesterone Receptor Modulators <ul style="list-style-type: none"> <li>• Hormonal Contraceptives, Androgens and Anabolic Steroids</li> </ul> </li> </ul> | A, B |

|    |   |                                                                                                                                                                                                                                                                                                                       |   |
|----|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 16 | 4 | <b>6. Dermatological Pharmacology</b> <ul style="list-style-type: none"> <li>• Drug Used for the Treatment of Skin Problems: Pediculicides and Scabicides</li> <li>Sunscreens and Agents Affecting Pigmentation</li> <li>• Drugs for the Treatment of Acne</li> <li>• Drugs for the Treatment of Psoriasis</li> </ul> | C |
|    |   | <b>FINAL EXAM</b>                                                                                                                                                                                                                                                                                                     |   |

**Course title: Clinical toxicology**

**Course code: Phar 3132**

**Module name: Pharmacology module II**

**Module code: Phar-M3131**

- Lecture: 32 hours
- Tutorial: 16 hours
- Home study: 20 hours
- Presentation : 8
- Assessment : 5

**Year/Semester Course is offered: Year III Semester II**

**Contact hours/ week: 2**

**Pre-requisite:** Pharmacology I and II

Schedule of contact time, contents/topics & reading/reference materials for each topic

**Course Description:**

Clinical toxicology course comprehensively provides the student with the fundamental concepts of clinical toxicology and provides students with knowledge about the toxic effects of a number of important therapeutic drug classes as well as additional insight into a number of organ systems. Specifically, the course covers general principles of toxicology; routes and types of exposure to toxicants and experimental toxicity testing methods. The course also describes the strategies in the management of poisoned patients starting from clinical stabilization to giving specific antidote. Then the course covers industrial and environmental toxicology in reference to, heavy metals, pesticides, air pollutants, organic solvents and vapours.. Moreover, the course emphasis to general and specific measures to be taken during poisoning in the management of common poisons substances, drugs, household toxicants and others.

**Course Objectives:**

To provide students with a conceptual framework for understanding the broad spectrum of toxicology with more emphasis on clinical toxicology

**Learning objectives**

After completing the course, students will acquire sound knowledge on toxicology, which will enable them to:

- a. Describe general toxicology principles and clinical management practice

- b. Describe drug adverse effects/toxicities particularly their recognition, prevention and treatment
- c. Decide on treatment of poisoned patients, and apply them effectively and safely for the benefit of the patient
- d. Describe the toxic effects and management of industrial and environmental toxicants
- e. Share the responsibilities to solve the emerging social, economic and medical problems of related to different toxicants.

**Course mode of delivery: Parallel**

**Course learning and teaching methods**

Active learning methods (brain storming, discussion, etc), Lecture, group and individual presentation, assignment, Case Studies and Practical visit to Emergency care unit to see treatment of poisoned patients.

**Assessment techniques:**

Continuous assessment & summative assessment

- Quiz (10%)
- Tests (20 %)
- Assignments with Presentation (10 %)
- Presentations on practical visit (Case presentations) (10 %)
- Final Exam (40%)

**Teachers and Students Role**

**Role of Instructor**

The instructor will be expected to:

- Facilities students' individual and group activities
- Organize students' hospital visit, laboratory work, workshop practices (if any) and project work presentation(s) and discussion sessions
- Assess students' performances (written and oral presentations)
- Provide timely feedback orally and in writing
- Make follow-up on developments made
- Plan and implement students' consultation program

**Role of Students**

Students are expected to:



- Attend sessions
- Carry out individual and group tasks
- Active participant
- Reflect on feedbacks and take actions
- Carry out reading assignment

## Reference Books

### Required Reading (Text)

A. Doull, J., Kalassen, C.D., and Amdur, M.D., (eds.) Casarett and Doull's Toxicology, the Basic sciences of poisons, 10th Ed, MCGraw Hill, 2010.

### Recommended Reading

B. Timbrell, J.A. Introduction to Toxicology, Taylor and Francis Ltd. 4<sup>th</sup> ed. 2009.

C. Peter Viccellio. Handbook of medical toxicology (2008)

D. Lester M. Haddad et al. Clinical management of poisoning and drug overdose (2008)

**Course Schedule:** contact time, contents/topics & reading/reference materials for each topic

| Week | Contact hrs | Topic/sub-topic/chapter/Assessment/Assignments                                                                                                                                                                                     | Reading materials |
|------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| 1    | 2           | <b>1. Introduction and General Principles of toxicology</b> <ul style="list-style-type: none"> <li>➤ Definition, History and Branches of toxicology</li> <li>➤ Scope and Application</li> </ul>                                    | A, B              |
| 2    | 2           | <ul style="list-style-type: none"> <li>○ Dose Response Relationship and LD 50</li> <li>○ Evaluation of Drug Toxicity in lower animals &amp; in man</li> </ul>                                                                      | A, B              |
| 3    | 2           | ❖ Categories of Toxicologic Tests (acute, sub-acute, sub-chronic and chronic Toxicity, toxicogenetics)                                                                                                                             | A, B              |
| 4    | 2           | <b>2. Management of the poisoned or overdosed patient</b> <ul style="list-style-type: none"> <li>➤ Common poisons and Epidemiology of Poisoning</li> <li>➤ General Principles in Management of Poisoned patients</li> </ul>        | A, B, D           |
| 5    | 2           | <ul style="list-style-type: none"> <li>○ Clinical Stabilization and Supportive Therapy (Cardiovascular, Respiratory, Neuropsychiatry)</li> <li>○ Physical Examination and Laboratory investigation of Poisoned patients</li> </ul> | A, B, D           |
| 6    | 2           | <ul style="list-style-type: none"> <li>○ GI decontaminations (Gastric Lavage, Activated Charcoal, Emesis, Laxatives and others)</li> </ul>                                                                                         | A, B, D           |
|      |             | <ul style="list-style-type: none"> <li>○ Enhancement of elimination of toxicants (PH Alteration, Dialysis)</li> <li>○ Specific Antidote</li> </ul>                                                                                 | A, B, D           |

|    |   |                                                                                                                                                                                                                                                                          |         |
|----|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| 7  | 2 | <b>3. Industrial and environmental toxicants</b><br>○ Heavy Metal Poisoning (Chemicals involved, Mechanisms of Toxicity and Toxic Manifestations , Heavy Metal Antagonists )                                                                                             | A, B, C |
| 8  | 2 | ○ Non-Metallic Environmental Toxicants (Air and water pollutants, Organic Solvents and Vapor, Utility gases Pesticides and other Agricultural Toxicants                                                                                                                  | A, B, C |
| 9  | 2 | <b>4. Teratogenic and Carcinogens</b><br>➤ Introduction on teratogenesis and Carcinogenesis<br>➤ Mechanisms of teratogenesis and Carcinogenesis<br>➤ Chemical having teratogenic and carcinogenic effects                                                                | A, B    |
| 10 | 2 | <b>5. Clinical Toxicology of some chemicals, drugs and household agents</b><br>➤ Narcotic analgesics, NSAIDs (Salicylates )<br>➤ Barbiturates<br>➤ Alkaloids                                                                                                             | A, E    |
| 11 | 2 | ➤ Phenothiazines<br>➤ Calcium Channel Blockers, $\beta$ -Adrenergic Antagonists, Cardioactive Steroids, Methylxanthines and Selective $\beta$ -2-Adrenergic Agonists                                                                                                     | A, B, D |
| 12 | 2 | ➤ Antipsychotics, Lithium, Cyclic Antidepressants and anesthetic drugs<br>➤ Monoamine Oxidase Inhibitors, Serotonin Reuptake Inhibitors and Atypical Antidepressants                                                                                                     | A, B, D |
| 13 | 2 | Forensic Toxicology<br>➤ Def. and Scope<br>➤ Examination Methods<br>Toxic effects of chemotherapeutic agent<br>Cyanide                                                                                                                                                   | A, B, D |
| 14 | 2 | ➤ Drugs of abuse ( <b>Presentation</b> )<br>➤ Poisons of Animal Origin (venoms) ( <b>Presentation</b> )<br>➤ Poisons of plant origin ( <b>Presentation</b> )<br>➤ Cleaning & polishing agents ( <b>Presentation</b> )<br>➤ Food & food additives ( <b>Presentation</b> ) | A, B, D |
| 15 | 2 | ➤ <b>Hospital Visit and Case Presentations</b>                                                                                                                                                                                                                           |         |
| 16 | 2 | ➤ <b>Hospital Visit and Case Presentations</b>                                                                                                                                                                                                                           |         |
|    |   | <b>FINAL EXAM</b>                                                                                                                                                                                                                                                        |         |

**Module 14: Medicinal Chemistry Module II**  
**Module Name: Medicinal Chemistry Module II**  
**Module Category: Core**

**Module Code:** Phar-M3141

**Module Number:** 14

**Module weight in ECTS:** 5 ECTS

**Courses:**

**Medicinal Chemistry II (Phar3141) (5 ECTS)**

### **Module description**

Medicinal chemistry is a chemistry-based discipline involving aspects of biological, medical and pharmaceutical sciences. It is the application of chemistry in the context of human medicine. The general purpose of this Module is to train highly qualified pharmacists who are competent in the invention, discovery, design, identification and preparation of biologically active compounds, the study of their metabolism, the interpretation of their mode of action at the molecular level and the construction of structure-activity relationships. Besides, this module helps the student in their future career especially in pharmaceutical industry drug research and development sections, in research institutions and in universities. Students will gain an appreciation for the drug development process, together with brief introduction to the drug discovery and designing methods, and also deals with the chemistry of various classes of drugs that act on different systems and organs of human body, and reviews the general principles of drug action and the pharmacological activities of various classes of drugs. The major focus is on the molecular mechanisms of drug action, with a detailed discussion of one or more prototypes of each drug class, which includes drugs acting on; autonomic nervous system, central nervous system, Histamine and histamine antagonists, non-narcotic analgesics, drugs used in gout; Antidiabetics, cardiovascular drugs; vitamins; pesticides; diagnostic agents; expectorants and antitussives; non-steroidal and steroidal hormones, local and general anesthetics, chemotherapeutic and products of biotechnology

**Module objective:** Upon completion of the module; students have concept of drug at molecular level to which they understand the effect of structure on the pharmacokinetics and Pharmacodynamics. Students are able to apply the knowledge in drug design, discovery and development.

### **Module competencies:**

- Understand and demonstrate principles and practice of medicinal chemistry
- Discuss and Practice on different methods employed in drug design that helps to drug discovery and development
- Ability to follow and critically interpret the latest advances in the theory and practice of medicinal Chemistry
- Describe, identify and classify drugs based on their chemical structure, pharmacological action and site of drug action
- Relate the relevance of structure to pharmacological action
- Explain the principles of drug action and the role of bonding in drug-target interactions
- Discuss and Analyze the structural activities relationships of different compounds
- Develop skill to identify and synthesize biologically active compounds using standard methods of synthesis
- Identify and Practice on naming of pharmaceutical products
- Understand the basic biotransformation of organic compounds

- Suggest chemistry based application of biologically active compounds in advance; evaluate the probable side effect and adverse reactions
- Participate in problem solving drug development strategies
- Describe the physicochemical properties of biologically active compounds and currently available drugs
- Apply the knowledge and skill in drug therapy decision making process
- Transfer knowledge obtained from medicinal chemistry

**Mode of Delivery:** Parallel

**Mode of Assessment:**

Quizzes (10%)

Laboratory (20%)

Tests (20%)

Assignments (10%)

Final Exam (40%)

### **Learning activities and teaching methods**

Interactive lectures, case studies, computer assisted learning, formative problem-solving exercises, self-directed learning through virtual learning environment and technologies.

### **Teachers' and students' role**

#### **Teacher's role**

Course instructors are expected to:

- Organize group discussions
- Provide lecture and guide students
- Providing assignments and feedbacks for students (reading, working)
- Prepare lecture note, Assignment topics and title for group discussions
- Select seminar title and advice students in preparation and presentations
- Prepare assessing questions and examine students
- Prepare cases

#### **Student's role**

Students are expected to:

- Attend each lecture classes and Be an active participant in class discussion (ask questions and answering questions)
- Read text books, lecture handouts and reference books
- Prepare and present seminar papers
- Analyze and evaluate different literatures, reference books and journal articles
- Present case studies
- Take exams

## Medicinal Chemistry Course Syllabus

|                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| <b>Course Title</b>                                                                                                                                                                                                                                        | <b>Medicinal Chemistry II</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                          |
| <b>Course Code</b>                                                                                                                                                                                                                                         | Phar3141                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                          |
| <b>Course EtCTS (Course hour)</b>                                                                                                                                                                                                                          | 5 (135 hrs)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                          |
| <b>Pre-requisite</b>                                                                                                                                                                                                                                       | Medicinal Chemistry I                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                          |
| <b>Co-requisite</b>                                                                                                                                                                                                                                        | Pharmacology II                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                          |
| <b>Course Description</b>                                                                                                                                                                                                                                  | Medicinal Chemistry II is a continuation of medicinal chemistry I & covers; Diuretics and Cardiovascular agents, Chemotherapeutic agents which includes Antiseptics and Disinfectants, Preservatives, Antifungal agents, Antitubercular and Antileprotic agents, Topical agents, Antiviral agents, Antiprotozoal drugs, Anthelmintics, Antiscabies & Antipedicular agents, Sulfonamides, Antibiotics & Anticancer drugs; and Steroidal and Non Steroidal and related drugs.                                                                                                                                    |                                          |
| <b>Course Objectives</b>                                                                                                                                                                                                                                   | After completion of this course, students will be able to understand drugs that will act on the kidney, cardiovascular system, drugs used for treatment of infectious diseases, vitamins and hormones.                                                                                                                                                                                                                                                                                                                                                                                                         |                                          |
| <b>Course Content</b>                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                          |
| <b>Diuretics and Cardiovascular Drugs</b>                                                                                                                                                                                                                  | <ul style="list-style-type: none"> <li>○ Diuretics <ul style="list-style-type: none"> <li>▪ Carbonic anydrase inhibitors</li> <li>▪ Loop diuretics</li> <li>▪ Thiazide diuretics</li> <li>▪ Potassium sparing diuretics</li> <li>▪ Osmotic diuretics</li> </ul> </li> <li>○ Antianginal agents and vasodilators</li> <li>○ Antihypertensive agents</li> <li>○ Drugs used for treatment of congestive heart failure</li> <li>○ Antiarrythmic drugs</li> <li>○ Anti-anemic Agents</li> <li>○ Drugs for Hypotensive States</li> <li>○ Antihyperlipidemic drugs</li> <li>○ Coagulant and anticoagulants</li> </ul> | <b>10 hrs</b>                            |
| <ul style="list-style-type: none"> <li>▪ <b>Oral hypoglycemic Agents</b></li> <li>▪ <b>Chemotherapeutic Agents</b> <ul style="list-style-type: none"> <li>○ Introduction (1 hr.)</li> <li>○ Antiseptics &amp; disinfectants (1 hrs)</li> </ul> </li> </ul> | <p>Definition, Classification, Alcohols and related compounds, Aldehydes, Phenols and their derivatives, Oxidizing agents, Halogen containing compounds, Cationic and anionic Surfactants, Dyes, Mercury compounds, Antiseptic nitro furan derivatives</p> <ul style="list-style-type: none"> <li>○ Preservatives (1 hr.)</li> </ul>                                                                                                                                                                                                                                                                           | <p><b>2 hrs</b></p> <p><b>30 hrs</b></p> |

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| <p>Definition, p-Hydroxybenzoic acid derivatives, p- Hydroxybenzoic acid derivatives, Other miscellaneous preservatives</p> <ul style="list-style-type: none"> <li>○ Antibacterial agents and related drugs (8 hr.) <ul style="list-style-type: none"> <li>▪ Introduction</li> <li>▪ <math>\beta</math>-lactam antibiotics and Glycopeptides</li> <li>▪ The amino glycosides</li> <li>▪ The tetracyclines</li> <li>▪ The macrolides</li> <li>▪ Chloramphenicol</li> <li>▪ Sulfonamides and related drugs</li> <li>▪ Quinolones</li> <li>▪ The polypeptides, polyenes and lincomycin</li> <li>▪ Antitubercular &amp; Antileprotic agents</li> <li>▪ Antimicrobial topical agents</li> <li>▪ Other antibacterial agents</li> </ul> </li> <li>○ Antifungal agents (2 hr.)</li> </ul> <p>Fatty acids, Substituted imidazoles and triazoles, Antifungal antibiotics (the polyenes and other), Miscellaneous antifungal agents</p> <ul style="list-style-type: none"> <li>○ Antiviral agents (6 hr.) <ul style="list-style-type: none"> <li>▪ Anti-HIV agents</li> <li>▪ Other antiviral drugs</li> </ul> </li> <li>○ Antiprotozoal agents (4 hr.) <ul style="list-style-type: none"> <li>▪ Antimalarial drugs</li> <li>▪ Drugs used in amebiasis, giardiasis &amp; trichomoniasis</li> <li>▪ Drugs used in other protozoal infections</li> </ul> </li> <li>○ Anthelmintic agents (2hr.)</li> </ul> <p>Antiscabies &amp; Antipedicular agents</p> <ul style="list-style-type: none"> <li>○ Antineoplastic agents (5 hr.) <ul style="list-style-type: none"> <li>▪ Chemotherapy</li> <li>▪ Alkylating agents</li> <li>▪ Antimetabolites</li> <li>▪ Antibiotics</li> </ul> </li> </ul> <ul style="list-style-type: none"> <li>▪ <b>Vitamins and Coenzymes</b> <ul style="list-style-type: none"> <li>○ Fat-soluble vitamins</li> <li>○ Water-soluble vitamins</li> <li>○ Co-enzymes</li> </ul> </li> <li>▪ <b>Non Steroidal Hormones and Related Drugs</b> <ul style="list-style-type: none"> <li>○ Hormones of the hypothalamus</li> <li>○ Pituitary hormones</li> <li>○ Thyroid hormones and anti-thyroid drugs</li> <li>○ Parathyroid hormones</li> <li>○ Pancreatic hormones and related drugs</li> </ul> </li> <li>▪ <b>Steroidal Hormones and Related Drugs</b> <ul style="list-style-type: none"> <li>○ Male and female sex hormones, derivatives and related drugs</li> </ul> </li> </ul> | <p>2 hrs</p> <p>2 hrs</p> |
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| ○ Adrenocorticoids        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2 hrs                                                                                                                                                                                                    |
| <b>Total</b>              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>48 hrs</b>                                                                                                                                                                                            |
| <b>Mode of Delivery</b>   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <ul style="list-style-type: none"> <li>▪ Lecture: 48hrs</li> <li>▪ Tutorial: 28hrs</li> <li>▪ Independent study hour: 40hrs</li> <li>▪ Seminar, Assignment: 12hrs</li> <li>▪ Assessment: 7hrs</li> </ul> |
| <b>Mode of Assessment</b> | <ul style="list-style-type: none"> <li>▪ Quizzes (10%)</li> <li>▪ Tests (30%)</li> <li>▪ Seminars and Assignments (20%)</li> <li>▪ Final Exam (40%)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                          |
| <b>Text Book</b>          | <i>Lemke, T.L. and Williams, D.A., Roche, V. F., Zito, W.S.Foye's Principles of Medicinal Chemistry, 6th. ed. Lippincott Williams &amp; Wilkins, 2008.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                          |
| <b>Reference Books</b>    | <ul style="list-style-type: none"> <li>- Wilson-Gisvold-Doerge; Text book of organic medicinal chemistry and pharmaceutical chemistry. 12<sup>th</sup> edn.; Lippincott (USA), 2011.</li> <li>- Carmen Avendano and J. Carlos Menendez, Medicinal Chemistry of Anticancer drugs, first edition, 2008</li> <li>- Donald J. Abraham (Ed.). Burgers's medicinal Chemistry and Drug Discovery, 2006, 6th edn., vol1-6, wiley-interscience (USA).</li> <li>- J.H. Block and J.M. Beale, Gisvold, O. Wilson &amp; Gisvold's Textbook of Organic Medicinal and Pharmaceutical Chemistry, 11th. ed. Lippincott Williams &amp; Wilkins, 2004.</li> <li>- Burger's Medicinal Chemistry and Drug discovery, Sixth edition, Volume 3: Cardiovascular Agents and Endocrines; 2003</li> <li>- Burger's Medicinal Chemistry and Drug discovery, Sixth edition, Volume 5: Chemotherapeutic Agents; 2003</li> </ul> |                                                                                                                                                                                                          |

**Module 15: Pharmaceutical Analysis****Module Name:** Pharmaceutical Analysis**Module category:** Core**Module Weight in ECTS:** 14 ECTS**Module Number:** 15**Module Code:** Phar-M3151**Course Code:**

Pharmaceutical Analysis I: Phar3151 (7 EtCTS)

Pharmaceutical Analysis II: Phar3152 (7 EtCTS)

**Module Description**

Pharmaceutical Analysis aims at cultivating students to the world of regulation, especially the areas of quality control, quality assurance and validation. It also encourages an interdisciplinary approach to problem solving in a modern analytical laboratory that is practiced at regulatory, pharmaceutical manufacturing and other setups. This module introduces students to different analytical techniques with a focus on the basic working principles of the instruments, applications and limitations of the techniques as well as data analysis. It also introduces the students to the principles of quality control and the regulatory process in assuring the quality, efficacy and safety of drug and drug products. In the module, students will be exposed to different techniques such as various classical methods including titration methods, simple physical and chemical tests for drugs, electro-analytical methods, biological methods, radiochemical methods, extraction and analysis techniques using chromatography, spectroscopic methods, mass spectrometry, and their applications in a wide number of key areas, particularly pharmaceutical, food, herbals and chemical analysis. The quality control aspect of diagnostic reagents and chemicals, medical instruments and medical supplies is also addressed.

**Module objective:** Upon completion of the module, the student should have: a mature understanding of the theory and application of modern analytical techniques for pharmaceutical analysis; the ability to select the analytical method of choice for a particular circumstance; a deeper understanding of the regulatory matters and quality assurance principles currently in practice for the manufacture and licensing of medicines; the ability to apply the modern knowledge gained in this course to provide practical solutions to real problems. It is intended to train highly qualified pharmacists with strong background in the quality assurance and quality control of pharmaceuticals and related products. It also familiarizes the student with the concepts of drug quality and QA in the supply chain system in routine pharmacy practice. It is also believed to supplement their understanding of therapeutic drug monitoring emphasizing on analytical aspects.

**Module Competencies:**

- Realize the importance of pharmaceutical quality control and the regulatory process in assuring the quality, efficacy and safety of drug and related products
- Reflect on the importance of standards for assuring the quality of drug products and consider the availability of pharmacopoeial standards and the importance of good manufacturing processes
- Propose and implement the sampling and sample preparation procedures required for pharmaceutical analysis, and evaluate the appropriate methodologies and protocols for the required analytical goals
- To provide an up-to-date understanding of the principles and application of current analytical techniques for the quantitative and qualitative measurement of drugs in a variety of real world matrices e.g. pharmaceutical, biological and herbal



- Able to select the appropriate analytical techniques for a given sample type to detect and quantify organic molecules and their application in the investigation of real-world problems
- Confidently operate a range of instrumentation used in the modern pharmaceutical quality control laboratories
- Apply laboratory skills to perform tests and prepare written laboratory reports that provide a description of the experiment, demonstrate clear and logical reasoning, and provide an appropriate conclusion

**Mode of delivery:** Parallel

- Total study hours in the module:  $14 \times 27 = 378$ hrs
- Lecture: 117
- Tutorial: 16
- Seminars, assignments and presentations: 20
- Practical/ Laboratory: 72
- Home study: 133
- Assessment: 20
- Visit pharmaceutical industry quality control unit: 6 hrs

**Mode of assessment**

- Seminar presentations
- Group assignments
- Laboratory works and report writing
- Practical exams
- Laboratory written exams
- Tests
- Quizzes
- Final Exam

**Learning Activities and Teaching Methods**

**A. Learning Activities**

Students are expected to undertake among others the following major learning activities:

- Engage in learning by doing (independent study, group assignments, presentation, report writing, and etc...)
- Participation and discussions in and outside the classroom
- Practical laboratory works including reagent and sample preparation, analysis, interpretation and report writing

**B. Teaching Methods**

- The course facilitator is expected to lecture for introducing concepts and topics, and give references, facilitate discussions, ask questions, give and correct assessments,
- Arrange and facilitate seminar sessions by inviting professionals for public lectures or practical works
- Facilitate and organize attachment to quality control laboratories
- Guide the students towards other specific/relevant sources of information

**Roles of Instructors and Students**

**A. Roles of Instructors**

The instructor is expected to:

- Be a facilitator (introduce the subject; give guidance, moderate discussions, etc.)

- Read, comment and provide feedback on assignments of students on time
- Prepare his/her lessons and deliver lectures
- Guide laboratory works
- Provide available and necessary reference materials
- Encourage active participation of students in the teaching learning process
- Assist students with learning difficulties
- Arrange attachment sites

#### **B. Roles of Students**

Students are expected to:

- Engage in learning by doing (independent study, project work, group work, etc.)
- Be active learners (participate effectively in group assignments, give presentations, write reports, etc.)
- Be active participants during the laboratory experiments and compile and submit laboratory reports on time Critically read relevant journal articles and related topics from book chapters

## Pharmaceutical Analysis Module syllabi

**Module Number:** 15

**Course Title:** Pharmaceutical Analysis I

**Course Code:** Phar3151

**Course ECTs:** 7

**Pre-requisite:** None

**Course Description:** This course covers pharmaceutical quality control aspects, various titrimetric methods such as acid-base, argentometric, complexometric, non-aqueous and miscellaneous titrimetry methods. Besides, it also gives the student theoretical knowledge on applications of radiochemical and electrochemical techniques in quality control of pharmaceuticals. The course has 16 weeks of practical classes in which different titrimetric methods will be studied as employed in the estimation of the constituents of drugs included in the national drug list.

**Course Objectives:** After completing this course, students will be able to describe the purpose of pharmacopoeia, importance of quality control, steps involved in drug analysis, sample pretreatment, different analytical methods and their use in pharmaceutical analysis. They will also be able to propose suitable analytical technique for a sample, carry out analysis for different pharmaceuticals as well as handle validation of analytical procedures, interpret and report data obtained from an analysis.

### Supporting objectives:

- Describe what is meant by pharmaceutical analysis
- Explain Purpose of pharmaceutical analysis
- Describe quality control and quality assurance
- Perform basic calculations in pharmaceutical analysis
- Describe sample preparation for analysis
- Describe titrimetry
- Perform different titration
- Describe electrochemical method of analysis

### Reading Materials:

**Text Book:** Beckett, A.H. and Stenlake, J.B. Practical Pharmaceutical Chemistry, Parts I & II, 4th edn., The Athlone Press, London, 2001.

### Reference Books:

1. Connors, K.A. Textbook of Pharmaceutical Analysis, 3rd edition., 2007.
2. David G. Watson. Pharmaceutical Analysis, A Textbook for Pharmacy Students and Pharmaceutical Chemists, 5th Edition, 2020, Elsevier.
3. Gary D. Christian; Analytical chemistry, 6th edition, John Wiley and Sons INC., USA, 2004
4. USP/NF (Latest edition). The United States Pharmacopoeial convention, Inc. Rockville, MD., USA
5. British Pharmacopoeia (Latest edition), Her Majesty stationery office, London.
6. David Harvey. Modern analytical chemistry. 1st ed, Mc Graw Hill, Boston, 2000

| Week | Contact hrs | Topic/sub-topic/chapter                                                                                      | Reading materials |
|------|-------------|--------------------------------------------------------------------------------------------------------------|-------------------|
| 1    | 4           | 1. Introduction to pharmaceutical analysis<br>1.1. Introduction to quality control and Quality Assurance (1) |                   |

|   |   |                                                                                                                                                                                                                                                                                                                                                                                                               |                  |
|---|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
|   |   | 1.2. The compendia (3 hrs)<br>1.2.1. Pharmacopoeia and other official methods<br>1.2.2. General notices<br>1.2.3. Monographs                                                                                                                                                                                                                                                                                  |                  |
| 2 | 4 | 1.3. Analytical Errors and validation of analytical procedures (2 hrs)<br>1.4. Basic calculations in pharmaceutical analysis (2 hrs)<br>1.4.1. Percentage:<br>- Percentage volume/volume (% V/V)<br>- Percentage weight/volume (% W/V)<br>- Percentage weight/weight (% W/W)<br>1.4.2. Parts per million (ppm) and parts per billion (ppb)<br>1.4.3. Molarity (M)<br>1.4.4. Normality (N)<br>1.4.5. Dilutions |                  |
| 3 | 4 | 1.5. Physical and chemical properties of drug molecules (4 hrs)<br>1.5.1. Calculation of pH<br>1.5.2. Acidic and basic strength and pKa<br>1.5.3. Buffers<br>1.5.4. Drug stability and stability study                                                                                                                                                                                                        | Reference: 1,4,5 |
| 3 | 4 | 2. Sample preparation for analysis (2 hours)<br>2.1. Steps in chemical analysis<br>2.2. Preliminary treatment of sample<br>3. Titrimetry<br>3.1. Introduction (1 hr)<br>3.1.1. End point and equivalence point<br>3.1.2. Direct titration, back titration and back titration with blank determination<br>3.1.3. Calculations in titrimetry<br>3.1.4. Titration curves                                         |                  |
| 4 | 4 | 3.2. Acid-Base titrations (4 hrs)<br>3.2.1. Acid-Base titrations in aqueous media<br>3.2.2. Acid-Base titrations in non-aqueous media<br>3.2.3. Applications                                                                                                                                                                                                                                                  |                  |

|   |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |
|---|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| 5 | 4 | <p>3.3. Precipitometric titrations (4 hrs)</p> <p>3.3.1. Limitations</p> <p>3.3.2. Solubility product</p> <p>3.3.3. Factors affecting solubility of precipitate</p> <p>3.3.4. Indicators</p> <p>3.3.5. Argentometric titrations</p> <ul style="list-style-type: none"> <li>- Mohr's method</li> <li>- Volhard's method</li> <li>- Fajan's method</li> </ul> <p>3.3.6. Mercurimetric titrations as substitutes for precipitation methods</p> <p>3.3.7. Applications</p>                                      |  |
| 6 | 4 | <p>3.4. Complexometric titrations (4 hrs)</p> <p>3.4.1. Introduction</p> <p>3.4.2. Basic principles of complexometric analysis</p> <p>3.4.3. General principles in disodium edetate titrations</p> <p>3.4.4. Indicators and end point detection</p> <p>3.4.5. Applications</p>                                                                                                                                                                                                                              |  |
| 7 | 4 | <p>3.5. Redox titrations (4 hrs)</p> <p>3.5.1. Basic principles of redox titrations</p> <p>3.5.2. Redox indicators</p> <p>3.5.3. Principles and procedures in different types of redox titration</p> <ul style="list-style-type: none"> <li>- Potassium permanganate methods</li> <li>- Ceric sulfate methods</li> <li>- Iodimetric and Iodometric methods</li> <li>- Bromination methods</li> <li>- Potassium iodate methods</li> <li>- Potassium dichromate methods</li> </ul> <p>3.5.4. Applications</p> |  |

|    |   |                                                                                                                                                                                                                                                                                                                                                                                                |  |
|----|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| 8  | 4 | <p>3.6 Diazotization titration (1 hrs)</p> <p>3.6.1 Theory of diazotization reaction</p> <p>3.6.2 Applications</p> <p>3.7 Microfluidic paper-based analytical devices (<math>\mu</math>PADs) and other automated titration techniques (3 hours)</p> <p>3.7.1 Introduction, principles, equipment and applications</p>                                                                          |  |
| 9  | 4 | <p>4. Gravimetry (4 hours)</p> <p>4.1. Steps in gravimetric analysis</p> <p>4.2. Selected gravimetric methods</p> <p>4.3. Applications</p>                                                                                                                                                                                                                                                     |  |
| 10 | 4 | <p>5. Limit tests (4 hours)</p> <p>5.1. Definition and importance</p> <p>5.2. Limits on ash values</p> <p>5.3. Limits on moisture content: Importance, Los on drying method, Azeotropic distillation method, Karl-Fischer titration method, Other methods</p> <p>5.4. Limit tests for some metals: Lead, Arsenic, and Iron</p> <p>5.5. Limit tests for some non-metals: Chloride, Sulphate</p> |  |
| 11 | 4 | <p>6. Electro-analytical methods</p> <p>6.1 Introduction (2 hrs)</p> <p>1.1.1. Electrochemical cells</p> <p>1.1.2. Various types of electrodes</p> <p>1.1.3. Instrumentation and measurement of cell e.m.f.</p> <p>6.2 Potentiometry (2 hrs)</p> <p>6.2.1 Principle</p> <p>6.2.2 Applications in pharmaceutical analysis</p>                                                                   |  |
| 12 | 4 | <p>6.3 Conductometry (1 hrs)</p> <p>6.3.1 Principles</p> <p>6.3.2 Apparatus and measurement</p>                                                                                                                                                                                                                                                                                                |  |

|    |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |
|----|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| 13 | 4 | <p>6.3.3 Applications of direct conductometric methods (1 hour)</p> <p>6.4 Coulometry (3 hrs)</p> <p>6.4.1 Principles</p> <p>6.4.2 Apparatus and measurement</p> <p>6.4.3 Applications of coulometric methods</p>                                                                                                                                                                                                                                                                                      |  |
| 14 | 4 | <p>6.5 Voltammetry (4 hours)</p> <p>6.5.1 Principles</p> <p>6.5.2 Apparatus and measurement</p> <p>6.5.3 Types of voltammetric methods</p> <p>6.5.4 Applications of voltammetric methods</p>                                                                                                                                                                                                                                                                                                           |  |
| 15 | 4 | <p>6.6 Electro-analytical sensors and their application in pharmaceutical analysis (4 hours)</p>                                                                                                                                                                                                                                                                                                                                                                                                       |  |
| 16 | 4 | <p>7 Radiochemical techniques (4 hours)</p> <p>7.1 Introduction</p> <p>7.1.1 Radioactivity</p> <p>7.1.2 Radioisotopes</p> <p>7.1.3 Radioactive decay</p> <p>7.1.4 Fate of different types of radiation</p> <p>7.1.5 Radiation limits</p> <p>7.1.6 Stability of radioactive compounds</p> <p>7.2 Measurement of radioactivity</p> <p>7.2.1 Ion collection method</p> <p>7.2.2 Scintillation technique</p> <p>7.2.3 Other techniques</p> <p>7.3 Analytical applications of radioisotopes in pharmacy</p> |  |
| 17 | 4 | <p>8 Introduction to quality control (QC) for different dosage forms</p>                                                                                                                                                                                                                                                                                                                                                                                                                               |  |

|           |   |                                                                                                                                           |  |
|-----------|---|-------------------------------------------------------------------------------------------------------------------------------------------|--|
|           |   | 8.1 General tests for Dosage forms:<br>Dissolution, Disintegration, Hardness,<br>Clarity etc...<br>8.2 Visual inspection as a tool for QC |  |
| 18        | 4 | 9. Identification of Counterfeit and Substandard products (4 hrs)                                                                         |  |
| 19        | 4 | 10. QRM in drug analysis, introduction, principles, WHO check list (4 hrs)                                                                |  |
| <b>20</b> |   | <b>FINAL EXAM</b>                                                                                                                         |  |

**Mode of Delivery:**

- Lecture: 59
- Tutorial: 10
- Seminars, assignments and Presentation: 10
- Practical/ Laboratory: 36
- Home study: 66
- Assessment: 10

**Mode of Evaluation**

- Seminar and assignments: 10%
- Laboratory written exams and report writing: 10%
- Practical exams: 10%
- Quizzes: 10%
- Tests: (10%)
- Final Exam: 50%



**Module Number: 15**

**Course Title: Pharmaceutical Analysis II**

**Course Code:** Phar3152

Course ECTs: 7

**Co-requisite: NONE**

**Pre-requisite:** Pharmaceutical Analysis I

**Course Description:** The course deals with the principles, instrumentation and applications of important instrumental analytical techniques such as spectroscopic methods including UV-Visible, atomic absorption/emission, fluorescence, IR and nuclear magnetic resonance spectroscopy; chromatographic methods including Gas Chromatography and High Performance Liquid Chromatography; as well as mass spectrometry in the quality control of pharmaceutical products. Biological methods of analysis; basic coverage on bio-analysis; the QC aspects of herbal drugs, medical equipment, medical supplies and diagnostic kits; and QA in supply chain systems have been included. The course has 16 weeks of practical classes in which different instrumental analytical techniques will be performed as employed in the estimation of the constituents of drugs included in the national drug list.

**Course Objectives:** After completing this course, students will be able to describe the principles and instrumentation of different modern instrumental analytical techniques and their use in pharmaceutical and biochemical analysis. They will also be able to propose suitable analytical technique for a sample, carry out analysis for different pharmaceuticals as well as handle, interpret and report data obtained from the analysis.

• **Supporting objectives:**

- Describe UV-Visible spectroscopy
- Describe infrared spectroscopy
- Describe fluorescence spectroscopy
- Describe atomic spectroscopy
- Describe mass spectroscopy
- Identify different chromatographic techniques
- Describe gas chromatography
- Describe high performance liquid chromatography
- Describe biological methods of analysis

**Reading Materials:**

**Text Book:** Beckett, A.H. and Stenlake, J.B. Practical Pharmaceutical Chemistry, Parts I & II, 4th edn., The Athlone Press, London, 2001.

**Reference Books:**

1. Connors, K.A. Textbook of Pharmaceutical Analysis, 3rd edition., 2007
2. David G. Watson. Pharmaceutical Analysis, A Textbook for Pharmacy Students and Pharmaceutical Chemists, 5th Edition, 2020, Elsevier.
3. Gary D. Christian; Analytical chemistry, 6th edition, John Wiley and Sons INC., USA, 2004
4. USP/NF (Latest edition). The United States Pharmacopoeial convention, Inc. Rockville, MD, USA
5. British Pharmacopoeia (Latest edition), Her Majesty stationery office, London.
6. David Harvey. Modern analytical chemistry. 1st ed, Mc Graw Hill, Boston, 2000.

7. Francis Rouessac and Annick Rouessac, Chemical Analysis, Modern instrumental methods and techniques, 2nd ed, John Wiley and Sons, LTD, England, 2013.
8. Satinder Ahuja and Michael W. Dong. Handbook of Pharmaceutical Analysis by HPLC. 1st ed, volume 6, Elsevier Academic Press, New York, 2005.

| Week | Contact hrs | Topic/sub-topic/chapter                                                                                                                                                                                                                                                                                                                                                         | Reading materials |
|------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| 1    | 4           | 1. Analytical instruments and their calibration; validation; regulatory requirements (2 hrs)<br>2. UV- Visible spectrophotometry<br>2.1. Introduction                                                                                                                                                                                                                           |                   |
| 2    | 4           | 2.2. Factors governing absorption of radiation in the UV/Visible region<br>2.2.1. The concept of chromophore and auxochrome<br>2.2.2. Absorption intensity shifts<br>2.2.3. Effect of pH on absorption<br>2.2.4. Conjugated dienes and the Woodward-Fieser rules                                                                                                                |                   |
| 3    | 4           | 2.3. Instrumentation<br>2.3.1. Radiation sources<br>2.3.2. Monochromators<br>2.3.3. Sample cells and compartments<br>2.3.4. Detectors<br>2.3.5. Recording systems<br>2.3.6. Double and single beam instruments<br>2.4. Qualitative spectrophotometry<br>2.5. Quantitative spectrophotometry<br>2.5.1. The Beer-Lambert law and its limitations<br>Spectrophotometric titrations | Reference: 1,4,5  |
| 3    | 4           | 2.6. Analysis of binary mixtures<br>2.7. Differential Spectrophotometry<br>2.8. Derivative spectra<br>2.9. Colorimetry<br>2.9.1. General requirements for colored substances<br>2.9.2. Chemistry in colorimetry<br>2.9.3. Applications                                                                                                                                          |                   |

|   |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |
|---|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| 4 | 4 | 3. Fluorescence spectrophotometry<br>3.1.Introduction<br>3.2.Instrumentation<br>3.3.Structural requirements of fluorescent compounds                                                                                                                                                                                                                                                                                                                                             |  |
| 5 | 4 | 3.4. Factors affecting fluorescence intensity<br>3.5. Applications in pharmaceutical analysis<br>4. Infrared Spectrophotometry<br>4.1. Introduction                                                                                                                                                                                                                                                                                                                              |  |
| 6 | 4 | 4.2. Instrumentation:<br>Dispersive, FTIR and NIR,<br>Radiation sources,<br>Monochromators, Detectors,<br>Recorders and display<br>4.3. Fundamental vibrations and factors affecting vibration frequency<br>4.4. Sample preparation<br>4.5. Scanning IR spectra.<br>4.6. Interpretation of the spectra<br>4.7. Applications<br>4.7.1. IR spectrophotometry as a fingerprint technique<br>4.7.2. Quantitative IR analysis<br>4.7.3. IR spectrophotometry in structure elucidation |  |
| 7 | 4 | 5. Atomic spectrophotometry<br>5.1. Introduction<br>5.2. Types of atomic spectrophotometric techniques<br>5.2.1. Atomic absorption spectrophotometry (AAS) techniques<br>5.2.2. Atomic emission spectrophotometry (AES) techniques<br>5.3. Instrumentation<br>5.4. Applications                                                                                                                                                                                                  |  |
| 8 | 4 | 5.5.Other elemental analysis techniques ( X-Ray Fluorescence, etc)<br>6. Nephelometry and Turbidimetry                                                                                                                                                                                                                                                                                                                                                                           |  |

|    |   |                                                                                                                                                                                                                                                                                                                      |  |
|----|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|    |   | 6.1. Introduction<br>6.2. Instrumentation<br>6.3. Pharmaceutical applications                                                                                                                                                                                                                                        |  |
| 9  | 4 | 7. Introduction to chromatography<br>7.1. History and principles<br>7.2. Classifications<br>7.3. Definition of terminologies<br>7.4. TLC<br>7.5. HPTLC<br>8. Gas Chromatography (GC)<br>8.1. Introduction                                                                                                            |  |
| 10 | 4 | 8.2. Instrumentation<br>8.3. Carrier gas cylinder, Injection port, column and column oven, detectors, recorders and integrators<br>8.4. Factors affecting choice of carrier gas<br>8.5. Temperature programming in GC<br>8.6. Pyrolysis and derivatization in GC<br>8.7. Qualitative and quantitative analysis by GC |  |
| 11 | 4 | 9. High performance liquid chromatography (HPLC)<br>9.1. Introduction and theory<br>9.2. Instrumentation<br>Pump, injection system, column, detectors, data system<br>9.3. Stationary and mobile phases<br>9.4. Structural factors governing rate of elution of compounds<br>9.5. Evaluation of column performance   |  |
| 12 | 4 | 9.6. Applications in: identification, quantitative analysis, chiral separation<br>9.7. Electrophoresis: high performance capillary electrophoresis                                                                                                                                                                   |  |
| 13 | 4 | 10. Mass Spectrometry<br>10.1. Introduction<br>10.2. Instrumentation<br>10.3. Molecular fragmentation patterns<br>10.4. GC-MS and LC-MS: Introduction, Instrumentation and Application                                                                                                                               |  |
| 14 | 4 | 10.5. Applications in pharmaceutical analysis                                                                                                                                                                                                                                                                        |  |

|    |   |                                                                                                                                                                                                                                                                                                                   |  |
|----|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|    |   | 10. Nuclear magnetic resonance spectroscopy<br>10.1. Introduction to $^1\text{H}$ NMR and $^{13}\text{C}$ NMR spectroscopy<br>10.2. Basic instrumentation.<br>10.3. The chemical shift<br>Shielding and de-shielding effects<br>10.4. Factors influencing the chemical shift<br><br>Peak area and proton counting |  |
| 15 | 4 | 10.1. Important tips for interpreting NMR spectra.<br>11. Biological methods of analysis<br>11.1. Introduction<br>11.2. Microbiological assay<br>11.3. Pyrogen testing (in vivo & in vitro)<br>11.4. Microbial limit test<br>11.5. Sterility test<br>11.6. Preservative efficacy test                             |  |
| 16 | 4 | 12. Introduction to bio-analysis (2 hours)<br>12.1. Biological samples and their preparation for analysis<br>12.2. Sample extraction<br>12.3. Biological sample analysis<br>12.4. Applications; Therapeutic Drug Monitoring and others                                                                            |  |
| 17 | 4 | 13. Introduction to herbal drugs quality control (3 hrs)<br><br>13.1. Introduction<br>13.2. Methods of herbal drugs quality control<br>Challenges in standardization of herbal drugs                                                                                                                              |  |
| 18 | 4 | 14. Quality assurance in supply chain systems (2 hrs)<br><br>14.1. Specifications preparation; bid evaluation that takes quality in to considerations<br>15. 15. Quality documents;<br>15.1. Certificate of analysis (COA) and its interpretation                                                                 |  |

|    |   |                                                                                                                                                                                                                                                                                                                                       |  |
|----|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| 19 | 4 | 16. Introduction to medical equipment, medical supplies and diagnostic kits quality control (2 hrs)<br>16.1. Introduction<br>16.2. Methods for quality control of medical equipment<br>16.3. Methods for quality control of medical supplies<br>16.4. Methods for the quality control of diagnostics<br>16.5. Challenges, limitations |  |
| 20 |   | <b>FINAL EXAM</b>                                                                                                                                                                                                                                                                                                                     |  |

**Mode of Delivery:**

- Lecture: 58
- Tutorial: 10
- Seminars, assignments and Presentation: 10
- Practical/ Laboratory: 36
- Home study: 59
- Assessment: 10
- Visits to pharmaceutical firms and quality control laboratories: (6 hrs)

**Mode of Evaluation**

- Seminar and assignments: 10%
- Laboratory written exams and report writing: 10%
- Practical exams: 10%
- Quizzes: 10%
- Final Exam: 50%
- Quality control visit report: 10%

## **Module 16: Pharmacotherapeutics**

**Module name:** Pharmacotherapeutics module I

**Module category:** Core

**Module code:** Phar-M3161

**Module weight in ECTS:** 16 (16x27) = 432hrs

**Courses:**

| Course name                | Course code | ECTS   |
|----------------------------|-------------|--------|
| Physical Assessment        | Intm3161    | 2 ECTS |
| Integrated therapeutics I  | Phar3162    | 7 ECTS |
| Integrated therapeutics II | Phar3163    | 7 ECTS |

### **Module description:**

Students will learn about the Pathophysiology and pharmacotherapy of various disease states that health care practitioners (pharmacists) may encounter in their practice settings. Courses in this module introduce essential therapeutic knowledge needed for providing pharmaceutical care in individual patient. These courses integrate the pathophysiologic abnormalities of disease state with concepts of drug selection, dose optimization and monitoring of therapeutic outcomes for safety and efficacy of medication. Courses discussed include: integrated therapeutics I-IV which extends from general principles of pharmacotherapy to detailed pharmacotherapy of each disease states (gastrointestinal, respiratory, cardiovascular, renal, hematologic, neurologic, psychiatric, endocrinologic, infectious diseases etc.)

### **Module objective:**

At the end of this module, the students are expected to:

- Explain the etiology, pathophysiology, clinical presentation and diagnosis of each disease states
- Set goals of treatment and select treatment options for the management of each disease states
- Formulate dose recommendations and pharmacokinetic considerations for individual patient management
- Monitor clinically significant adverse drug reactions and drug interactions
- Evaluate therapeutic outcomes for effectiveness, safety and patient adherence

- Develop and exercise pharmaceutical care planning for managing a specific patient condition
- Provide patient medication counseling and drug information

**Module competency:**

Provide patient centered Pharmaceutical care services

**Mode of delivery: Parallel**

**Mode of Assessment:**

Continuous assessment & summative assessment: Class attendance, Continuous assessment, Assignments, Hospital attachment Report, Final Exam

**Module learning teaching methods**

Illustrated lectures and group discussions, Individual and group exercise and assignments, Role plays and case studies, Simulation, Audiovisuals, Clinical scenarios, Tutorials, demonstration



## **Physical Assessment course guide book/syllabus**

**Course Name: Physical Assessment**

**Course code:** Intm3161

**Module name:** Pharmacotherapeutics Module I

**Module code:** Intm3161

**Course ECTS:** 2 ECTS Non Credit (NC)

**Totally required hours for the module:** 2 X 27 = 54

**Lecture hours:** 13 hrs

**Home Study:** 13 hrs

**Demonstration/ clinical simulation:** 13 hrs

**Bedside presentation:** 10 hrs

**Assessment:** 5 hrs

**Year/Semester Course is offered:** Year III/Semester I

**Course Co-requisite/s:** Integrated Therapeutics I

### **Course Description**

This physical diagnosis course for year five UG students is to have basic knowledge in history taking and to demonstrate how to do physical examinations. Major physical assessment skills (inspection, palpation, percussion and auscultation) will be discussed and demonstrated. The purpose of this course for pharmacy student is to enable pharmacy graduates make physical examination and monitor drug effectiveness and toxicity.

### **Course Objectives**

- To be able to take medical history and perform proper physical examination with effective communication.

#### **Specific Course Objectives:**

At the end of this course the students will be able to

- Describe the common terminologies used in normal and abnormal findings in the history and physical examination
- Communicate effectively and sensitively with patient and the relative to extract relevant information about the patient's problem at the background of his whole life
- Demonstrate a methodological approach to perform physical examination
- Analyze and interpret the history and physical findings in scientifically sound way

**Course of delivery:** Block

**Course Teaching and Learning Methods:**

- Interactive presentation
- Role play
- Demonstration
- Patient interview and physical examination
- Bedside teaching

#### **Assessment Techniques:**

- |                                                        |     |
|--------------------------------------------------------|-----|
| • Tests and quizzes                                    | 10% |
| • Assignments                                          | 10% |
| • Simulated patient interview and physical examination | 40% |
| • Written exam:                                        | 40% |

#### **Teachers' and students' role**

##### **Roles of Instructors**

Instructors should be well versed in the outcomes expected of students and the pedagogical methods that best enhance learning. Accordingly, they will be expected to:

- Be a facilitator (introduce the subject; give guidance, moderate discussions, etc.)
- Closely supervise in actual practice in different clinical and community settings
- Assists students in resolving ever-increasing ethical dilemma in the service delivery
- Foster and assess self-initiated student learning
- Read and comment assignments of students on time;
- Prepare his/her lessons and deliver lectures;
- Provide available and necessary reference materials
- Encourage active participation of students in the teaching learning process;
- Strive to be the role model ethically and professionally

##### **Roles of Students**

Students are expected to:

- Make every effort to apply valid research findings and theoretical teachings into practice in various practice settings
- Engage in learning by doing independent study, project work; group work, etc..
- Be active learners (participate effectively in group assignments, perform practice actively, make presentations, write reports, prepare seminars etc.);
- Critically assess laws, regulations, journal articles and related topics from different sources

#### **References:**

**Required readings (Text)**

1. Lynn Bickley (2012). Bates' Guide to Physical Examination and History-Taking, 11<sup>th</sup>, Lippincott Williams & Wilkins.
2. Karen J. Tietze (2011). Clinical Skills for Pharmacists: A Patient-Focused Approach, 3<sup>rd</sup> edition, Mosby publisher.

**Recommended readings**

3. Henry M. Seidel, Rosalyn W. Stewart, Jane W. Ball, Joyce E. Dains, John A. Flynn, Barry S. Solomon, Rosalyn W. Stewart (2010). Mosby's Guide to Physical Examination, 7<sup>th</sup> edition, Mosby publisher.
4. Jane W. Ball, Joyce E. Dains, John A. Flynn, Barry S. Solomon, Rosalyn W. Stewart (2014). Seidel's Guide to Physical Examination, 8<sup>th</sup> edition, Mosby publisher.

**Course Schedule**

| Week | Contact hr | Topic/sub-topic/chapter/Assessments/Assignments                                                                 | Reading Material  |
|------|------------|-----------------------------------------------------------------------------------------------------------------|-------------------|
| 1    | 2          | Introduction: (patient history taking, the process of physical assessment, equipments for physical examination) | Reference No. 1,2 |
|      | 2          | Vital signs, skin,                                                                                              |                   |
|      | 2          | Breast, head and neck                                                                                           |                   |
|      | 2          | Chest and lungs, abdomen, genitourinary system                                                                  |                   |
|      | 2          | Musculo-skeletal, neurologic system                                                                             |                   |
| 2    | 16         | Demonstration/ clinical simulation                                                                              |                   |
|      | 12         | Bedside presentation                                                                                            |                   |
|      | 6          | Assessment:                                                                                                     |                   |

## Course syllabus

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |          |                 |          |                         |          |                  |          |                    |          |                    |          |                                   |          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------------|----------|-------------------------|----------|------------------|----------|--------------------|----------|--------------------|----------|-----------------------------------|----------|
| <b>Integrated therapeutics I Course syllabus</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |                 |          |                         |          |                  |          |                    |          |                    |          |                                   |          |
| <b>Course name:</b> Integrated Therapeutics I                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |          |                 |          |                         |          |                  |          |                    |          |                    |          |                                   |          |
| <b>Course code:</b> Phar3162                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |          |                         |          |                  |          |                    |          |                    |          |                                   |          |
| <b>Module name:</b> Pharmacotherapeutics module                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |          |                 |          |                         |          |                  |          |                    |          |                    |          |                                   |          |
| <b>Module code:</b> Phar-M3161                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |          |                 |          |                         |          |                  |          |                    |          |                    |          |                                   |          |
| <b>Course ECTS:</b> 7 ECTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |          |                 |          |                         |          |                  |          |                    |          |                    |          |                                   |          |
| <p><b>Totally required hours for the module:</b> 7x27 = 189 hours</p> <table> <tr> <td><b>Lecture:</b></td><td>60 hours</td></tr> <tr> <td><b>Ward attachment:</b></td><td>30 hours</td></tr> <tr> <td><b>Tutorial:</b></td><td>34 hours</td></tr> <tr> <td><b>Home study:</b></td><td>46 hours</td></tr> <tr> <td><b>Assessment:</b></td><td>14 hours</td></tr> <tr> <td><b>Project work/presentation:</b></td><td>15 hours</td></tr> </table>                                                                                                                                                                                                                                                                                                                                                            |          | <b>Lecture:</b> | 60 hours | <b>Ward attachment:</b> | 30 hours | <b>Tutorial:</b> | 34 hours | <b>Home study:</b> | 46 hours | <b>Assessment:</b> | 14 hours | <b>Project work/presentation:</b> | 15 hours |
| <b>Lecture:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 60 hours |                 |          |                         |          |                  |          |                    |          |                    |          |                                   |          |
| <b>Ward attachment:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 30 hours |                 |          |                         |          |                  |          |                    |          |                    |          |                                   |          |
| <b>Tutorial:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 34 hours |                 |          |                         |          |                  |          |                    |          |                    |          |                                   |          |
| <b>Home study:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 46 hours |                 |          |                         |          |                  |          |                    |          |                    |          |                                   |          |
| <b>Assessment:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 14 hours |                 |          |                         |          |                  |          |                    |          |                    |          |                                   |          |
| <b>Project work/presentation:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 15 hours |                 |          |                         |          |                  |          |                    |          |                    |          |                                   |          |
| <b>Year/Semester course offered:</b> Year III/Semester I                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |          |                 |          |                         |          |                  |          |                    |          |                    |          |                                   |          |
| <b>Pre-requisite if any:</b> Pharmacology I                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |          |                 |          |                         |          |                  |          |                    |          |                    |          |                                   |          |
| <b>Co-requisite:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |                 |          |                         |          |                  |          |                    |          |                    |          |                                   |          |
| <p><b>Course description:</b></p> <p>This course is designed to introduce the pharmacy student to the study of integrated therapeutics. It will provide introductory information designed to assist the student to begin understanding the rationale upon which many drug therapy decisions are based. Principles, concepts, processes, and skills in pharmacotherapy will be emphasized. Therapeutic topics and case studies will be used to provide students with the opportunity to apply these skills.</p> <p>This course will also enable students to understand and interpret the common diagnostic tests. Gastrointestinal and dermatological disorders will be addressed in the therapeutics section</p>                                                                                           |          |                 |          |                         |          |                  |          |                    |          |                    |          |                                   |          |
| <p><b>Course objectives:</b></p> <p>At the completion of this course the student should be able to:</p> <ol style="list-style-type: none"> <li>1. Apply the Pharmacists' Patient Care Process</li> <li>2. Perform a brief patient assessment, including: <ul style="list-style-type: none"> <li>○ Interviewing a patient</li> <li>○ Obtaining current medication list</li> </ul> </li> <li>3. Identify drug therapy problems by evaluating drugs for indication, effectiveness, safety, convenience</li> <li>4. Develop individualized and clinically appropriate care plans for a patient with one to three simple conditions</li> <li>5. Educate patients on their drug therapy and assess for patient understanding</li> <li>6. Communicate effectively with all patients and their families</li> </ol> |          |                 |          |                         |          |                  |          |                    |          |                    |          |                                   |          |

7. Accept responsibility of providing continuing care to patients
8. understand the various factors that may influence drug therapy in a patient
9. Understand how to gather relevant patient information during drug therapy
10. Interpret common diagnostic tests used
11. Understand the etiology, pathophysiology, diagnosis, treatment and monitoring parameters of therapy outcomes in the management of common gastrointestinal and dermatological disorders

**Course mode of delivery:** Parallel

**Course learning and teaching methods:**

During this course the following mode of teaching will be employed:

- Illustrated lectures and group discussions
- Individual and group exercise and assignment
- Ward attachment at least 2 hours/day, one afternoon a week
- Role plays and case studies
- Simulation
- Audiovisuals
- Clinical scenarios
- Tutorials
- Demonstration

**Assessment mechanisms:**

- |                          |     |
|--------------------------|-----|
| • Quizzes:               | 10% |
| • Seminar presentations: | 10% |
| • Mid Exam:              | 20% |
| • Bedside presentation:  | 15% |
| • Assignments            | 5%  |
| • Final Exam             | 40% |

**Teachers' and students' role**

**Roles of Instructors:**

Instructors should be well versed in the outcomes expected of students and the pedagogical methods that best enhance learning. Accordingly, they will be expected to:

- Be a facilitator (introduce the subject; give guidance, moderate discussions, etc.)
- Closely supervise in actual practice in different clinical and community settings
- Assists students in resolving ever-increasing ethical dilemma in the service delivery
- Foster and assess self-initiated student learning
- Read and comment assignments of students on time
- Prepare his/her lessons and deliver lectures

- Provide available and necessary reference materials
- Encourage active participation of students in the teaching learning process;
- Strive to be the role model ethically and professionally

### **Roles of Students**

Students are expected to:

- Make every effort to apply valid research findings and theoretical teachings into practice in various practice settings
- Engage in learning by doing independent study, project work; group work, etc.
- Be active learners (participate effectively in group assignments, perform practice actively, make presentations, write reports, prepare seminars etc.);
- Critically assess laws, regulations, journal articles and related topics from different sources

### **References:**

#### **Required readings (Text)**

1. Lee, Mary. American Society of Health-System Pharmacists. Basic skills in interpreting laboratory data Ashp Publication, 2017
2. Dipiro JT, Talbert RL, Yee GC, et.al. Pharmacotherapy, a Pathophysiologic Approach. 10<sup>th</sup> or later edition.
3. A Practical Guide to Pharmaceutical Care, American Pharmacists Association, 3<sup>rd</sup> edition.

#### **Recommended readings**

4. Koda - Kimble MA, Young LY, Kradjan WA, et.al. Applied Therapeutics, The Clinical Use of Drugs. 11<sup>th</sup> or later edition.
5. Walker R and Edwards C. Clinical Pharmacy and Therapeutics. 3rd or later edition.
6. Kasper, Braunwald, et al. Harrison's Principles of Internal Medicine, 20<sup>th</sup> or later edition
7. Tierney, McPhee, Papadakis. Current Medical Diagnosis and Treatment 2020 or later edition
8. Conn's Current therapy 2020
9. Washington Manual of Medical Therapeutics 32<sup>nd</sup> edition
10. Jacobs & DeMott Laboratory Test Handbook, 5th edition
11. Handouts including copies of PowerPoint slides from lectures
12. Guidelines and articles as specified by the instructor

### **Course Schedule**

| Week # | Contact hrs | Topic/sub-topic/chapter/Assessments/Assignments                                                                                                                                                                                        | Reading Material       |
|--------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| 1      | 2 hrs       | <b>Introduction to Pharmacotherapy</b>                                                                                                                                                                                                 | Reference No. 1, 3, 10 |
|        | 2 hrs       | <b>An overview of pharmaceutical care:</b> The pharmacist in patient care, medical terminologies & common medical abbreviations, pharmaceutical care as a general practice                                                             |                        |
|        | 2 hrs       | <i>Bed-side/ward attachment</i>                                                                                                                                                                                                        |                        |
| 2      | 2 hrs       | <b>An overview of pharmaceutical care:</b> Clinical & economic impact of pharmaceutical care practice, Practice responsibilities, The practitioner's philosophy of practice                                                            |                        |
|        | 2 hrs       | <b>An overview of pharmaceutical care: <i>Pharmaceutical care process</i></b> (Assessment: drug related needs, components of drug therapy problems, identification of drug therapy problems)                                           |                        |
|        | 2 hrs       | <i>Bed-side/ward attachment</i>                                                                                                                                                                                                        |                        |
| 3      | 2 hrs       | <b>An overview of pharmaceutical care: <i>Pharmaceutical care process</i></b> (care plan: establishing goals of therapy, interventions, schedule and plan for follow-up)                                                               |                        |
|        | 2 hrs       | <b>An overview of pharmaceutical care: <i>Pharmaceutical care process</i></b> (follow-up evaluation: determining the clinical outcome status, evaluation for new drug therapy problems, schedule for continuous follow-up evaluations) |                        |
|        | 2 hrs       | <i>Bed-side/ward attachment</i>                                                                                                                                                                                                        |                        |
| 4      | 2 hrs       | <b>An overview of pharmaceutical care:</b> documentation in pharmaceutical care, ethical considerations in practice                                                                                                                    |                        |
|        | 2 hrs       | <b>Diagnostic Tests:</b> General principles                                                                                                                                                                                            |                        |
|        | 2 hrs       | <i>Bed-side/ward attachment</i>                                                                                                                                                                                                        |                        |
| 4      | 2 hrs       | <b>Diagnostic Tests:</b> Electrolytes                                                                                                                                                                                                  |                        |
|        | 2 hrs       | <b>Diagnostic Tests:</b> Electrolytes                                                                                                                                                                                                  |                        |
|        | 2 hrs       | <i>Bed-side/ward attachment</i>                                                                                                                                                                                                        |                        |
| 5      | 2 hrs       | <b>Diagnostic Tests:</b> Hematology                                                                                                                                                                                                    |                        |
|        | 2 hrs       | <b>Diagnostic Tests:</b> Hematology                                                                                                                                                                                                    |                        |
|        | 2 hrs       | <i>Bed-side/ward attachment</i>                                                                                                                                                                                                        |                        |
| 5      | 2 hrs       | <b>Diagnostic Tests:</b> Renal function tests                                                                                                                                                                                          |                        |
|        | 2 hrs       | <b>Diagnostic Tests:</b> Renal function tests                                                                                                                                                                                          |                        |
|        | 2 hrs       | <i>Bed-side/ward attachment</i>                                                                                                                                                                                                        |                        |
| 6      | 2 hrs       | <b>Diagnostic Tests:</b> Liver function tests                                                                                                                                                                                          |                        |
|        | 2 hrs       | <b>Diagnostic Tests:</b> Liver function tests                                                                                                                                                                                          |                        |
|        | 2 hrs       | <i>Bed-side/ward attachment</i>                                                                                                                                                                                                        |                        |
| 7      | 2 hrs       | <b>Diagnostic Tests:</b> Urinalysis                                                                                                                                                                                                    |                        |
|        | 2 hrs       | <b>Diagnostic Tests:</b> Cardiovascular tests                                                                                                                                                                                          |                        |
|        | 2 hrs       | <i>Bed-side/ward attachment</i>                                                                                                                                                                                                        |                        |
| 8      | 2 hrs       | <b>Diagnostic Tests:</b> Endocrine function tests                                                                                                                                                                                      |                        |

|                                            |                |                                                                                                                       |               |                                                |
|--------------------------------------------|----------------|-----------------------------------------------------------------------------------------------------------------------|---------------|------------------------------------------------|
|                                            | 2 hrs          | <b>Diagnostic test:</b> Lipid Panel                                                                                   |               |                                                |
|                                            | 2 hrs          | <i>Bed-side/ward attachment</i>                                                                                       |               |                                                |
| 9                                          | 2 hrs          | <b>Diagnostic Tests:</b> Pulmonary Function Testing                                                                   |               |                                                |
|                                            | 2 hrs          | <b>Diagnostic Tests:</b> Microbiology                                                                                 |               |                                                |
|                                            | 2 hrs          | <i>Bed-side/ward attachment</i>                                                                                       |               |                                                |
| 10                                         | 2 hrs          | Diagnostic imaging and common bed-side procedures                                                                     |               |                                                |
|                                            |                | Basic electrocardiography, echocardiography                                                                           |               |                                                |
|                                            | 2 hrs          | <i>Bed-side/ward attachment</i>                                                                                       |               |                                                |
|                                            | 2 hrs          | 50% Assessment                                                                                                        |               |                                                |
| 10                                         | 2 hrs          | <b>Drug Therapy in Specific Patient Groups:</b> Neonates and Pediatrics                                               |               | Reference No. 1, 3, 5<br>Reference No. 1, 3, 6 |
|                                            | 2 hrs          | <b>Drug Therapy in Specific Patient Groups:</b> Geriatrics                                                            |               |                                                |
| 11                                         | 2 hrs          | <b>Drug Therapy in Specific Patient Groups:</b> Pregnancy and lactation                                               |               |                                                |
|                                            | 2 hrs          | <b>Gastrointestinal Disorders therapeutics:</b> Gastrointestinal tract evaluation & GERD                              |               |                                                |
|                                            | 2 hrs          | <i>Bed-side/ward attachment</i>                                                                                       |               |                                                |
| 12                                         | 2 hrs          | <b>Gastrointestinal Disorders therapeutics:</b> IBD, IBS,                                                             |               |                                                |
|                                            | 2 hrs          | <b>Gastrointestinal Disorders therapeutics:</b> Peptic ulcer disease                                                  |               |                                                |
|                                            | 2 hrs          | <i>Bed-side/ward attachment</i>                                                                                       |               |                                                |
| 13                                         | 2 hrs          | <b>Gastrointestinal Disorders therapeutics:</b> Nausea, Vomiting, Constipation and Diarrhea                           |               |                                                |
|                                            | 2 hrs          | <b>Gastrointestinal Disorders therapeutics:</b> Drug induced liver disease, Pancreatitis                              |               |                                                |
|                                            | 2 hrs          | <i>Bed-side/ward attachment</i>                                                                                       |               |                                                |
| 14                                         | 2 hrs          | <b>Gastrointestinal Disorders therapeutics:</b> Portal hypertension & cirrhosis                                       |               |                                                |
|                                            | 2 hrs          | <b>Gastrointestinal Disorders therapeutics:</b> Viral hepatitis                                                       |               |                                                |
|                                            | 2 hrs          | <i>Bed-side/ward attachment</i>                                                                                       |               |                                                |
| 15                                         | 2 hrs          | Respiratory disorders Pharmacotherapy: Asthma                                                                         |               |                                                |
|                                            | 2 hrs          | Respiratory disorders Pharmacotherapy: COPD                                                                           |               |                                                |
|                                            | 2 hrs          | <i>Bed-side/ward attachment</i>                                                                                       |               |                                                |
| 16                                         | 2 hrs          | Respiratory disorders Pharmacotherapy: ARDS & Neonatal Respiratory distress syndrome –with pediatrics pharmacotherapy | Reference1, 6 |                                                |
|                                            | 2 hrs          | Respiratory disorders Pharmacotherapy: Drug-induced pulmonary diseases and cystic fibrosis                            |               |                                                |
|                                            | 2 hrs          | <i>Bed-side/ward attachment</i>                                                                                       |               |                                                |
|                                            | 50% Assessment |                                                                                                                       |               |                                                |
| Integrated therapeutics II Course syllabus |                |                                                                                                                       |               |                                                |



|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Course Name: Integrated Therapeutics II</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Course code:</b> Phar3163                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Module name:</b> Pharmacotherapeutics module                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Module code:</b> Phar-M3161                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Course ECTS:</b> 7 ECTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <p><b>Totally required hours for the module:</b> 7x27 = 189 hours</p> <p><b>Lecture:</b> 60 hours</p> <p><b>Ward attachment:</b> 30 hours</p> <p><b>Tutorial:</b> 34 hours</p> <p><b>Home study:</b> 46 hours</p> <p><b>Assessment:</b> 14 hours</p> <p><b>Project work/presentation:</b> 15 hours</p>                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Year/Semester Course is offered:</b> Year III/Semester II                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Pre-requisite if any:</b> Integrated Therapeutics I                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <p>Course description:</p> <p>This course is a continuation of Integrated therapeutics-I. The purpose of this course is to provide didactic framework for the therapeutic management of a number of common diseases, including renal diseases, cardiovascular diseases, endocrine diseases, pulmonary diseases, and Eye and ENT. With a thorough background established in physiology, pharmacology, pharmacokinetics and other courses in the curriculum, the goal of this course is to prepare students to develop rational drug therapy plans for patients, identify conditions for monitoring pharmacotherapy in patients, and identify conditions associated with these common diseases that require referral.</p> |
| <p><b>Course objectives:</b></p> <p>After completion of this course, students will be able to explain, practice and choose appropriate treatment strategies for cardiovascular, respiratory, endocrine, and eye and ENT diseases so as to improve patient outcomes.</p> <p>To meet these objectives, students will:</p> <ul style="list-style-type: none"> <li>• Describe the pathophysiologic processes underlying the diseases</li> <li>• Analyze and interpret diagnostic findings</li> <li>• Recommend appropriate treatment regimen</li> <li>• Provide follow up and monitor outcome</li> </ul>                                                                                                                    |
| <b>Course mode of delivery:</b> Parallel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

**Course learning and teaching methods**

During this course the following mode of teaching can be used:

- Illustrated lectures and group discussions
- Individual and group exercise and assignments
- Role plays and case studies
- Problem based learning
- Simulation
- Audiovisuals
- Clinical scenarios
- Tutorials
- Demonstration
- Ward attachment

**Assessment mechanisms:**

- |                          |     |
|--------------------------|-----|
| • Quizzes:               | 10% |
| • Seminar presentations: | 10% |
| • Mid Exam:              | 20% |
| • Bedside presentation:  | 15% |
| • Assignments            | 5%  |
| • Final Exam             | 40% |

**Teachers' and students' role****Roles of Instructors:**

Instructors should be well versed in the outcomes expected of students and the pedagogical methods that best enhance learning. Accordingly, they will be expected to:

- Be a facilitator (introduce the subject; give guidance, moderate discussions, etc.)
- Closely supervise in actual practice in different clinical and community settings
- Assists students in resolving ever-increasing ethical dilemma in the service delivery
- Foster and assess self-initiated student learning
- Read and comment assignments of students on time
- Prepare his/her lessons and deliver lectures
- Provide available and necessary reference materials
- Encourage active participation of students in the teaching learning process;
- Strive to be the role model ethically and professionally

**Roles of Students**

Students are expected to:

- Make every effort to apply valid research findings and theoretical teachings into practice in various practice settings
- Engage in learning by doing independent study, project work; group work, etc..
- Be active learners (participate effectively in group assignments, perform practice actively, make presentations, write reports, prepare seminars etc.);
- Critically assess laws, regulations, journal articles and related topics from different sources

## References:

### Required readings (Text)

1. Dipiro JT, Talbert RL, Yee GC, et.al. Pharmacotherapy, a Pathophysiologic Approach. 10<sup>th</sup> or later edition.

### Recommended readings

2. Koda - Kimble MA, Young LY , Kradjan WA, et.al. Applied Therapeutics, The Clinical Use of Drugs. 11<sup>th</sup> or later edition.
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6. Conn's Current therapy 2020
7. Washington Manual of Medical Therapeutics 32<sup>nd</sup> edition
8. Jacobs & DeMott Laboratory Test Handbook, 5th edition
9. Handouts including copies of PowerPoint slides from lectures
10. Guidelines and articles as specified by the instructor

## Course Schedule

| Week | Contact hrs | Topic/sub-topic/chapter/Assessments/Assignments               | Reading Material     |
|------|-------------|---------------------------------------------------------------|----------------------|
| 1    | 2 hrs       | Renal disorders Pharmacotherapy: Acute Kidney Injury (AKI)    | Reference No. 1,3    |
|      | 2 hrs       | Renal disorders Pharmacotherapy: Chronic Kidney Disease (CKD) | Reference No. 1, 3   |
|      | 2 hrs       | <i>Bed-side/ward attachment</i>                               |                      |
| 2    | 2 hrs       | Renal disorders Pharmacotherapy: Drug induced Renal Disease   | Reference No. 1, 3   |
|      | 2 hrs       | Renal disorders Pharmacotherapy: Glomerulonephritis           | Reference No. 1, 3,6 |
|      | 2 hrs       | <i>Bed-side/ward attachment</i>                               |                      |

|    |       |                                                                                                                                                                                    |                       |
|----|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| 2  | 2 hrs | Renal disorders Pharmacotherapy: Acid-base disorders                                                                                                                               |                       |
|    | 2 hrs | Renal disorders Pharmacotherapy: Disorders of fluid and electrolyte homeostasis                                                                                                    |                       |
|    | 2 hrs | <i>Bed-side/ward attachment</i>                                                                                                                                                    |                       |
| 3  | 2 hrs | Renal disorders Pharmacotherapy: Hemodialysis and peritoneal dialysis                                                                                                              | Reference No.1, 3     |
|    | 2 hrs | Case studies on acute renal failure, chronic renal failure, drug induced kidney disease glomerulonephritis, acid-base disorders and disorders of fluid and electrolyte homeostasis | Reference 1,3, 6      |
|    | 2 hrs | <i>Bed-side/ward attachment</i>                                                                                                                                                    |                       |
| 4  | 2 hrs | Endocrine Disorder Pharmacotherapy: Thyroid disorder                                                                                                                               | Reference No. 1, 3, 6 |
|    | 2 hrs | Endocrine disorder Pharmacotherapy: Diabetes mellitus                                                                                                                              |                       |
|    | 2 hrs | <i>Bed-side/ward attachment</i>                                                                                                                                                    |                       |
| 5  | 2 hrs | Endocrine disorder Pharmacotherapy: Diabetes mellitus                                                                                                                              |                       |
|    | 2 hrs | Endocrine disorder Pharmacotherapy: Pituitary and adrenal gland Disorders                                                                                                          |                       |
|    | 2 hrs | <i>Bed-side/ward attachment</i>                                                                                                                                                    |                       |
| 6  | 2 hrs | Case studies on Endocrine Disorder                                                                                                                                                 |                       |
|    | 2 hrs | Cardiovascular disorders Pharmacotherapy: Cardiopulmonary Resuscitation                                                                                                            |                       |
|    | 2 hrs | <i>Bed-side/ward attachment</i>                                                                                                                                                    |                       |
| 7  | 2 hrs | Cardiovascular disorders Pharmacotherapy: Hyperlipidemia                                                                                                                           |                       |
|    | 2 hrs | Cardiovascular disorders Pharmacotherapy: Hypertension                                                                                                                             |                       |
|    | 2 hrs | <i>Bed-side/ward attachment</i>                                                                                                                                                    |                       |
| 8  | 2 hrs | Cardiovascular disorders Pharmacotherapy: Heart failure                                                                                                                            |                       |
|    | 2 hrs | Cardiovascular disorders Pharmacotherapy: Rheumatoid Valvular Heart Disease                                                                                                        |                       |
|    | 2 hrs | <i>Bed-side/ward attachment</i>                                                                                                                                                    |                       |
| 8  | 2 hrs | Case Studies on Cardiovascular Disorders                                                                                                                                           |                       |
|    | 2 hrs | <b>Mid Exam</b>                                                                                                                                                                    |                       |
| 9  | 2 hrs | Congenital Heart Diseases in Pediatrics                                                                                                                                            |                       |
|    | 2 hrs | Cardiovascular disorders Pharmacotherapy: Ischemic Heart Disease & Acute coronary syndromes                                                                                        |                       |
|    | 2 hrs | <i>Bed-side/ward attachment</i>                                                                                                                                                    |                       |
| 10 | 2 hrs | Cardiovascular disorders Pharmacotherapy: Ischemic Heart Disease & Acute coronary syndromes                                                                                        |                       |

|    |       |                                                                                                                                                             |                       |
|----|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
|    | 2 hrs | Cardiovascular disorders Pharmacotherapy: Cardiac arrhythmia                                                                                                |                       |
|    | 2 hrs | <i>Bed-side/ward attachment</i>                                                                                                                             |                       |
| 11 | 2 hrs | Cardiovascular disorders Pharmacotherapy: VTE                                                                                                               |                       |
|    | 2 hrs | Cardiovascular disorders Pharmacotherapy: peripheral Arterial Disease                                                                                       |                       |
|    | 2 hrs | <i>Bed-side/ward attachment</i>                                                                                                                             |                       |
| 11 | 2 hrs | Cardiovascular disorders Pharmacotherapy: Stroke                                                                                                            |                       |
|    | 2 hrs | <b>50% continuous assessment report</b>                                                                                                                     |                       |
| 12 | 2 hrs | Cardiovascular disorders Pharmacotherapy: Shock                                                                                                             | Reference No. 1, 3, 5 |
|    | 2 hrs | Case studies on selected cardiovascular disorders Pharmacotherapy                                                                                           |                       |
|    | 2 hrs | <i>Bed-side/ward attachment</i>                                                                                                                             |                       |
| 13 | 2 hrs | <b>Pharmacotherapy of Dermatologic disorders:</b> dermatological drug reactions, allergic and pseudo-allergic drug reactions, self-treatable skin disorders |                       |
|    | 2 hrs | <b>Pharmacotherapy of Dermatologic disorders:</b> psoriasis, atopic dermatitis                                                                              |                       |
| 14 | 2 hrs | <b>Pharmacotherapy of Dermatologic disorders:</b> acne                                                                                                      | Reference No. 1, 3, 6 |
|    | 2 hrs | <i>Bed-side/ward attachment</i>                                                                                                                             |                       |
| 15 | 2 hrs | Case studies on Asthma, COPD                                                                                                                                |                       |
|    | 2 hrs | Eye & ENT disorders Pharmacotherapy: Glaucoma                                                                                                               |                       |
|    | 2 hrs | <i>Bed-side/ward attachment</i>                                                                                                                             |                       |
| 16 | 2 hrs | Eye & ENT disorders Pharmacotherapy: Allergic Rhinitis, Case studies on glaucoma and Allergic rhinitis                                                      |                       |
|    | 2 hrs | <i>Bed-side/ward attachment</i>                                                                                                                             |                       |
|    |       | <b>Final Exam</b>                                                                                                                                           |                       |

**Module Name: Pharmaceutical Technology I****Module Category:** Core**Module Code:** Phar-M3171**Module Number:** 17**Module Weight:** 3 ECTS**Courses:**

| S/N | Course name                           | Course code | ECTS |
|-----|---------------------------------------|-------------|------|
| 1   | Immunological and Biological Products | Phar3122    | 3    |

**Module Description**

The module comprehends the role of genetic engineering and allied technologies that have underpinned the development of a range of pharmaceutical products of modern biotechnology, collectively termed biopharmaceuticals (mainly immunological and biological products) that can be used in the pharmaceutical care of a patient. The emphasis will be to understand the application of recombinant DNA derived drugs (immunological and biological products) in pharmaceutical care of a patient. It also deals with handling of immunological and biological products and with the different therapeutic approaches such as gene therapy, antisense therapy, cell therapy and immunological principles (immunotechnology) that are used in prevention and diagnosis of diseases.

**Module Objective**

This module aims to introduce the students with the role of biotechnology and allied technologies in the development of a range of pharmaceutical products of modern biotechnology, and application of recombinant DNA derived drugs (immunological and biological products) in pharmaceutical care of a patient.

**Module Competency**

Upon a successful completion of this module/course, students will be capable of developing formulation and manufacture various pharmaceutical dosage forms (conventional and biopharmaceutical products) and evaluate their qualities.

**Module Mode of Delivery:**

- Block and Parallel

**Module teaching/learning methods****Learning Activities**

- Active participation during class lectures
- Engage in learning by doing
- Laboratory group work

**Teaching Methods**

- The instructor is expected to introduce concepts and topics, and give references, facilitate discussions, ask questions, correct assignments

**Module mode of Assessment****Formative and Summative assessments**

- Quizzes
- Laboratory reports
- Seminar Presentations
- Assignments
- Final exam

**Course Name:** Immunological and Biological Products

**Course code:** Phar3171

**Module Name:** Pharmaceutical Technology I

**Module Code:** Phar-M3171

**Course ECTS:** 3

**Totally required hours for the course:** 81 hrs

**Lecture hours:** 26

**Study hours:** 30

**Presentation:** 10

**Tutorial:** 5

**Assignment:** 10

**Year/Semester Course is offered:** Year III Semester II

**Course prerequisite/s:** Integrated Physical Pharmacy and Pharmaceutics I&II

**Course Description:**

This course designed to introduce the student the application of biotechnology in drug discovery and development. This course comprehends the role of genetic engineering and allied technologies that have underpinned the development of a range of pharmaceutical products of modern biotechnology, collectively termed biopharmaceuticals (mainly immunological and biological products) that can be used in the pharmaceutical care of a patient. The emphasis will be to understand the application of recombinant DNA derived drugs (immunological and biological products) in pharmaceutical care of a patient. It also deals with handling of immunological and biological products and with the different therapeutic approaches such as gene therapy, antisense therapy, cell therapy and immunological principles (immunotechnology) that are used in prevention and diagnosis of diseases.

**Course Objectives:**

- This course is designed to introduce the students with the role of biotechnology and allied technologies in the development of a range of pharmaceutical products of modern biotechnology, and application of recombinant DNA derived drugs (immunological and biological products) in pharmaceutical care of a patient.

**Course mode of delivery:** Parallel

**Course learning and teaching methods:**

- Active participation during class lectures
- Engage in learning by doing



- The course instructor is expected to introduce concepts and topics and give references, facilitate discussions, ask questions, correct assignments
- Arrange and facilitate excursions

#### **Assessment techniques:**

- Tests: 20%
- Quizzes: 15%
- Assignments: 15%
- Presentation: 10%,
- Final exam 40%

#### **Teachers' and students' role**

##### **Roles of Instructors**

##### **The instructor will be expected to:**

- Be a facilitator (introduce the subject; give guidance, moderate discussions, etc.)
- Read and comment assignments & exercises of students on time;
- Prepare his/her lessons and deliver lectures;
- Provide available and necessary reference materials;
- Encourage active participation of students in the teaching learning process;
- Assist students with learning difficulties

##### **Roles of Students**

##### **Students are expected to:**

- Engage in learning by doing (independent study, group works, etc.)
- Be active learners (participate effectively in group assignments, class activities, etc.);
- Should submit all group and/or individual assignments on due date
- Attend classes regularly

#### **References:**

1. Zink G. L., Biological Products, In Remington's Pharmaceutical Sciences, 23<sup>rd</sup> ed. Mack Publishing Co., Easton, 2020.
2. Arnold L. Demain, Julian E. Davies, Ronald M. Manuals of industrial microbiology and Biotechnology, 2nd ed. ASM press, Washington D.C.
3. Daan JA Crommelin and Robert D Sindelar: Pharmaceutical biotechnology: An introduction for Pharmacists and Pharmaceutical scientists. Harwood academic Publishers

4. Hugo, W. B. and Russell, A. D. Pharmaceutical Microbiology, 7<sup>th</sup> ed. 2004
5. S S Kori and M A Halakaia. Pharmaceutical biotechnology: Fundamentals and Applications. 2<sup>nd</sup> ed. 2005
6. B.D. Singh. Biotechnology. 2<sup>nd</sup> ed. Kaliani Publishers.2005
7. Internet

### Course Schedule

| Week | Contact Hours | Topic/sub-topic/chapter/Assessments/Assignments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | 2             | <ul style="list-style-type: none"> <li>• Introduction to Biotechnology and Pharmaceutical Biotechnology</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 1    | 2             | <ul style="list-style-type: none"> <li>• Introduction to genetic engineering/rDNA technology               <ul style="list-style-type: none"> <li>○ Concepts in rDNA technology</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 2-3  | 4             | <ul style="list-style-type: none"> <li>• Introduction to genetic engineering/rDNA technology (Continued...)               <ul style="list-style-type: none"> <li>○ Tools of genetic engineering (enzymes, cloning vectors, cloning hosts)</li> <li>○ Basic techniques (gene cloning, protein expression)</li> <li>○ Application of rDNA technology                   <ul style="list-style-type: none"> <li>▪ Polymerase chain reaction (PCR) and other techniques of modern biotechnology</li> </ul> </li> </ul> </li> </ul>                                                                                                                                                                                                              |
| 4-5  | 4             | <ul style="list-style-type: none"> <li>• Immunological products and biological products               <ul style="list-style-type: none"> <li>○ General introduction                   <ul style="list-style-type: none"> <li>▪ Conventional Immunological and Biological products</li> <li>▪ Biopharmaceuticals / Biologics/ or biotech drugs</li> </ul> </li> <li>○ The different classes of immunological and biological products:</li> <li>○ Immunological products                   <ul style="list-style-type: none"> <li>▪ Vaccines (definition, ideal properties, types, adjuvant and delivery systemsprophylaxis application, handling (including transportation), storage and administration)</li> </ul> </li> </ul> </li> </ul> |

|       |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6-7   | 4 | <ul style="list-style-type: none"> <li>Immunological products and biological products (Continued...) <ul style="list-style-type: none"> <li>Vaccines (definition, ideal properties, types, adjuvant and delivery systemsprophylaxis application, handling (including transportation), storage and administration) (Continued...)</li> <li>Antibodies (polyclonal vs monoclonal antibodies (MAbs), hybridoma technology, mouse MAbs, humanized MAbs, MAb- drug conjugates, application of MAbs (therapeutic, prophylactic, diagnostic, targeted drug delivery, affinity chromatography)</li> </ul> </li> </ul> |
| 8-9   | 4 | <ul style="list-style-type: none"> <li>Immunological products and biological products (Continued...) <ul style="list-style-type: none"> <li>Antibodies (polyclonal vs monoclonal antibodies (MAbs), hybridoma technology, mouse MAbs, humanized MAbs, MAb- drug conjugates, application of MAbs (therapeutic, prophylactic, diagnostic, targeted drug delivery, affinity chromatography) (Continued...)</li> </ul> </li> <li>Biological products <ul style="list-style-type: none"> <li>Cytokines (Interleukins, Interferones, Tumor necrosis factors, Hematopoietic growth factors)</li> </ul> </li> </ul>   |
| 10-11 | 4 | <ul style="list-style-type: none"> <li>Immunological products and biological products (Continued...)</li> <li>Biological products (Continued...) <ul style="list-style-type: none"> <li>Cytokines (Interleukins, Interferones, Tumor necrosis factors, Hematopoietic growth factors) (Continued...)</li> <li>Other growth factors</li> <li>Recombinant hormones</li> </ul> </li> </ul>                                                                                                                                                                                                                        |
| 10-11 |   | <ul style="list-style-type: none"> <li>Test I</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 12-13 | 4 | <ul style="list-style-type: none"> <li>Immunological products and biological products (Continued...)</li> <li>Biological products (Continued...) <ul style="list-style-type: none"> <li>Recombinant blood products (clotting factors, thrombolytic agents, anticoagulating agents)</li> <li>Therapeutic enzymes</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                        |

|       |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       |   | <ul style="list-style-type: none"> <li>○ Nucleic acid based biological products</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                      |
| 14-15 | 4 | <ul style="list-style-type: none"> <li>• Immunological products and biological products (Continued...)</li> <li>• Biological products (Continued...) <ul style="list-style-type: none"> <li>○ Nucleic acid based biological products (Continued...)</li> <li>○ Gene based products (gene therapy, gene delivery systems)</li> <li>○ Antisense based products (antisense oligonucleotides, siRNAs, microRNAs, aptamers, peptide nucleic acids, ribozymes)</li> </ul> </li> </ul> |
| 16    | 4 | <ul style="list-style-type: none"> <li>• Immunological products and biological products (Continued...)</li> <li>• Biological products (Continued...) <ul style="list-style-type: none"> <li>○ Introduction to cell based therapy (stem cell therapy, transplant rejection and cell encapsulation technology)</li> <li>○ Production, Formulation &amp; Manufacturing, Handling and Dispensing of rDNA derived drugs (Biotech drugs)</li> </ul> </li> </ul>                       |
| 16    |   | <ul style="list-style-type: none"> <li>• Test II</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                     |
|       |   | <b>FINAL EXAM</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

## **Module 14: Social and Administrative Pharmacy Module I**

**Module Name:** Social and administrative pharmacy I

**Module Category:** Core

**Module Code:** Phar-M3181

**Module Number:** 18

**Module Weight:** 5 ECTS

**Courses:**

| <b>Course name</b>                     | <b>Course code</b> | <b>ECTS</b> |
|----------------------------------------|--------------------|-------------|
| Health service management and policies | Comh 3181          | 5           |

### **Module description:**

The social and administrative pharmacy module is designed for undergraduate pharmacy students aiming at cultivating their ability to apply socio-behavioural, health economics and supply chain management disciplines in various pharmacy practice settings starting from interactions with patients and other professionals to health systems-level decision making.

### **Module objective:**

After completion of this module students will be able to:

- Explain the basic concepts in public health services management
- Explain the health care delivery system in Ethiopia

### **Module competencies:**

Upon a successful completion of this module, students will be able to

- Render pharmaceutical services in the context of the Ethiopian health care system and national drug policy
- Actively participate in managing the supply of medical supplies, equipment and reagents
- Effectively communicate with other health professionals to promote and market pharmaceuticals
- Demonstrate management skills to lead a health care program

### **Mode of delivery (Parallel/Block):** Block

- Totally required hours for the module:  $5 \times 27 =$  135hrs

**Module teaching/learning method:**

- Lecture
- Active learning methods (brain storming, buzz group, discussion, group and individual presentation, assignment etc),

**Module mode of assessment:**

The assessment criteria are based on continuous assessment of class activities, individual and group assignment, report writing, test and final exams. This in turn can be broken down into;

- Quizzes: 15%
- Assignment: 15%
- Tests-15%%
- Seminar presentations:10%
- Project work: 15%
- Final written exam-30%

## **Course syllabus**

### **Health Service Management and Policies**

**Course title:** Health Service Management and Policies

**Course code:** PUBH 3181

**Course ECTS credits:** 5 ECTS

**Module Number:** 18

**Course pre-requisite if any:** None

**Status of the course (compulsory/supportive/common/elective course):** supportive

**Course description:** This course is designed to give the student a basic concept of public health services management. It begins by introducing the trainee to the lay and professional concept of health. It then goes on to deal with factors affecting health. It will also give students the knowledge about the relationship of health and development, health systems, Primary health care and health for all, definition and applicability to public health of subjects taught under public health, identifying community health problems, indicators of health status of a population, and major strategies of improving public health in developing countries

#### **Specific course learning objectives:**

Upon completion of this course, the student will be able to:

- Describe basic principles , concepts and methods in health management
- Demonstrate management skills so as to plan ,implement & evaluate effectively & efficiently PHC programs
- Identify planning decisions relating to objectives ,activities & resources
- List & relate functions of management dealing with the execution of activities, and the use of human and physical resources.
- Evaluate the different programs and components of services.
- Apply the basic principles involved in management of resources including drugs
- Establish and manage a working health tam
- Describe the organization and administration of health services in Ethiopia to correlate planning and management to the existing situation
- Describe what PHC is, its component and approaches so and to implement PHC activities to reach at social-health goals of the world

Schedule of contact time, contents/topics & reading/reference materials for each topic

| Week  | Contact hrs | Topic/sub-topic/chapter                                                                                                                                                                                                                                                                                                                                                                             | Reading materials      | Remark |
|-------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|--------|
| 1-2   | 4           | Chapter 1: introduction to Health service management <ul style="list-style-type: none"> <li>• Rationale for the course</li> <li>• Definitions, principles and concepts of management</li> <li>• Management and environment</li> <li>• Types, skills and roles of managers</li> <li>• Main functions of management</li> </ul>                                                                        | Reference: 1,1         |        |
| 3-4   | 6           | Chapter 2: Health delivery system in Ethiopia <ul style="list-style-type: none"> <li>• Organization of Ministry of health</li> <li>• National health policy</li> <li>• National health plan</li> <li>• Specific programs within the health service</li> </ul>                                                                                                                                       | Reference: 1           |        |
| 5 - 6 | 6           | chapter 3: Primary health care <ul style="list-style-type: none"> <li>• Historical development</li> <li>• The PHC approach</li> <li>• PHC as part of socio-economic development</li> <li>• PHC as level of health care</li> <li>• PHC strategies</li> <li>• Essential components of PHC</li> </ul>                                                                                                  | Reference: 1,4,5       |        |
| 7-8   | 4           | Chapter 4: Health planning <ul style="list-style-type: none"> <li>- General overview</li> <li>- Steps in planning <ul style="list-style-type: none"> <li>• situation analysis</li> <li>• identifying and selecting priority problems</li> <li>• setting objectives and targets</li> <li>• setting strategies, review obstacles and limitation</li> <li>• prepare action plan</li> </ul> </li> </ul> | Reference: 1,2,4,4.6,7 |        |
| 9-11  | 4           | Chapter 5: Implementation <ul style="list-style-type: none"> <li>• principles of organization</li> <li>• organizational structure</li> <li>• coordination</li> <li>• Monitoring and control</li> <li>• Supervision</li> </ul>                                                                                                                                                                       | Reference: 1,2,3       |        |



|       |   |                                                                                                                                                                                                                                                                                            |                    |  |
|-------|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|--|
| 12-13 | 2 | Chapter 6: Evaluation <ul style="list-style-type: none"> <li>• Effectiveness</li> <li>• Efficiency <ul style="list-style-type: none"> <li>○ How to evaluate work progress monitoring</li> <li>○ Appraising staff performance</li> <li>○ Evaluating use of resources</li> </ul> </li> </ul> | Reference: 1,2,4,5 |  |
| 14    | 2 | Chapter 7: Managing a health team <ul style="list-style-type: none"> <li>• what a health team means</li> <li>• how to lead a health team</li> <li>• organizing a health team</li> <li>• controlling and assessing the work</li> </ul>                                                      | Reference: 1,1     |  |
| 15-16 | 2 | Chapter 8: Managing resources                                                                                                                                                                                                                                                              | Reference: 1, 2,   |  |

1. Delivery mode/methodology:

- Lecture
- Active learning methods (brain storming, buzz group, discussion, group and individual presentation, assignment etc),

2. Assessment mechanisms:

- Quizzes: 15%
- Assignment: 15%
- Tests-15%%
- Seminar presentations:10%
- Project work: 15%
- Final written exam-30%

3. Course policies:

- Lecture is mandatory
- Student should submit assignments on due date
- Student should take all continuous assessments as scheduled. If he/she misses quiz or assignment, no make-up will be arranged for her/him.
- Student should do his/her own work. If he/she is caught red-handed while cheating, he/she will get zero for that particular work

4. References

Text Book

1. On Being In charge - A Guide for Middle - Level Management in PHC, WHO, Geneva, 1980.1992.

#### Other References

1. Challi Jira, Amsalu Feleke, Getnet Mitike (2003) Health Science Management for Health Science Students. Lecture Note Series. Jimma University: JU.
2. MOH (1993), Health Policy of Transitional Government of Ethiopia, Addis Ababa.
3. Health and Health Related indicators, By the Federal Ministry Of Health ,
4. WHO “Health For All” series 1-7. WHO Geneva.
5. Review of PHC (National). 1985
6. Alma Ata Declaration. WHO/UNICEF, 1978.
7. 20 Year health sector plan (HSDP) and HSTP.

**Module Name: Biopharmaceutics and Clinical Pharmacokinetics Modules****Module Category:** Core**Module Code:** Phar-M3191**Module Number:** 19**Module Weight:** 7 ECTS**Course:**

| S/N | Course name                                    | Course code | ECTS |
|-----|------------------------------------------------|-------------|------|
| 1   | Biopharmaceutics and Clinical Pharmacokinetics | Phar4151    | 7    |

**Module Description**

This module deals with mechanisms of drug absorption, effect of pH on drug absorption and the pH partition principle, role of dosage forms in the absorption of drugs, bioavailability and bioequivalence, factors affecting bioavailability, and evaluation of the bioavailability of a drug. It also deals with the pharmacokinetics aspect of drug molecules i.e. how drugs are absorbed, distributed, metabolized and eliminated in the body. This is essential for pharmacists to provide patients the appropriate drug regimen that will reduce the chance of under-treatment, inadvertent poisoning, and dose related adverse effects.

**Module Objective:**

This module aims to develop the ability to logically apply the interrelationship of the physicochemical properties of the drug, the dosage form in which the drug is given and the route of administration on the rate and extent of drug absorption. It also enable to develop a graduate with good practical knowledge and understanding of pharmacokinetics and the ability to logically apply relatively simple pharmacokinetic principles in everyday clinical pharmacy practice.

**Module Competency**

After completion of this module the students will be able to use the principle of pharmacokinetics in dose adjustment, therapeutic drug monitoring and decision making with regard to rational drug use.

**Module Mode of Delivery:**

- Parallel

**Module teaching/learning methods****Learning Activities**

- Engage in learning by doing (independent study, group assignments, presentation, report, writing, and etc...)
- Participation and note takings during class lectures and debates and discussions;
- Analysis, summarization and presentations of chapter/article, scientific reports (and be able present or submit in a concise and summarized form)
- Liver and kidney function data collection, interpretation, identifying appropriate formula and accordingly adjust the dose for that patient

**Teaching Methods**

- The course facilitator is expected to lecture for introducing concepts and topics, and give references, facilitate discussions, ask questions, give and correct assessments,

- Arrange and facilitate seminar sessions by inviting other health care professionals for public lectures.
- Inform the physician and nurse utilize the already adjusted dose and  $r$  calculated loading dose to their client.

**Module mode of Assessment**

- Group assignments
- Case study presentation
- Tests/quizzes
- Final Exam

**Course Name:** Biopharmaceutics and Clinical Pharmacokinetics  
**Course code:** Phar3191  
**Module Name:** Biopharmaceutics and Pharmacokinetics Module  
**Module Code:** Phar-M3191  
**Course ECTS:** 7

**Totally required hours for the course:** 189 hrs

|                         |     |
|-------------------------|-----|
| <b>Lecture hours:</b>   | 64  |
| <b>Study hours:</b>     | 100 |
| <b>Group work:</b>      | 00  |
| <b>Project work:</b>    | 00  |
| <b>Presentation(s):</b> | 15  |
| <b>Tutorial:</b>        | 10  |
| <b>Assessment:</b>      |     |

**Year/Semester Course is offered:** Year III Semester II

**Course prerequisite/s:** Physiology II and Pharmacology II

**Course Description:**

This module deals with mechanisms of drug absorption, effect of pH on drug absorption and the pH partition principle, role of dosage forms in the absorption of drugs, bioavailability and bioequivalence, factors affecting bioavailability, and evaluation of the bioavailability of a drug. It also deals with the pharmacokinetics aspect of drug molecules i.e. how drugs are absorbed, distributed, metabolized and eliminated in the body. This is essential for pharmacists to provide patients the appropriate drug regimen that will reduce the chance of under-treatment, inadvertent poisoning, and dose related adverse effects.

**Course Objectives:**

After completion of this course students will be able to:

- To develop the ability to logically apply the interrelationship of the physicochemical properties of the drug, the dosage form in which the drug is given and the route of administration on the rate and extent of drug absorption.
- To develop a graduate with good practical knowledge and understanding of pharmacokinetics and the ability to logically apply relatively simple pharmacokinetic principles in everyday clinical pharmacy practice.

**Course mode of delivery:** Parallel

**Course learning and teaching methods:**

- Active participation during class lectures and excursions

- Engage in learning by doing
- The course instructor is expected to introduce concepts and topics and give references, facilitate discussions, ask questions, correct assignments
- Arrange and facilitate excursions

#### **Assessment techniques:**

- Group and assignments:10%
- Case study presentation: 10%
- Tests: 15%
- Quizzes: 15%
- Seminar presentation: 10%
- Final Exam: 40%

#### **Teachers' and students' role**

##### **Roles of Instructors**

##### **The instructor will be expected to:**

- Be a facilitator (introduce the subject; give guidance, moderate discussions, etc.)
- Read and comment assignments & case studies & presentations of students on time;
- Prepare his/her lessons and deliver lectures;
- Provide available and necessary reference materials;
- Encourage active participation of students in the teaching learning process;
- Assist students with learning difficulties

##### **Roles of Students**

##### **Students are expected to:**

- Engage in learning by doing (independent study, group works/exercises, etc.)
- Be active learners (participate effectively in group assignments, class activities/presentations, etc.);
- Attend classes regularly (Both theory and presentations)

#### **References:**

1. Robin L. Southwood      Virginia H. Fleming      Gary Huckaby. Concept in clinical pharmacokinetics. 7<sup>th</sup> edition, American Society of Health-System Pharmacists, 2018
2. M. E. Aulton, Pharmaceutics: the science of dosage form design, 7<sup>th</sup> ed., Churchill Livingstone, Edinburgh.

3. Shargel, L, Yu, ABC. Wu-Pong, S, Applied Biopharmaceutics and Pharmacokinetics, 7<sup>th</sup> ed. McGraw Hills, 2004
4. Washington N, Washington C, Wilson Physiological pharmaceuticals-Barriers to drug absorption, 2<sup>nd</sup> ed., Taylor & Francis, London.
5. Gibaldi, M., Biopharmaceutics and clinical Pharmacokinetics 4<sup>th</sup> ed. Lea and Febiger, Philadelphia, 1992.
6. Rowland, M. and Tozer, T.N., Clinical Pharmacokinetics, 5<sup>th</sup> ed., Lea and Febiger, New Delhi, 2019.
7. Curry, S.H., Drug Disposition and pharmacokinetics, 3<sup>rd</sup> ed., Blackwell Scientific Publications, Oxford, 1980.
8. Bauer, L A. Applied Clinical Pharmacokinetics, 2<sup>nd</sup> ed., McGraw-Hill, New York, 2008
9. Atkinsons AJr, Abernethy, DR, Daniels, CE, Dedrick, RL, Markey, SP, Principles of Clinical Pharmacology, 2<sup>nd</sup> ed., London, Elsevier Inc., 2007
10. Notari, R.E. Biopharmaceutics and clinical Pharmacokinetics, 4<sup>th</sup> ed. Marcel Dekker, Inc., New York, 1987.

### Course Schedule

| Week                            | Contact Hours | Topic/sub-topic/chapter/Assessments/Assignments                                                                                                                                                                                                                                                                                                                                                 | Reading Materials | Remark |
|---------------------------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|--------|
| <b>Part I: Biopharmaceutics</b> |               |                                                                                                                                                                                                                                                                                                                                                                                                 |                   |        |
| 1                               | 4             | <ul style="list-style-type: none"> <li>• Introduction</li> <li>• Definitions, concepts and applications</li> <li>• Barriers of drug transport (epithelia and plasma membrane)</li> <li>• Mechanisms of drug transport (Paracellular and transcellular routes, passive diffusion, carrier-mediated transport (Active transport, facilitated diffusion), vesicular transport)</li> </ul>          |                   |        |
| 2                               | 4             | <ul style="list-style-type: none"> <li>• Factors affecting oral drug absorption               <ul style="list-style-type: none"> <li>○ GIT anatomy and physiology</li> <li>○ Physiologic factors (blood flow, GIT motility; emptying and transit times, gastrointestinal pH, pre-systemic metabolism, Stability in the GIT, other drugs, effect of food, disease states)</li> </ul> </li> </ul> |                   |        |

|   |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |               |
|---|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|---------------|
|   |   | <ul style="list-style-type: none"> <li>Quiz</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |  |               |
| 3 | 4 | <ul style="list-style-type: none"> <li>Factors affecting oral drug absorption (Continued...)               <ul style="list-style-type: none"> <li>Physiochemical factors (Drug dissolution and Noyes-Whitney equation, particle size and surface area, crystal forms, salt formation, PKa and PH, lipid solubility, pH-partition hypothesis)</li> <li>Formulation factors (Drug release from Solution, suspension, capsules and tablets; effects of excipients) (3hrs)</li> </ul> </li> </ul>                                                                                          |  |               |
| 4 | 3 | <ul style="list-style-type: none"> <li>Factors affecting oral drug absorption (Continued...)               <ul style="list-style-type: none"> <li>Formulation factors (Drug release from Solution, suspension, capsules and tablets; effects of excipients)</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                     |  |               |
| 4 |   | <ul style="list-style-type: none"> <li>Test I</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  | Chapters I&II |
| 5 | 3 | <ul style="list-style-type: none"> <li>Drug absorption from other routes of administration               <ul style="list-style-type: none"> <li>Percutaneous drug absorption (anatomy &amp; physiology, process of absorption, factors influencing absorption)</li> <li>Parenteral drug absorption (Intramuscular, Subcutaneous, Intradermal) (Anatomy &amp; physiology, Process of absorption, Factors influencing absorption)</li> <li>Rectal drug absorption (anatomy &amp; physiology, process of absorption, factors influencing absorption)</li> </ul> </li> <li>Quiz</li> </ul> |  |               |
| 6 | 3 | <ul style="list-style-type: none"> <li>Drug absorption from other routes of administration (Continued...)               <ul style="list-style-type: none"> <li>Vaginal drug absorption (anatomy &amp; physiology, process of absorption, factors influencing absorption)</li> <li>Pulmonary drug absorption (anatomy &amp; physiology, process of absorption, Factors influencing absorption)</li> </ul> </li> </ul>                                                                                                                                                                   |  |               |



|                                           |   |                                                                                                                                                                                                                                                                                                                                                                                 |  |                 |
|-------------------------------------------|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-----------------|
|                                           |   | <ul style="list-style-type: none"> <li>○ Nasal drug absorption (Anatomy &amp; physiology, process of absorption, factors influencing absorption)</li> <li>○ Ophthalmic drug absorption (Anatomy &amp; physiology, Process of absorption, Factors influencing absorption)</li> </ul>                                                                                             |  |                 |
| 7                                         | 3 | <ul style="list-style-type: none"> <li>• Bioavailability and Bioequivalence               <ul style="list-style-type: none"> <li>○ Introduction and terminologies</li> <li>○ Types of bioavailability (absolute, relative)</li> <li>○ Methods of Assessing Bioavailability (in vivo methods, in vitro methods)</li> <li>○</li> </ul> </li> </ul>                                |  |                 |
| 8                                         | 3 | <ul style="list-style-type: none"> <li>• Bioavailability and Bioequivalence (Continued...)               <ul style="list-style-type: none"> <li>○ In Vitro/ in Vivo Correlations and Biopharmaceutical classification Scheme</li> <li>○ Bioequivalence studies</li> </ul> </li> </ul>                                                                                           |  |                 |
| 8                                         |   | <ul style="list-style-type: none"> <li>• Test II</li> <li>• Assignment</li> </ul>                                                                                                                                                                                                                                                                                               |  | Chapters III&IV |
| <b>Part II: Clinical Pharmacokinetics</b> |   |                                                                                                                                                                                                                                                                                                                                                                                 |  |                 |
| 9                                         | 2 | <ul style="list-style-type: none"> <li>• Introduction</li> <li>• Definitions, applications and types of pharmacokinetics               <ul style="list-style-type: none"> <li>○ Pharmacokinetic and Pharmacodynamic relationships</li> </ul> </li> </ul>                                                                                                                        |  |                 |
| 10                                        | 3 | <ul style="list-style-type: none"> <li>• Basic pharmacokinetics               <ul style="list-style-type: none"> <li>○ Order and rate constants</li> <li>○ Pharmacokinetic models                   <ul style="list-style-type: none"> <li>▪ Physiologic (perfusion models)</li> <li>▪ Model independent methods (statistical moment theory)</li> </ul> </li> </ul> </li> </ul> |  |                 |
| 11                                        | 4 | <ul style="list-style-type: none"> <li>• Basic pharmacokinetics (Continued...)               <ul style="list-style-type: none"> <li>○ Non-linear pharmacokinetics</li> <li>○ Drug distribution (Tissue permeability, distribution co-efficient, binding of drugs, volume of distribution)</li> </ul> </li> </ul>                                                                |  |                 |

|    |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |                        |
|----|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|------------------------|
|    |   | <ul style="list-style-type: none"> <li>Quiz</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |                        |
| 12 | 4 | <ul style="list-style-type: none"> <li>Basic pharmacokinetics (Continued...) <ul style="list-style-type: none"> <li>Drug elimination (drug metabolism, renal and extra-renal excretion, concept of clearance)</li> <li>Pharmacokinetics of IV bolus injection (plasma conc. vs time profile, determination of kinetic parameters)</li> <li></li> </ul> </li> </ul>                                                                                                                                                               |  |                        |
| 13 | 4 | <ul style="list-style-type: none"> <li>Basic pharmacokinetics (Continued...) <ul style="list-style-type: none"> <li>Pharmacokinetics of constant rate infusion (plasma conc. vs time profile, steady state, loading dose)</li> <li>Pharmacokinetics of IV extravascular dose (plasma conc. vs time profile, Parameters)</li> <li></li> </ul> </li> </ul>                                                                                                                                                                         |  |                        |
| 14 | 4 | <ul style="list-style-type: none"> <li>Basic pharmacokinetics (Continued...) <ul style="list-style-type: none"> <li>Dosage regimen (Introduction, multiple dosing and the therapeutic window, drug accumulation and steady state; maintenance and loading doses, designing a dosage regimen)</li> </ul> </li> </ul>                                                                                                                                                                                                              |  |                        |
| 14 |   | <ul style="list-style-type: none"> <li>Assignment</li> <li>Test III</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  | Basic pharmacokinetics |
| 15 | 4 | <ul style="list-style-type: none"> <li>Clinical pharmacokinetics <ul style="list-style-type: none"> <li>Definition and Applications</li> <li>Individualization and optimization of drug therapy <ul style="list-style-type: none"> <li>Dosage regimen adjustment in renal impairment</li> <li>Dosage regimen adjustment in hepatic impairment</li> <li>Dosage regimen adjustment in Pediatrics</li> <li>Dosage regimen adjustment in Geriatrics</li> <li>Dosage regimen adjustment in Obesity</li> </ul> </li> </ul> </li> </ul> |  |                        |

|    |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |  |  |
|----|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|
|    |   | <ul style="list-style-type: none"> <li>▪ Pharmacokinetic drug interactions in combination therapy</li> <li>▪</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |  |
| 16 | 4 | <ul style="list-style-type: none"> <li>• Clinical pharmacokinetics (Continued...)               <ul style="list-style-type: none"> <li>○ Therapeutic drug monitoring                   <ul style="list-style-type: none"> <li>▪ Principles and applications</li> <li>▪ Drugs requiring therapeutic drug monitoring</li> <li>▪ Antibiotics – Aminoglycosides and vancomycin</li> <li>▪ Anticonvulsants – Phenytoin, carbamazepine, valproic acid, Phenobarbital/primidone, ethosuximide</li> <li>▪ Cardiovascular drugs – Digoxin, lidocaine, procainamide and N-acetyl procainamide and quinidine</li> <li>▪ Immunosuppressants – cyclosporine and tacrolimus</li> <li>▪ Other drugs - Lithium, theophylline</li> </ul> </li> </ul> </li> </ul> |  |  |
|    |   | <b>FINAL EXAM</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |  |

**Module Name: Pharmaceutical Technology II****Module Category:** Core**Module Code:** Phar-M4201**Module Number:** 20**Module Weight:** 7 ECTS**Courses:**

| S/N | Course name         | Course code | ECTS |
|-----|---------------------|-------------|------|
| 1   | Industrial Pharmacy | Phar3121    | 7    |

**Module Description**

The module covers preformulation, formulation, manufacturing and packaging of oral liquids, capsules, tablets, aerosols and sterile products (parenterals & ophthalmic); equipment and instruments used for production, quality control and the overall quality assurance and current good manufacturing practices. It also covers unit operations in pharmaceutical technology.

**Module Objective**

This module aims to equip the student with theoretical and practical aspects of manufacturing of pharmaceuticals; the necessary skills required for processing dosage forms at industrial scale; and the fundamentals of quality assurance of pharmaceuticals and current good manufacturing practices.

**Module Competency**

Upon a successful completion of this module/course, students will be capable of developing formulation and manufacture various pharmaceutical dosage forms (conventional and biopharmaceutical products) and evaluate their qualities.

**Module Mode of Delivery:**

- Parallel

**Module teaching/learning methods****Learning Activities**

- Active participation during class lectures
- Engage in learning by doing
- Laboratory group work

**Teaching Methods**

- The instructor is expected to introduce concepts and topics, and give references, facilitate discussions, ask questions, correct assignments

**Module mode of Assessment****Formative and Summative assessments**

- Quizzes
- Laboratory reports

- Seminar Presentations
- Assignments
- Final exam

## **Pharmaceutical Technology Module Syllabi**

**Module Number:** 20

**Course Title:** Industrial Pharmacy

**Course Code:** Phar4201

**Course EtCTS:** 7

**Course Hours:** 189

**Prerequisite:** Integrated Physical Pharmacy and Pharmaceutics I and II

**Co-requisite:** None

**Totally required hours for the course:** 189hrs

**Lecture hours:** 64

**Study hours:** 54

**Excursion:** 18

**Practical:** 18

**Report writing:** 9

**Tutorial:** 8

**Assessment:** 18

**Year/Semester Course is offered:** Year IV Semester I

**Course prerequisite/s:** Integrated Physical Pharmacy and Pharmaceutics I&II

### **Course Description:**

This course covers the theoretical and practical considerations of pertinent unit operations in pharmacy, namely milling, mixing, drying, filtration, centrifugation, crystallization. It also addresses the manufacturing and packaging of oral liquids, capsules, conventional tablets, coated tablets, and sterile products (parenterals, ophthalmic, irrigating solutions). Equipment and machinery used for production, quality control and the overall quality assurance and good manufacturing aspects of these dosage forms are also discussed. The practical sessions include granulation and characterization, tablet pressing and assessing the qualities of the tablets.

### **Course Objectives:**

After completion of this course students will be able to:

- Perform pertinent unit operations employed in the production of dosage forms.
- Apply the theoretical and practical aspects of manufacturing of pharmaceuticals;
- Acquire the necessary skills required for processing dosage forms at industrial scale;
- Perform of quality assurance of pharmaceuticals and
- Be familiar with Current good manufacturing practices.

**Course mode of delivery:** Parallel

**Course learning and teaching methods:**

- Active participation during class lectures and excursions
- Engage in learning by doing
- The course instructor is expected to introduce concepts and topics and give references, facilitate discussions, ask questions, correct assignments
- Arrange and facilitate excursions

**Assessment techniques:**

- Tests/quizzes: 25%
- Written exam for the practical session: 15%
- Assignments (Group and/or individual): 10%
- Laboratory and excursion reports: 10%
- Written final exam: 40%

**Teachers' and students' role****Roles of Instructors****The instructor will be expected to:**

- Be a facilitator (introduce the subject; give guidance, moderate discussions, etc.)
- Read and comment assignments & exercises of students on time;
- Prepare his/her lessons and deliver lectures;
- Provide available and necessary reference materials;
- Encourage active participation of students in the teaching learning process;
- Assist students with learning difficulties

**Roles of Students****Students are expected to:**

- Engage in learning by doing (independent study, group works/exercises, etc.)
- Be active learners (participate effectively in group assignments, class activities/exercises, etc.);
- Should submit all group and/or individual assignments on due date
- Attend classes regularly

**References:****Required readings (Text)**

1. The Theory and Practice of Industrial Pharmacy, L. Lachman, H. A. Liberman and J. L. Kanig, 4th ed., Lea & Febiger, Philadelphia, 2015.
2. Unit Processes in Pharmacy, D. Ganderton, William Heinemann Medical Books Ltd., London, 1968.

### Recommended readings

3. Unit Operations in Chemical Engineering, W. L. McCabe and J. C. Smith, 3rd Ed., McGraw-Hill, Inc. USA, 1976.
4. M. E. Aulton, Pharmaceutics: the science of dosage form design, 7<sup>th</sup> ed., Churchill Livingstone, Edinburgh.
5. Bentley's Textbook of Pharmaceutics, Edited by E. A. Rawlins, 8th Edition, Bailliere Tindall, London, 2002.
6. L. V. Allen, N. G Popovich, H. C Ansel, Ansel's Pharmaceutical Dosage Forms and Drug Delivery Systems, 11<sup>th</sup> edition, Lippincott Williams & Wilkins, 2017.
7. Modern Pharmaceutics by Gilbert S. Banker (Editor), Christopher T. Rhodes (Editor) 4th edition, 2002, Marcel Dekker
8. Merck Index: An Encyclopedia of Chemicals, Drugs, & Biologicals by Merck, Co, Maryadele J. Oneil (Editor), Ann Smith (Editor) 15th edition, 2013, Merck & Co
9. Physical Pharmacy: Physical Chemical Principles in the Pharmaceutical Sciences by Alfred Martin, Pilar Bustamante, A.H.C. Chun (Illustrator) , 4th edition, 1993, Lea & Febiger
10. Handbook of Pharmaceutical Excipients by Arthur H. Kibbe (Editor), Ainley Wade, Paul J. Weller, 3rd edition Vol 3, 2000, Amer. Pharmaceutical Assoc.

### Course Schedule

| Week | Contact Hours | Topic/sub-topic/chapter/Assessments/Assignments                                                                                                                                                                                                                                                                                                                                                                                                      |
|------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | 3             | <ul style="list-style-type: none"><li>• Unit operations<ul style="list-style-type: none"><li>○ Size reduction<ul style="list-style-type: none"><li>▪ Introduction (Definition and applications)</li><li>▪ Mechanisms of size reduction</li><li>▪ Equipments</li></ul></li><li>○ Size separation<ul style="list-style-type: none"><li>▪ Introduction (definition and applications)</li><li>▪ Size separation techniques</li></ul></li></ul></li></ul> |
| 2    | 3             | <ul style="list-style-type: none"><li>• Unit operations (Continued...)<ul style="list-style-type: none"><li>○ Mixing<ul style="list-style-type: none"><li>▪ Introduction (definition and applications)</li></ul></li></ul></li></ul>                                                                                                                                                                                                                 |



|   |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |   | <ul style="list-style-type: none"> <li>▪ Fluid/semisolid/solid mixing</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 3 | 3 | <ul style="list-style-type: none"> <li>• Unit operations (Continued...)               <ul style="list-style-type: none"> <li>○ Drying                   <ul style="list-style-type: none"> <li>▪ Introduction</li> <li>▪ Drying of wet solids, mechanisms and equipment</li> <li>▪ Dryers for dilute solution and suspension</li> <li>▪ Spray drying</li> <li>▪ Freeze drying</li> </ul> </li> </ul> </li> </ul>                                                                                                                                                                                                      |
| 4 | 3 | <ul style="list-style-type: none"> <li>• Unit operations (Continued...)               <ul style="list-style-type: none"> <li>○ Filtration and clarification                   <ul style="list-style-type: none"> <li>▪ Introduction (principles and applications)</li> <li>▪ Types and mechanism of filtration</li> <li>▪ Factors affecting the rate of filtration</li> <li>▪ Equipments</li> <li>▪ Centrifugation</li> </ul> </li> <li>○ Crystallization                   <ul style="list-style-type: none"> <li>▪ Introduction (concepts)</li> <li>▪ Crystallization techniques</li> </ul> </li> </ul> </li> </ul> |
| 5 | 4 | <ul style="list-style-type: none"> <li>• Tablets               <ul style="list-style-type: none"> <li>○ Introduction (Rationale, quality attributes and classification)</li> <li>○ Tablet formulation (API properties, excipients)</li> <li>○ Tablet manufacturing by direct compression</li> </ul> </li> <li>• Quiz</li> </ul>                                                                                                                                                                                                                                                                                       |
| 6 | 4 | <ul style="list-style-type: none"> <li>• Tablets (Continued...)               <ul style="list-style-type: none"> <li>○ Tablet manufacturing by granulation</li> <li>○ Reasons for granulation, mechanisms of granule formation, methods of granulation, granulators, characterization</li> <li>○ Tablet compression machines (types, basic parts and auxiliaries)</li> </ul> </li> </ul>                                                                                                                                                                                                                              |

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|    |   | <ul style="list-style-type: none"> <li>○ Stages of tablet formation</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 7  | 4 | <ul style="list-style-type: none"> <li>● Tablets (Continued...) <ul style="list-style-type: none"> <li>○ Problems in tableting and troubleshooting (capping and lamination, picking and sticking, mottling, chipping and friability, weight/content variation)</li> <li>○ Quality evaluation (general appearance, thickness, hardness, friability, disintegration time, dissolution, weight variation, content uniformity)</li> </ul> </li> </ul>                                                              |
| 7  |   | <ul style="list-style-type: none"> <li>● Test I</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 8  | 3 | <ul style="list-style-type: none"> <li>● Tablet Coating <ul style="list-style-type: none"> <li>○ Tablet coating principles</li> <li>○ Coating process and equipment</li> <li>○ Sugar coating and film coating</li> <li>○ Coating formula optimization</li> <li>○ Quality control</li> </ul> </li> </ul>                                                                                                                                                                                                        |
| 9  | 4 | <ul style="list-style-type: none"> <li>● Capsules <ul style="list-style-type: none"> <li>○ Hard Gelatin Capsules <ul style="list-style-type: none"> <li>▪ Introduction</li> <li>▪ Raw materials for empty capsules</li> <li>▪ Empty capsule manufacturing</li> <li>▪ Formulation (powder formulation, excipients, semisolid, solid, liquid)</li> <li>▪ Formulation for filling properties, formulation for release of API, formulation for position release</li> </ul> </li> </ul> </li> <li>● Quiz</li> </ul> |
| 10 | 4 | <ul style="list-style-type: none"> <li>● Capsules (Continued...) <ul style="list-style-type: none"> <li>○ Hard Gelatin Capsules (Continued...) <ul style="list-style-type: none"> <li>▪ Capsule Filling Machine: Filling Mechanism</li> <li>▪ Quality control of filled HGC</li> <li>▪ Storage, packaging and stability of HGC</li> </ul> </li> <li>○ Soft gelatin capsules</li> </ul> </li> </ul>                                                                                                             |

|    |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |   | <ul style="list-style-type: none"> <li>▪ Introduction (Description of the soft gel, Rationale for selection)</li> <li>▪ Capsule shell</li> <li>▪ Formulation (fill materials)</li> <li>▪ Manufacturing techniques</li> <li>▪ Quality control tests</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 11 | 4 | <ul style="list-style-type: none"> <li>• Liquid Dosage Forms: Solutions and Coarse dispersions               <ul style="list-style-type: none"> <li>○ Solutions                   <ul style="list-style-type: none"> <li>▪ Introduction</li> <li>▪ Formulation considerations (solvents and other excipients, stability)</li> <li>▪ Manufacturing processes: Equipments and machines, filling and packaging</li> <li>▪ Quality evaluations</li> </ul> </li> <li>○ Suspension and Emulsion                   <ul style="list-style-type: none"> <li>▪ Introduction (physical properties)</li> <li>▪ Formulation considerations, stability and preservation</li> </ul> </li> </ul> </li> </ul> |
| 12 | 1 | <ul style="list-style-type: none"> <li>• Liquid Dosage Forms: Solutions and Coarse dispersions (Continued...)               <ul style="list-style-type: none"> <li>○ Suspension and Emulsion (Continued...)                   <ul style="list-style-type: none"> <li>▪ Manufacturing processes: equipments and machines, filling and packaging</li> <li>▪ Quality evaluations</li> </ul> </li> </ul> </li> </ul>                                                                                                                                                                                                                                                                             |
| 12 | 3 | <ul style="list-style-type: none"> <li>• Sterile Products               <ul style="list-style-type: none"> <li>○ Parenteral products</li> <li>○ Types of parenterals</li> <li>○ Product development, vehicles, solutes, containers</li> <li>○ Production design facilities, steps in processing, packaging</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                            |
| 12 |   | <ul style="list-style-type: none"> <li>• Test II</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

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|----|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 13 | 3 | <ul style="list-style-type: none"> <li>• Sterile Products (Continued...) <ul style="list-style-type: none"> <li>○ Production design facilities, steps in processing, packaging (Continued...)</li> <li>○ Aseptic room &amp; processing, quality control and quality assurance</li> <li>○ Ophthalmic and other sterile preparations</li> </ul> </li> </ul> |
| 14 | 4 | <ul style="list-style-type: none"> <li>• Pharmaceutical Aerosols <ul style="list-style-type: none"> <li>○ Physicochemical principles of aerosol science and technology</li> <li>○ Components of aerosols: propellants, containers, valve and actuator systems</li> <li>○ Types of aerosol drug delivery systems</li> </ul> </li> </ul>                    |
| 15 | 2 | <ul style="list-style-type: none"> <li>• Pharmaceutical Aerosols (Continued...) <ul style="list-style-type: none"> <li>○ Product development, manufacturing and quality control of pharmaceutical aerosols</li> <li>○ Recent developments in pharmaceutical aerosols</li> </ul> </li> </ul>                                                               |
| 15 | 2 | <ul style="list-style-type: none"> <li>• Modified Release Dosage Forms <ul style="list-style-type: none"> <li>○ Introduction: general mechanisms of drug release from dosage forms</li> <li>○ Types of modified release dosage forms: controlled release, sustained release, delayed release, repeated release formulation</li> </ul> </li> </ul>         |
| 16 | 4 | <ul style="list-style-type: none"> <li>• Modified Release Dosage Forms (Continued...) <ul style="list-style-type: none"> <li>○ Design, development and characterization of modified release dosage forms</li> <li>○ In vitro/In vivo evaluation of modified release dosage forms</li> </ul> </li> </ul>                                                   |
| 16 | 6 | <ul style="list-style-type: none"> <li>• Current Good Manufacturing Practices <ul style="list-style-type: none"> <li>○ Building and facilities</li> <li>○ Organization and personnel</li> <li>○ Material, packaging, labeling control</li> </ul> </li> </ul>                                                                                              |

|  |  |                                                                                                                              |
|--|--|------------------------------------------------------------------------------------------------------------------------------|
|  |  | <ul style="list-style-type: none"> <li>○ Production &amp; process controls</li> <li>○ Handling &amp; distribution</li> </ul> |
|  |  | <ul style="list-style-type: none"> <li>● Test III</li> </ul>                                                                 |
|  |  | <b>FINAL EXAM</b>                                                                                                            |

## **Module 21: Social and Administrative Pharmacy Module**

**Module Name:** Social and administrative pharmacy

**Module Category:** Core

**Module Code:** Phar-M4211

**Module Number:** 21

**Module Weight:** 17 ECTS

**Courses:**

| <b>Course name</b>                       | <b>Course code</b> | <b>ECTS</b> |
|------------------------------------------|--------------------|-------------|
| Introduction to pharmacoeconomics        | Phar 4141          | 5           |
| Pharmaceutical supply chain management   | Phar4142           | 7           |
| Medical supplies, equipment and reagents | Phar3143           | 3           |
| Pharmaceuticals promotion and marketing  | Phar3144           | 2           |

### **Module description:**

The social and administrative pharmacy module is designed for undergraduate pharmacy students aiming at cultivating their ability to apply socio-behavioural, health economics and supply chain management disciplines in various pharmacy practice settings starting from interactions with patients and other professionals to health systems-level decision making. It helps students to be able to ensure and manage the supply chains of pharmaceuticals in cost efficient ways (drugs, medical supplies, equipment and reagents) in the various health care settings. The module also equips students with appropriate health service and pharmaceutical management skills. Moreover, it equips students with the basics of cost and outcome analysis of drug therapy. It essentially enables prospective graduate pharmacists to competently involve in pharmacoeconomic decision making of drug treatments in communication with all the relevant parties. In addition, this module introduces students to the concept of essential drugs, rational drug use and drug policy: need, development process, objectives and component strategies. The structure of the Ethiopian Health Care System and the National Drug Policy will also be dealt. It also sheds light on monitoring and evaluation of pharmaceutical programs as part of drug policy. It elaborates drug supply management cycle, namely, selection, quantification procurement, distribution, and rational use of pharmaceuticals. The module deals about effective ways of drug use information gathering to investigate drug use-related problems. And finally, the module also deals with planning, implementation and evaluation of health activities.

**Module objective:**

After completion of this module students will be able to:

- Develop and apply the principles and theory of pharmacoeconomics for health care decision-making.
- Describe the concept of essential drugs, the national drug policy including the core objectives and key strategies.
- Discuss the importance of drug management in controlling costs and preventing morbidity and mortality.
- Explain the basic concepts in public health services management
- Explain the health care delivery system in Ethiopia
- Describe the different types of medical supplies, equipment and reagents and their uses
- Describe the environmental factors that affect marketing of pharmaceuticals and
- Understand and demonstrate the successful application of marketing principles in response to environmental factors.

**Module competencies:**

Upon a successful completion of this module, students will be able to

- Render pharmaceutical services in the context of the Ethiopian health care system and national drug policy
- Manage the development of essential drug list, standard treatment guideline and national formulary based on the essential medicines concept
- Actively participate in managing the supply of medical supplies, equipment and reagents
- Manage the selection, procurement, distribution and use of pharmaceuticals
- Effectively communicate with other health professionals to promote and market pharmaceuticals
- Appraise published pharmacoeconomics studies and effectively use them for decision making process
- Identify the type of economic evaluation most appropriate for a particular decision-making context.
- Identify the economics models appropriate for different scenarios
- Demonstrate management skills to lead a health care program

**Mode of delivery (Parallel/Block):** Parallel

- Totally required hours for the module:  $17 \times 27 =$  459hrs
- Lecture hours: 101hrs (25%)
- Study hours: 122 hrs (30%)
- Group work: 61 hrs (15%)
- Project work: 20 hrs (5%)
- Presentation(s): 41hrs (10%)
- Tutorial: 20 hrs (5%)
- Assessment: 41hrs (10%)

**Module teaching/learning method:**

## Learning Activities

Students are expected to undertake among others the following major learning activities:

- Engage in learning by doing (independent study, group assignments, presentation, report writing, etc...)
- Participation and note takings during class lectures and debates and discussions; Analysis, summarization and presentations of journals and cases studies

## Teaching Methods

- The course facilitator is expected to lecture for introducing concepts and topics, and give references, facilitate discussions, ask questions, give and correct assessments, forming groups, encouraging students for peer learning
- Arrange and facilitate seminar sessions prepared by students

**Module mode of assessment:**

The assessment criteria are based on continuous assessment of class activities, individual and group assignment, report writing, test and final exams. This in turn can be broken down into;

- Group/individual assignments: 15%
- Presentation: 10%
- Journal clubs 10%
- Tests/quizzes 15%
- Case studies 10%
- Final Exam 40%
- Total 100%



|                     |                                              |
|---------------------|----------------------------------------------|
| <b>Course Name:</b> | Introduction to pharmacoeconomics            |
| <b>Course code:</b> | Phar4211                                     |
| <b>Module Name:</b> | Social and administrative pharmacy module II |
| <b>Module Code:</b> | Phar-M4211                                   |
| <b>Course ECTS:</b> | 3                                            |

- Totally required hours for the course: 135hours (5ECTS x 27)
- Lecture: 48
- Project work: 14 hours
- Presentations= 14
- Tutorial: 13 hours
- Home study: 36 hours
- Assessment= 10 hours

**Year/Semester Course is offered:** Year IV Semester I

**Course prerequisite/s:** None

#### **Course description:**

This introductory course in pharmacoeconomics is designed to prepare pharmacists who can competently involve in pharmacoeconomic decision-making. Students will be able to describe and use different methods of pharmacoeconomic evaluation and effectively analyse and evaluate different pharmacoeconomic studies.

#### **Course objective:**

After completion of this course students will be able to describe the different methods of pharmacoeconomic analysis and evaluate pharmacoeconomic studies and effectively use them for decision making.

#### **Supporting objectives:**

To meet this objective, students will:

- Define pharmacoeconomics
- Identify measures of direct and indirect costs based on data on charges, expenditures, and treatment algorithms.

- Determine and use standard costs in economic evaluations.
- Discount costs and benefits appropriately.
- Describe pharmacoeconomics analytical models
- Discuss the importance of specification/selection of perspectives to be included in the analysis.
- Identify the strengths and weaknesses of different evaluation designs(cost-effectiveness/cost-utility/cost benefit/cost minimization).
- Identify measures of outcomes and understand the appropriate use of HRQOL, QALY, and utility measures.
- Discuss the difference between efficacy and effectiveness data.
- Discuss the implications of choice of endpoints for the analysis, including the use of (a) intermediate outcomes measures, (b) utilities and quality of life measures, (c) projected final outcomes based on trial data, and (d) summary of findings by meta-analysis.
- Understand sensitivity analysis, including choice of variables and one- and two-way analysis.
- Critique current Pharmacoeconomics literature.
- Describe the rationale and importance of pharmacoeconomic (PE) analyses.
- Describe the importance of Model in Pharmacoeconomics
- Identify the types of Decision Models Used in Economic Evaluation
- Discuss the advantages and disadvantages of Decision Trees
- Determine the advantages of Markov models and its application in Pharmacoeconomics

**Course mode of delivery:** Parallel

This course is thought using a variety of instructional methods including

- Illustrated Lectures and case studies
- Active learning methods (brain storming, buzz group, discussion, etc)
- Individual and group exercises and assignments
- Presentations and participation in class discussion

**Assessment techniques:**

Continuous assessment & summative assessment

- Quizzes: 15%
- Test: 15%

- Journal club presentation: 10%
- Case studies: 10%
- Assignments (group or individual): 15%
- Written final exam: 35%

## **Teachers' and students' role**

### **Roles of Instructors**

The instructor is expected to:

- Be a facilitator (introduce the subject; give guidance, moderate discussions, etc.)
- Read and comment assignments of students on time;
- Prepare his/her lessons and deliver lectures;
- Provide available and necessary reference materials;
- Encourage active participation of students in the teaching learning process;
- Assist students with learning difficulties;

### **Roles of Students**

Students are expected to:

- Engage in learning by doing (independent study, project work; group work, etc.)
- Be active learners (participate effectively in group assignments, make presentations, prepare and present seminars, write reports, etc.);
- Critically assess journal articles and related topics from book chapters.

## **References:**

### **Required readings (Text)**

1. Drummond MF, Bernie O, Stoddard GL, Torrance GW (2005). Methods for the Economic Evaluation of Health Care Programs, 3<sup>rd</sup> ed. or latter edition, Oxford University Press Inc, New York.

### **Recommended readings**

2. Bootman JL, Townsend RJ, McGhan WF (2002). Principles of Pharmacoeconomics, 2nd ed. or later edition, Harvey Whitney Books Company, United States of America.
3. Haddix AC, Teutsch SM, Shaffer PA, Dunet DO (1996 or later edition). Prevention Effectiveness: A Guide to Decision Analysis and Economic Evaluation, Oxford University Press Inc, New York.

4. Online Lecture Notes – the Economics of Health Care:  
<http://www.oheschools.org/> produced by the UK Office of Health Economics.
5. Renee JG. Pharmacoeconomics from theory to practice drug discovery (2010).  
CRC Press Taylor & Francis Group 6000 Broken Sound Parkway NW, Suite 300  
Boca Raton, FL 33487-2742
6. Ceri J Phillips. Health economics an introduction for health professionals (2010).  
Published by Blackwell Publishing Ltd
7. Andrew M. Jones, Nigel Rice, Teresa Bago d'Uva and Silvia Balia (2007).  
Applied health economics. Routledge 2 Park Square, Milton Park, Abingdon OX  
14 4RN

### Course schedule\*

| Week | Contact Hours | Topic/sub-topic/chapter/Assessments/Assignments                                                                                                                | Reading Materials |
|------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| 1    | 3             | Applications of economic principles in health care and pharmacy <ul style="list-style-type: none"> <li>Health Care and Market Failure</li> </ul>               | References 1,2,7  |
| 1    | 3             | 1. Introduction to Pharmacoeconomics<br>1.1 Definition and Importance of Pharmacoeconomic studies                                                              | Reference 2,3 & 4 |
| 2    | 3             | 2. Decision Analysis and Pharmacoeconomic Evaluations<br>3.1. Basic Concepts of Probability<br>Definition and Theoretical Basis of Decision Analysis in Health | Reference 2,3 & 4 |
| 2    | 3             | 3.3 The Basic Steps of Decision Analysis<br>Evaluation of the Result of Decision Analysis                                                                      | Reference 2,3 & 4 |
| 2    | 3             | 3.4 Possible Benefits and Common Criticisms of Decision Analysis in Health                                                                                     | Reference 2,3 & 4 |
| 3    | 3             | 4. Costs and Time Preference.<br>4.1. Types of Intervention Cost Studies<br>Collection and Types of Costs                                                      | Reference 2,3 & 4 |
| 4    | 3             | 4.3 Discounting of Costs<br>Adjusting for Inflation and Annuitizing Capital Expenditures                                                                       | Reference 2,3 & 4 |
| 5    | 3             | 5. Cost Benefit Analysis (6hrs)<br>5.1. Introduction to CBA<br>5.2. Steps in Conducting CBA                                                                    | Reference 2,3 & 4 |
| 6    | 3             | 5.3 Group discussions and Exercises about CBA                                                                                                                  | Reference 2,3 & 4 |
| 7    | 3             | 6. Cost Effectiveness Analysis<br>6.1 Introduction to CEA                                                                                                      | Reference 2,3 & 4 |
| 8    | 3             | 6.2 Principles of CEA<br>6.3 Conducting CEA                                                                                                                    | Reference 2,3 & 4 |

|    |   |                                                                     |                    |
|----|---|---------------------------------------------------------------------|--------------------|
| 9  | 3 | 6.4 Group Discussions and Exercises about CEA                       | Reference 2,3 & 4  |
| 10 | 3 | 7. Cost Utility Analysis<br>7.1. Introduction to CUA                | Reference 2,3 & 4  |
| 11 | 3 | 7.2. QALY and DALY<br>7.3 Measuring QALY and DALY                   | Reference 2,3 & 4  |
| 12 | 3 | 7.4 Conducting Cost Utility Analysis                                |                    |
| 13 | 3 | 8. Miscellaneous Topics<br>8.1. Markov Models of Chronic Conditions | Reference 2,3 & 4  |
| 14 | 3 | 8.2. Screening and Pharmacoeconomics                                | Reference 2,3 & 4w |
| 15 | 3 | 8.3. Assessing Articles and Critique of Pharmacoeconomic Evaluation |                    |
| 16 | 3 | 8.3. Assessing Articles and Critique of Pharmacoeconomic Evaluation |                    |
|    |   | <b>FINAL EXAM</b>                                                   |                    |

**Course Name:** Pharmaceutical Supply Chain Management  
**Course code:** Phar4212  
**Module Name:** Social and administrative pharmacy module II  
**Module Code:** Phar-M4211  
**Course ECTS:** 7 ECTS

**Totally required hours for the module: 189 hours**

**Lecture=** 4hrs

**Practical=** 16hrs

**Assignment=** 25hrs

**Presentations=** 24hrs

**Tutorial=** 13hrs

**Home study=** 36hrs

**Hospital pharmacy visit=** 10hrs

**PFSA visit=** 15hrs

**Assessment=** 10hrs

**Year/Semester Course is offered:** Year IV Semester I

**Course prerequisite/s:** Health Service Management and Policies (PUBH3141)

**Course description:**

The drug supply management course introduces students with the concept of drug policy, and its objectives, and strategies to ensure access to medicine and other health commodities to the end user (patient) at an affordable price. It also gives an insight on how the drug supply management cycle, namely, selection, quantification, procurement, distribution, and rational use, function. Moreover, the students will have a site visit in hospital and PFSA to have exposure how pharmaceutical supply chain operation looks on the ground.

**Course objective:**

After completion of this course students will be able to:

- Source, store and distribute the right medicines and other health commodities to the end user (patient) in the right place at the right time and affordable price.
- Use the appropriate tools to investigate appropriateness of medicine use;
- Assess the performance of the supply chain and take the appropriate corrective measures to ensure continuous improvement.

**Supporting objectives:**

To meet this objective, students will:

- Explain what national drug policy is and why countries need to have a drug policy.
- Describe the principles of drug supply management system including selection, quantification procurement, distribution and usage at health facility or national level.
- Adopt methods of community drug needs assessment and drug use evaluation.
- Adopt good storage practice and standard operating procedures for drug management at health facilities.
- Prepare list of essential drugs and supplies at health facilities or national level.
- Perform estimation (forecasting) of the quantities of drugs and supplies needed.
- Fill records and generate various reports to the appropriate persons and/or organizations.

- Demonstrate how to dispose expired and other unfit-for-use products based on national guideline
- Identify challenges for drug supply management.
- Explain drug selection, its rationale, and criteria for drug selection.
- Describe drug storage and stock management, stock rotation medical stock security and importance of stock management.
- Describe rational drug use: promotion of rational prescribing, dispensing and use by patients.
- Manage, monitor and evaluate the performance of the supply chain in sourcing, stocking and delivering medicines, medical equipment, medical supplies and other health commodities.
- Assume responsibility and accountability for wastage (damage or expiry) of medicines or supplies resulting from negligence.

**Course mode of delivery:** Parallel

**Course learning and teaching methods:**

This course is thought using a variety of instructional methods including

- Illustrated Lectures and case studies
- Active learning methods (brain storming, buzz group, discussion, etc)
- Individual and group exercises and assignments
- Presentations and participation in class discussion
- Hospital and PFSA visit

**Assessment techniques:**

- Assignments: 15%
- Practical reports 10%
- Quizzes: 15%
- Test: 20%
- Final exam: 40%

**Teachers' and students' role**

**Roles of Instructors**

The instructor will be expected to:

- Be a facilitator (introduce the subject; give guidance, moderate discussions, etc.)
- Read and provide feedback to students' assignment submissions on time;
- Prepare his/her lessons and deliver lectures;
- Provide available and necessary reference materials;
- Encourage active participation of students in the teaching learning process;
- Assist students with learning difficulties;

**Roles of Students**

Students are expected to:

- Have a minimum of 85% class attendance
- Read all reading assignments in advance
- Submit all group and/or individual assignments on due date
- Take all continuous assessments as scheduled.

## References:

### Required readings (Text)

1. Supply MD. MDS-3: Managing Access to Medicines and Health Technologies. Arlington, VA: Management Sciences for Health. 2012.
2. Ministry of Health of Ethiopia (1997). The National Drug Policy of the transitional government of Ethiopia, Addis Ababa, Ethiopia.
3. Prashant Yadav (2015) Health Product Supply Chains in Developing Countries: Diagnosis of the Root Causes of Underperformance and an Agenda for Reform, Health Systems & Reform, 1:2, 142-154, DOI: 10.4161/23288604.2014.968005
4. Chopra S, Meindl P, Kalra DV. Supply chain management: strategy, planning, and operation. Boston, MA: Pearson; 2013 Nov.

### Recommended readings

5. Holloway K. (ed.) and Terry Green (2003). Drug and Therapeutics Committees. WHO/MSH.
6. WHO (2001). How to develop and implement a national drug policy. Second edition. Geneva, Switzerland.
7. Smith, H.A. (2003). Principles and Methods of Pharmacy Management, 2nd ed. Lea & Febiger, Philadelphia, 1980.
8. Hardon A., Hodgkin C., and Fresle D. (2004). How to investigate the use of medicines by consumers, WHO/University of Amsterdam/Royal Tropical Institute.
9. WHO (1997). The use of Essential Drugs: Seventh Report of the WHO Expert Committee (including the revised Model List of Essential Drugs), TRS 867, WHO, Geneva, Switzerland.
10. WHO (1988). Estimating Drug Requirements, a practical Manual, WHO/DAP/88.2
11. WHO (1994). Indicators for Monitoring National Drug policies, a Practical Manual, WHO/DAP/94.12, WHO Geneva, Switzerland.
12. WHO (1990) Management of Drugs at Health Facilities, WHO/DAP/90.10 WHO, Geneva, Switzerland.
13. WHO (1997). Public-private Roles in the Pharmaceutical sector. Implications for equitable Access and Rational Drug use. Health Economics and Drugs, DAP series N. 5, WHO/DAP/97.12, WHO, Geneva, Switzerland.
14. Kermit D. Larsen: Fundamental Accounting Principles: 12th ed.; Richard Irwin Inc., 1990.
15. Hermanson, Edwards and Salmon son: Accounting Principles, 4th-7th ed. Richard D. Irwin Inc.1989.
16. Needles, Belverd E. Jr., Henry R. Anderson, James C. Caldwell, and Sherry K. Mills. Principles of Accounting. Houghton Mifflin Company.
17. International Pharmaceutical Federation. Pharmacists in the supply chain: The role of the medicines expert in ensuring quality and availability. The Hague, the Netherlands:



- International Pharmaceutical Federation, 2017Min H. Healthcare supply chain management: basic concepts and principles. Business Expert Press; 2014 Sep 30.
18. McCabe A, Seiter A, Diack A, Herbst CH, Dutta S, Saleh K. Private sector pharmaceutical supply and distribution channels in Africa: a focus on Ghana, Malawi and Mali.

### Course schedule\*

| Week    | Contact Hours<br>Theory/<br>Practice | Topic/sub-topic/chapter/Assessments/Assignments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 and 2 | 4/<br>3                              | 1. Introduction<br>1.1. Introduction to National Drug Policy<br>1.2. Development and implementation of National Drug Policy<br>1.3. Objectives and key strategies of NDP<br>1.4. Concept of essential medicine<br>1.5. Formulary process<br>1.6. Pharmaceutical Sector Transformation Plan of Ethiopia                                                                                                                                                                                                                                                                                                                    |
| 3       | 2/<br>3                              | 2. Health commodity security<br>2.1. What is health commodity security?<br>2.2. Strategic pathway to health commodity security<br>2.3. Steps involved in health commodity security<br>2.4. The purpose of logistics system<br>2.5. Major activities of the logistics system<br>2.6. Components of a logistics cycle<br>2.7. Key logistics terms                                                                                                                                                                                                                                                                           |
| 4 and 5 | 3                                    | 3. Essentials of Healthcare Supply Chain Management<br>3.1. What Is Supply Chain and Supply Chain Management?<br>3.2. Key Supply Chain Terminology and Concepts (value, push strategy, pull strategy, postponement strategy, bullwhip effect, strategic alliance, core competency,<br>3.3. Developing Healthcare Supply Chain Maps<br>3.4. Globalization and supply chains<br>3.5. The importance and integrity of the supply chain to global health (universal health coverage; sustainable development goal; medicines availability, access and shortage; substandard and falsified medicines and supply chain security |
| 6       | 2/<br>2                              | 4. Selection of pharmaceuticals<br>4.1 Introduction to selection of pharmaceuticals<br>4.2. Selection criteria<br>4.3. Approach to developing medicine list<br>4.4. Contribution of supply chain personnel in selection<br>4.5. Exercise                                                                                                                                                                                                                                                                                                                                                                                  |
| 7 - 8   | 10/<br>6                             | 5. Quantification<br>5.1 Introduction to quantification, forecasting and supply planning<br>5.2 Importance of quantification in supply chain management<br>5.3 Key steps in quantification<br>5.4 Process for initiating, reviewing and updating quantification<br>5.5 Exercise                                                                                                                                                                                                                                                                                                                                           |
| 9       | 4/<br>6                              | 6. Procurement of pharmaceuticals<br>6.1 Procurement<br>6.2 Procurement process and Management<br>6.3 Common procurement challenges.<br>6.4 Procurement rules and regulations in Ethiopia<br>6.5 Exercise                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 10      | 6/<br>3                              | 7. Inventory management<br>7.1 Introduction to and key terms of inventory management<br>7.2 Purpose and guideline of good storage practice                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

|         |         |                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         |         | 7.3 Storage System<br>7.4 Steps in storing pharmaceuticals<br>7.5 Designing pharmaceutical store<br>7.6 Pharmaceutical Disposal<br>7.7 Good distribution practice and its purpose<br>7.8 Designing/Redesigning distribution system<br>7.8.1 Transportation Management                                                                                                                                               |
| 11      | 6/<br>6 | 8. Inventory control system<br>8.1 Introduction to and Key terms of an inventory control system<br>8.2 Types of inventory control system and determining order quantities using the three versions of inventory control system<br>8.3 Setting minimum and maximum levels and emergency order point<br>8.4 Comparison of the three versions of inventory control systems<br>8.5 Inventory control system in Ethiopia |
| 12 - 13 | 8/<br>4 | 9 Logistics management information system<br>9.1 Logistics Management Information Systems<br>9.2 Essential Data for Decision Making<br>9.3 Logistics Records<br>9.4 Reporting Systems and Summary Reports<br>9.5 Feedback Reports<br>9.6 Ethiopian logistics management information system–the integrated pharmaceutical logistics system.<br>9.6.1 Auditable Pharmaceutical Transaction                            |
| 14      | 2/<br>3 | 10. Stock status assessment<br>10.1 Assessing stock status<br>10.2 When to assess stock status<br>10.3 How to assess stock status at any level in the system                                                                                                                                                                                                                                                        |
| 14      | 3/<br>3 | 11. Managing for rational use medicines<br>11.1 The medicine use process<br>11.2 Rational use of medicines<br>11.3 Irrational use of medicines and examples<br>11.4 Causes of irrational use of medicines<br>11.5 Impacts of irrational medicine use<br>11.6 Strategies to promote rational use of medicines                                                                                                        |
| 15 - 16 | 5/<br>6 | 12. Monitoring and evaluation<br>12.1 Introduction to Monitoring and Evaluation<br>12.2 Tools to investigate supply chain practice and performance<br>12.3 Tools to investigate medicine use practice and performance<br>12.4 Data collection methods                                                                                                                                                               |
| 13      |         | <b>FINAL EXAM</b>                                                                                                                                                                                                                                                                                                                                                                                                   |

**Course title: Medical Supplies, Equipment & Reagents**

**Module category: Core**

**Module Code: Phar-M4211**

**Course code:** Phar 4213

**Course ECTS:** 3

**ECTS credits:** 81 hours

- Lecture:32
- Practical: 18 hours
- Presentations=5 hours
- Tutorial: 10 hours
- Home study: 8 hours
- Assessment=8

Pre-requisite if any: - None

***Course Description:***

- This introductory course is designed to prepare graduate pharmacists who can competently involve in managing the supply of medical supplies, equipment and reagents. Students will be able to differentiate between the different classes of medical supplies and equipment. This course also introduces students with the basic types of diagnostic drugs and reagents. This in turn helps the trainees for ease selection, quantification and procurement of different classes of medical supplies, equipments and reagents which are required by the health establishment or academic institution they shall join.

***Course objectives:***

After completion of this course students will be able to:

- Identify the different types of Medical Supplies and Equipments commonly used at different settings
- Describe how these Medical Supplies and Equipments function and used.
- Explain how these Medical Supplies and Equipments should be handled, transported and stored safely.
- Identify the different types of Diagnostic Supplies and Reagents
- Describe how these Diagnostic Supplies and Reagents are clinically applied or used.
- Identify the common Medical Supplies and Equipments used in Veterinary Medicine.

| <b>Week</b> | <b>Contact hrs</b> | <b>Topic/sub-topic/chapter</b>                                                                                                                                       | <b>Reading materials</b> |
|-------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| 1           | 4                  | 1. Medical Supplies and Equipment (10hr<br>1.1. Definition of Terms (Medical Supplies, Medical Equipment or Instrument, Reagents)                                    | Reference 1&2            |
| 1           | 4                  | 1.2.Surgical Dressings (Fibres, Fabrics, Bandages, Self-adhesive plasters, Compound dressings etc.)                                                                  | Reference 1&2            |
| 2           | 4                  | 1.3 Sutures and Ligatures (absorbable sutures, non-absorbable sutures, surgical needles etc.)                                                                        | Reference 1&2            |
| 2           | 4                  | 1.4 Medicinal Gases (classifications, uses & applications, Safety precaution, and different components, etc.                                                         | Reference 1&2            |
| 3           | 4                  | 1.5 Other Medical Supplies (Needles and syringes, Gloves, Masks, Surgical blades, Scissors, Forceps, Catheter, Nasogastric tubes, Endotracheal tubes, rectal tubes). | Reference 1&2            |
| 3           | 4                  | 1.6 Equipments Used in Surgery, Anesthesia, Orthopedics, Ophthalmology, Dentistry, ENT.                                                                              | Reference 1&2            |
| 4           | 4                  | 1.6 Equipments Used in Surgery, Anesthesia, Orthopedics, Ophthalmology, Dentistry, ENT.....                                                                          | Reference 1&2            |
| 4           | 4                  | 1.7 Infection Control, Sterilization and Care of Surgical Instruments                                                                                                | Reference 1&2            |
| 5           | 4                  | 2. Diagnostic Supplies and Reagents (4hrs)                                                                                                                           | Reference 1&2            |
| 5           | 4                  | 2.1.Different Diagnostic Supplies (laryngoscope, otoscope, thermometers, sphygmomanometers, glucometers, X-ray supplies, microscope, stethoscope, etc.               | Reference 1&2            |
| 6           | 4                  | 2.1.Diagnostic Imaging Drugs (X-ray contrast agents, Magnetic resonance contrast agents, Ultrasound contrast agents, etc.)                                           | Reference 1&2            |

|   |   |                                                                                                                                                                                                                         |                  |
|---|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 6 | 4 | 2.1 Non-Imaging In-Vivo Diagnostic Drugs<br>(Cardiovascular System, Endocrine System, GIT,<br>Lymphatic System, Reproductive System,<br>Ophthalmic, Urinary Tract, Miscellaneous Skin<br>Antigen Tests, etc.)           | Reference<br>1&2 |
| 7 | 4 | 2.1.Reagents Used in the Medical Laboratory<br>(mycobacterium testing (AFB), pregnancy tests,<br>enteric fever tests, uric acid tests, blood grouping<br>tests, VDRL tests, HIV tests, clinical chemistry test,<br>etc. | Reference<br>1&2 |
| 7 | 4 | 2.2.Self-Care Diagnostic Devices                                                                                                                                                                                        | Reference<br>1&2 |
| 8 | 4 | 3. Medical Supplies and Equipments used in Veterinary<br>Medicine (2hrs)<br>3.1. Peculiar characteristics of Supplies Used in<br>Veterinary Medicine                                                                    | Reference 3      |
| 8 | 4 | 3.2.Equipments Used for Oral Administration of Drugs<br>3.3.Equipments Used for Intravenous Administration<br>3.4. Materials Used for Administration of Topical<br>Medication                                           | Reference 3      |

### **Mode of delivery: Block**

This course is thought using a variety of instructional methods including

- Illustrated Lectures
- Active learning methods (brain storming, buzz group, discussion, etc)
- Individual and group exercises and assignments
- Presentations and participation in class discussion
- Demonstration and visit

### **Assessment mechanisms**

Continuous assessment & summative assessment

- Assignments: 10%
- Quizzes: 10%
- Presentations: 10%
- Test: 15%
- Visits and report: 15%

- Final exam: 40%

References:

1. Troy BD (ed) (2006). Remington: The Science and Practice of Pharmacy, 21<sup>st</sup> ed, Lippincott Williams and Wilkins, Philadelphia
2. Kapur MM (2005). A Complete Hospital Manual of Instruments and Procedures, Jaypee Brothers Medical Publishers Ltd, New Delhi
3. Wanamaker BP, Pettes CL (2000). Applied Pharmacology for the Veterinary Technician, 2<sup>nd</sup> ed., W.B. Saunders Company, USA.

**Course Name:** Pharmaceutical Marketing and promotion  
**Course code:** Phar 4214  
**Module Name:** Social and administrative pharmacy module  
**Module Code:** Phar-M4211  
**Course ECTS:** 3

**Totally required hours for the module: 81 hrs**

**Lecture hours:**

**Study hours:**

**Group work:**

**Project work:**

**Presentation(s):**

**Tutorial:**

**Assessment:**

**Year/Semester Course is offered:** Year IV Semester II

**Course prerequisite/s:** None

**Course description:**

This course will discuss on pharmaceutical marketing principles, environment and practice. The pharmaceutical marketing area will be covered from its history and development through the vast array of environmental forces and to the marketing mixes, known as the 4Ps. The 4Ps, considered as vital elements of pharmaceutical marketing include product development, pricing, place and promotion strategies. Due emphasis will be made to the promotion practice to reflect local contexts. Case studies will be used to illustrate the concepts.

**Course objective:**

After completion of this course students will be able to

- Apply the principles and concepts of pharmaceutical marketing in the pharmaceutical service delivery
- Integrate pharmaceutical marketing practice with the professional ethics as per the Ethiopian national guidelines
- Apply effective communication skill in the pharmaceutical service delivery
- Employ integrated pharmaceutical distribution networking and inventory management skills

**Course mode of delivery:** Block

**Course learning and teaching methods**

- Interactive presentation and discussion
- Group discussions and presentation
- Individual and group exercises
- Role play and case study

**Assessment techniques:**

Continuous assessment & summative assessment

- Assignments: 10%
- Quizzes: 10%
- Presentations: 10%
- Test: 15%
- Visits and report: 15%

**Teachers' and students' role**

**Roles of Instructors**

The instructor is expected to:

- Be a facilitator (introduce the subject; give guidance, moderate discussions, etc.)
- Read and comment assignments of students on time;
- Prepare his/her lessons and deliver lectures;
- Provide available and necessary reference materials;
- Encourage active participation of students in the teaching learning process;
- Assist students with learning difficulties;

**Roles of Students**

Students are expected to:

- Engage in learning by doing (independent study, project work; group work, etc.)
- Be active learners (participate effectively in group assignments, make presentations, prepare and present seminars, write reports, etc.);

**References:**



1. Mickey C. Smith, E. M. Mick Kolassa, Greg Perkins, Bruce Siecker (Eds). Pharmaceutical Marketing: Principles, Environment and Practice. Informa Healthcare, New York, 2002.
2. Mickey C. Smith. Pharmaceutical Marketing: Strategy and Cases 1st Edition. CRC Press, 1991.
3. Lidstone J. Marketing Planning for the Pharmaceutical Industry 2nd Edition. Routledge publisher 1999.
4. Dimitri sDogramatzis. Pharmaceutical Marketing: A Practical Guide. Taylor and Francis, Boca Raton, 2002.
5. Philip Kotler, Joel Shalowitz, Robert J. Stevens. Strategic Marketing for Health Care Organizations: Building a Customer-Driven Health System. Jossey-Bass, San Francisco, 2008.
6. Remington's Pharmaceutical Science, 21st ed., Lippincott Williams & Wilkins, Pennsylvania, 2006.

#### Course schedule\*

| Week | Contact Hours | Topic/sub-topic/chapter/Assessments/Assignments                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    |               | <b>1. Introduction to Principles of Marketing</b><br><b>1.1. Marketing</b><br>1.1.1. Introduction<br>1.1.2. Definition<br>1.1.3. Marketing Concept<br>1.1.4. Marketing Versus selling<br>1.1.5. Evolution/philosophies of Marketing<br>1.1.6. Functions of Marketing<br>1.1.7. Market Research<br>1.1.8. Marketing strategies                                                                                                                                                          |
| 2    |               | <b>2. Pharmaceutical Marketing</b><br><b>2.1. Introduction</b><br><b>2.2. Definition</b><br><b>2.3. Purposes</b><br><b>2.4. Pharmaceutical Marketing Environment</b><br>2.4.1. The internal Environment<br>241.1. Patients and Customers<br>241.2. Marketing Mixes (The Four Ps)<br>241.3. Analysis of the Internal Environment<br>2.4.2. The external Environment<br>242.1. The micro environment<br>2.4.2.1.1.The Competitive Environment<br>2.4.2.1.2.Porter's Five-Factor Analysis |

|   |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |  | 2.4.2.2. The macro environment (PESTLE)<br>2.4.2.2.1. Political Environment<br>2.4.2.2.2. Economic Environment<br>2.4.2.2.3. Social Environment<br>2.4.2.2.4. Technological Environment<br>2.4.2.2.5. Ethical Environment<br>2.4.2.2.6. Legal Environment<br>2.4.2.2.7. Environmental<br>2.4.2.2.8. Health care environment                                                                                                                           |
| 3 |  | <b>2.5. Market situational analysis</b><br>2.5.1. SWOT analysis<br>2.5.2. PESTEL analysis<br>2.5.3. Market feasibility study                                                                                                                                                                                                                                                                                                                          |
| 4 |  | <b>2.6. Pharmaceutical marketing mix (4Ps)</b><br><b>2.6.1. Product</b><br>2.6.1.1. New Product Strategy<br>2.6.1.2. Product Positioning Strategy<br>2.6.1.3. Product Repositioning Strategy<br>2.6.1.4. Product Elimination Strategy<br>2.6.1.5. Diversification Strategy<br>2.6.1.6. Product Life Cycle                                                                                                                                             |
| 5 |  | <b>2.6.2. Price</b><br>2.6.2.1. Elements of price<br>2.6.2.2. Determinants of pricing<br>2.6.2.3. Pricing Approaches<br>2.6.2.4. Pricing Decision<br>2.6.2.5. Pharmaceutical financing<br>2.6.2.6. The Reimbursement Environment                                                                                                                                                                                                                      |
| 6 |  | <b>2.6.3. Place</b><br>2.6.3.1. Overview of pharmaceutical distribution<br>2.6.3.2. Distribution strategy<br>2.6.3.3. Challenges of managing place<br>2.6.3.4. Place Factors<br>2.6.3.5. Key distribution channel decisions<br>Inventory Management                                                                                                                                                                                                   |
| 7 |  | <b>2.6.4. Promotion</b><br>2.6.4.1. Introduction to promotion<br>2.6.4.1.1. Objectives<br>2.6.4.1.2. Environmental Factors<br>2.6.4.1.3. Characteristics of pharmaceutical promotion<br>2.6.4.2. Targets of Promotion<br>2.6.4.3. Promotional appeals<br>2.6.4.4. Sources of promotional information<br>2.6.4.5. Promotional mix<br>2.6.4.5.1. Personal selling<br>2.6.4.5.2. Advertising<br>2.6.4.5.3. Public relation<br>2.6.4.5.4. Sales promotion |

|    |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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|    |  | 2.6.4.5.5. Internet<br>2.6.4.6. Life cycle consideration<br>2.6.4.7. Promotional Planning<br>Promotional budget                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 8  |  | <b>3. Effective Pharmaceutical marketing communication</b><br><b>3.1.</b> Introduction to effective communication<br><b>3.2.</b> Communication channels<br><b>3.3.</b> Factors affecting effective communication<br><b>3.4.</b> Pharmaceutical marketing Communication process<br><b>3.5.</b> Integration of marketing communication with the marketing mix<br>Digital: Media, Pharmaceutical Marketing opportunities and Challenges                                                                                                                                                                               |
| 9  |  | <b>4. Ethical pharmaceutical marketing and promotion practice</b><br><b>4.1.</b> Overview of ethical pharmaceutical marketing<br>4.1.1. Ethics in pharmaceutical marketing<br>4.1.2. Current ethical challenges in pharmaceutical marketing<br><b>4.2.</b> Applying compassionate, respectful and caring (CRC) approach to the pharmaceutical service<br><b>4.3.</b> International and national guidelines on ethical pharmaceutical marketing<br>4.3.1. Ethiopian Pharmaceutical promotion guideline<br>4.3.2. Code of ethics for pharmacists practicing in Ethiopia<br>4.3.3. WHO pharmaceutical promotion guide |
| 10 |  | <b>FINAL EXAM</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

## **Module 22: Pharmacotherapeutics**

**Module name:** Pharmacotherapeutics Module II

**Module category:** Core

**Module code:** Phar-M4221

**Module weight in ECTS:** 14 (14x27) = 378hrs

**Courses:**

| <b>Course name</b>          | <b>Course code</b> | <b>ECTS</b> |
|-----------------------------|--------------------|-------------|
| Integrated therapeutics III | Phar4221           | 7 ECTS      |
| Integrated therapeutics IV  | Phar4222           | 7 ECTS      |

### **Module description:**

Students will learn about the Pathophysiology and pharmacotherapy of various disease states that health care practitioners (pharmacists) may encounter in their practice settings. Courses in this module introduce essential therapeutic knowledge needed for providing pharmaceutical care in individual patient. These courses integrate the pathophysiologic abnormalities of disease state with concepts of drug selection, dose optimization and monitoring of therapeutic outcomes for safety and efficacy of medication. Courses discussed include: integrated therapeutics I-IV which extends from general principles of pharmacotherapy to detailed pharmacotherapy of each disease states (gastrointestinal, respiratory, cardiovascular, renal, hematologic, neurologic, psychiatric, endocrinologic, infectious diseases etc.)

### **Module objective:**

At the end of this module, the students are expected to:

- Explain the etiology, pathophysiology, clinical presentation and diagnosis of each disease states
- Set goals of treatment and select treatment options for the management of each disease states
- Formulate dose recommendations and pharmacokinetic considerations for individual patient management
- Monitor clinically significant adverse drug reactions and drug interactions
- Evaluate therapeutic outcomes for effectiveness, safety and patient adherence

- Develop and exercise pharmaceutical care planning for managing a specific patient condition
- Provide patient medication counseling and drug information

**Module competency:**

Provide patient centered Pharmaceutical care services

**Mode of delivery: Parallel**

**Mode of Assessment:**

Continuous assessment & summative assessment: Class attendance, Continuous assessment, Assignments, Hospital attachment Report, Final Exam

**Module learning teaching methods**

Illustrated lectures and group discussions, Individual and group exercise and assignments, Role plays and case studies, Simulation, Audiovisuals, Clinical scenarios, Tutorials, demonstration

## Course Syllabus

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |          |                         |          |                  |          |                    |          |                    |          |                                   |          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------------|----------|-------------------------|----------|------------------|----------|--------------------|----------|--------------------|----------|-----------------------------------|----------|
| <b>Integrated therapeutics III Course syllabus</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |          |                 |          |                         |          |                  |          |                    |          |                    |          |                                   |          |
| <b>Course name: Integrated Therapeutics III</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |          |                         |          |                  |          |                    |          |                    |          |                                   |          |
| <b>Course code:</b> Phar4221                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |          |                 |          |                         |          |                  |          |                    |          |                    |          |                                   |          |
| <b>Module name:</b> Pharmacotherapeutics module                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |          |                         |          |                  |          |                    |          |                    |          |                                   |          |
| <b>Module code:</b> Phar-M4221                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |          |                 |          |                         |          |                  |          |                    |          |                    |          |                                   |          |
| <b>Course ECTS:</b> 7 ECTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |          |                 |          |                         |          |                  |          |                    |          |                    |          |                                   |          |
| <p><b>Totally required hours for the module:</b> 7x27 = 189 hours</p> <table> <tr> <td><b>Lecture:</b></td><td>60 hours</td></tr> <tr> <td><b>Ward attachment:</b></td><td>30 hours</td></tr> <tr> <td><b>Tutorial:</b></td><td>34 hours</td></tr> <tr> <td><b>Home study:</b></td><td>46 hours</td></tr> <tr> <td><b>Assessment:</b></td><td>14 hours</td></tr> <tr> <td><b>Project work/presentation:</b></td><td>15 hours</td></tr> </table>                                                                                                                                                                                                                                                                               |          | <b>Lecture:</b> | 60 hours | <b>Ward attachment:</b> | 30 hours | <b>Tutorial:</b> | 34 hours | <b>Home study:</b> | 46 hours | <b>Assessment:</b> | 14 hours | <b>Project work/presentation:</b> | 15 hours |
| <b>Lecture:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 60 hours |                 |          |                         |          |                  |          |                    |          |                    |          |                                   |          |
| <b>Ward attachment:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 30 hours |                 |          |                         |          |                  |          |                    |          |                    |          |                                   |          |
| <b>Tutorial:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 34 hours |                 |          |                         |          |                  |          |                    |          |                    |          |                                   |          |
| <b>Home study:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 46 hours |                 |          |                         |          |                  |          |                    |          |                    |          |                                   |          |
| <b>Assessment:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 14 hours |                 |          |                         |          |                  |          |                    |          |                    |          |                                   |          |
| <b>Project work/presentation:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 15 hours |                 |          |                         |          |                  |          |                    |          |                    |          |                                   |          |
| <b>Year/Semester Course is offered:</b> Year IV/Semester I                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |          |                 |          |                         |          |                  |          |                    |          |                    |          |                                   |          |
| <b>Pre-requisite if any:</b> Integrated Therapeutics I                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |                 |          |                         |          |                  |          |                    |          |                    |          |                                   |          |
| <p><b>Course description:</b></p> <p>This is the third course in a sequence of four integrated therapeutics courses in the curriculum. The areas of therapeutic focus in integrated therapeutics-III include: psychiatric diseases, neurological diseases, gynecologic and obstetric disorders, urological disorders, immunological and musculoskeletal disorders. The course will utilize the case-assisted student-centered learning format to enhance the student's ability to apply and utilize information in solving problems and/or enhancing patient care with medications.</p>                                                                                                                                       |          |                 |          |                         |          |                  |          |                    |          |                    |          |                                   |          |
| <p><b>Course objectives:</b></p> <p>After completion of this course, the student will be able to describe, analyze and identify various psychiatric, neurologic, immunological, gynecology and obstetrics, urologic and musculoskeletal disorders; and manage drug therapy.</p> <p>To meet this objective, students will:</p> <ul style="list-style-type: none"> <li>• Describe the pathophysiologic processes underlying, psychiatric, neurologic, musculoskeletal, gynecology and obstetrics, urologic and immunological disorders.</li> <li>• Analyze and interpret diagnostic findings relevant to, psychiatric, neurologic, musculoskeletal, gynecology and obstetrics, urologic and immunological disorders.</li> </ul> |          |                 |          |                         |          |                  |          |                    |          |                    |          |                                   |          |

- Recommend appropriate treatment regimen for patients suffering from psychiatric, neurologic, musculoskeletal, gynecology and obstetrics, urologic and immunological disorders.

**Course mode of delivery:** Parallel

**Course learning and teaching methods:**

- During this course the following mode of teaching can be used:
- Illustrated lectures and group discussions
- Individual and group exercise and assignments
- Role plays and case studies
- Problem-based learning
- Simulation
- Audiovisuals
- Clinical scenarios
- Tutorials
- Demonstration

**Assessment mechanisms:**

- |                          |     |
|--------------------------|-----|
| • Quizzes:               | 10% |
| • Seminar presentations: | 10% |
| • Mid Exam:              | 20% |
| • Bedside presentation:  | 15% |
| • Assignments            | 5%  |
| • Final Exam             | 40% |

**Teachers' and students' role**

**Roles of Instructors:**

Instructors should be well versed in the outcomes expected of students and the pedagogical methods that best enhance learning. Accordingly, they will be expected to:

- Be a facilitator (introduce the subject; give guidance, moderate discussions, etc.)
- Closely supervise in actual practice in different clinical and community settings
- Assists students in resolving ever-increasing ethical dilemma in the service delivery
- Foster and assess self-initiated student learning
- Read and comment assignments of students on time
- Prepare his/her lessons and deliver lectures

- Provide available and necessary reference materials
- Encourage active participation of students in the teaching learning process;
- Strive to be the role model ethically and professionally

### **Roles of Students**

Students are expected to:

- Make every effort to apply valid research findings and theoretical teachings into practice in various practice settings
- Engage in learning by doing independent study, project work; group work, etc..
- Be active learners (participate effectively in group assignments, perform practice actively, make presentations, write reports, prepare seminars etc.);
- Critically assess laws, regulations, journal articles and related topics from different sources

### **References:**

#### **Required readings (Text)**

1. Dipiro JT, Talbert RL, Yee GC, et.al. Pharmacotherapy, a Pathophysiologic Approach. 10<sup>th</sup> or later edition.

#### **Recommended readings**

2. Koda - Kimble MA, Young LY , Kradjan WA, et.al. Applied Therapeutics, The Clinical Use of Drugs. 11<sup>th</sup> or later edition.
3. Walker R and Edwards C. Clinical Pharmacy and Therapeutics. 3rd or later edition.
4. Kasper, Braunwald, et al. Harrison's Principles of Internal Medicine, 20<sup>th</sup> or later edition
5. Tierney, McPhee, Papadakis. Current Medical Diagnosis and Treatment 2020 or later edition
6. Conn's Current therapy 2020
7. Washington Manual of Medical Therapeutics 32<sup>nd</sup> edition
8. Jacobs & DeMott Laboratory Test Handbook, 5th edition
9. Handouts including copies of PowerPoint slides from lectures
10. Guidelines and articles as specified by the instructor

### **Course Schedule:**

| Week | Contact Hr | Topic/sub-topic/chapter/Assessments/Assignments                          | Reading Material      |
|------|------------|--------------------------------------------------------------------------|-----------------------|
| 1    | 2 hrs      | Psychiatric disorders Pharmacotherapy: Childhood disorders               | Reference No. 1, 3    |
|      | 2 hrs      | Psychiatric disorders Pharmacotherapy: Sleep disorders, Eating disorders | Reference No. 1, 2, 3 |



|   |       |                                                                                             |                    |
|---|-------|---------------------------------------------------------------------------------------------|--------------------|
|   | 2 hrs | <i>Bed-side/ward attachment</i>                                                             |                    |
| 2 | 2 hrs | Psychiatric disorders Pharmacotherapy: Anxiety Disorders                                    | Reference No. 1, 3 |
|   | 2 hrs | Psychiatric disorders Pharmacotherapy: Schizophrenia                                        | Reference No. 1, 3 |
|   | 2 hrs | <i>Bed-side/ward attachment</i>                                                             |                    |
| 2 | 2 hrs | Psychiatric disorders Pharmacotherapy: Eating Disorders                                     | Reference No. 1, 3 |
|   | 2 hrs | Psychiatric disorders Pharmacotherapy: Mood disorders I (Major depressive disorder)         | Reference No. 1, 3 |
|   | 2 hrs | <i>Bed-side/ward attachment</i>                                                             |                    |
| 3 | 2 hrs | Psychiatric disorders Pharmacotherapy: Mood disorders I (Major depressive disorder)         |                    |
|   | 2 hrs | Psychiatric disorders Pharmacotherapy: Mood disorders II (Bipolar Disorders I)              | Reference No. 1, 3 |
|   | 2 hrs | <i>Bed-side/ward attachment</i>                                                             |                    |
| 3 | 2 hrs | Psychiatric disorders Pharmacotherapy: Substance abuse-related disorder                     | Reference No. 1, 3 |
|   | 2 hrs | Case studies on Schizophrenia, Major depression, bipolar disorders                          |                    |
|   | 2 hrs | <i>Bed-side/ward attachment</i>                                                             |                    |
| 4 | 2 hrs | Neurological disorder Pharmacotherapy: Pain management                                      | Reference No. 1, 3 |
|   | 2 hrs | Neurological disorder Pharmacotherapy: Headache disorders                                   | Reference No. 1, 3 |
|   | 2 hrs | <i>Bed-side/ward attachment</i>                                                             |                    |
| 5 | 2 hrs | Neurological disorder Pharmacotherapy: Epilepsy                                             | Reference No. 1, 3 |
|   | 2 hrs | Neurological disorder Pharmacotherapy: Epilepsy                                             |                    |
|   | 2 hrs | <i>Bed-side/ward attachment</i>                                                             |                    |
| 6 | 2 hrs | Neurological disorder Pharmacotherapy: status epileptics                                    | Reference No. 1, 3 |
|   | 2 hrs | Neurological disorder Pharmacotherapy: Parkinsonism                                         | Reference No. 1, 3 |
|   | 2 hrs | <i>Bed-side/ward attachment</i>                                                             |                    |
| 7 | 2 hrs | Neurological disorder Pharmacotherapy: Alzheimer's disease                                  | Reference No. 1, 3 |
|   | 2 hrs | Neurological disorder Pharmacotherapy: Acute management of brain injury, Multiple Sclerosis | Reference No. 1, 3 |
|   | 2 hrs | <i>Bed-side/ward attachment</i>                                                             |                    |
| 7 | 2 hrs | Case studies on Seizure and epilepsy, HA, pain management)                                  |                    |
|   | 2 hrs | 50% continuous assessment report                                                            |                    |
|   | 2 hrs | <i>Bed-side/ward attachment</i>                                                             |                    |
| 8 | 2 hrs | Musculoskeletal disorders Pharmacotherapy: Osteoporosis/osteomalacia                        |                    |
|   | 2 hrs | Musculoskeletal disorders Pharmacotherapy: Osteoarthritis                                   |                    |

|    |       |                                                                                                                                  |                    |
|----|-------|----------------------------------------------------------------------------------------------------------------------------------|--------------------|
|    | 2 hrs | <i>Bed-side/ward attachment</i>                                                                                                  |                    |
| 9  | 2 hrs | Musculoskeletal disorders Pharmacotherapy: Rheumatoid Arthritis                                                                  |                    |
|    | 2 hrs | Musculoskeletal disorders Pharmacotherapy: Gout and Hyperuricemia                                                                |                    |
|    | 2 hrs | <i>Bed-side/ward attachment</i>                                                                                                  |                    |
| 10 | 2 hrs | Gynecologic and obstetric disorders Pharmacotherapy: Pregnancy associated diseases (eclampsia/pre-eclampsia)                     | Reference No. 1, 3 |
|    | 2 hrs | Gynecologic and obstetric disorders Pharmacotherapy: Pregnancy associated diseases (GDM)                                         |                    |
|    | 2 hrs | <i>Bed-side/ward attachment</i>                                                                                                  |                    |
| 11 | 2 hrs | Gynecologic and obstetric disorders Pharmacotherapy: Pregnancy associated diseases (Nausea and vomiting, hyperemesis gravidarum) |                    |
|    | 2 hrs | Gynecologic and obstetric disorders Pharmacotherapy: Contraception                                                               |                    |
|    | 2 hrs | <i>Bed-side/ward attachment</i>                                                                                                  |                    |
| 12 | 2 hrs | Gynecologic and obstetric disorders Pharmacotherapy: Contraception                                                               |                    |
|    | 2 hrs | Gynecologic and obstetric disorders Pharmacotherapy: Menstruation-related disorders                                              |                    |
|    | 2 hrs | <i>Bed-side/ward attachment</i>                                                                                                  |                    |
| 13 | 2 hrs | Gynecologic and obstetric disorders Pharmacotherapy: Hormone therapy in women                                                    |                    |
|    | 2 hrs | Gynecologic and obstetric disorders Pharmacotherapy: Female infertility                                                          |                    |
|    | 2 hrs | <i>Bed-side/ward attachment</i>                                                                                                  |                    |
| 14 | 2 hrs | Case studies on Gynecology and obstetrics disorders                                                                              |                    |
|    | 2 hrs | Urologic disorders Pharmacotherapy: Erectile Dysfunction                                                                         |                    |
|    | 2 hrs | <i>Bed-side/ward attachment</i>                                                                                                  |                    |
| 15 | 2 hrs | Urologic disorders Pharmacotherapy: BPH, Urinary incontinence                                                                    |                    |
|    | 2 hrs | Case studies on BPH and ED                                                                                                       |                    |
|    | 2 hrs | <i>Bed-side/ward attachment</i>                                                                                                  |                    |
| 16 | 2 hrs | Immunological Disorders Pharmacotherapy: Systemic Lupus Erythematosus and Other Collagen-Vascular diseases                       |                    |
|    | 2 hrs | Immunological Disorders Pharmacotherapy: Solid-Organ Transplantation                                                             |                    |
|    | 2 hrs | <i>Bed-side/ward attachment</i>                                                                                                  |                    |
| 20 | 2 hrs | Case studies on SLE, Solid organ transplantation                                                                                 |                    |
|    |       | Final exam                                                                                                                       |                    |

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| <b>Integrated Therapeutics IV Course syllabus</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |          |                 |          |                         |          |                  |          |                    |          |                    |          |                                   |          |
| <b>Course name:</b> Integrated Therapeutics IV                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |          |                 |          |                         |          |                  |          |                    |          |                    |          |                                   |          |
| Course code: Phar4222                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |          |                 |          |                         |          |                  |          |                    |          |                    |          |                                   |          |
| <b>Module name:</b> Pharmacotherapeutics module II                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |          |                 |          |                         |          |                  |          |                    |          |                    |          |                                   |          |
| <b>Module code:</b> Phar-M4222                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |          |                 |          |                         |          |                  |          |                    |          |                    |          |                                   |          |
| <b>Course ECTS:</b> 7 ECTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |          |                 |          |                         |          |                  |          |                    |          |                    |          |                                   |          |
| <p><b>Totally required hours for the module:</b> 7x27 = 189 hours</p> <table> <tr> <td><b>Lecture:</b></td><td>60 hours</td></tr> <tr> <td><b>Ward attachment:</b></td><td>30 hours</td></tr> <tr> <td><b>Tutorial:</b></td><td>34 hours</td></tr> <tr> <td><b>Home study:</b></td><td>46 hours</td></tr> <tr> <td><b>Assessment:</b></td><td>14 hours</td></tr> <tr> <td><b>Project work/presentation:</b></td><td>15 hours</td></tr> </table>                                                                                                                                                                                                                                                                                                                |          | <b>Lecture:</b> | 60 hours | <b>Ward attachment:</b> | 30 hours | <b>Tutorial:</b> | 34 hours | <b>Home study:</b> | 46 hours | <b>Assessment:</b> | 14 hours | <b>Project work/presentation:</b> | 15 hours |
| <b>Lecture:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 60 hours |                 |          |                         |          |                  |          |                    |          |                    |          |                                   |          |
| <b>Ward attachment:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 30 hours |                 |          |                         |          |                  |          |                    |          |                    |          |                                   |          |
| <b>Tutorial:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 34 hours |                 |          |                         |          |                  |          |                    |          |                    |          |                                   |          |
| <b>Home study:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 46 hours |                 |          |                         |          |                  |          |                    |          |                    |          |                                   |          |
| <b>Assessment:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 14 hours |                 |          |                         |          |                  |          |                    |          |                    |          |                                   |          |
| <b>Project work/presentation:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 15 hours |                 |          |                         |          |                  |          |                    |          |                    |          |                                   |          |
| <b>Year/Semester Course is offered:</b> Year IV/Semester II                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |          |                 |          |                         |          |                  |          |                    |          |                    |          |                                   |          |
| <b>Pre-requisite if any:</b> Integrated Therapeutics I                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |          |                 |          |                         |          |                  |          |                    |          |                    |          |                                   |          |
| <b>Course description:</b> <p>This course is a continuation of integrated therapeutics III. It is designed to prepare graduate pharmacy students to manage a number of common diseases, including infectious,, oncologic and hematologic, and nutritional disorders. It also prepares students to develop rational drug therapy plans, identify conditions for monitoring pharmacotherapy, and conditions that require referral.</p>                                                                                                                                                                                                                                                                                                                           |          |                 |          |                         |          |                  |          |                    |          |                    |          |                                   |          |
| <b>Course objectives:</b> <p>After completion of this course, the student will be able to describe, analyze and identify various infectious, oncologic, hematologic, and nutritional disorders; and manage drug therapy.</p> <p>Specific Objectives</p> <p>To meet this objective, students will:</p> <ul style="list-style-type: none"> <li>• Describe the pathophysiologic processes underlying infectious diseases, oncologic, immunologic, nutritional disorders.</li> <li>• Analyze and interpret diagnostic findings relevant to infectious, oncologic, hematologic, and nutritional disorders.</li> <li>• Recommend appropriate treatment regimen for patients suffering from infectious, oncologic, hematologic, and nutritional disorders.</li> </ul> |          |                 |          |                         |          |                  |          |                    |          |                    |          |                                   |          |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |            |     |                          |     |             |     |                         |     |               |    |              |     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----|--------------------------|-----|-------------|-----|-------------------------|-----|---------------|----|--------------|-----|
| <ul style="list-style-type: none"><li>• Provide follow up and monitor outcomes in patients who have infectious, oncologic, hematologic, and nutritional disorders</li><li>• Perform research and activities in pharmacotherapy of infectious, oncologic, hematologic, and nutritional disorders.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                           |            |     |                          |     |             |     |                         |     |               |    |              |     |
| <b>Course mode of delivery:</b> Parallel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |            |     |                          |     |             |     |                         |     |               |    |              |     |
| <b>Course learning and teaching methods</b> <p>During this course the following mode of teaching can be used:</p> <ul style="list-style-type: none"><li>• Illustrated lectures and group discussions</li><li>• Individual and group exercise and assignments</li><li>• Role plays and case studies</li><li>• Simulation</li><li>• Audiovisuals</li><li>• Clinical scenarios</li><li>• Tutorials</li><li>• Demonstration</li></ul>                                                                                                                                                                                                                                                                                                                                    |            |     |                          |     |             |     |                         |     |               |    |              |     |
| <b>Assessment mechanisms:</b> <table><tr><td>• Quizzes:</td><td>10%</td></tr><tr><td>• Seminar presentations:</td><td>10%</td></tr><tr><td>• Mid Exam:</td><td>20%</td></tr><tr><td>• Bedside presentation:</td><td>15%</td></tr><tr><td>• Assignments</td><td>5%</td></tr><tr><td>• Final Exam</td><td>40%</td></tr></table>                                                                                                                                                                                                                                                                                                                                                                                                                                        | • Quizzes: | 10% | • Seminar presentations: | 10% | • Mid Exam: | 20% | • Bedside presentation: | 15% | • Assignments | 5% | • Final Exam | 40% |
| • Quizzes:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 10%        |     |                          |     |             |     |                         |     |               |    |              |     |
| • Seminar presentations:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 10%        |     |                          |     |             |     |                         |     |               |    |              |     |
| • Mid Exam:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 20%        |     |                          |     |             |     |                         |     |               |    |              |     |
| • Bedside presentation:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 15%        |     |                          |     |             |     |                         |     |               |    |              |     |
| • Assignments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 5%         |     |                          |     |             |     |                         |     |               |    |              |     |
| • Final Exam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 40%        |     |                          |     |             |     |                         |     |               |    |              |     |
| <b>Teachers’ and students’ role</b> <p><b>Roles of Instructors:</b></p> <p>Instructors should be well versed in the outcomes expected of students and the pedagogical methods that best enhance learning. Accordingly, they will be expected to:</p> <ul style="list-style-type: none"><li>▪ Be a facilitator (introduce the subject; give guidance, moderate discussions, etc.)</li><li>▪ Closely supervise in actual practice in different clinical and community settings</li><li>▪ Assists students in resolving ever-increasing ethical dilemma in the service delivery</li><li>▪ Foster and assess self-initiated student learning</li><li>▪ Read and comment assignments of students on time</li><li>▪ Prepare his/her lessons and deliver lectures</li></ul> |            |     |                          |     |             |     |                         |     |               |    |              |     |

- Provide available and necessary reference materials
- Encourage active participation of students in the teaching learning process;
- Strive to be the role model ethically and professionally

### **Roles of Students**

Students are expected to:

- Make every effort to apply valid research findings and theoretical teachings into practice in various practice settings
- Engage in learning by doing independent study, project work; group work, etc..
- Be active learners (participate effectively in group assignments, perform practice actively, make presentations, write reports, prepare seminars etc.);

Critically assess laws, regulations, journal articles and related topics from different sources

### **References:**

#### **Required readings (Text)**

1. Dipiro JT, Talbert RL, Yee GC, et.al. Pharmacotherapy, a Pathophysiologic Approach. 10<sup>th</sup> or later edition.

#### **Recommended readings**

2. Koda - Kimble MA, Young LY , Kradjan WA, et.al. Applied Therapeutics, The Clinical Use of Drugs. 11<sup>th</sup> or later edition.
3. Walker R and Edwards C. Clinical Pharmacy and Therapeutics. 3rd or later edition.
4. Kasper, Braunwald, et al. Harrison's Principles of Internal Medicine, 20<sup>th</sup> or later edition
5. Tierney, McPhee, Papadakis. Current Medical Diagnosis and Treatment 2020 or later edition
6. Conn's Current therapy 2020
7. Washington Manual of Medical Therapeutics 32<sup>nd</sup> edition
8. Jacobs & DeMott Laboratory Test Handbook, 5th edition
9. Handouts including copies of PowerPoint slides from lectures

Guidelines and articles as specified by the instructor

### **Course Schedule:**

| Week | Contact hour | Topic/sub-topic/chapter/Assessments/Assignments                                    | Reading Material      |
|------|--------------|------------------------------------------------------------------------------------|-----------------------|
| 1    | 2 hrs        | Infectious diseases Pharmacotherapy: Principles of antimicrobial regimen selection | Reference No. 1, 3, 6 |
|      | 2 hrs        | Infectious diseases Pharmacotherapy: Upper respiratory tract infections            |                       |
|      | 2 hrs        | <i>Bed-side/ward attachment</i>                                                    |                       |

|   |       |                                                                                               |                   |
|---|-------|-----------------------------------------------------------------------------------------------|-------------------|
| 1 | 2 hrs | Infectious diseases Pharmacotherapy: Lower respiratory tract infections in adults             |                   |
|   | 2 hrs | Infectious diseases Pharmacotherapy: lower respiratory tract infections in pediatrics         |                   |
|   | 2 hrs | <i>Bed-side/ward attachment</i>                                                               |                   |
| 2 | 2 hrs | Infectious diseases Pharmacotherapy: Infective Endocarditis                                   |                   |
|   | 2 hrs | Infectious diseases Pharmacotherapy: Skin and soft tissue infections                          |                   |
|   | 2 hrs | <i>Bed-side/ward attachment</i>                                                               |                   |
| 3 | 2 hrs | Infectious diseases Pharmacotherapy: Urinary tract infections and Prostatitis                 |                   |
|   |       | Infectious diseases Pharmacotherapy: Gastrointestinal infections                              |                   |
|   | 2 hrs | Infectious diseases Pharmacotherapy: Intra-abdominal infections                               |                   |
| 4 | 2 hrs | <i>Bed-side/ward attachment</i>                                                               |                   |
|   | 2 hrs | Infectious diseases Pharmacotherapy: Parasitic infections; Osteomyelitis and Septic arthritis |                   |
|   | 2 hrs | Infectious diseases Pharmacotherapy: Central Nervous System infections                        |                   |
| 5 | 2 hrs | <i>Bed-side/ward attachment</i>                                                               |                   |
|   | 2 hrs | Infectious diseases Pharmacotherapy: Tuberculosis                                             |                   |
|   | 2 hrs | Infectious diseases Pharmacotherapy: Tuberculosis                                             |                   |
| 6 | 2 hrs | <i>Bed-side/ward attachment</i>                                                               |                   |
|   | 2 hrs | Infectious diseases Pharmacotherapy: Sepsis and septic shock, Neonatal Sepsis                 |                   |
|   | 2 hrs | Infectious diseases Pharmacotherapy: Neonatal Sepsis                                          |                   |
| 6 | 2 hrs | <i>Bed-side/ward attachment</i>                                                               |                   |
|   | 2 hrs | Infectious diseases Pharmacotherapy: Sexually Transmitted Infections, Eye infections          |                   |
|   | 2 hrs | Infectious diseases Pharmacotherapy: Human Immunodeficiency Virus infection and OIs           |                   |
| 7 | 2 hrs | <i>Bed-side/ward attachment</i>                                                               |                   |
|   | 2 hrs | Infectious diseases Pharmacotherapy: Human Immunodeficiency Virus infection and OIs           |                   |
|   | 2 hrs | Infectious diseases Pharmacotherapy: Superficial fungal infections                            |                   |
| 8 | 2 hrs | <i>Bed-side/ward attachment</i>                                                               |                   |
|   | 2 hrs | Infectious diseases Pharmacotherapy: Invasive fungal infections                               |                   |
|   | 2 hrs | Infectious diseases Pharmacotherapy: Surgical antibiotic prophylaxis                          |                   |
| 8 | 2 hrs | <i>Bed-side/ward attachment</i>                                                               |                   |
|   | 2 hrs | Case Studies on pneumonia, meningitis, infective endocarditis, STI                            |                   |
|   | 2 hrs | Case studies on tuberculosis, HIV/AIDS, neonatal sepsis and Intra-abdominal infections        |                   |
| 9 | 2 hrs | <i>Bed-side/ward attachment</i>                                                               |                   |
|   | 2 hrs | 50% continuous assessment report                                                              |                   |
|   | 2 hrs | Hematological disorders Pharmacotherapy: Anemia                                               | Reference No. 1,3 |

|    |       |                                                                                                              |                                                                                  |
|----|-------|--------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| 10 | 2 hrs | Hematological disorders Pharmacotherapy: Coagulation disorders                                               | Reference<br>No. 1, 2<br>Reference<br>No. 1, 4<br>Reference<br>No. 1, 2, 3,<br>4 |
|    | 2 hrs | Hematological disorders Pharmacotherapy: Sickle Cell disease                                                 |                                                                                  |
|    |       | Hematological disorders Pharmacotherapy: Drug induced Hematological disorders                                |                                                                                  |
|    | 2 hrs | <i>Bed-side/ward attachment</i>                                                                              |                                                                                  |
| 11 | 2 hrs | Case study on anemia, coagulation disorders                                                                  |                                                                                  |
|    | 2 hrs | Oncologic disorders Pharmacotherapy: Principles of cancer treatment and chemotherapy                         |                                                                                  |
|    | 2 hrs | <i>Bed-side/ward attachment</i>                                                                              |                                                                                  |
| 12 | 2 hrs | Oncologic disorders Pharmacotherapy: Breast cancer                                                           |                                                                                  |
|    | 2 hrs | Oncologic disorders Pharmacotherapy: Lung cancer                                                             |                                                                                  |
|    | 2 hrs | <i>Bed-side/ward attachment</i>                                                                              |                                                                                  |
| 13 | 2 hrs | Oncologic disorders Pharmacotherapy: GI malignancies (Major focus on colorectal cancer)                      |                                                                                  |
|    | 2 hrs | Oncologic disorders Pharmacotherapy: Head and neck cancer, prostate cancer                                   |                                                                                  |
|    | 2 hrs | <i>Bed-side/ward attachment</i>                                                                              |                                                                                  |
| 14 | 2 hrs | Oncologic disorders Pharmacotherapy: Gynecological malignancies (major focus on Ovarian and cervical cancer) |                                                                                  |
|    | 2 hrs | Hematologic malignancies Pharmacotherapy: Lymphomas and Multiple myeloma                                     |                                                                                  |
|    | 2 hrs | <i>Bed-side/ward attachment</i>                                                                              |                                                                                  |
| 15 | 2 hrs | Hematologic malignancies Pharmacotherapy: Acute Leukemias; Chronic Leukemias                                 |                                                                                  |
|    | 2 hrs | Pharmacotherapy of oncologic emergencies                                                                     |                                                                                  |
|    | 2 hrs | <i>Bed-side/ward attachment</i>                                                                              |                                                                                  |
| 16 | 2 hrs | Case studies on Breast CA, Colorectal Ca, Cervical Ca, Hematologic malignancies                              |                                                                                  |
|    | 2 hrs | Nutritional disorders Pharmacotherapy: Moderate acute malnutrition and severe acute malnutrition             |                                                                                  |
|    | 2 hrs | <i>Bed-side/ward attachment</i>                                                                              |                                                                                  |
|    | 2 hrs | Nutritional disorders Pharmacotherapy: Obesity; Nutritional considerations in major organ failure            |                                                                                  |
|    |       | Final exam                                                                                                   |                                                                                  |

## **Pharmacognosy and Alternative Medicine module II**

**Module Name: Pharmacognosy and Alternative Medicine module**

**Module Category: Core**

**Module Code: Phar-M4231**

**Module Number: 23**

**Module Weight: 3 ECTS**

**Courses:**

| <b>Course name</b>                     | <b>Course code</b> | <b>ECTS</b> |
|----------------------------------------|--------------------|-------------|
| Complementary and Alternative Medicine | (Phar4231)         | (3 ECTS)    |

**Module description:** The module studies various alternative and complementary medicine practices including the Ethiopian traditional medicine.

**Module objective:** At the end of this module students will explain different forms of complementary & alternative medicines.

**Module competencies:**

- Describe & compare the role of various forms of complementary & alternative medicines in primary health care service
- Display rational usage of natural products (as drugs, foods, alternative medicines)
- Ready to provide service that ensure rational usage of natural products

**Mode of delivery (Parallel/Block): Parallel**

- Total study hour: 81 hours
- Illustrated Lecture: 32 hours
- Tutorial: 18 hours
- Seminars, assignments and presentation: 5 hours
- Assessment (continuous & final): 5 hours
- Independent study (alone or in groups): 11 hours
- Field visit = 10 hours

**Module teaching/learning method:**

Learning Activities:

- Attend lectures and demonstrations, take notes, and ask questions



- Engage in learning by doing (independent study, group assignments, presentation, report writing, and etc.)
- Participation and discussions

#### Teaching Methods

- The course facilitator is expected to lecture for introducing concepts and topics, and give references, facilitate discussions, ask questions, give and correct assessments, guide practical sessions, correct and give feedbacks of reports of practical sessions.
- Arrange and facilitate seminar sessions, discussions and give comments and feedbacks.
- Field visit and visiting traditional practitioners

#### **Module mode of assessment:**

- Seminars and assignments
- Quizzes
- Final exam

## **Course syllabus**

|                                               |                                           |
|-----------------------------------------------|-------------------------------------------|
| <b>Course Name:</b>                           | Complementary and Alternative Medicine    |
| <b>Course code:</b>                           | Phar 4231                                 |
| <b>Module Name:</b>                           | Pharmacognosy and Alternative Medicine II |
| <b>Module Code:</b>                           | <b>Phar-M4231</b>                         |
| <b>Course ECTS:</b>                           | 3 ECTS                                    |
| <b>Totally required hours for the module:</b> | 81 hrs                                    |
| <b>Year/Semester Course is offered:</b>       | Year IV Semester I                        |

## **Course prerequisite/s: Pharmacognosy**

### **Course description:**

The course is designed in such a way that the trainee gets well acquainted with the various alternative and complementary medicine practices including the Ethiopian traditional medicine. In addition, the trainee will be able to understand and make use of complementary medicine in primary health care.

### **Course objective:**

After completion of this course students will be able to:

- Familiarize with the different forms of complementary medicine and also to be well oriented about the use of complementary medicine in primary health care. Furthermore, the trainee is expected to apply the knowledge gained in this course in providing health education about the potential benefits and risk associated with a given herb, herb-drug interaction to other health care professionals and patients, monitoring, and reporting of adverse drug reaction associated with herbal medicine.

### **Course mode of delivery:** Block

### **Course learning and teaching methods**

- Illustrated Lecture: 32 hours
- Tutorial: 8 hours
- Seminars and assignments: 10 hours
- Field visit and visiting traditional practitioners: 10 hours

- Assessment : 8 hours
- Independent study hour: 13hours Assessment

#### **Assessment techniques:**

- Seminar/Assignments: 25%
- Continuous assessment = 30%,
- Quizzes = 5%)
- Final Exam: 40%

#### **Teachers' and students' role**

##### **Roles of Instructors**

- The instructor will be expected to:
- Be a facilitator (introduce the subject; give guidance, moderate discussions, etc.)
- Read and comment assignments of students on time;
- Prepare his/her lessons and deliver lectures;
- Provide available and necessary reference materials;
- Encourage active participation of students in the teaching learning process;
- Assist students with learning difficulties and
- Arrange and follow up practical sessions
- Roles of Students

##### **Students are expected to:**

- Engage in learning by doing (independent study, project work; group work, etc.)
- Be active learners (participate effectively in laboratory activities, in group assignments, make presentations, write reports, etc.);

#### **References:**

1. WHO, Traditional Medicine and Health Care Coverage, WHO, Geneva, 1983
2. Steven B Kayne. Complementary and Alternative Medicine. Pharmaceutical Press 2009.

## Course schedule

| Week   | Contact Hours | Topic/sub-topic/chapter/Assessments/Assignments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1      | 2hrs          | 1. Introduction<br>1.1. Definition of Terms<br>- Traditional medicine, Ethnomedicine, Allopathy, CAM etc.<br>1.2. Traditional Medicine versus modern medicine<br>1.3. Traditional medicine policy (national and global perspectives)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 2 -3   | 4hrs          | 2. Ethiopian Traditional Medicine<br>2.1. Historical background of Ethiopian Traditional Medicine<br>2.2. Principles<br>2.3. Methods of practice and practices<br>2.4. Documentation<br>2.5. Clinical trials/ scientific investigations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 4      | 3hrs          | 3. African Traditional Medicine<br>3.1. Historical development of African Traditional Medicine<br>3.2. Principles<br>3.3. Methods and areas of practice<br>3.4. Practitioners<br>3.5. Documentation<br>3.6. Clinical trials                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 5 - 14 | 21hrs         | 4. Some popular complementary and alternative medical practices<br>4.1. Medical herbalism (4 hrs)<br>Herbal Medicine, herbs as therapeutic agents, various ways of preparing herbs as therapeutics in traditional medicine, areas of practice, significant herbs used in CAM (including some commonly used herbal medicines in Ethiopia, Moringa stenopetala. Ocimum lamifolium and other Ocimum species, garlic, fenugreek seed, ginger, etc), quality control and standardization of herbal medicine,safety and pharmacovigilance of herbal medicines (herbavigilance)<br>4.2. Aromatherapy (2 hrs)<br>Definition, Historical background, Principles and laws of cure, Areas of practice, Methods and materials used in the healing process<br>4.3. Nutraceuticals (1 hrs)<br>4.4. Homeopathy (4 hrs)<br>Definition, Historical background, Principles and laws of cure, Areas of practice, Methods and materials used in the healing process, Practitioners, Clinical trials/current scientific investigations<br>4.5. Traditional Chinese Medicine (3 hrs) |

|    |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |      | <p>Definition, historical development, principles, examples of Chinese Traditional Medicine, Acupuncture, herbal medicine in China</p> <p>4.6. Traditional Indian Medicine (3 hrs)</p> <p>Definition, Historical development, Examples of Indian Complementary Medicine,</p> <p>4.6.1. Ayurveda</p> <p>Principles of Ayurvedic medicine, Methods and materials used, Areas of practice, Clinical trials, Current scientific investigations</p> <p>4.6.2. Unani</p> <p>Principles of Unani, Methods and materials used in the healing process, Areas of practice, Scientific investigations,</p> <p>4.7. Oriental Medicine (2 hrs)</p> <p>Definition of terms and introduction, Historical development, Principles of Oriental Medicine, Methods and materials used for diagnosis/healing etc., Regions of practice, Scientific investigations</p> <p>4.8. Naturopathy (2 hrs)</p> <p>Definition of terms and introduction, Historical development, Principles of Naturopathy, Methods and materials used for healing/diagnosis etc., Regions of practice, Scientific investigations</p> |
| 15 | 2hrs | <p>5. Traditional medicine and primary health care (PHC)</p> <p>5.1. Brief overview of PHC.</p> <p>5.2. Methods of using traditional medicine in PHC.</p> <p>5.3. Training the practitioners.</p> <p>5.4. Advantages of using traditional medicine in PHC.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

## **Module 24: Pharmacy Practice Module**

**Module Name:** Pharmacy Practice Module

**Module category:** core

**Module Number:** 24

**Module Code:** Phar-M4241

**Module Weight in ECTS:** 22 ECTS

### **Courses**

| <b>Course name</b>                   | <b>Course code</b> | <b>ECTS</b> |
|--------------------------------------|--------------------|-------------|
| Drug Informatics                     | Phar4241           | 3 ECTS      |
| Communication skills for pharmacists | Phar4172           | 3 ECTS      |
| Pharmacy Law & Ethics                | Phar4173           | 3 ECTS      |
| Pharmacy Practice                    | Phar4244           | 7 ECTS      |
| First Aid                            | Nurs4245           | 3 ECTS      |
| Nutrition                            | Comh4246           | 3 ECTS      |

### **Module description:**

This module focuses on the fundamentals of Drug Informatics, Communication skills for pharmacists, Pharmacy Law & Ethics, The regulatory and routine aspects of Pharmacy Practice, First Aid and Nutrition as it applies for pharmacy practice

### **Module objective:**

After completion of this module students will be able to:

- Provide appropriate drug information for queires
- Describe the importance and strategies of effective communication skills in meeting pharmacists' professional responsibilities.
- Discuss the legal and ethical principles which are applied in pharmacy practices and develop responsible attitude.
- Describe the main areas of pharmacy practice and pharmacists' roles in each setting
- Provide adequate counselling on nutritrional issues for the purpose of improving outcomes of care for patients and preventing disease in the general public
- Provide basic first aid when the need arises

**Module competencies:**

By the end of this module, students should be able to:

- Develop effective communication skill with patients, physicians and other healthcare professionals
- Discuss the legal and ethical principles that apply in pharmacy practices and develop responsible attitude.
- Resolve ever-increasing ethical dilemma in the service delivery
- Provide appropriate drug information for queires
- Provide adequate counselling on nutritrional issues for the purpose of improving outcomes of care for patients and preventing disease in the general public
- Provide basic first aid when the need arises

**Mode of delivery:** Parallel

**Module teaching/learning method:**

Totally required hours for the module:  $22 \times 27 = 594$  hrs

**Mode of Assessment:**

The assessment criteria are based on continuous assessment of class activities, individual and group assignment, case presentations, role play, journal club presentations, practical attachments, report writing, tests/quizzes and final exams. This in turn can be broken down into;

|                                            |             |
|--------------------------------------------|-------------|
| Group assignments and presentations        | 30%         |
| Evaluations of onsite practice performance | 20%         |
| Tests/quizzes                              | 20%         |
| Final Exam                                 | 30%         |
| <b>Total</b>                               | <b>100%</b> |

**Module learning and teaching methods****A. Learning Activities**

Students are supposed to involve in the following major learning activities:

- Learning by doing independent study, practices, group assignments, presentation, report writing, preparing seminars, resolving ethical dilemma etc...

- Participating actively in class lectures;
- Critical Analysis, summarization and presentations of journal articles and relevant documents

#### **B. Teaching Methods**

- The course instructor is expected to lecture for introducing concepts and topics, and give references, facilitate discussions, ask questions, give and correct assessments, forming groups, encouraging students for peer learning
- Arrange and facilitate seminar sessions prepared by students



## **Drug Informatics Syllabus**

**Course name:** Drug Informatics

**Course code:** Phar 4241

**Module name:** Pharmacy Practice Module

**Module code:** Phar-M4241

Course ECTS: 3

**Totally required hours for the module:** 3 x 27 =81

The distribution of these hours will be as follows

- |                              |          |
|------------------------------|----------|
| ▪ Lecture:                   | 20 hours |
| ▪ Home study:                | 22 hours |
| ▪ Tutorial:                  | 8 hours  |
| ▪ Project work/presentation: | 8 hours  |
| ▪ Assessment:                | 8 hours  |

**Year/Semester Course is offered:** Year IV/Semester I

**Pre-requisite:** Computer literacy

### **Course Description:**

This course is designed to provide pharmacy students with an overview of drug information resources used in healthcare system. Students will learn the advantages and disadvantages of primary, secondary, and tertiary literatures and will also gain experience of extracting information from these types of literature. The students will learn how to evaluate the biomedical literature using a systematic approach and will assess the statistical analyses reported to determine whether the interpretation and conclusions are valid. Students will also have a hands-on training at the Drug Information Center, SOP on the various computer based drug information resources.

### **Course objectives:**

After completion of this course students will be able to:

- ❖ Rapidly locate and evaluate drug information sources
- ❖ Systematically manage and communicate drug information
- ❖ Apply drug information knowledge for preparation and management of formularies, guidelines and bulletins.
- ❖ Provide drug information to health care professionals and patients on the rational use of drugs.
- ❖ Compare and contrast online resources to printed resources.
- ❖ Differentiate between primary, secondary and tertiary literature.

**Course mode of delivery:** Parallel

**Module teaching/learning method:**

### **Learning Activities**

Students are supposed to involve in the following major learning activities:

- Learning by doing independent study, practices, group assignments, presentation, report writing, preparing seminars, resolving ethical dilemma etc...
- Participating actively in class lectures;
- Critical Analysis, summarization and presentations of journal articles and relevant documents

### **Teaching Methods**

- The course instructor is expected to lecture for introducing concepts and topics, and give references, facilitate discussions, ask questions, give and correct assessments, forming groups, encouraging students for peer learning
- Arrange and facilitate seminar sessions prepared by students

### **Assessment techniques:**

The assessment criteria are based on continuous assessment of class activities, individual and group assignment, case presentations, role play, journal club presentations, practical attachments, report writing, tests/quizzes and final exams. This in turn can be broken down into;

Assignments (monograph development, critical appraisal & drug information response) and presentations 30%

Evaluations of onsite practice performance 20%

Tests/quizzes 20%

Final Exam 30%

Total 100%

### **Teachers' and students' role**

#### **Roles of Instructors**

Instructors should be well versed in the outcomes expected of students and the pedagogical methods that best enhance learning. Accordingly, they will be expected to:

- Be a facilitator (introduce the subject; give guidance, moderate discussions, etc.)
- Closely supervise in actual practice in different clinical and community settings
- Assists students in resolving ever-increasing ethical dilemma in the service delivery
- Foster and assess self-initiated student learning
- Read and comment assignments of students on time;
- Prepare his/her lessons and deliver lectures;
- Provide available and necessary reference materials

- Encourage active participation of students in the teaching learning process;
- Strive to be the role model ethically and professionally

### **Roles of Students**

Students are expected to:

- Make every effort to apply valid research findings and theoretical teachings into practice in various practice settings
- Engage in learning by doing independent study, project work; group work, etc..
- Be active learners (participate effectively in group assignments, perform practice actively, make presentations, write reports, prepare seminars etc.);
- Critically assess laws, regulations, journal articles and related topics from different sources

### **References:**

#### **Required readings (Text)**

1. Malone PM et al. Drug Information: A Guide for pharmacists, 6<sup>th</sup> edition, McGraw-Hill Companies, Inc., 2018 USA.

#### **Recommended readings**

2. Online Drug Information Databases
3. American Hospital Formulary Service: Drug Information
4. Drug Facts and Comparison
5. Slaughter RL, Edwards D. Evaluating Drug Literature: A Statistical Approach, McGraw-Hill Companies, Inc., 2001, USA.
6. MSH/WHO. Managing Drug Supply, Kumarian Press, 1997, USA.

### **Course schedule**

| Week | Contact<br>1 hr | Topic/sub-topic/chapter/Assessments/Assignments                                                                                                         | Reading<br>Material  |
|------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| 1    | 1 hr            | 1. <b>Introduction to the concept of drug Information</b>                                                                                               | Reference No.<br>1,2 |
|      |                 | 1.1. Definitions of basic terms<br>1.2. The evolution of DI                                                                                             |                      |
| 1    | 1 hr            | 1.3. Medication information services and skills<br>1.4. Factors influencing the evolution of the pharmacist's role as a medication information provider |                      |

|   |      |                                                                                                                                                                                                                                             |                       |
|---|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
|   |      | 1.5.Opportunities in specialty practice                                                                                                                                                                                                     |                       |
| 2 | 1 hr | <b>2. Types of Drug Information Resources</b>                                                                                                                                                                                               | 1, 2                  |
|   |      | 2.1.Primary sources                                                                                                                                                                                                                         |                       |
| 2 | 1 hr | 2.2.Secondary sources                                                                                                                                                                                                                       |                       |
|   |      | 2.3.Tertiary sources                                                                                                                                                                                                                        |                       |
| 3 | 1 hr | 2.4.Other internet-based source                                                                                                                                                                                                             |                       |
|   |      | 2.5.Core Drug Information Resources                                                                                                                                                                                                         |                       |
| 3 | 1 hr | 2.6.Specialized Drug Information Sources                                                                                                                                                                                                    |                       |
|   |      | 2.7.Evaluation of drug information sources                                                                                                                                                                                                  |                       |
| 4 | 1 hr | <b>3. Systematic approach to receiving and answering questions on drugs</b><br>2.1.Steps in the modified systematic approach:<br><ul style="list-style-type: none"> <li>• Requestor Demographics</li> <li>• Background Questions</li> </ul> | Reference No.<br>1, 2 |
| 4 | 1 hr | <ul style="list-style-type: none"> <li>• Ultimate Question/Categorization of Question</li> <li>• Search Strategy</li> </ul>                                                                                                                 |                       |
| 5 | 1 hr | <ul style="list-style-type: none"> <li>• Data Evaluation, Analysis, and Synthesis</li> <li>• Formulation and Provision of Response</li> </ul>                                                                                               |                       |
| 6 | 1 hr | <ul style="list-style-type: none"> <li>• Follow-Up, Follow-Through, and Documentation</li> </ul>                                                                                                                                            |                       |
| 7 | 1 hr | <b>6. Introduction to literature evaluation</b><br>6.1. Controlled Clinical Trial                                                                                                                                                           | Reference No.<br>1, 2 |
|   |      | 6.2.Observational studies                                                                                                                                                                                                                   |                       |
| 7 | 1 hr | 6.3.Review articles evaluation                                                                                                                                                                                                              | Reference No.<br>1, 2 |
|   |      | <b>7. Evidence based clinical practice guidelines</b>                                                                                                                                                                                       |                       |
| 8 | 1 hr | <b>8. Ethical/Legal Issues in Drug Information</b><br>8.1. What Is Ethics and What Is Not                                                                                                                                                   | Reference No.<br>1, 2 |
|   |      | 8.2.Ethical Dilemmas in Pharmacy Practice                                                                                                                                                                                                   |                       |
| 8 | 1 hr | 8.3.Basics of Ethics Analysis                                                                                                                                                                                                               |                       |
|   |      | 8.4.Labeling and Advertising                                                                                                                                                                                                                |                       |
| 9 | 1 hr | 8.5.Liability Concerns for Internet Information                                                                                                                                                                                             |                       |

|                   |      |                                                                                   |                       |
|-------------------|------|-----------------------------------------------------------------------------------|-----------------------|
|                   |      | 8.6.Intellectual Property Rights                                                  |                       |
| 10                | 1 hr | <b>9. Professional writing</b>                                                    | Reference No.         |
| 11                | 1 hr | <b>10. Drugs &amp; Therapeutics Committee</b>                                     | 1, 2                  |
|                   |      | 10.1. Introduction                                                                |                       |
|                   |      | 10.2. Organizational Background                                                   |                       |
|                   |      | 10.3. Clinical Guidelines                                                         |                       |
|                   |      | 10.4. Standard Order Set Development                                              |                       |
|                   |      | 10.5. Communication within an Organization                                        |                       |
| 12                | 1 hr | <b>11. Formulary management</b>                                                   | Reference No.<br>1, 2 |
| 13                | 1 hr | <b>12. Drug evaluation monographs</b>                                             | Reference No.<br>1, 2 |
|                   |      | <b>13. Medication misadventures:</b> Adverse drug reactions and medication errors | Reference No.<br>1, 2 |
| <b>Final exam</b> |      |                                                                                   |                       |

## **Communication skills for Pharmacists course syllabus**

**Course title:** Communication skills for pharmacists

**Course code:** Phar 4242

**Module Name:** Pharmacy Practice Module

**Module Code:** Phar-M4241

**Course EtCTS:** 3ECTS

**Total required hours for the course:** 3 x 27 = 81

- Lecture: 32 hours
- Group work 10 hours
- Home study: 20 hours
- Tutorial: 10 hours
- Presentation: 10 hours
- Assessment: 8 hours

**Year/Semester Course is offered:** Year IV Semester I

**Pre-requisite if any:** None

### **Course Description:**

This course introduces students with the basic concepts of communications, establishing pharmacist-patient relationship, practical skills in communication with patients, collaborative working relationship with other healthcare professionals, conflict management, and written communication skills.

### **Course objectives:**

After completion of this course students will be able to:

- ❖ Identify the importance of communication skills in meeting pharmacists' patient care responsibilities.
- ❖ Demonstrate competency in the use of interpersonal communication skills of listening, interviewing, providing feedback, and relationship development.
- ❖ Apply appropriate communication strategies to address barriers and handle sensitive issues in interactions with patients and health care professionals.
- ❖ Analyze the impact of elements of written, verbal, and e-communication on the practitioner image.
- ❖ Collaborate with peers in developing effective interpersonal communication skills required of a pharmacist.

**Course mode of delivery:** Block

## **Course learning and teaching methods**

This course is thought using a variety of instructional methods including:

- Illustrated Lectures
- Active learning methods (brain storming, buzz group, discussion, etc),
- Individual and group exercises and assignments
- Presentations
- Role play

### **Assessment techniques:**

- |                 |     |
|-----------------|-----|
| • Quizzes       | 10% |
| • Assignment    | 10% |
| • Tests         | 20% |
| • Presentations | 20% |
| • Final Exam    | 40% |

## **Teachers' and students' role**

### **Roles of Instructors:**

Instructors should be well versed in the outcomes expected of students and the pedagogical methods that best enhance learning. Accordingly, they will be expected to:

- Be a facilitator (introduce the subject; give guidance, moderate discussions, etc.)
- Closely supervise in actual practice in different clinical and community settings
- Assists students in resolving ever-increasing ethical dilemma in the service delivery
- Foster and assess self-initiated student learning
- Read and comment assignments of students on time
- Prepare his/her lessons and deliver lectures
- Provide available and necessary reference materials
- Encourage active participation of students in the teaching learning process;
- Strive to be the role model ethically and professionally

### **Roles of Students**

Students are expected to:

- Make every effort to apply valid research findings and theoretical teachings into practice in various practice settings
- Engage in learning by doing independent study, project work; group work, etc..
- Be active learners (participate effectively in group assignments, perform practice actively, make presentations, write reports, prepare seminars etc.);

- Critically assess laws, regulations, journal articles and related topics from different sources

## References:

### Required readings (Text)

1. Beardsley, R. S., Kimberlin C., Tindall, W. N., Communication Skills in Pharmacy Practice: A Practical Guide for Students and Practitioners, 6th Ed., Lippincott Williams & Wilkins, Pennsylvania, 2012.

### Recommended readings

2. Motivational Interviewing in Health Care: Helping Patients Change Behavior (2007)
3. To Err is Human: Building a Safer Health System (2000) see link for reading below.
4. Whalley B. J., Fletcher K.E., Weston S.E., Howard R.L. and Rawlinson C.F., Foundation in Pharmacy Practice, Pharmaceutical Press, London, 2008.
5. Wiedenmayer K., Summers R.S., Mackie C.A., Gous A.G.S., Everard M. and Tromp D. Developing pharmacy practice, WHO/IPF, 2006.
6. Winfield, A. J. and Richards, R. M. E. (eds.), Pharmaceutical Practice, 4th ed., Churchill Livingstone, London, 2009.
7. Berger B. A., Communication Skills for Pharmacists: Building Relationships, Improving Patient Care, 3rd edition, American Pharmacists Association, Washington DC ,2009.
8. Cipolle Robert J., Strand Linda M., Morley Peter C., Pharmaceutical Care Practice 3rd ed., the McGraw-Hill Companies Inc., New York, 2012.

## Course schedule

| Week | Contact hour | Topic/sub-topic/chapter/Assessments/Assignments                       | Reference |
|------|--------------|-----------------------------------------------------------------------|-----------|
| 1    | 2            | 1. Patient-Centered Communication and Elements of Basic Communication | 1         |
|      |              | 1.1. What Is Communication?                                           |           |
| 2    | 2            | 1.2. Principles and Elements of Interpersonal Communication           |           |
|      |              | 1.3. Perception in Professional Communication                         |           |
| 2    | 2            | 1.4. Nonverbal Communication in Pharmacy                              |           |
|      |              | 1.5. Barriers in Communication                                        |           |
| 3    | 2            | 2. Establishing the pharmacist-patient relationship                   | 1         |



|    |   |                                                                                                 |        |
|----|---|-------------------------------------------------------------------------------------------------|--------|
|    |   | 2.1. Patient counseling                                                                         |        |
|    |   | 2.1. Patient counseling                                                                         |        |
| 4  | 2 | 2.2. Interviewing patients                                                                      |        |
|    |   | <i>Role play on patient counselling</i>                                                         |        |
| 4  | 2 | 2.3 Educating patients                                                                          |        |
| 5  | 2 | 2.4.Non-prescription medication consults                                                        |        |
| 6  | 2 | 2.5. Communication about medicines with special patients and children                           |        |
| 6  | 2 | 2.6. Ethical issues in patient counseling                                                       |        |
| 7  | 2 | 3. Practical Skills for Pharmacists                                                             | 1-4    |
|    |   | 3.1. Listening and Empathic Responding                                                          |        |
| 8  | 2 | 3.2. Helping Patients Manage Therapeutic Regimens and Communication Regarding medication safety |        |
|    |   | 3.3. Assertiveness                                                                              | 1      |
| 8  | 2 | 3.4. Interviewing and Assessment                                                                |        |
| 9  | 2 | 4. Interaction with other health professionals                                                  |        |
|    |   | 4.1. Communication with Physicians                                                              |        |
|    |   | 4.2.Communication with nurses                                                                   |        |
|    |   | 4.3.Communication with other pharmacists                                                        |        |
|    |   | <i>Role play: communication with physicians</i>                                                 |        |
| 10 | 2 | 4.4.Communications in Organizations                                                             |        |
|    |   | 4.5.Interviewing and being interviewed                                                          | 1, 4   |
|    |   | 4.6.Small Group Communications                                                                  |        |
| 11 | 2 | 4.7.Public Communication                                                                        |        |
| 12 | 2 | 5. Conflict management                                                                          | 1-3, 9 |
|    |   | 5.1. Managing conflict in organizations                                                         |        |
|    |   | <i>Case study on conflict management</i>                                                        |        |
|    |   | 6. Written communication skills                                                                 | 1-3, 9 |
|    |   | 6.1. Note taking and documentation practices                                                    |        |
| 13 | 2 | 6.2. Correspondences                                                                            |        |
|    |   | 6.3. Developing patient education materials, newsletters, etc                                   |        |

|    |   |                                            |  |
|----|---|--------------------------------------------|--|
| 14 | 2 | 6.4. Preparing a patient case presentation |  |
| 15 | 2 | 6.5. Preparing a journal club presentation |  |
|    |   | Final exam                                 |  |

## **Pharmacy Law & Ethics Course Syllabus**

**Course title:** Pharmacy Law & Ethics

**Course code:** Phar 4243

**Course ECTS:** 3 ECTS

**Course hours:** 3\*27 = 81

- Lecture: 20 hours
- Group work: 10 hours
- Presentations: 12 hours
- Tutorial: 14 hours
- Home study: 14 hours
- Assessment: 8 hours

Pre-requisite if any: - None

### **Course Description:**

The course offers topics on principles of ethical decision making, health professional patient relationship, frameworks for ethical analysis, ethical theories, ethical principles and moral values, and ethical codes, laws, regulations and directives pertaining to pharmaceutical services internationally and in Ethiopia, product registration and licensing requirements, regulations related to narcotic and psychotropic drugs. Case-study practice scenarios will also be presented to allow students make pharmaceutical care decisions based upon ethico-legal reasoning.

### **Course Objectives:**

Upon completion of the course, students will be able to:

- Explain the process of policy development and evaluation
- Identify laws governing the practice of pharmacy
- Describe the ethical principles in pharmacy practice of Ethiopia
- Identify standards and guidelines governing pharmacy practice in Ethiopia

Supporting objectives:

To meet this objective, student will be able to:

- Define pharmaceutical jurisprudence and Professional Ethics
- Discuss Laws Governing the practice of pharmacy
- Describe the Pharmaceutical legal system
- Describe Drug nomenclature and patents
- Explain about the requirements to practice pharmacy, and also to run health institutions involved in use and dispensing of pharmaceuticals

- Explain the Control of manufacture, import, export, whole sale, distribution, labeling and packaging, utilization, administration and quality assurance of pharmaceuticals
- Describe management and use of controlled substances, poisons and radiopharmaceuticals
- Identify prohibitions; legal procedures for offences and penalties for violation of Pharmaceutical laws and regulations
- Discuss the Code of ethics for pharmacists in Ethiopia

| Week | Conta<br>ct hrs | Topic/sub-topic/chapter                                                                                                                                                                                      | Reading<br>materials  |
|------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| 1    | 2               | Part 1: Ethics<br>1. What is ethics?<br>2. Ethical theories <ul style="list-style-type: none"> <li>• Teleological (consequentialist) and</li> <li>• deontological (non-consequentialist) theories</li> </ul> | Reference 1,2, 8 & 16 |
| 2    | 2               | 3. Ethical principles and moral rules<br>☞ Autonomy; informed consent; confidentiality; beneficence/nonmaleficence; fidelity; distributive justice.                                                          | Reference 1,2 & 8     |
| 3    | 2               | 4. Ethical issues in health care: Law and ethics; rationing; assisted suicide; human drug experimentation; drug formularies<br>5. Framework for ethical analysis                                             | Reference 1,2 & 16    |
| 4    | 2               | 5. Framework for ethical analysis .....                                                                                                                                                                      | Reference 1,2 & 16    |
| 5    | 2               | 6. Professional ethics                                                                                                                                                                                       | Reference 1,2 8 & 16  |
| 6    | 2               | 7. Ethical codes FIP standards of ethical practices; code of ethics for pharmacists practicing in Ethiopia                                                                                                   | Reference 1,2 8 & 16  |
| 7    | 2               | 8. Standards of practice for pharmacists practicing in Ethiopia                                                                                                                                              | Reference 8           |

|    |   |                                                                                                                                                                                                                                                                                                                                              |                                  |
|----|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| 8  | 2 | <p>9. The health professional–patient relationship: Consumerism versus paternalism; patients’ rights; moral rights versus legal rights to health care; health care practitioners’ duty to their patients.</p> <p>10. Introduction to Compassionate respectful and caring pharmacy professionals</p> <p>11. Importance of CRC to patients</p> | Reference 1,2 8 & 16             |
| 9  | 2 | <p>Part 2: Laws and Regulations</p> <p>1. Laws Governing the Practice of Pharmacy</p> <p>10.1 Food, Drugs and Cosmetics Act;</p> <p>10.2 Narcotic Drugs and Psychotropic Substances Act.</p>                                                                                                                                                 | Reference 4,7 & 17               |
| 10 | 2 | <p>10.3 Drug Abuse Prevention;</p> <p>10.4 Poisons Act;</p>                                                                                                                                                                                                                                                                                  | Reference 4, & 17                |
| 11 | 2 | <p>10.5 Tort Law: negligence, international torts, privacy, business premises liability,</p> <p>10.6 Commercial Law: business, contract, agency, private product ownership, insurance plans and antitrust, advertising, etc.</p>                                                                                                             | Reference 1,2, 3, 4&15           |
| 12 | 2 | <p><b>11. Pharmaceutical Laws and Regulations</b></p> <p>11.1Pharmaceutical legal systems (legislative, executive, judicial systems);</p> <p>11.2Drug nomenclature;</p> <p>11.3 Labeling and packaging requirements of pharmaceuticals;</p> <p>11.4 Administration/quality assurance;</p>                                                    | Reference 1,2, 3 4,13,14,15 & 16 |
| 13 | 2 | <p>11.1Control of fulfillment of requirements to practice pharmacy:</p> <p>11.5.1 Pharmaceutical retail outlets,</p> <p>11.5.2. Utilization of pharmaceuticals,</p> <p>11.5.3 Manufacture of pharmaceuticals,</p> <p>11.5.4 Import/export of pharmaceuticals,</p>                                                                            | Reference 4,5,9,11 & 12          |

|    |   |                                                                                                                                                                                                                                                                                                                              |                              |
|----|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
|    |   | 11.5.5 Wholesale/ distribution of pharmaceuticals,                                                                                                                                                                                                                                                                           |                              |
| 14 |   | 11.5.6 Health institutions involved in use and dispensing of pharmaceuticals,<br>11.5.7 Dispensing of proprietaries and manufacture of nostrums;<br>11.5.8 Management and use of controlled substances, poisons, radiopharmaceuticals; 11.5.8.1. Prohibitions; clinical trials; legal procedures for offenses and penalties. | Reference 1,4, 7,9,10 & 12   |
| 15 | 2 | 12. Laws, regulations and directives pertaining to pharmaceutical services in Ethiopia:<br>12.1 product registration and marketing,<br>12.2 manufacturing,                                                                                                                                                                   | Reference 1,4, 7,9,10 & 12   |
| 16 | 2 | 12.3 pharmaceutical promotions,<br>12.4 clinical trials,<br>12.5 herbal remedies,<br>12.6 veterinary drugs<br>12.7 professional licensing requirements                                                                                                                                                                       | Reference 1,4, 6,7,9,10 & 12 |
| 17 |   |                                                                                                                                                                                                                                                                                                                              |                              |
|    |   |                                                                                                                                                                                                                                                                                                                              |                              |

**Mode of delivery: Block**

- Illustrated Lectures
- Active learning methods (brain storming, buzz group, discussion, etc),
- Individual and group exercises and assignments
- Presentations
- Case study

**Mode of Assessment:**

- Quiz: 10%,
- Tests: 15%
- Assignment: 25%
- Presentation: 10%

- Final Exam: 40%

### Reference Materials:

1. Remington's: The Science and Practice of Pharmacy, 21<sup>st</sup> edition, University of The Sciences in Philadelphia, 2005, USA.
2. Dale and Appelbe's Pharmacy Law and Ethics, 8<sup>th</sup> edition, Pharmaceutical Press, 2005, London.
3. De. Marco, C. T. Pharmacy & the law, Aspen Systems Corp., Rochville, MD, 1984.
4. Drug Administration and Control Proclamation No. No. 661/2009
5. Standards for the Establishment and Practice of Pharmaceutical Manufacturing Plant, Drug Administration and Control Authority, 2001, Addis Ababa.
6. Directive for the Regulation of Promotion and Advertisement of Drugs, Drug Administration and Control Authority, 2005, Addis Ababa.
7. Guideline to Control and Promote Proper Use of Narcotic Drugs and Psychotropic Substances, Drug Administration and Control Authority, 2004, Addis Ababa.
8. Code of Ethics and Standards of Practice for Pharmacists Practicing in Ethiopia, 2<sup>nd</sup> edition, Ethiopian Pharmaceutical Association, 2006, Addis Ababa.
9. Guidelines on the Requirements for the Registration of Pharmaceutical Manufacturers, Drug Administration and Control Authority, Addis Ababa.
10. Requirements and Guidelines for the Registration of Human Drugs, Drug Administration and Control Authority, Addis Ababa.
11. Drug Import and Wholesale Guidelines, DACA.
12. Drug Retail sale guidelines, DACA.
13. Fink III, J.I., Marquard K.W & Simonsmeir, KM, Pharmacy Law Digest, Facts and Comparison, St. Souio, MD 1998.
14. Pharmacoethics: A Problem-Based Approach (Pharmacy Education Series), David A. Gettman and Dean Arneson, CRC, 2003
15. Law, Liability & Ethics for the Medical Office Professional, Myrtle R. Flight, 4th edition, Delmar Cengage Learning, 2003.
16. Mappes, T.A., and Zembaty, J.S. (1991). Biomedical ethics (3rd ed.). New York, NY: McGraw-Hill.
17. US Food, Drug and Cosmetic Act , URL: <http://www.fda.gov/RegulatoryInformation/Legislation/FederalFoodDrugandCosmeticActFDCA/default.htm>





## **Pharmacy practice course syllabus**

**Course name:** Pharmacy Practice

**Course code:** Phar 4244

**Module Number:** 24

**Module Name:** Pharmacy practice module

**Module code:** Phar-M4241

**Course ECTS:** 7

**Course hours:** 7 ECTS

**Totally required hours for the module**  $7 \times 27 = 189$

|                                                    |                         |
|----------------------------------------------------|-------------------------|
| Community pharmacy practice                        | 45 hours (9hrs x 5 wks) |
| Hospital pharmacy practice                         | 45 hours (9hrs x 5 wks) |
| Pharmaceutical industry and Drug quality assurance | 36 hours (9hrs x 4 wks) |
| Regulatory Pharmacy and Quality control practice   | 18 hours (9hrs x 2 wks) |
| Drug supplies management                           | 18 hours (9hrs x 2 wks) |
| R&D of Natural/Herbal medicines                    | 9 hours (9hrs x 1 wks)  |
| Home study                                         | 18 hours                |

**Year/Semester Course is offered:** Year IV Semester II

**Pre-requisite:** None

### **Course Description:**

This course provides series of practical attachments on pharmacy practice, including: community and hospital pharmacies, pharmaceutical industry, and drug regulatory body, Drug supplies management, and R&D of natural/herbal medicine. The course enables the student to assume the duties and responsibilities of various settings of pharmacy practice. It also enables the student to develop good working relationships with other health care professionals, and participate in public health education.

### **Course objectives:**

- To enable students enhance their skill level in the different areas of community pharmacy;
- To deliver different pharmaceutical services at hospital pharmacy settings;
- To enable students develop competence in the production and quality assurance of pharmaceuticals;
- Develop the capability to comfortably and confidently provide drug information to individuals and groups;

- To control the quality of drugs, cosmetics, food and drinks;
- To enable students aware of R&D of Natural/Herbal Medicine;
- To enable students develop competence in inventory control and store management, quantification and procurement process, the use and implementation of LMIS and APTS.

**Course mode of delivery:** Parallel

**Course learning and teaching methods**

- Introductory lectures prior o detachment to the respective practice sites
- *Practice/Visit (3 hours per day, 3 days in a week)*
- Presentations and discussions

**Assessment Technique:**

- Competence at practice including oral exam: 45%
- Seminar presentations: 15%
- Assignment: 10%
- Written exam: 30%

**Teachers' and students' role**

**Roles of Instructors**

Instructors should be well versed in the outcomes expected of students and the pedagogical methods that best enhance learning. Accordingly, they will be expected to:

- Be a facilitator (introduce the subject; give guidance, moderate discussions, etc.)
- Closely supervise in actual practice in different clinical and community settings
- Assists students in resolving ever-increasing ethical dilemma in the service delivery
- Foster and assess self-initiated student learning
- Read and comment assignments of students on time;
- Prepare his/her lessons and deliver lectures;
- Provide available and necessary reference materials
- Encourage active participation of students in the teaching learning process;
- Strive to be the role model ethically and professionally

**Roles of Students**

Students are expected to:

- Make every effort to apply valid research findings and theoretical teachings into practice in various practice settings
- Engage in learning by doing independent study, project work; group work, etc..
- Be active learners (participate effectively in group assignments, perform practice actively, make presentations, write reports, prepare seminars etc.);

- Critically assess laws, regulations, journal articles and related topics from different sources

## References:

### Recommended readings

1. Winfield, A. J. and Richards, R. M. E. (eds.), Pharmaceutical Practice, 4<sup>th</sup> ed., Churchill Livingstone, London, 1998.
2. Smith, H. A. Principles and Methods of Pharmacy Management, 2nd ed. Lea & Febiger, Philadelphia, 1980.
3. Lawson, D. H. and Richards, R.H., (eds.) Clinical Pharmacy and Hospital Drug Management, 2<sup>nd</sup> ed. Chapman and Hall. London. 1982.
4. Remington's Pharmaceutical Science, 21<sup>st</sup> ed., Lippincott Williams & Wilkins, Pennsylvania, 2006.
5. National standard treatment guidelines and formularies.
6. USP/NF (Latest edition). The United States Pharmacopoeia convention, Inc. Rockville, MD., USA

### Course Schedule

| Time       | 10:00am-12:20pm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 1:30-5:30pm, 3 days per week                                                                                                                               | Tasks                                                                                                                                                                                                                                                                                                                                                                          |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Week 1 - 5 | Interactive lecture x 1 wk <ul style="list-style-type: none"> <li>• Organization of a community pharmacy</li> </ul> Services provided by the community pharmacy                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Community pharmacy attachment x 5 wk                                                                                                                       | <ul style="list-style-type: none"> <li>• Dispensing</li> <li>• Good dispensing practice</li> <li>• Stock management</li> <li>• Compounding extemporaneous preparations</li> <li>• Patient counseling</li> <li>• Inventory control</li> </ul>                                                                                                                                   |
| Week 6-10  | Interactive lecture x 1 wk <ul style="list-style-type: none"> <li>• Organization of a typical hospital pharmacy</li> <li>• Traditional roles of the hospital pharmacist</li> <li>• Specializations in hospital pharmacy:</li> <li>• Pharmacy and therapeutics committee; teaching</li> <li>• Pharmacy and therapeutics committee; teaching</li> <li>• Hospital Manufacturing Activities               <ul style="list-style-type: none"> <li>○ Necessary facilities and equipments</li> <li>○ manufacturing sterile products: Enteral and parenteral nutrition (TPN), Cytotoxic admixture, Parenteral infusions</li> </ul> </li> </ul> | Attachment at Hospital pharmacies x 5 wk (Adult OPD Pharmacy, Pediatric Pharmacy, ART Pharmacy, Emergency Pharmacy, Oncology Pharmacy, Inpatient Pharmacy) | <ul style="list-style-type: none"> <li>• Prescription evaluation</li> <li>• Dispensing</li> <li>• Rational drug use</li> <li>• Compounding extemporaneous preparations and TPN</li> <li>• Patient counseling</li> <li>• ART pharmacy dispensing</li> <li>• Distribution of drugs to the different wards</li> <li>• Practice unit dose drug dispensing to inpatients</li> </ul> |

|            |                                                                                                                                                   |                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|            | <ul style="list-style-type: none"> <li>Nonsterile products: Compounding, Handrub preparation</li> </ul> <p>Documentation in pharmacy practice</p> |                                                               | <ul style="list-style-type: none"> <li>Monitoring drug/food/disease interactions</li> <li>Calculate different parenteral doses to be administered to patients</li> <li>Monitoring appropriateness of therapy</li> <li>Participate in morning sessions and ward visits</li> <li>Ward visits and rounds; chart review, assessment, care plan and follow up evaluation of hospitalized patients.</li> <li>Patient care</li> <li>Discharge medications counseling</li> </ul> |
| Week 11-14 |                                                                                                                                                   | Pharmaceutical industry and Drug quality assurance<br>x 4 wks | <ul style="list-style-type: none"> <li>Tablet Production line</li> <li>Capsule Production line</li> <li>Oral Liquid Production line</li> <li>Parenteral Liquids Production line</li> <li>Topical dosage forms/Ointments, Creams/ Production line</li> </ul> <p>Quality Control Laboratory</p>                                                                                                                                                                            |
| Week 15-16 |                                                                                                                                                   | Regulatory Pharmacy and Quality control practice              | <ul style="list-style-type: none"> <li><b>Participate in</b> evaluating the safety, efficacy and quality of pharmaceuticals</li> <li>Participate in disposal pharmaceuticals unfit for use</li> </ul> <p><b>Perform physicochemical and microbiological drug quality analysis</b></p>                                                                                                                                                                                    |
| Week 17    |                                                                                                                                                   | Research and Development                                      | <b>Participate in natural/herbal medicine R&amp;D sites</b>                                                                                                                                                                                                                                                                                                                                                                                                              |
| Week 18-19 |                                                                                                                                                   | Drug supplies management                                      | <ul style="list-style-type: none"> <li><b>Inventory control</b></li> <li><b>Store management</b></li> <li><b>Participate in quantification and procurement process,</b></li> </ul>                                                                                                                                                                                                                                                                                       |

|  |  |  |                                                                                                                                                                                                    |
|--|--|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  |  |  | <ul style="list-style-type: none"> <li>• Assess rational use drugs in hospitals</li> <li>• Conduct satisfaction surveys in hospitals</li> </ul> <p>Participate in LMIS and APTS implementation</p> |
|--|--|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## **First Aid Course Syllabus**

**Course title:** First aid

**Course code:** Nurs4245

**Course EtCTS:** 3

**Course hours:** 3 \*27 = 81

Pre-requisite: Anatomy and physiology

### *Course Description:*

This First aid and accident prevention course is designed for BSc degree pharmacy students as a competent provider of victims in emergency situation and also helps the students to avoid hazards to self and others. The basic first aid and accident prevention skill and knowledge and attitude required to sustain human body function and/or prevent premature death will be discussed, demonstrated and through independent home study and applied in and out of health care settings.

### *Course objectives:*

After completion of this course, the student will be able to apply knowledge and skill of first and accident prevention anywhere.

Supportive objectives

At the end of each topic the student will be able to

- Define first aid and accident prevention.
- Explain principles of first Aid
- Identify the respiratory emergency and artificial respiration
- Discuss cardiac arrest and cardiopulmonary resuscitation
- Identify the wound, types of bleeding and arrest bleeding
- Explain dressing and bandages
- Discuss injuries and caring for the causality with shock, suddenly illness
- Determine the importance of lifting and transporting of casualty safety
- Determine first aid approach for fracture, poisoning and disaster.
- Demonstrate artificial respiration, CPR, bleeding control.
- Differentiate between what to do and what not to do

| WEEK/DATE  | TOPIC                                                                                                                                                                                                                                                                                                                                                                                                                              | Contact hours | Assignments due              |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|------------------------------|
| Week 1     | <ul style="list-style-type: none"> <li>▪ Define first aid and accident prevention</li> <li>▪ Identify reasons for First Aid</li> <li>▪ Explain principles of first aid</li> <li>▪ List value of First Aid Training</li> <li>▪ Define infection prevention and patient safety</li> <li>▪ Adopt general directions for given first aid</li> </ul>                                                                                    | 2             | Assignment 1                 |
| Week 2-4   | <ul style="list-style-type: none"> <li>✓ Definition respiratory Emergency</li> <li>✓ Cause of Respiratory failure <ul style="list-style-type: none"> <li>♣ Anatomic obstruction</li> <li>♣ Mechanical obstruction</li> <li>♣ Air depleted of oxygen or containing toxic gases</li> </ul> </li> <li>✓ Artificial respiration</li> <li>✓ Cardiac arrest</li> <li>✓ Cardiopulmonary resuscitation</li> </ul>                          | 6             | Case study, Re-demonstration |
| Week 5-6   | <ul style="list-style-type: none"> <li>♣ Define wound</li> <li>♣ Wound classification based on skin integrity and cause</li> <li>♣ List types of open wound</li> <li>♣ Common causes and symptoms wound infection</li> <li>♣ First Aid for open wounds</li> <li>♣ First aid for severe bleeding</li> <li>♣ Prevention of contamination and infection of wounds</li> </ul>                                                          | 4             | Assignment.2                 |
| Week 7     | <ul style="list-style-type: none"> <li>➤ Definition of dressings</li> <li>➤ Principles of clean dressing</li> <li>➤ Bandages</li> <li>➤ Types of commercially available bandages</li> <li>➤ Application of bandages</li> <li>➤ First Aid kits and supplies</li> </ul>                                                                                                                                                              | 2             | Re-Demonstration.            |
| Week 8-11  | <ul style="list-style-type: none"> <li>➤ Eye injuries</li> <li>➤ Head injuries</li> <li>➤ Neck injuries</li> <li>➤ Open Wounds of the chest</li> <li>➤ Abdominal injuries</li> <li>➤ Burns <ul style="list-style-type: none"> <li>♣ Definition</li> <li>♣ Causes and effects</li> <li>♣ Classification based on Extent and location</li> <li>♣ First aid measures</li> <li>♣ Prevention of heat emergencies</li> </ul> </li> </ul> | 8             | Case study 2                 |
| Week 12-13 | <ul style="list-style-type: none"> <li>☞ Definition of shock</li> <li>☞ Cause of shock</li> <li>☞ Sign and symptoms</li> <li>☞ Treatment objectives</li> <li>☞ First aid measures</li> <li>☞ Sudden illness</li> </ul>                                                                                                                                                                                                             | 4             | Case study                   |

|         |                                                                                                                                                                                                                                                   |   |                                 |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---------------------------------|
|         | <ul style="list-style-type: none"> <li>❖ Heart attack</li> <li>❖ Stroke</li> <li>❖ Fainting</li> <li>❖ Epilepsy</li> <li>❖ Prevention of heart attack</li> <li>○ Unconsciousness</li> </ul>                                                       |   |                                 |
| Week 14 | <ul style="list-style-type: none"> <li>▪ Definitions</li> <li>▪ Fractures</li> <li>▪ Dislocation</li> <li>▪ Sprains</li> <li>▪ Prevention of Accidents resulting in skeleton &amp; muscular injuries</li> </ul> Pro.8.                            | 2 | Assignment<br>Re-Demonstration. |
| Week 15 | <ul style="list-style-type: none"> <li>☞ Definition</li> <li>☞ Causes</li> <li>☞ Sign and Symptoms</li> <li>☞ Objective in treatment of first aid</li> <li>☞ Contact poisons</li> <li>☞ Prevention of Accidental poisoning</li> </ul> Procedure.9 | 2 | Case study,                     |
| Week 16 | <ul style="list-style-type: none"> <li>♣ Define disaster</li> <li>♣ Types of disaster</li> <li>♣ Prevent disaster</li> </ul>                                                                                                                      | 2 | Assignment                      |
|         |                                                                                                                                                                                                                                                   |   |                                 |

### Mode of delivery:

- Lecture
- Discussion
- Demonstration and role play
- Video show

### Teaching aids and materials (course logistics)

- Human Anatomic Models/dolls
- Demonstration equipment and Instruments
- Chalk and board, white board
- Audiovisual aid (LCD, OHP, Laptop)

### Mode of Assessment:

#### Formative assessment

- Attendance and class activity
- Practical exam (skill lab)



- Assignment
- Quiz

Summative assessment

|                            |     |
|----------------------------|-----|
| Assignments                | 10% |
| Test                       | 30% |
| Written final exam         | 40% |
| Practical exam (skill lab) | 20% |

Course policy:

- A student who is unable to pass 50% of the continuous assessment should not be allowed to sit for final exam
- Attendance:
  - The student who is absent from over 20% of the contact hours should not be eligible for final examination and is enforced to repeat the course
  - 100% attendance for practical/skill/lab hours

*Reference*

1. Skeet, M. First Aid for Community health worker to developing countries. Macmillan/tong Kong 1984.
2. American Red Cross standard first Aid and Personal Safety, 2<sup>nd</sup> ed. New York 1979.
3. Caroline L. Nancy. Emergency care in the streets U.S.A. 19979.
4. Warner. C. Germanie. Emergency cares Assessment and intervention 3<sup>rd</sup> Ed. The C.V Mosey Comp. London 1983
5. Infection prevention and patient safety guideline of Ethiopia, February 2005

## **Nutrition Course Syllabus**

**Course title:** Nutrition

**Course code:** ComH4246

**Course ECTS:** 3

**Course hours:** 3\*27 = 81

Pre-requisite if any: None

### *Course Description:*

This human nutrition course is designed to prepare Bachelor of pharmacy students in order to be competent in nutrition related to health and disease. The course is designed to introduce students to normal nutrition, diet therapy, infant & child and maternal nutrition. It helps the students to identify different nutrients and to be competent in assessing and managing nutrition and nutrition related problems in the community and for women, children and PLWHA in particular. It also helps students to recognize public importance of ensuring food safety and quality.

### **Course Objectives:**

After completion of this course, the student will be able to recognize essential nutrients for life function, develop skill on nutritional assessment methods, recognize nutritional intervention methods and also able to apply them in promotion of health and in the care of the sick in an effective and integrated manner.

### **Supporting objectives:**

At the end of this course, the students will be able to:

- Explain the historical development of nutrition
- Describe the characteristics and types, physiological functions, and food sources of essential nutrients
- Explain recommended intakes and the adverse effects of both inadequate and excessive intake of nutrients
- Describe the epidemiology, population at risk, classification, clinical feature and management of malnutrition
- Analyze and develop skill on the major nutritional assessment methods
- Discuss infant and young child feeding options
- Describe the public health importance nutritional deficiency states in Ethiopia
- Integrate maternal nutrition with other programs and services
- Describe the management algorithm for HIV patients with malnutrition

- Describe the importance and application of nutritional surveillances
- Prevent micronutrient deficiencies through active participation in micronutrient supplementation programs and control of common infections such as malaria and helmentiasis
- Control micronutrient deficiencies through proper therapeutic supplementation with micronutrients
- Discuss the public health importance of ensuring food safety and quality in terms of protection from microbiological hazards, pesticide residues, misuse of food additive, chemical contaminants, biological toxins (national toxins in foods), and adulteration.

## Course schedule

| Week/date | Lesson/ Topics                                                                                                                                                                                                                                                  | Teaching methods                                                                       | Duties expected from the instructor                                                                                                                                                                                                                                            | Duties expected from students                                                                                                                                                                                                                                                       | Assignments due/evaluation           | Required readings                                                                                                                                                 |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Day 1     | Introduction to nutrition <ul style="list-style-type: none"> <li>Historical development of nutrition</li> <li>Terminologies</li> <li>Growth and development</li> <li>Consequences of malnutrition</li> <li>Major factors contribute to malnutrition.</li> </ul> | Exercise<br>Brain storming<br>Illustrated and interactive lectures<br>Group discussion | <ul style="list-style-type: none"> <li>Introducing the objective of the lesson</li> <li>Give class ,home and library works</li> <li>Monitor students activities</li> <li>Give gap lectures</li> <li>Give concluding remarks</li> </ul>                                         | Grasping lesson objectives<br>Participating in activities<br><b>In class room:</b><br>Discussing with groups on factors that can affect nutrition<br><b>Outside class room:</b><br>Library work: read books on historical development of nutrition, terminologies used in nutrition | Project work<br>Group assignment     | <ul style="list-style-type: none"> <li>Melkie E, Human nutrition lecture note</li> <li>Tefera B. nutrition lecture note</li> </ul>                                |
| Day 2     | <b>Carbohydrates, Proteins</b><br><b>Lipids</b><br>Functions, Types, Food Sources, Digestion, Absorption, metabolism, RDA                                                                                                                                       | Exercise<br>Brain storming<br>Illustrated and interactive lectures<br>Group discussion | <ul style="list-style-type: none"> <li>Introducing the objective of the lesson</li> <li>Asking brainstorming questions</li> <li>Give class ,home and library works</li> <li>Monitor students activities</li> <li>Give gap lectures</li> <li>Give concluding remarks</li> </ul> | Grasping lesson objectives<br>Participating in activities<br><b>In class room:</b><br>Discussing with groups on food sources of CHO and proteins<br><b>Outside class room:</b><br>Library work: read books on carbohydrates and proteins                                            | Quiz 1<br>On previous lesson         | <ul style="list-style-type: none"> <li>Tefera B &amp; Melkie E, human nutrition lecture notes</li> <li>Dudek, nutrition hand book for nursing practice</li> </ul> |
| Day 3     | <b>Vitamins, Minerals</b><br>Functions, Types, Food Sources, Digestion, Absorption, metabolism, RDA                                                                                                                                                             | Exercise<br>Brain storming<br>Illustrated and interactive lectures<br>Group discussion | <ul style="list-style-type: none"> <li>Introducing the objective of the lesson</li> <li>Asking brainstorming questions</li> <li>Give class ,home and library works</li> <li>Monitor students activities</li> <li>Give gap lectures</li> <li>Give concluding remarks</li> </ul> | Grasping lesson objectives<br>Participating in activities<br><b>In class room:</b><br>Discussing with groups on types and functions of lipids<br><b>Outside class room:</b><br>Library work: read books on lipids and vitamins                                                      | <b>Exam one</b>                      | <ul style="list-style-type: none"> <li>Melkie Edris, Tefera B. Human nutrition lecture notes</li> <li>Dudek, nutrition hand book for nursing practice</li> </ul>  |
| Day 4     | <b>Nutrition requirement:</b> <ul style="list-style-type: none"> <li>Methods of calculating normal food requirements</li> <li>Influence of age sex and occupation</li> <li>Nutrition value of common foods</li> </ul>                                           | Exercise<br>Brain storming<br>Illustrated and interactive lectures<br>Group discussion | <ul style="list-style-type: none"> <li>Introducing the objective of the lesson</li> <li>Give class ,home and library works</li> <li>Monitor students activities</li> <li>Give gap lectures</li> <li>Give concluding remarks</li> </ul>                                         | Grasping lesson objectives<br>Participating in activities<br><b>In class room:</b><br>Do exercise on calculation of nutritional requirement<br><b>Outside class room:</b><br>Library work: read books on nutritional requirement                                                    | Reading assignment<br><br>Exercise 2 | Dudek, nutrition hand book for nursing practice<br><br>Human energy requirements Report of a Joint FAO/WHO/UNU Expert Consultation Rome, October 2001             |

|           |                                                                                                                                                                                                                                                  |                                                                                        |                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                     |                       |                                                                                                                                                                                                                                                                         |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|           |                                                                                                                                                                                                                                                  |                                                                                        |                                                                                                                                                                                                                                                                                 | <b>Home take assignment:</b><br>Producing sample Menu                                                                                                                                                                                                               |                       |                                                                                                                                                                                                                                                                         |
| Day 5     | Nutritional deficiency states<br><ul style="list-style-type: none"> <li>Chronic deficiency.</li> <li>Micronutrient deficiencies</li> </ul>                                                                                                       | Exercise<br>Brain storming<br>Illustrated and interactive lectures<br>Group discussion | <ul style="list-style-type: none"> <li>Introducing the objective of the lesson</li> <li>Asking brainstorming questions</li> <li>Give class ,home and library works</li> <li>Monitor students activities</li> <li>Give gap lectures</li> <li>Give concluding remarks</li> </ul>  | Grasping lesson objectives<br>Participating in activities<br><b>In class room:</b><br>Discussing about PEM,IDA,IDD and VAD<br><b>Outside class room:</b><br>Library work: read books on nutritional deficiencies common in Ethiopia                                 | <b>Exam two</b>       | Tefera B. Human nutrition lecture note<br><br>National guideline for control and prevention of micronutrient deficiency ,FMOH, June 2004<br><br>Management of SAM: A manual for physicians and other senior health Workers WHO,Geneva,1999                              |
| Day 6-7   | Nutritional deficiency states continued<br>Nutritional assessment:<br><ul style="list-style-type: none"> <li>Anthropometry</li> <li>Biochemical method</li> <li>Clinical method</li> <li>Dietary survey method</li> </ul> Prevention and control | Exercise<br>Brain storming<br>Illustrated and interactive lectures<br>Group discussion | <ul style="list-style-type: none"> <li>Introducing the objective of the lesson</li> <li>Give class ,home and library works</li> <li>Monitor students activities</li> <li>Give gap lectures</li> <li>Give concluding remarks</li> </ul>                                          | Grasping lesson objectives<br>Participating in activities<br><b>In class room:</b><br>Discussing on methods of nutritional assessment<br>Practice methods of assessment<br><b>Outside class room:</b><br>Library work: read books on nutritional assessment         | Quiz 3<br>(Summative) | Teferra.B Human nutrition lecture note<br><br>Gibson, principles of nutritional assessment ,oxford,1990                                                                                                                                                                 |
| Day 8-9   | Nutritional deficiency states continued<br><ul style="list-style-type: none"> <li>Diet as therapeutic agent</li> <li>Diet and the patient</li> <li>Hospital diets</li> </ul>                                                                     | Exercise<br>Brain storming<br>Illustrated and interactive lectures<br>Group discussion | <ul style="list-style-type: none"> <li>Introducing the objective of the lesson</li> <li>Asking brain storming questions</li> <li>Give class ,home and library works</li> <li>Monitor students activities</li> <li>Give gap lectures</li> <li>Give concluding remarks</li> </ul> | Grasping lesson objectives<br>Participating in activities<br><b>In class room:</b><br>List and discuss fluid diets<br>Special diets with related to disease<br><br><b>Outside class room:</b><br>Library work: read books on factors affecting human food selection | <b>Exam three</b>     | <ul style="list-style-type: none"> <li>Melkie Edris, Human nutrition lecture note</li> <li>Dudek, nutrition hand book for nursing practice</li> </ul>                                                                                                                   |
| Day 10-11 | Nutritional care and support for PLHIV                                                                                                                                                                                                           | Brain storming<br>Illustrated and interactive lectures<br>Group discussion             | <ul style="list-style-type: none"> <li>Introducing the objective of the lesson</li> <li>Asking brain storming questions</li> <li>Give class ,home and library works</li> <li>Monitor students activities</li> <li>Give gap lectures</li> <li>Give concluding remarks</li> </ul> | Grasping lesson objectives<br>Participating in activities<br><b>In class room:</b><br>Discussing on HIV and Nutrition<br><b>Outside class room:</b><br>Home work: read updated national guideline for HIV/AIDS and nutrition                                        | Quiz 4<br>formative   | <ul style="list-style-type: none"> <li>Melkie Edris and Tefera B Human nutrition lecture notes</li> <li>FMOH: National guidelines for HIV/AIDS and Nutrition,2008</li> <li>Nutrition and HIV/AIDS A Training Manual For Nurses and Midwives, updated on 2010</li> </ul> |

|            |                                                                                                                                                                                        |                                                                                                                             |                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                             |                                                                                                                                                       |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Day 12-13  | <b>Nutritional interventions</b> for major nutritional problems in Ethiopia<br>Methods, mechanisms and criteria,<br>Essential Nutrition Actions(ENA)<br><b>Teaching Good nutrition</b> | Brain storming<br>Illustrated and interactive lectures<br>Group discussion                                                  | <ul style="list-style-type: none"> <li>• Introducing the objective of the lesson</li> <li>• Asking brain storming questions</li> <li>• Give class ,home and library works</li> <li>• Monitor students activities</li> <li>• Give gap lectures</li> <li>• Give concluding remarks</li> </ul>                                | Grasping lesson objectives<br>Participating in activities<br><b>In class room:</b><br>Discussing with groups about ENA<br><b>Outside class room:</b><br>Library work: read books on nutritional intervention methods                                                                                                                                                                                                                                          | Home take assignment and Group presentation | <ul style="list-style-type: none"> <li>• Tefera B. Human nutrition lecture note</li> <li>• ENA counselor guide, FMOH, revised January 2005</li> </ul> |
| Day 14- 15 | The quality and safety of nutrition related products                                                                                                                                   | Exercise<br>Brain storming<br>Illustrated and interactive lectures<br>Group discussion<br>Role play<br>Seminar presentation | <ul style="list-style-type: none"> <li>• Introducing the objective of the lesson</li> <li>• Asking brain storming questions</li> <li>• Give class ,home and library works</li> <li>• Monitor students activities</li> <li>• Give gap lectures</li> <li>• Check students work</li> <li>• Give concluding remarks</li> </ul> | Grasping lesson objectives<br>Participating in activities<br><b>In class room:</b><br>Discussing with groups on the public health importance of ensuring food safety and quality in terms of protection from microbiological hazards, pesticide residues, misuse of food additive, chemical contaminants, biological toxins (national toxins in foods), and adulteration<br><b>Outside class room:</b><br>Library work: read books on food safety and quality | Quiz 5 (Summative) presentation             | <ul style="list-style-type: none"> <li>• Melkie Edris and Tefera B Human nutrition lecture notes</li> </ul>                                           |
| Day 16     |                                                                                                                                                                                        |                                                                                                                             |                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Final Exam                                  |                                                                                                                                                       |

**Mode of delivery:** Block

- ❖ Brain storming
- ❖ Illustrated and interactive lectures
- ❖ Group discussion
- ❖ Case study
- ❖ Individual and group exercises
- ❖ Seminar/ presentation

**Mode of Assessment:**

- Exercises
- Quizzes
- Individual assignment
- Presentation

**Summative assessment**

|                                 |     |
|---------------------------------|-----|
| Quizzes                         | 10% |
| Group and individual assignment | 20% |
| Tests                           | 20% |
| Presentation                    | 10% |
| Final exam                      | 40% |

**Learning materials:**

- Printed materials (text books, manual exercises, learning guides, handout)

**Text Books:**

1. Melkie Edris, Human Nutrition for Health Science students, Gondar University, 2004.
2. Tefera Belachew. Human Nutrition for health science students. Lecture note series. Jimma University, Faculty of Public Health, January 2003.

**References:**

1. Dudek S.G., Nutrition hand book for nursing practice, third ed Lippincott, Newyork, 1997
2. FMOH: Protocol for management of sever acute malnutrition(SAM) in Ethiopia, 2007
3. Human energy requirements Report of a Joint FAO/WHO/UNU Expert Consultation Rome, 17–24 October 2001
4. Management of severe Malnutrition: A manual for physicians and other senior health Workers WHO,Geneva,1999
5. FMOH: National guidelines for HIV/AIDS and Nutrition,2008

6. Nutrition and HIV/AIDS A Training Manual For Nurses and Midwives, updated on 2010
7. National nutrition guideline
8. ENA counselor guide, FMOH, revised January 2005
9. National guideline for control and prevention of micronutrient deficiency ,FMOH, June 2004
10. Gibson, principles of nutritional assessment ,oxford,1990



## Module 25: Professional Elective Courses

**Module Name:** Professional Elective

**Module Category:** Elective

**Module code:** Phar-M4252

**Module Number:** 25

**Module weight in ECTS:** 5 ECTS

Courses

| Course                                               | Course code | ECTS   |
|------------------------------------------------------|-------------|--------|
| Introduction to Pharmacoepidemiology                 | Phar4251    | 5 ECTS |
| Phytochemistry                                       | Phar4252    | 5 ECTS |
| Pharmaceutical Manufacturing                         | Phar4253    | 5 ECTS |
| Pharmacogenetics                                     | Phar4254    | 5 ECTS |
| Pharmaceutical Quality control and quality assurance | Phar4255    | 5 ECTS |
| Drug design and synthesis                            | Phar4256    | 5 ECTS |
| Warehouse management                                 | Phar4257    | 5 ECTS |
| Research in pharmacology                             | Phar4258    | 5 ECTS |

## **Introduction to Pharmacoepidemiology Course Syllabus**

**Module name:** Professional elective

**Module Number in which the course exists:** 25

**Course title:** Introduction to Pharmacoepidemiology

**Course code:** Phar 4251

**Course ECTS:** 5

**ECTS credits:** 5 (This course needs a total of  $5 \times 27 = 135$  working hours to spend in each teaching and learning as well as assessment activities). The distribution of these hours will be as follows

- Lecture: 48 hours
- Project work: 14 hours
- Presentations=10 hours
- Case studies/journal club=10 hours
- Tutorial: 8 hours
- Home study: 38 hours
- Assessment=7 hours

Pre-requisite if any:

### *Course Description*

The goal of the course is to introduce pharmacoepidemiology and drug safety and research application for post-marketing drug safety surveillance. The course will describe how to develop a research protocol and conduct a research, describe various health care data sources used for research, and discuss how pharmacoepidemiology contribute to pharmacy practice, such as, drug utilization review, assessment of drug therapy, and adverse drug reaction monitoring. A series of case studies from thalidomide to cisapride to cerivastatin will be also discussed in class. Students can have a better understanding of Pharmacoepidemiologic research, drug safety regulatory, pregnancy registry, and risk management.

q. Course Objective:

Upon completion of this course, the students will be able to:

- ☞ Describe the purposes and scope of pharmacoepidemiology
- ☞ Describe and explain basic concepts in pharmacoepidemiology and its relevance to public health and for health policy making.
- ☞ Describe the relationship between national drug policies and Pharmacoepidemiology
- ☞ Describe the basic pharmacoepidemiologic concepts and measures of drug-related occurrence and its effect in population;
- ☞ Discuss common study designs and methods used in pharmacoepidemiological studies.

- ☞ Explain the applications of pharmacoepidemiological methods for studies of effects and adverse effects of drugs and economic consequences.
- ☞ Assess the relevance and limitations of various pharmacoepidemiological research designs
- ☞ Describe systems for the reporting of adverse effects and their use for pharmacoepidemiology.
- ☞ Apply pharmacoepidemiologic principles in practice.
- ☞ Discuss Pharmacovigilance in drug development
- ☞ Evaluate drug safety case studies and policy implications based on the medical and pharmacy literature.

Skills and abilities:

On successful completion of the course, the student should be able to:

- ☞ Review and evaluate pharmacoepidemiological studies.

| Week | Contact hrs | Topic/sub-topic/chapter                                                                                                                        | Reading materials  | Remarks |
|------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|---------|
| 1    | 4           | 1. Introduction<br>1.1. What is Pharmacoepidemiology?<br>1.2. Contributions of Pharmacoepidemiology                                            | Reference 2 & 4    |         |
| 2    | 4           | 2. National medicinal drug policies: their relationship to Pharmacoepidemiology                                                                | Reference 2, 3 & 4 |         |
| 3    | 4           | 3. National medicinal drug policies: their relationship to Pharmacoepidemiology.....<br>3.1. Premarketing applications of Pharmacoepidemiology | Reference 1&2      |         |
| 4    | 4           | 2. Study Designs<br>4.1 Observational studies                                                                                                  | Reference 2,3 & 4  |         |
| 5    | 4           | 4.1.1 Descriptive studies                                                                                                                      | Reference 2,3 & 4  |         |
| 6    | 4           | 4.1.2 Analytical studies                                                                                                                       | Reference 2,3 & 4  |         |
| 7    | 4           | 4.2 Experimental studies<br>4.2.1 Randomized Clinical Trial (RCT)                                                                              | Reference 2,3 & 4  |         |
| 8    | 4           | 4.2.2 Community Intervention Trials (CITs)<br>4.3 Selection of study designs                                                                   | Reference 2,3 & 4  |         |

|    |   |                                                                                                                                                         |                     |  |
|----|---|---------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|--|
| 9  | 4 | 5. Drug Utilization<br>5.1 Definition<br>5.2 Drug-centered and patient-centered approach in drug use studies                                            | Reference 2,3 & 4   |  |
| 10 | 4 | 5.3 Indicator based approach in drug use studies<br>5.3.1 Prescribing indicators<br>5.3.2 Patient care indicators<br>5.3.3 Facility specific indicators | Reference 2,3 & 4   |  |
| 11 | 4 | 5.4 The social aspects of drug use                                                                                                                      | Reference 2,3 & 4   |  |
| 12 | 4 | 5.5 The economic aspects of drug use                                                                                                                    | Reference 1,2,3 & 4 |  |
| 13 |   | 5.6 Studies of patient compliance                                                                                                                       | Reference 1,2,3 & 4 |  |
| 14 | 4 | 6 Pharmacovigilance<br>6.2 What is pharmacovigilance?                                                                                                   | Reference 2,3 & 4   |  |
| 15 |   | 6.3 Pharmacovigilance methods                                                                                                                           | Reference 1,2,3 & 4 |  |
| 16 | 4 | 6.4 The need for effective drug safety programs<br>6.4 Elements of drug safety programs                                                                 | Reference 1,2 & 3   |  |

**Mode of delivery:**

- Illustrated Lectures and case studies
- Active learning methods (brain storming, buzz group, discussion, etc)
- Individual and group exercises and assignments
- Presentations and participation in class discussion
- Case studies

**Mode of assessment:**

- Quizzes and tests: 30%
- Attendance : 5%
- Case studies:5%
- Assignments (group or individual):20%
- Written final exam: 40%

**LEARNING MATERIALS:**

- Recommended Readings:
  1. Pharmacoepidemiology, 4<sup>th</sup> edition, Storm B. L. (Ed), John Wiley and Sons Ltd, England, 2005.
  2. Textbook of Pharmacoepidemiology, Storm B. L. And Kimmel S.E. (Eds), 2007, John Wiley, New Jersey.
  3. Pharmacoepidemiology – An Introduction, 3<sup>rd</sup> edition, Hartzema A.G., Porta M., Tilson H.H., (Eds), 1998, Cincinnati OH, Harvey Witney Books Company.
  4. Remington's: The Science and Practice of Pharmacy, 21<sup>st</sup> edition, University of The Sciences in Philadelphia, 2005, USA.

**Module name:** Professional electives

**Module Number in which the course exists:** 25

**Course title:** Photochemistry

**Course code:** Phar4252

**Course ECTS:** 5

A course guide for Phytochemistry course

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Year/ semester:             | Year IV, Semester II Regular B. Pharm                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Pre-requisite:              | Pharmacognosy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Status of the course:       | Elective                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Course description:         | The course covers some selected topics in phytochemistry. Classes of phytochemistry along with their extraction and isolation methods.                                                                                                                                                                                                                                                                                                                                                                            |
| Course objective:           | By the end of this course students will be able to describe the scientific methods in the investigations of phytochemicals. Moreover, students can able to acquire the application of phytochemistry in the health care.                                                                                                                                                                                                                                                                                          |
| Delivery mode/ methodology: | During this course the following mode of teaching can be used: <ul style="list-style-type: none"><li>• Illustrated lectures</li><li>• Brain storming</li><li>• Buzz group</li><li>• Gapped lecture</li><li>• Questions and answers</li><li>• Self reflection</li><li>• Individual and group exercise and assignments</li><li>• Group discussions</li><li>• Seminar presentation</li></ul>                                                                                                                         |
| Assessment mechanisms:      | Formative assessment <ul style="list-style-type: none"><li>• Participation</li><li>• Question and answer</li><li>• Pair work</li><li>• Peer assessment</li><li>• Self assessment</li><li>• Quizzes</li><li>• Performance on buzz group</li><li>• Self- reflection</li></ul> Summative assessment <ul style="list-style-type: none"><li>• Test/s.....20%</li><li>• Seminar presentation and paper work..... 20%</li><li>• Report (lab visit, garden and TM clinic)----- 20%</li><li>• Final exam.....40%</li></ul> |
| Grading                     | As per the university's regulation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Teaching aids:              | <ul style="list-style-type: none"><li>• Smart board</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                  | <ul style="list-style-type: none"> <li>• LCD</li> <li>• Laptop</li> <li>• Black board &amp; chalk/white board &amp; marker</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Course policies: | <p>Ground rules:</p> <ul style="list-style-type: none"> <li>• Students are expected to attend all classes and consideration pertaining to class room attendance is as per the senate legislation of the university.</li> <li>• Active participation is required at most</li> <li>• Punctuality in class and assignment is mandatory</li> <li>• Misbehaving at class is highly forbidden</li> <li>• Disabling a cellular phone is a must</li> <li>• The students should submit the assignments and presentation manuals before the deadline.</li> <li>• The students should actively participate in group discussions which by itself is part of the student evaluation.</li> </ul> <p>Late work</p> <ul style="list-style-type: none"> <li>• Late assignments/homework will be penalized depending on circumstances.</li> <li>• Make up exam should be performed within reasonable time of absence per university policy and provide with required documents.</li> </ul> <p>Academic dishonesty policy/plagiarism</p> <ul style="list-style-type: none"> <li>• Academic dishonesty includes , but is not limited to, cheating, plagiarizing, fabricating of information or citations, facilitating acts of academic dishonesty by others, having unauthorized possession of examinations, submitting work of another person or work previously used without informing the instructor, or tampering with the academic work of other students. Students found in violation of such policy are subjected to disciplinary actions as per university policy.</li> </ul> <p>Disclaimer</p> <ul style="list-style-type: none"> <li>• This syllabus represent a best plan for the course, but, as with most plans, it is subject to changes made necessary by time, space and personal constraints as the course progresses.</li> </ul> |
| References :     | <p>Your reading materials for the course:</p> <ol style="list-style-type: none"> <li>1. Sarker D, Latif Z, Gray A. Methods in Biothechnology Natural Products Isolation, 2<sup>nd</sup> edition, Human Press, Totowa, New Jersey, 2006.</li> <li>2. Dewick PM. Medicinal Natural Products: A biosynthetic Approach, 3<sup>rd</sup> edition. Jhon Wiley and Sons, LTD, England 2009.</li> <li>3. <a href="#">Bruneton, J.</a> Pharmacognosy, phytochemistry, medicinal plants</li> <li>4. <input type="checkbox"/> Monika Waksmundzka-Hajnos, Joseph Sherma, Teresa Kowalska (2008).</li> </ol> <p>Thin Layer Chromatography in Phytochemistry. CRC Press<br/>Taylor &amp; Francis Group</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

#### Course schedule

| Week | Topic to be covered | Time |
|------|---------------------|------|
|------|---------------------|------|

|         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |       |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 1.      | 2. General introduction about phytochemistry<br>2.1. Biological activity of phytochemicals<br>2.2. Major Classes of phytochemicals                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 4 hr  |
| 2- 3    | 3. Major secondary metabolic pathway <ul style="list-style-type: none"> <li>○ The acetate pathway</li> <li>○ The shikimate pathway</li> <li>○ The mevalonate pathway</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                    | 4 hr  |
| 4 -7    | 4. Steps in phytochemical screening <ul style="list-style-type: none"> <li>➤ Selection and collection of plant materials</li> <li>➤ Initial and bulk extraction methods</li> <li>➤ Preliminary phytochemical screening</li> <li>➤ Bioassay methods for phytochemicals <ul style="list-style-type: none"> <li>• <i>In vitro</i> evaluation (Anti-microbial evaluation methods, Anti-oxidant evaluation methods)</li> <li>• <i>In vivo</i> evaluation (Anti-malarial evaluation methods, Anti-diabetic evaluation methods)</li> </ul> </li> <li>➤ Isolation techniques</li> <li>➤ Structural elucidations</li> </ul> | 12 hr |
| 8-10    | 5. Extraction and Isolation of Plant Secondary Metabolites<br>5.1. General extraction and Isolation methods for alkaloidal extracts from crude plant material<br>5.2. General extraction and Isolation methods to obtain saponin from plants<br>5.3. General extraction and Isolation methods to obtain Flavonoids from plants<br>5.4. General extraction and Isolation methods to obtain tannins from plants<br>5.5. General extraction and Isolation methods to obtain volatile oils from plants                                                                                                                 | 8     |
| 11      | 6. Marine origin Natural Products                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 4 hr  |
| 12 - 14 | 7. Laboratory demonstration <ul style="list-style-type: none"> <li>➤ Extraction (simple maceration, steam distillation)</li> <li>➤ Evaporation techniques (Oven, Rotary vapor)</li> <li>➤ Isolation techniques (Types of TLC, TLC jar, sample preparation and spotting, Column chromatography, Column packing, Silica gel, Alumina), HPLC</li> </ul>                                                                                                                                                                                                                                                               | 9 hr  |
| 14 - 15 | ➤ Visit to small medicinal plant garden (Identification of medicinal plants and their medicinal values)<br>Visit to Traditional Medicine clinic                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 6 hr  |



|    |                                                                                                                                                                                                                                                                                                |      |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 16 | Seminar presentation <ol style="list-style-type: none"> <li>1. Nutraceuticals and their roles in health and disease prevention</li> <li>2. Cytotoxic and carcinogenic compounds of plant origin</li> <li>3. Some selected Ethiopian herbal medicine and their drug-herb interaction</li> </ol> | 3 hr |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|

**Course Name:** Pharmaceutical Manufacturing (Elective)

**Course code:** Phar4253

**Module Name:** Professional Elective Module

**Module Code:** 25

**Course ECTS:** 5

**Totally required hours for the course: 135hrs**

**Lecture hours:** 48

**Study hours:** 42

**Field visit:** 15

**Project work:** 20

**Presentation(s):** 10

**Year/Semester Course is offered:** IV/II

**Course prerequisite/s:** Pharmaceutical Technology Module

**Course Description:**

This elective course is designed to prepare graduate pharmacists towards the practical aspects of manufacturing of dosage forms with particular emphasis on formulation, processing and regulatory affairs. The course also introduces some advanced/novel drug delivery systems.

**Course Objectives:**

After completion of this course students will be able to:

- Select appropriate pharmaceutical excipients for a specific dosage form
- Apply principles and techniques of aseptic processing in sterile pharmaceutical manufacturing
- Prepare different sterile and non-sterile preparation at hospital settings
- Understand the regulatory affairs related with product manufacturing and marketing
- Describe some advanced/novel drug delivery systems

**Course mode of delivery:** Block

**Course learning and teaching methods:**

- Illustrated lectures and discussions, student research project, field trip, individual and group exercises and assignments presentation, guided reading

**Assessment techniques:**

- Group and assignments:10%
- Journal club: 10%
- Tests: 15%

- Quizzes: 15%
- Seminar presentation: 10%
- Final Exam: 40%

### **Teachers' and students' role**

#### **Roles of Instructors**

##### **The instructor will be expected to:**

- Be a facilitator (introduce the subject; give guidance, moderate discussions, etc.)
- Read and comment assignments & presentations of students on time;
- Prepare his/her lessons and deliver lectures;
- Provide available and necessary reference materials;
- Encourage active participation of students in the teaching learning process;
- Assist in a field trip,
- Assist students with learning difficulties

#### **Roles of Students**

##### **Students are expected to:**

- Engage in learning by doing (independent study, group works/exercises, etc.)
- Be active learners (participate effectively in group assignments, class activities/presentations, field study etc.);
- Attend classes regularly (Both theory, field study and presentations)

**References:**

1. The Theory and Practice of Industrial Pharmacy, L. Lachman, H. A. Liberman and J. L. Kanig, 4th ed., Lea & Febiger, Philadelphia, 2015.
2. Handbook of Pharmaceutical Excipients: The American Pharmaceutical Association and the Pharmaceutical Society of Great Britain, Washington DC and London, 1986.
3. M. E. Aulton, Pharmaceutics: the science of dosage form design, 7th ed., Churchill Livingstone, Edinburgh.
4. Pharmaceutical Manufacturing Handbook: Production and Processes, S. C. Gad, John Wiley & Sons, Inc., Hoboken, New Jersey, 2008.
5. Pharmaceutical Manufacturing Handbook: Regulations and Quality, S. C. Gad, John Wiley & Sons, Inc., Hoboken, New Jersey, 2008.
6. Excipient Development for Pharmaceutical, Biotechnology, and Drug Delivery Systems, A. Katdare, and M. V. Chaubal, Informa Healthcare USA, Inc, New York, 2006.
7. WHO Guidelines: GMP

**Course Schedule**

| Week | Contact Hours | Topic/sub-topic/chapter/Assessments/Assignments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | 4             | <b>Pharmaceutical Excipients</b> <ul style="list-style-type: none"><li>○ Definition and goals (why excipients?)</li><li>○ Type of pharmaceutical excipients</li></ul> <b>Excipients for solid dosage forms</b> <ul style="list-style-type: none"><li>○ Desirable characteristics</li><li>○ Diluents, directly compression fillers, granulating agent (binder), disintegrant, lubricant/glidant/antiadherent, coating excipients (polymers, colorants, opaquants, polishing agents)</li></ul> <b>Excipients for liquid/semisolids dosage forms</b> <ul style="list-style-type: none"><li>○ Desirable characteristics</li><li>○ Solvents/vehicles, cosolvents, solubilizers, preservatives, antioxidants, chelating agents,</li></ul> |

|   |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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|   |   | <p>suspending agents, wetting agents, surfactants, viscosity modifying agents, emulsifying agents, buffers, tonicity agents, sweeteners/ flavors, coloring agents, ointment base, emollients, gelling agents, suppository bases and others</p> <ul style="list-style-type: none"> <li>• Drug - Excipient Compatibility</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 2 | 4 | <p><b>Manufacturing of Sterile Pharmaceuticals</b></p> <ul style="list-style-type: none"> <li>• Introduction</li> <li>• Sterile pharmaceutical products: types and formulation aspects</li> <li>• Aseptic Processing <ul style="list-style-type: none"> <li>○ Facility design, GMP requirements and process flow, Clean room: design and qualification standards, HVAC systems, Principles of aseptic processing and media fill, environment and personnel monitoring</li> </ul> </li> <li>• Quiz</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 3 | 4 | <p><b>Manufacturing of Sterile Pharmaceuticals (Continued...)</b></p> <ul style="list-style-type: none"> <li>• Production Activities <ul style="list-style-type: none"> <li>○ Water treatment, Material and component entry, Cleaning and preparation, Compounding, Filling and Stoppering/Crimping/Sealing</li> </ul> </li> <li>• Terminal Sterilization/Depyrogenation <ul style="list-style-type: none"> <li>○ Techniques for sterilization (Thermal methods: steam sterilization, dry heat sterilization, Chemical sterilization: gas sterilization, other methods: radiation sterilization, sterilization by filtration)</li> <li>○ Techniques for depyrogenation (Acid base hydrolysis, oxidation, alkylation, thermal and ionization radiation depyrogenation)</li> </ul> </li> <li>• Quality Control Tests <ul style="list-style-type: none"> <li>○ Particle size and clarity test, pH and osmolality test, sterility test, pyrogen test</li> </ul> </li> </ul> |

|    |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| 4  | 3 | <ul style="list-style-type: none"> <li>• Hospital Manufacturing <ul style="list-style-type: none"> <li>○ Introduction</li> <li>○ Water treatment</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                |
| 5  |   | <ul style="list-style-type: none"> <li>• Test I</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 6  | 3 | <ul style="list-style-type: none"> <li>• Hospital Manufacturing (Continued...) <ul style="list-style-type: none"> <li>○ Sterile and non-sterile preparations (TPN, radiopharmaceuticals,</li> </ul> </li> <li>• Quiz</li> </ul>                                                                                                                                                                                                                                                                        |
| 7  | 4 | <ul style="list-style-type: none"> <li>• Good Manufacturing Practices and Validation <ul style="list-style-type: none"> <li>○ Good Manufacturing Practices <ul style="list-style-type: none"> <li>▪ Basic principles</li> <li>▪ Quality management</li> <li>▪ Sanitation and hygiene, premises, equipment, material, personnel, product, QC, compliant handling and product recall, documentation</li> <li>▪ Heating Ventilation and Air Conditioning (HVAC) system</li> </ul> </li> </ul> </li> </ul> |
| 8  | 4 | <ul style="list-style-type: none"> <li>• Good Manufacturing Practices and Validation (Continued...) <ul style="list-style-type: none"> <li>○ Validation and Qualification <ul style="list-style-type: none"> <li>▪ Introduction, Regulatory basis (principle)</li> <li>▪ Validation and QA, Process validation (Prospective, concurrent and retrospective validation), Cleaning Validation, Sterilization validation, Analytical method validation, Revalidation,</li> </ul> </li> </ul> </li> </ul>   |
| 9  | 2 | <ul style="list-style-type: none"> <li>• Good Manufacturing Practices and Validation (Continued...) <ul style="list-style-type: none"> <li>○ Validation and Qualification (Continued...) <ul style="list-style-type: none"> <li>▪ Facility and equipment qualification (DQ, IQ, OQ and PQ), Change control</li> </ul> </li> </ul> </li> </ul>                                                                                                                                                          |
| 10 |   | <ul style="list-style-type: none"> <li>• Test II</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                            |

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|    |   | <ul style="list-style-type: none"> <li>• Assignment</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 11 | 4 | <ul style="list-style-type: none"> <li>• Regulatory Affairs <ul style="list-style-type: none"> <li>○ General Guidance and Principle</li> <li>○ Pharmaceutical plant inspection</li> <li>○ Inspection for premises licensing, Inspection for GMP certification, Inspection of API plants</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                          |
| 12 | 4 | <ul style="list-style-type: none"> <li>• Regulatory Affairs (Continued...) <ul style="list-style-type: none"> <li>○ Registration/Marketing authorization of medicines</li> <li>○ General</li> <li>○ Administrative and Product Information (Application form, agency agreement etc), Dossier Overall Summary (DOS), Quality (Drug substance and drug product), Non-Clinical Study Reports (Pharmacology, Pharmacokinetics and Toxicology), Clinical Study Reports (Reports of biopharmaceutic study, BA, Comparative BA and BE, Invitro-Invivo Correlation, human pharmacokinetic study, human pharmacodynamic study, report of efficacy and safety studies, study of controlled clinical study)</li> </ul> </li> </ul> |
| 13 | 4 | <ul style="list-style-type: none"> <li>• Regulatory Affairs (Continued...) <ul style="list-style-type: none"> <li>○ Quality Control Testing</li> <li>○ Inspection and Post Marketing Surveillance</li> <li>○ Bioequivalence and Product Interchangeability</li> <li>○ Principles of interchangeability testing, Design of BE studies, Selection of comparators</li> <li>○ Basic statistical and analytical considerations, Regulatory requirements for bioequivalence and existing guidelines, Presentation of BE data in product dossier, BE study assessment - practical issues</li> </ul> </li> </ul>                                                                                                                |
| 14 | 4 | <ul style="list-style-type: none"> <li>• Advanced Pharmaceutical Dosage Forms <ul style="list-style-type: none"> <li>○ Microencapsulation</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

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|----|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |  | <ul style="list-style-type: none"> <li>○ Liposomes and micelles, nanoparticles, hydrogel based drug delivery, Introduction to novel drug delivery systems: overview, classifications and structure, physicochemical properties and applications</li> </ul> |
| 15 |  | <ul style="list-style-type: none"> <li>• Test III</li> </ul>                                                                                                                                                                                               |
| 18 |  | <b>FINAL EXAM</b>                                                                                                                                                                                                                                          |



**Course syllabus Pharmacogenetics**

**Module name:** Professional electives

**Module Number in which the course exists:** 25

**Course title:** Pharmacogenetics

**Course code:** Phar 4254

**Module name:** Elective module

**Module code:** Phar-M4252

**Course ECTS:** 5 ECTS (135 hrs)

- Lecture: 48 hours
- Practical: 20 hours
- Tutorial: 24 hours
- Home study: 30 hours
- Assignment and presentation: 16 hours
- Assessment : 7 hours

**Year/Semester Course is offered:** Year IV Semester II

**Contact hours/ week:** 135- 40= 95 hours/ 16 weeks= 6 hours

**Pre-requisite:** Pharmacology I and II

**Course Description:**

Pharmacogenetics is aimed at advancing knowledge of the genetic basis for variable drug response. The ultimate goal of offering this course is to enable students to understand and identify clinically significant variations to predict the right choice and dose of medications for individuals--“personalizing medicine” drugs with narrow therapeutic index and a large proportion of patients do not respond. The course starts by introducing the concepts of Pharmacogenetics, the scope and its application. It also describes an introduction about the human genome, genetic variations and genetic code. In addition the course tries to emphasize on the principles of Pharmacogenetics in relation to individualizing therapy and the techniques used in Pharmacogenetic studies. Pharmacogenetics of pharmacokinetics and Pharmacodynamics will also be discussed. Finally as pharmacogenomic advances allow for individualized drug therapies based on genotypic information, future directions in personalized medicine will be discussed with examples on Pharmacogenetic of Cardiovascular, Psychiatry, Infectious, hematology and Oncology.

**Course Objectives**

To understand genetic factors underlying efficacy/toxicity of drug therapy; to assess the value of phenotyping/genotyping in guiding drug therapy of individual patients

### **Learning Objectives:**

Upon completion of this course, the student will be able to:-

- Explain the basic principles of human genetics and Pharmacogenetics with its application
- Apply the principles of molecular and cellular biology to explain the genetic basis of variability in drug response.
- Describe the various biochemical/molecular biology methods used to determine genotype and polymorphic variability.
- Discuss how genetic variability in genes encoding drug metabolizing enzymes, drug transporting proteins, and drug receptors (targets) can contribute to variability in drug disposition and action, leading to changes in pharmacokinetics, pharmacodynamics and clinical outcome.
- Apply pharmacogenomic concepts to a particular drug therapy to solve relevant problems in pharmaceutical care.
- Critically evaluate the current and future literature in the area of pharmacogenomics.

### **Course EtCTS: 5 (135 hours)**

- Lecture: 48 hours
- Tutorial: 16 hours
- Home study: 20 hours
- Presentation : 8

### **Course mode of delivery: Block/Parallel**

### **Course learning and teaching methods**

Active learning methods (brain storming, discussion, etc), Lecture, group and individual presentation, assignment, and Practical visit to Emergency care unit to see treatment of poisoned patients.

### **Assessment techniques:**

Continuous assessment & summative assessment

- Quiz (10%)
- Tests (20 %)
- Assignments (10 %)
- Presentations on practical visit (10 %)
- Final Exam (40%)

### **Teachers and Students Role**

#### **Role of Instructor**

The instructor will be expected to:

- Facilitate students' individual and group activities
- Organize students' hospital visit, laboratory work, workshop practices (if any) and project work presentation(s) and discussion sessions
- Assess students' performances (written and oral presentations)
- Provide timely feedback orally and in writing
- Make follow-up on developments made
- Plan and implement students' consultation program

## Role of Students

Students are expected to:

- Attend sessions
- Carry out individual and group tasks
- Active participant
- Reflect on feedbacks and take actions
- Carry out reading assignment

## References

### Required reading (text)

1. Pharmacogenomics. Applications to Patient Care. American College Clinical Pharmacy, Kansas City, MO. 2004.

### Recommended Reading

2. Strachan T, Read AP. Human Molecular Genetics. 3rd ed. New York, NY: Garland Science; 2004.
3. Pharmacogenetics and Individualized Therapy. Ed. Maitland-van der Zee & Ann Daly; (2011).

**Course Schedule:** contact time, contents/topics & reading/reference materials for each topic

| Week | Contact hrs | Topic/sub-topic/chapter/Assessment/Assignments                                                                                                                                                                                                       | Reading materials |
|------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| 1    | 3           | <b>1. Introduction</b> <ul style="list-style-type: none"> <li>➤ Definitions of pharmacogenetics and pharmacogenomics</li> <li>➤ History of pharmacogenetics and pharmacogenomics</li> <li>➤ Scope and application in the biomedical field</li> </ul> | A, B              |
| 2    | 3           | <ul style="list-style-type: none"> <li>○ Introduction to human genome, evolving concepts of genes/locus.</li> </ul>                                                                                                                                  | A, B              |

|    |   |                                                                                                                                                                                                                                                            |         |
|----|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
|    |   | <ul style="list-style-type: none"> <li>○ Introduction to genetic variation (Mechanism/cause of genetic and epigenetic variations, types of variants, SNPs, coding and cis/trans regulatory variants, insertion/deletions, copy number variants)</li> </ul> |         |
| 3  | 3 | <ul style="list-style-type: none"> <li>➤ The genetic code (Information flow in biological system, Replication processes, Transcription, Translation, Regulation of gene expression)</li> </ul>                                                             | A, B, C |
| 4  | 3 | <b>2. Principles of Pharmacogenetics</b> <ul style="list-style-type: none"> <li>○ Genetic diversity</li> <li>✓ Overview on sources of variability in drug disposition and response</li> </ul>                                                              | A, B    |
| 5  | 3 | <ul style="list-style-type: none"> <li>✓ Individualization and optimizing drug therapy for phenotypes and genotypes.</li> <li>✓</li> </ul>                                                                                                                 | A, B    |
| 6  | 3 | <ul style="list-style-type: none"> <li>✓ Clinical Implementation of Pharmacogenetics (PGx) and Dosing Guidelines</li> </ul>                                                                                                                                | A, B, C |
| 7  | 3 | <b>3. Pharmacogenetic techniques</b> <ul style="list-style-type: none"> <li>➤ Genotyping (DNA sequencing, , Microarrays, Polymerase chain reaction (PCR),</li> </ul>                                                                                       | A, B, C |
| 8  | 3 | <ul style="list-style-type: none"> <li>➤ SNP Identification (RFLP techniques, Allele specific amplifications, Blotting techniques, Gene cloning</li> <li>➤ Phenotyping</li> </ul>                                                                          | A, B, C |
| 9  | 3 | <b>4. Pharmacogenetics of drug pharmacokinetic profile</b> <ul style="list-style-type: none"> <li>➤ Pharmacogenetics of drug metabolism and activation</li> <li>➤ Overview of CYP450 enzyme families</li> </ul>                                            |         |
| 10 | 3 | <ul style="list-style-type: none"> <li>✓ Pharmacogenetics of phase I drug metabolism</li> </ul>                                                                                                                                                            | A, B, C |
| 11 | 3 | <ul style="list-style-type: none"> <li>➤ Pharmacogenetics of phase I drug metabolism Con...</li> </ul>                                                                                                                                                     | A, B,C  |
| 12 | 3 | <ul style="list-style-type: none"> <li>➤ Pharmacogenetics of phase I drug metabolism Con...</li> <li>➤ Pharmacogenetics of Phase II drug metabolism</li> </ul>                                                                                             | A, B,C  |
| 13 | 3 | <ul style="list-style-type: none"> <li>➤ Pharmacogenetics of drug transporters</li> </ul>                                                                                                                                                                  | A, B,C  |
| 14 | 3 | <b>6. Pharmacogenetics of drug response</b> <ul style="list-style-type: none"> <li>➤ Pharmacogenetics of drug targets</li> </ul>                                                                                                                           | A, B,C  |
| 15 | 3 | <ul style="list-style-type: none"> <li>➤ Pharmacogenetics of Cellular signaling pathways</li> </ul>                                                                                                                                                        | A, B, C |

|    |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |         |
|----|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| 16 | 3 | Pharmacogenetics in Drug Discovery and Drug Development                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | A, B, C |
| 17 | 3 | <b>5. Future Directions in Personalized Medicine</b><br>✓ Pharmacogenetics: Cardiovascular Diseases (PGx guided dosing of statins, PGx guided dosing of Warfarin and Warfarin Pharmacogenetics PGx guided dosing of clopidogrel.                                                                                                                                                                                                                                                                                                            | A, B, C |
| 18 | 3 | ✓ Pharmacogenetics: Central Nervous System and Psychiatry (PGx guided dosing of antidepressant and antipsychotic therapy. *PGx guided dosing in pain management.<br>✓ Pediatric Pharmacogenomics (Variants affecting common drugs used in children and special considerations of genetics in this population (e.g ADHD, PPIs, codeine, asthma)<br>✓ Genetic Counseling (Counseling on genetic findings, tools to assist in counseling, how to explain genetic and genetic variation to patients, disease risk vs pharmacogenomic variants ) | B, D    |
| 19 | 3 | ✓ Pharmacogenetics: Transplantation ( <b>Presentation</b> )<br>✓ Pharmacogenetics: Oncology and Hematology (PGx guided dosing of anticancer agents) ( <b>Presentation</b> )                                                                                                                                                                                                                                                                                                                                                                 |         |
| 20 |   | <b>FINAL EXAM</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |         |

**Module name:** Professional electives

**Module Number in which the course exists:** 25

**Course title:** Pharmaceutical Quality Control and Quality Assurance

**Course code:** Phar4255

**Course EtCTS:** 5

**EtCTS credits:** 5 (This course needs a total of  $5 \times 27 = 135$  working hours to spend in each teaching and learning as well as assessment activities). The distribution of these hours will be as follows

- Lectures =48 hours
- Assignment (group and individual) =10 hours
- Practical/laboratory sessions: 16hrs
- Home based study=31 hours
- Group discussion and presentation=10 hours
- Tutorial=10 hours

Pre-requisite if any: Pharmaceutical Analysis I & II

**Course description:**

The course deals with different quality aspects of pharmaceutical products starting from their production to consumption by the customers. The course mainly covers areas of GMP and different quality control measures taken after the drug is released to market.

**Course objectives:**

At the end of the course, the student will be able to:

- ✓ Describe the concepts and philosophies of TQM AND GMP
- ✓ Explain the different manufactures and controls taken on dosage forms
- ✓ Describe good laboratory practice and standard operating procedures
- ✓ Describe the major concepts of packaging and labeling of pharmaceuticals

**References:**

1. Quality Assurance Guide by Organisation of Pharmaceutical products of Ethiopia.
2. Good Laboratory Practice Regulations, 2nd Edition, Sandy Weinberg, Vo. 69, Decker Series
3. Quality Assurance of Pharmaceuticals – A compendium of guidelines and related materials – Vol. I – WHO Publications
4. A guide to Total Quality Management – Kaushik Maitra and Sedhan K. Ghosh.
5. How to practice GMPs – P. P. Sharma

6. ISO 9000 and Total Quality Management – Sadhank. G. Ghosh.
7. The International Pharmacopoeia Vol. 1,2,3,4 - 3rd Edition, General Methods of Analysis and Quality specification for Pharmaceutical Substances, Excipients and Dosage forms

**Schedule of chapters/topics/subtopics, allotted time and reference materials for each topic**

| <b>Week</b> | <b>Chapters and topics/subtopics</b>                                                                                                                                                                                                                                                           | <b>Time allotted (hrs)</b> |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| 1           | <b>Chapter One:</b> Introduction to pharmaceutical regulation World, Africa, Ethiopia)                                                                                                                                                                                                         | 4                          |
| 1           | <b>Chapter Two:</b> Drug registration Process, Import and Export permit in Ethiopia<br>Pharmaceutical establishments ;<br>Brief over view about Setting and minimum quality requirments for a pharmacy, whole sale, import, small scale compounding and manufacturing, Pharmaceutical industry | 4                          |
| 2           | <b>Chapter three:</b><br>Concepts and Philosophy of TQM, GMP (orange guide) and ISO-90004                                                                                                                                                                                                      | 4                          |
| 3           | <b>Chapter four:</b><br>Organization and personnel responsibilities, training, hygiene                                                                                                                                                                                                         | 4                          |
|             | Chapter Five: Premises                                                                                                                                                                                                                                                                         | 4                          |
|             | Location, Design, Plan Layout, Construction, Maintenance and Sanitations.                                                                                                                                                                                                                      |                            |
| 3           | Environmental control, Sterile areas, control of contamination                                                                                                                                                                                                                                 | 4                          |
|             | <b>Chapter six: Equipments :</b>                                                                                                                                                                                                                                                               |                            |
|             | Selection, purchase specifications, maintenance, sterilization of an area (TP & STP)                                                                                                                                                                                                           |                            |

|   |                                                                                                                                                                                                |   |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 4 | Selection, purchase specifications, maintenance, sterilization of an area (TP & STP)                                                                                                           | 4 |
|   | <b>Chapter seven: Raw Materials :</b>                                                                                                                                                          |   |
|   | Purchase specifications, Maintenance of stores, Selection of vendors, Controls on Rawmaterials                                                                                                 |   |
|   | <b>Chapter eight: Manufacture and controls on dosage forms</b>                                                                                                                                 |   |
|   | Manufacturing Documents, Master Formula, Batch Formula                                                                                                                                         |   |
| 5 | Records, Standard operating procedure, Quality audits of manufacturing processes and facilities                                                                                                | 4 |
|   | <b>Chapter Nine:</b>                                                                                                                                                                           |   |
|   | Standard operating procedures for various operations like cleaning, filling, drying,compression, coating, disinfection, sterilisation, membrane filtration                                     |   |
| 6 | Standard operating procedures for various operations like cleaning, filling, drying,compression, coating, disinfection, sterilisation, membrane filtration                                     | 4 |
|   | <b>Chapter Ten: Packaging and labeling controls</b>                                                                                                                                            |   |
|   | Line clearance, reconciliation of labels; cartons and other packagingmaterial; typesandtestsassuringquality of glass.                                                                          |   |
| 7 | Types of plastics used, permeation, leaching, sorption, chemical reactions, biological tests, modification of plastics by drugs;                                                               | 4 |
|   | Different types of closures and closure liners; film wrapper; Blisterpacks, Bubble packs, shrink handling; foil / plastic pouches, bottle seals, tape seals, breakable seals and sealed tubes; |   |
| 8 | Quality control of packaging material and filling equipment                                                                                                                                    | 4 |
|   | <b>Chapter Eleven: Quality control Laboratory :</b>                                                                                                                                            |   |



|      |                                                                                                                              |   |
|------|------------------------------------------------------------------------------------------------------------------------------|---|
|      | Responsibilities, Good Laboratory Practices, Routine controls, Instruments, Protocols, Non-clinical testing                  |   |
| 8    | Controls on animal house, Application of Computers in Quality control laboratory.                                            | 4 |
|      | <b>Chapter Twelve:</b> Finished product release :                                                                            |   |
|      | Quality review, Quality audits, Batch release document                                                                       |   |
| 9-10 | Quality review, Quality audits, Batch release document                                                                       | 4 |
|      | <b>Chapter Thirteen:</b> Warehousing :                                                                                       |   |
|      | Good warehousing practice, Materials, Managements.                                                                           |   |
|      | <b>Chapter Fourteen:</b><br>Waste disposal, Scrap disposal procedure and records.                                            | 4 |
| 11   | Waste disposal, Scrap disposal procedure and records.                                                                        | 4 |
|      | <b>Chapter Fifteen:</b>                                                                                                      |   |
|      | Regulatory aspects of Pharmaceuticals and Bulk drug Manufacturing<br>Regulatory drug analysis                                |   |
| 12   | <b>Chapter Sixteen:</b> WHO Certification, Globalisation of Drug Industry, Introduction to Export of Drugs and Import Policy | 4 |
| 13   | Chapter Seventeen: Harmonization process in pharmaceuticals and over view                                                    | 4 |
| 14   | Chapter Eighteen: <b>Good Documentation Practice</b>                                                                         | 4 |
| 15   | <b>Pharmaceutical waste and its handling process, Regulatory requirements</b>                                                | 4 |
|      |                                                                                                                              |   |

**Delivery mode/methodology:**

- ✓ Active learning methods (brain storming, group discussions, etc),
- ✓ Lecture,
- ✓ group and individual presentation,
- ✓ assignment

✓ Laboratory practice

**Assessment mechanisms:**

#### Continuous assessment & summative assessment

- Quiz (10%)
- Tests (10%)
- Assignments (15%)
- Presentation (10%)
- Lab reports and exam (20%)
- Final Exam (35%)

## **Drug Design and Synthesis (Phar4256) (Elective)**

### **Course Title: Drug Design and Synthesis**

**Prerequisite:** - Organic Chemistry, Medicinal Chemistry I and II

**Course description:** Drug design and synthesis is one of professional elective course of Pharmacy students that covers laboratory safety rules, drug discovery, drug development drug synthesis and drug designing methods, lead identification and modifications, stereochemistry, chemical reactions and different drug design techniques. In addition, this course will study about the process of drug delivery and phases in drug discovery and development. Moreover, this course covers the synthesis of different class of drugs and structure characterization using different spectroscopic techniques.

**Course Objective:** After completion of this course, students will be able to understand drug discovery, development, designing, drug targets and synthesis of various class of drugs that act on different systems and organs of the body.

### ***Teaching Methods:***

Total hours 114hrs

Lecture 48 hrs

Seminar 12 hrs

Reading assignment 54 hrs

### ***Assessment methods***

Assignments (20%)

Seminar (20%)

Written Examination (Tests 30%; Final Exam 30%)

## Course Content

- 1. Safety in Medicinal Chemistry Laboratory ..... 2hrs**
- 2. Introduction to Drug Discovery and Development ..... 8hrs**
  - 2.1. History of drug discovery
  - 2.2. Strategies in drug discovery, lead discovery, pharmacophore identification, lead development, Bioassays, screening of compounds
  - 2.3. Definition of terms (drug discovery, drug design, development and synthesis....)
  - 2.4. Pharmacokinetics and Pharmacodynamics of drug action
  - 2.5. Pre-clinical and Clinical Testing
  - 2.6. Stages in Drug Discovery and Development
- 3. Drug targets (Membrane Proteins, DNA, RNA, Enzymes) ..... 2hrs**
- 4. Lead Identification and Modification ..... 2hrs**
  - 4.1. Lead Identification and High Throughput Screening
  - 4.2. Lead Identification and Modification Practical
- 5. Fundamentals of Rational Design ..... 6hrs**
  - 5.1. Structure-Activity Relationships and Elements of Structure-Activity Relationship
  - 5.2. Quantitative Structure-Activity Relationships
- 6. Computer-Aided Drug Design (CADD) ..... 8hrs**
  - 6.1. Molecular Modeling (MD)
  - 6.2. Ligand-Based Drug Design (LBDD)
  - 6.3. Structure Determination
  - 6.4. Structure-Based Drug Design (SBDD)
- 7. Drug Delivery ..... 4hrs**
  - 7.1. Bioavailability
  - 7.2. Pro-drugs and Drug Delivery
- 8. Drug Synthesis Approaches ..... 4hrs**
  - 8.1. The Synthon approaches
  - 8.2. The Retro approach
  - 8.3. Synthesis of Some Common Drugs
- 9. Common Reactions in Medicinal Chemistry ..... 8hrs**
  - 9.1. Amide Formation Reactions
  - 9.2.  $S_NAr$  Reactions
  - 9.3.  $E1$  and  $E2$  Reactions
  - 9.4. Boc Protection/ Deprotection Reactions

9.5. Ester Hydrolysis Reactions

9.6. Suzuki–Miyaura Coupling Reactions

**10. Stereochemistry .....4hrs**

### Reference Books

1. **Kristian Strømgaard** Textbook of Drug Design and Discovery, Fifth Edition CRC Press; (2016)
2. Rama Rao Nadendla (2005). Principles of Organic Medicinal Chemistry. New Age International (p) Limited, Publishers
3. Donald J. Abraham (Ed.). Burgers's medicinal Chemistry and Drug Discovery, 2003, 6<sup>th</sup> edn., vol1-6, wiley-interscience (USA)
4. Gareth Thomas (2003). Fundamentals of Medicinal Chemistry. John Wiley & Sons Ltd, University of Portsmouth, UK
5. Thomas, L.Lemeke and David, A. Williams. Principle of Medicinal Chemistry, 2002, 5<sup>th</sup> edn. A Lea and Febiger book, Williams and Wilkins
6. Strategies for Organic Drug Synthesis and Design by Daniel Lednicer. Hardcover Wiley-Interscience; 1<sup>st</sup> edition (1998)
7. King, F.D. Medicinal Chemistry, Principles and Practice. The Royal Society of Chemistry, 1997.

## **Introduction to Warehouse management course syllabus**

**Module name:** Professional elective

**Module Number in which the course exists:** 25

**Course title:** Introduction to Ware house management

**Course code:** Phar 4257

**Course EtCTS:** 5(This course needs a total of  $5 \times 27 = 135$  working hours to spend in each teaching and learning as well as assessment activities). The distribution of these hours will be as follows

- Lecture: 48 hours
- Project work: 14 hours
- Presentations=10 hours
- Case studies/journal club=10 hours
- Tutorial: 8 hours
- Home study: 38 hours
- Assessment=7 hours

Pre-requisite if any: Pharmaceutical supply chain management

### **Course Description**

This course aims to introduce the students to the fundamental nature of inventory from a financial, physical, forecasting, and operational standpoint. The ultimate goal of this course is to present immediately usable information in the areas of forecasting, physical control and layout, and problem recognition and resolution. The course materials should enable students to:

- Understand that modern practice discourages holding large quantities of inventory.
- Grasp the significance of controlling actual, on-hand inventory as both a physical object (shelf count) and as an intangible object (record count and monetary worth).
- Understand the fundamental differences between finished goods inventories in the retail/distribution sectors and raw materials and work-in-process inventories found in the manufacturing environment.
- Understand basic formulas to calculate inventory quantities.
- Employ basic problem-solving techniques toward issue resolution

### **COURSE OBJECTIVES**

This course aims to help the supply chain professional to understand and apply four (4) major and significant aspects of inventory management: - Optimize Inventory Levels - Build an Inventory Management Plan - Design & Manage Warehouse Operations - Increase Accuracy, Traceability & Reduce Parts Variety Participants should be able to understand the financial impacts of inventory and the risks in both over and under holding of inventory - the management of inventory, including lead time management, demand planning and interfacing with other functional groups directly and indirectly involved in inventory planning and operations. They should also be able to understand the importance of effective Warehouse Management in minimizing the cost associated with the storing, moving and transporting goods into and out of the warehouse storage locations; the importance of reconciled physical count balances and system records and most importantly,

the efficient, fast, precise and perfectly-timed issuances of the right quality & quantity of stocks to its intended users.

| <b>Week</b> | <b>Contact hrs</b> | <b>Topic/sub-topic/chapter</b>                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1           | 4                  | 2.1. Introduction – Review of the “Fundamentals”<br>1.1 What is Inventory Management - Working Capital Cycle<br>1.2 Why is Inventory Management important<br>1.3 Why keep Inventory<br>1.4 How much Inventory to Keep<br>1.5 The Financial Implications of Holding Inventory - Inventory Carrying Cost - Effect on Financial<br>1.6 The Cost of not holding enough Inventory<br>1.7 The Role of the Inventory Manager<br>1.8 Exercises / Practical Application to Workplace |
| 2           | 4                  | 2.Setting the Stage for Effective Inventory Management<br>2.1 Introduction to Effective Inventory Management<br>2.2 Inventory Management& the Supply Chain Strategy<br>2.3 Demand Forecasting<br>2.4 Lead time Management<br>2.5 Exercises / Practical Application to Work / Questions & Answers<br>2.6 Understanding SAP Fundamentals & Terminology                                                                                                                        |
| 3           | 4                  | 3.Inventory Planning<br>3.1 Introduction to Inventory Planning<br>3.2 Service Level Policies - OTIF<br>3.3 Inventory Categorization Techniques - ABC Analysis - Fast & Slow Moving, Excess, Obsolete & Defective Stocks<br>3.4 Traceability and Variety Reduction<br>3.5 Inventory Coding Systems<br>3.6 The Inventory Management Plan<br>3.7 Group Discussion / Practical Application to Work                                                                              |
| 4           | 4                  | 4 – Inventory Operations<br>4.1 Introduction to Inventory Operations<br>4.2 Monitoring Movements - Inventory Accuracy<br>4.3 Measuring and Valuation of Inventory<br>4.4 Receipt & Issuance of Inventory<br>4.5 Systems to Replenish Inventory<br>4.6 How Much to Order – EOQ,<br>4.7 When to Place an Order – ROP, JIT<br>4.8 Exercises / Practical Application to Work                                                                                                    |
| 5           | 4                  | 5 – Warehouse Planning & Systems<br>5.1 Introduction to Warehouse Planning & Systems<br>5.2 Warehouse Location & Acquisition Options<br>5.3 Warehouse Design<br>5.4 Warehouse Layout<br>5.5 Materials Handling & Equipment<br>5.6 Warehouse Operations<br>5.7 Record Keeping & Communication                                                                                                                                                                                |

|  |  |                                                                                                                                                                                      |
|--|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  |  | 5.8 Perpetual Systems/Continuous Review Systems<br>5.9 International Quality Standards<br>5.10 Physical Inventory & Cycle Counting<br>5.11 Exercises / Practical Application to Work |
|--|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Mode of delivery:**

- Illustrated Lectures and case studies
- Active learning methods (brain storming, buzz group, discussion, etc)
- Individual and group exercises and assignments
- Presentations and participation in class discussion
- Case studies

**Mode of assessment:**

- Quizzes and tests: 30%
- Attendance : 5%
- Case studies:5%
- Assignments (group or individual):20%
- Written final exam: 40%

**LEARNING MATERIALS:**

Ackerman, K.B. *Practical Handbook of Warehousing*. New York, NY: Van Nostrand Reinhold, 1993.

Albright, B. "Recession Impacts Supply Chain Markets: WMS Growth Slows; Transportation and Events Software Picks Up Speed." *Frontline Solutions* 3, no. 6 (2002): 10–12.

Bolten, E.F. *Managing Time and Space in the Modern Warehouse*. New York, NY: American Management Association, 1997.

Forger, G. "Leading Trends in Manufacturing, Warehousing & Distribution." *Modern Materials Handling* 59, no. 13 (December 2004): 38.

Friedman, D. "How to Select the Best Warehouse Management System." *Material Handling Management* 60, no. 1 (January 2005): 28–29.

Harrington, L.H. "How to Solve the Warehousing Puzzle." *Logistics Today* 44, no. 9 (September 2003): 32–38.

Johnson, J.R. "Warehousing's Crystal Ball." *Warehousing Management* 9, no. 6 (July 2002): 24–28.

"An Overview of Warehousing in North America—Market Size, Major 3PLs, Benchmarking Prices and Practices." *North America Warehousing Market Report 2004*. Stoughton, WI: Armstrong & Associates, Inc., 2004.

Singer, T. "Trends in Warehousing and Distribution." *Industrial Maintenance & Plant Operation* 65, no. 11 (November 2004): 12–18.



## **Research in pharmacology Course syllabus**

**Module title:** Elective

**Course code:** Phar-M4258

**Credit:** 5 ECTS

**Pre-requisite (s):** Pharmacology I and II, Clinical toxicology

**Co-requisite (s):** none

### **Course Description:**

This course is designed to enable students recognize research methodology including various methods used for evaluation of drugs acting on central nervous system, autonomic nervous system, cardiovascular system, gastrointestinal tract, blood, endocrine system, kidney and liver, screening for wound healing, and anti-microbial screening techniques. It also enables students evaluate structural activity relationships, perform acute, sub-acute and chronic toxicity studies and blinded and planned screening of molecules. It allows students describe laboratory animal science, *in-vivo* and *in-vitro* pharmacological experiments and bioinformatics. In addition it allows students demonstrate ability to provide animal care and comply with ethical considerations in animal experimentations. After successful completion of the course, the learners will be able to describe the processes of drug development and conduct actual tests using animal models to answer a research question.

### **Learning outcomes**

- Up on successful completion of this module, students will be able to:
- Explain the process of new drug development
- Describe experimental models of drug action evaluation for different pharmacological effects
- Analyze the structure activity relationship of biologically active compounds
- Execute preclinical toxicity studies
- Recognize different models of *in-vivo* and *in-vitro* pharmacological experiments
- Provide appropriate care for experimental animals
- Comply with ethics of animal experimentation

- Conduct pharmacological experiments in various models

### **Course contents**

1. Guideline for the care and use of animals
2. Common laboratory animals
3. Standard techniques (bleeding & IV injection, ventilation rate, intragastric administration, procedures for rendering animal unconscious, and chemical euthanasia)
4. Standard drugs and chemicals (physiological salt solutions, drugs and vehicles)
5. Basic equipments (recording of BP, contraction of isolated tissues)
6. Study on isolated muscle preparation
7. Quantitative study of agonists and antagonists on isolated preparation
8. Identification and estimation of biologically active substances
9. Assays on anesthetized animal preparations
10. Common evaluation techniques
  - Methods to induce experimental hypertension
  - Analgesic agents
  - Anti-inflammatory agents
  - Agents acting on GIT (Anti-ulcer, anti-secretory, those affecting intestinal motility)
  - Antifertility agents
  - Antidiuretic agents
  - Antitussive activity
  - Antidiabetic activity
  - Neuropharmacological agents

### **Teaching strategy/Methods**

- Illustrated and interactive lecture
- Collaborative learning through brainstorming, question and answer, and group discussion
- Classroom student presentation and discussion
- Audiovisual simulation and animation
- Laboratory demonstrations and practice

### **Assessment strategy**

Continues assessment shall be carried out

- Formative assessment:
  - Quizzes, Drills and class room inquiry
  - Peer assessment: comments on assignment presentation
- Summative assessment:
  - Continuous assessment (20)
  - Practical examination [30%]
  - Final examination [50%]

## Module 26 Pharmaceutical Research I

**Module Name:** Pharmaceutical Research I

**Module Category:** Core

**Module Code:** Phar-M4261

**Module Number:** 26

**Module Weight:** 3 ECTS

**Courses:**

| Course name      | Course code | ECTS |
|------------------|-------------|------|
| Research methods | Phar 4261   | 3    |

### Module description:

This module is intended to equip pharmacy students with a basic working knowledge of pharmaceutical research methods. It also gives the trainee an acquaintance with research proposal writing, critical appraisal of scientific paper and application of common statistical packages. The module includes hands on research experience in the form of a directed studies course, which offers the student to perform research which culminates in the submission of substantial research work in the form of a senior essay/directed studies report paper.

### Module objective:

After successful completion of this Module the students will be able to:

1. Differentiate the major types of study designs
2. Identify the main issues in the design, conduct and presentation of a research
3. Explain the major elements that need to be examined when making a critical assessment of a research paper.
4. Demonstrate how to deal with each of these elements with reference to a published paper
5. Demonstrate a basic understanding of common statistical packages useful for data processing and analysis
6. Explain the major components of research in the pharmaceutical sciences
7. Prepare a research protocol and conduct pharmaceutical research

### Module competencies:

Upon a successful completion of this module, students will be able to conduct with minimal supervision by a senior pharmacist (researcher) research in the different areas of pharmacy practice and/or pharmaceutical sciences..

**Mode of delivery (Parallel/Block):** Block

### Module teaching/learning method:

Learning Activities

Students are expected to undertake among others the following major learning activities:

- Engage in learning by doing (independent study, group assignments, presentation, report writing, etc. )
- Participation and note takings during class lectures and debates and discussions;
- Reviewing the literature
- Data collection/conducting experiment (as appropriate), analyse results, report write-up and presentaiton

Teaching Methods

The course facilitator is expected to:

- Lecture for introducing concepts and topics, and give references, facilitate discussions, ask questions, give and correct assessments, forming groups, encouraging students for peer learning
- Monitor student progress and provide feedback

**Module mode of assessment:**

- Quizzes
- Exam
- Assignments
- A directed studies report
- A directed studies presentation and defense

**Course Name:** Research Methods

**Course code:** Phar 4211

**Module Name:** Pharmaceutical Research

**Module Code:** Phar-M5211

**Course ECTS:** 3

**Totally required hours for the module:** 81

**Lecture hours:** 32

**Study hours:** 22

**Group work:** 4

**Project work:** 10

**Presentation(s):** 6

**Tutorial:** 4

**Assessment:** 3

**Year/Semester Course is offered:** Year IV Semester II

**Course prerequisite/s:** Epidemiology and biostatistics courses

#### **Course description:**

This module is intended to equip pharmacy students with a basic working knowledge and skills in the proper conduct of research in pharmacy practice and the pharmaceutical sciences. It also gives the trainee an acquaintance with research proposal writing, critical appraisal of scientific paper and application of common statistical packages.

#### **Course objective:**

After completion of this course students will be able to:

1. Differentiate the major types of study designs
2. Identify the main issues in the design, conduct and presentation of a research
3. Explain the major elements that need to be examined when making a critical assessment of a research paper.
4. Demonstrate how to deal with each of these elements with reference to a published paper
5. Demonstrate a basic understanding of common statistical packages useful for data processing and analysis
6. Explain the major components of research in the pharmaceutical sciences

**Course mode of delivery:** Block

#### **Course learning and teaching methods**

Learning Activities

Students are expected to undertake among others the following major learning activities:

- Engage in learning by doing (independent study, group assignments, presentation, report writing, etc...)
- Participation and note takings during class lectures and debates and discussions;

Teaching Methods

The course facilitator is expected to:

- Lecture for introducing concepts and topics, and give references, facilitate discussions, ask questions, give and correct assessments, forming groups, encouraging students for peer learning

**Assessment techniques:**

## Formative / continuous and summative assessment

| Assessment methods             | Marks in % | Assessment time |
|--------------------------------|------------|-----------------|
| Test 1                         | 10         | Week 2          |
| Group work with presentation 1 | 5          | Week 4          |
| Test 2                         | 10         | Week 6          |
| Assignment                     | 10         | Week 8          |
| Test 3                         | 10         | Week 10         |
| Group work with presentation 2 | 5          | Week 12         |
| Practice (proposal writing)    | 5          | Week 13-15      |
| Final                          | 40         | Week 16         |
| Total                          | 100%       |                 |

### Teachers' and students' role

#### Roles of Instructors

The instructor is expected to:

- Be a facilitator (introduce the subject; give guidance, moderate discussions, etc.)
- Read and comment assignments of students on time;
- Prepare his/her lessons and deliver lectures;
- Provide available and necessary reference materials;
- Encourage active participation of students in the teaching learning process;
- Assist students with learning difficulties;

#### Roles of Students

Students are expected to:

- Engage in learning by doing (independent study, project work; group work, etc.)
- Be active learners (participate effectively in group assignments, make presentations, prepare and present seminars, write reports, etc.);

### References:

#### Required readings (Text)

#### Recommended readings

- Degu G. and Tessema F. Biostatistics for health science students, Lecture Note series Universty of Gondar. 2005.
- Kebede Y, Weldemichael K, Lulu K. Lectre note of epidemiology for health sciences. 2003.
- Degu G, Yigzaw T. Research Methodology Lecture Notes for Health Sciences Students. University of Gondar. 2006.
- Training module on Health and Health Related Research proposal writing, module I. The Ethiopian Science and Technology Commission in Collaboration with RegionalState Health Bureaus, Nov. 2004.
- Training module on Health Research Methods, module II, ESTC in collaboration with RHBs & EPHA, November 2004.
- Training Module on Data Processing, Analysis and Interpretation for Health Research, module III, ESTC in collaboration I RHB & EPHA November 2004.

- Training module on Health Research Ethics , module IV , ESTC in coll. with RHBs and EPHA, November 2004
- Training Module on Communicating Research Findings: oral & written communication techniques. ESTC in coll. with EPHA and CDC - Ethiopia, Nov. 2004.

**Course schedule\***

| <b>Week</b> | <b>Contact Hours</b> | <b>Topic/sub-topic/chapter/Assessments/Assignments</b>                                                                                                                  | <b>Reading Materials</b> | <b>Remark</b> |
|-------------|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|---------------|
| 1           | 4                    | 2. Introduction to research methodology: definition, types and importance of research<br>a. Study designs:<br>i. Observational studies<br>ii. Descriptive study design, |                          |               |
| 2           | 4                    | iii. Analytic study design<br>iv. Experimental study design<br>o Randomized Clinical Trial (RCT)<br>v. Community Intervention Trails (CITs)                             |                          |               |
| 3           | 4                    | 3. Sampling: definition, types & errors<br>a. Sample size determination                                                                                                 |                          |               |
| 4           | 4                    | 4. Types of data<br>a. Methods of data collection<br>b. Data processing, analysis and presentation                                                                      |                          |               |
| 5           | 4                    | 5. Research proposal and report writing<br>6. Research in pharmaceutical sciences<br>a. Experimental methodologies                                                      |                          |               |
| 6           | 4                    | b. Experiments involving the use of in vitro systems<br>c. Experiments involving the use of lab animal<br>d. Experiments involving human beings                         |                          |               |
| 7           | 4                    | e. Research Pharmacognosy<br>f. Research in pharmaceuticals<br>g. Research in pharmacology                                                                              |                          |               |
| 8           | 4                    | h. Research in Pharmacotherapy<br>7. Data documentation                                                                                                                 |                          |               |



## **Module 27 and 28: Pharmacy clerkship module I and II**

**Module name:** Pharmacy clerkship module

**Module category:** Core

**Module code:** Phar-M5271 and Phar-M5281

**Module Number:** 27 and 28

**Module weight in EtCTS:** = 22 and 20

**Prerequisite:** successful accomplishment of all core modules courses Year I – Year IV  
**Courses:**

| <b>Course name</b>                                 | <b>Course code</b> | <b>ECTS</b>           |
|----------------------------------------------------|--------------------|-----------------------|
| Ambulatory care clerkship                          | Phar5271           | <b>5 ECTS (3 wks)</b> |
| Drug information service clerkship                 | Phar5272           | <b>3 ECTS (2 wks)</b> |
| Internal medicine clerkship                        | Phar5273           | <b>7 ECTS (4 wks)</b> |
| Hospital pharmacy clerkship                        | Phar5274           | <b>7 ECTS (4 wks)</b> |
| Pediatric clerkship                                | Phar5281           | <b>7 ECTS (4 wks)</b> |
| Gynecology, obstetrics & family planning clerkship | Phar5282           | <b>3 ECTS (2 wks)</b> |
| Pharmaceutical manufacturing clerkship             | Phar5283           | <b>5 ECTS (3 wks)</b> |
| Community pharmacy clerkship                       | Phar5284           | <b>5 ECTS (3 wks)</b> |

### **Module description:**

This year-based clerkship module provides students with a structured, supervised program of participation in the practice of clinical pharmacy. Students gain experience in problem solving and providing patient care services while applying the basic and applied pharmaceutical sciences learned in the didactic courses & practical laboratories. This module should also provide the means by which the students will extend their clinical knowledge and skills. It emphasizes problem solving in the everyday milieu of patient care with in a setting of integrated inter-disciplinary patient care.

### **Module objective:**

At the end of this module students will be able to:

- Processing prescriptions/medication orders
- Identifying and resolving drug related problems through patient information retrieval and assessment
- Development of patient specific Pharmacotherapy care plans
- Monitor drug therapy

- Communicate with patients and other health care providers
- Provide patient education and training
- Demonstrate drug and literature information retrieval, evaluation, application, and related verbal and written communication skills
- Develop a practical and functional understanding of pharmacy services/systems in different practice settings, to include related patient safety and management responsibilities
- Describe the traditional and innovative roles of pharmacy practitioners in a variety of practice settings
- Apply and further evolve knowledge and skills in practice environments.
- Formulate career direction and strategy, in conjunction with the efforts of the student's academic advisor and preceptors.

**Module competency:**

- Provide Pharmaceutical Care and ensure the optimal use of medicines by the patient;
- Provide pharmacist-initiated care to patients and ensure the optimal use of medicines;
- Provide training and information on health care and medicines;
- Promote community health and provide related information and advice; and
- Conduct research to ensure the optimal use of medicines.
- Be able to work with members of the health team.
- Maintain pharmacy ethical code of conduct
- Demonstrate respect and compassion to patients, to their relatives and other professions

**Mode of delivery:** Year based/Parallel

**Mode of Assessment:**

Daily activity at rotation site

Clinical pharmacy related presentation: patient case presentation, seminar

Portfolio preparation & submission

Journal club presentation

Internal and external oral exam

Written exam

**Module learning teaching methods**

Case presentation/morning session, patient chart review, ward rounds, journal club & seminar, and project work, bedside teaching

**Clerkship Title: Ambulatory Care Pharmacy Clerkship**

**Clerkship Code: Phar5271**

**Clerkship ECTS: 5 (3 weeks)**

**Pre-requisites: accomplishment of all didactic courses Year I – Year IV**

**Co-requisites: None**

**Clerkship Description:**

Offered in a variety of ambulatory care settings to include outpatient specialist clinics including HIV/AIDS clinic, TB clinic, chronic illness clinic (e.g. cardiac, Hypertension, Diabetic and epilepsy clinic) and Chest Clinic, this experience emphasizes primary care using a case management approach with an out-patient population, to include: patient data collection, organization, and assessment; development of care plans that correspond to desired therapeutic objectives; patient monitoring, to include physical and laboratory assessment; communication with patients (and care givers) to acquire patient data, assess target outcomes and provide education and communication with prescribers, and other health care providers to optimize outcomes. Provision of preventive health education and screening services is expected in most settings.

**Clerkship Objectives:**

The student's learning goal for this clerkship is to develop the essential skills necessary to provide patient- specific care to patients in the ambulatory care

- Evaluate pathophysiology, clinical presentation, treatment goals, drug therapy, monitoring parameters, outcome measures, prognosis, and long-term management of common medical conditions in the ambulatory care setting.
- Identify drug-related problems; formulate and implement patient-specific, evidence-based patient care plans, and follow up to determine patient progress.
- Succinctly and clearly present oral and written outlines of patient work-ups.
- Synthesize succinct, evidence-based answers to drug information questions posed by patients or health care colleagues.
- Evaluate patient understanding of provided information about medical conditions, drug therapy, outcome goals, potential side effects (and what to do if side effects occur), and other medication-related information.
- Demonstrate professional conduct and demeanor that is ethical and responsible displaying integrity, compassion, empathy, and respect.

**Mode of delivery:**

Case presentation/morning session, patient chart review, attending ambulatory care team, journal club presentation/project work and seminar

**Mode of assessment:**

|      |                      |     |
|------|----------------------|-----|
| i.   | Case presentation    | 15% |
| ii.  | Seminar with pretest | 15% |
| iii. | Activities in OPD    | 20% |
| iv.  | Portfolio            | 10% |
| v.   | Internal oral exam   | 20% |
| vi.  | Written exam         | 20% |

### **Task List for Ambulatory Care Pharmacy Practice Rotations**

A list of specific tasks has been established for each of the core rotations to serve as a guide to the minimum requirements necessary to demonstrate competency. Reasonable effort should be made to accomplish these tasks when possible.

- For all assigned patients, review the patient's medical record and when possible conduct an interview. Evaluate current medication regimens and monitoring history. Be prepared to make therapeutic & monitoring recommendations. Be prepared to recommend & assess all new medication prescriptions.
- On selected patients, complete a detailed medication history. Assess achievement of treatment outcomes, ADRs, compliance etc. Make recommendations regarding drug therapy assessment and justify any proposed changes or interventions to preceptor or primary care provider.
- Assess and optimize doses for all applicable medications regarding renal function, pharmacokinetic evaluation of serum concentrations (if available) and other quantitative monitoring parameters (e.g. INR, blood glucose, etc): make dose or drug selection recommendations based on assessments.
- Prepare and present patients to preceptor (Pharmacotherapy Rounds): List patient problems, drug therapy, monitoring parameters, therapeutic end-points, dosage, potential ADRs and interactions. Discuss appropriateness of current and alternate medication therapies.
- Drug information: provide concise, up to date and evidence based drug information responses. Submit at least 4 written drug information responses given in the portfolio
- Submit 2 adverse drug events and 6 drug therapy problems/interventions identified during the rotation
- Submit 2 Pharmaceutical care services given during the rotation using 2-page SOAP format

**Clerkship Title: Drug Information Service Clerkship**

**Clerkship Code: Phar5272**

**Clerkship EtCTS: 3 (2 weeks)****Pre-requisites: accomplishment of all didactic courses Year I – Year IV****Co-requisites: none****Clerkship Description:**

- The Drug information service clerkship will be offered at Drug Information Centre and this experience emphasizes the use of drug literatures in the promotion of safe, effective and rational drug therapy. This clerkship provides students to develop the skills for retrieving, evaluating and communicating drug information to health care professionals and the patient.

**Clerkship Objectives:** At the end of the rotation in the Drug Information Centre students should be able to:

- Develop the capability to comfortably and confidently provide drug information to individuals and groups
- Demonstrate the ability to research drug information independently using multiple sources in a timely manner
- Develop a logical, step-wise approach to searching the drug information resources
- Develop critical decision-making skills relative to the selection, retrieval, and evaluation of appropriate literature resources

**Mode of delivery:** Journal Club/ Case presentation/morning session, patient education, attending drug information center, project work and seminar

**Mode of assessment:**

- |                                                      |       |
|------------------------------------------------------|-------|
| • Daily activities at drug information center        | 20%   |
| • Portfolio preparation and submission               | (50%) |
| ○ Newsletter                                         | 10%   |
| ○ Query request and response                         | 20%   |
| ○ Poison information                                 | 10%   |
| ○ Monograph preparation                              | 10%   |
| • Presentation – literature appraisal (journal club) | 10%   |
| • Written Examination                                | 20%   |

**Activities to be performed by students during the rotation**

- Receive drug information queries and fill each question on the Drug Information Centre (DIC) request form

- Provide concise, applicable, comprehensive, and timely responses to requests for drug information from patients, health care providers, and the public
  - Perform literature searches on Medline, International Pharmaceutical Abstracts, PubMed, Micromedex and related Databases
  - Identify potential evidence based solutions/answers to queries from literature;
  - Obtain Approval from the preceptor before communicating the requested information
  - Communicate the information professionally
  - Clearly express and justify their recommendation(s) in both written and oral forms
- Accurately document the drug information request
- There shall be a case presentation and discussion session twice per week in the afternoon.

The students will present a journal club on selected article in the area of drug information. The student is expected to submit handouts for the journal club and a portfolio for filled drug information response provided during the clerkship.

**Clerkship Title: Internal Medicine Clerkship Rotation Syllabus****Clerkship Code: Phar5273****Clerkship EtCTS: 7 (4 weeks)****Pre-requisites: accomplishment of all didactic courses Year I – Year IV****Co-requisites: None****Clerkship Description:**

- Taking place on adult internal medicine services located in acute care sites including medical intensive care and adult emergency units, this experience emphasizes primary patient care using a case-management approach, to include: patient data collection, organization, and assessment; development of plans that respond to desired therapeutic plans; patient monitoring to include physical and laboratory assessment; communication with patients (and care givers) to acquire patient data, assess target outcomes, and provide education; communication with prescribers, and other health care providers, to seek clarification and provide observations and recommendations consistent with the care plan; and provision of drug information to health care professionals in the hospital.

**Clerkship Objectives**

Upon successful completion of the experiential in Internal Medicine inpatient care, the student will be able to:

- Demonstrate an understanding of the pharmacotherapy of the most common acute and chronic disease states encountered in the institutional setting
- Discuss disease management including pathophysiology and drug therapy as it relates to patient specific characteristics and disease states
- Review the mechanism of action, indications, contraindications, adverse effects, and drug-interactions for each of the patient's medications
- Identify laboratory tests necessary to diagnose and monitor various disease states, describing the usefulness and limitations of each test
- Describe how certain disease states affect laboratory values and how they alter the interpretation of these laboratory values
- Assess findings to determine real and potential pharmacotherapeutic problems, rank them in order of acuity; describes probable causes/effects, and gathers additional information to define/clarify the problems
- Demonstrate ability to perform discharge/follow-up medication teaching and/or instruct patients in medication education classes

Retrieve and interpret patient chart information

- Obtain a complete medical and surgical history (including family history)
- Obtain allergy status of the patient including drug, food, other (E.g. dye, latex) and types of reactions
- Obtain a complete medication history including prescribed medications, over the-counter, and complementary medications (herbals, supplements, etc.). Include dose, route of administration, and frequency of use. Document in the appropriate manner
- Obtain information regarding whether or not the patient administers his/her own medication(s) or if a caregiver shares this responsibility, and if compliance aids are utilized, needed or appropriate; document and report noncompliance issues to preceptor
- Obtain information regarding behavioral issues such as tobacco, alcohol, and illicit drug use, diet, and exercise
- Obtain information regarding the patient's use of services outside of the primary care setting, including mental health, chiropractic, acupuncture, etc.
- Obtain information regarding the patient's financial situation (e.g. prescription costs, insurance, eligibility for indigent care)
- Obtain most recent information from medical record including laboratory data, vital signs, physician's orders, and consult notes
- Identify drug-related problems from patient chart and document evidence of the problem
- Recognize the need for pharmacist intervention

#### Design a Pharmaceutical Care Plan

- Retrieve and interpret patient chart information
- Construct and maintain patient problem list
- Assess each problem on patient's problem list appropriately (i.e. need for therapy, current therapy, potential therapy)
- Apply the knowledge in pathophysiology and drug therapy to patient care situations
- Formulate recommendation plan for each problem on patient's problem list (i.e. recommended pharmacological and non-pharmacological therapy, drugs to be avoided, further tests, follow-up)
- Provide rational drug therapy recommendations based on information obtained from the patient interview, physical assessment/examination, laboratory data, medical record, etc.
- Design medication regimens that are convenient, affordable, and will produce optimal patient outcomes (e.g. side effect profile, dosage form, etc.)



- Perform prospective drug regimen reviews to evaluate contraindications and drug-drug and food-drug interactions utilizing knowledge of medicinal chemistry, biochemistry, pharmacokinetics, herbal/nutritional supplements/over-the-counter medicines
- Evaluate the primary literature and national treatment guidelines and its utility in meeting patient needs (including case reports, if appropriate) to make a reasonable decision based on available information
- Identify goals of therapy, including effects on quality of life (QOL)
- List and obtain monitoring parameters (i.e. toxic, therapeutic)
- Write clear and concise consultation notes or progress notes (e.g. SOAP notes)
- Communicate to patient, physician, and preceptor therapeutic plan both verbally and written as needed
- Complete and update the patient medication record and other pharmacy notes/documentation systems as needed
- Modify recommendations as needed

#### Provide drug information

- Describe the pharmacist's role in providing health care information within the Internal medicine setting
- Obtain necessary background information to accurately answer drug information questions
- Identify and utilize, both efficiently and effectively, appropriate drug information sources (including local Poison Information Center)
- Effectively retrieve and evaluate medical information for patients and health care providers
- Effectively evaluate, interpret, and summarize pharmaceutical and medical primary literature; recognize uses and limitations of different information resources
- Retrieve the standards of care/disease management protocols for various disease states (e.g. AHCPR, APhA, Chest, NHLBI, NCEP, JNCIV, FMHACCA etc.)
- Demonstrate effective communication skills, written and verbal, to preceptors, patients, physicians, and other health professionals
- Provide health care professionals with accurate, concise, and timely drug or drug therapy information
- Document and reference all drug information responses

- Identify, define, and report adverse drug reactions; reports to FMHCCA, and when appropriate, the P&T Committee and Board of Pharmacy
- Prepare a drug monograph for P & T Committee meeting
- Prepare a well-researched article for the institutional newsletter
- Prepare an in-service to the RPh/RN/MD staff.
- Prepare a drug utilization review/medication guideline/protocol

Describe intradepartmental and interdepartmental dynamics to the facility

- Participate interdisciplinary continuing education activities with other health care practitioners (e.g. grand rounds, clinical conferences, in-service lectures)
- Attend and participate in journal club using an article published in the primary literature when opportunities exist
- Visit other departments in the internal medicine setting involved in diagnosis and treatment (e.g. E.R., critical care units, radiology, laboratory, anesthesiology, O.R., respiratory therapy, etc.) and describe the basic responsibilities of the department, its drug use policies and process, and its relationship with pharmacy
- Describe the pharmacist role in various Internal medicine committees (e.g. P&T, infection control, nursing, and pharmacy). The student should be scheduled to attend a committee meeting when possible and participate in providing background information for the committee to review
- Describe the various means the pharmacy department seeks reimbursement for its distributive and cognitive services. The student should discuss the procedures to follow in the institution in justifying reimbursement for pharmaceutical care

Demonstrate professional attitude and conduct

- Exhibit neatness and professionalism in appearance and work
- Accept constructive criticism, demonstrate receptiveness to feedback
- Demonstrate dependability, punctuality, courteousness, and tactfulness when dealing with patients and members of the health care team
- Maintain professional and ethical standards- compliance with laws and regulations, good professional judgment, reliability, and credibility when dealing with colleagues, patients, and other health care professionals
- Display self-directed (independent) learning, conduct self-assessment, develop a personal learning plan, and pursue knowledge independently

- Demonstrate competency in organizing and planning, establish management skills, set meaningful and attainable goals and be consistently well prepared
- Maintain confidentiality
- Display a patient and empathetic attitude towards patients including appropriate body language showing genuine interest in the well-being of the patient
- Respond to assignments in a timely manner and is consistently on time and ready for work upon arrival, with no unexcused absences

### **Task List for internal medicine inpatient care Rotations**

A list of specific tasks has been established for each of the required rotations to serve as a guide to the minimum requirements necessary to demonstrate competency. Provided below are samples of a few of the tasks defined for each specific core experiential rotation. Reasonable effort should be made to accomplish these tasks when possible.)

- **Actively participate in rounds on a daily basis.** Attend conferences required of the medical team (e.g. Grand Rounds, Teaching Rounds, Case Conference, etc.).
- Discuss at least 2 therapeutic guidelines relevant to the acute care site's patient population.
- Assess all potential ADEs. Complete adverse drug events and medication error report as per national or school designed ADEs/medication error reporting form and guidelines.
- Develop/Document written Drug Information responses from preceptor, patients, caregivers or prescribers (4).
- Interface with pharmacy staff regarding unusual medication orders, patient issues, non-formulary needs, etc.
- Present patients to preceptor (and others) (Pharmacotherapy Rounds): List patient problems, drug therapy, monitoring parameters, therapeutic end-points, dosage, potential ADRs and interactions. Discuss appropriateness of current or alternate medication/doses and nutritional therapies.
- Assess and monitor applicable doses and medication therapy outcomes (including potential ADEs) in relation to renal function, pharmacokinetic analysis of serum concentrations and other lab or quantitative or clinical monitoring parameters, if available.
- Perform medication dosage form conversion on medications that are typically converted from intravenous to oral dosing whenever possible or prior to patient discharge. It also involves review of situations during which conversion is appropriate, benefits of such conversion, and appropriate conversion guidelines.

- Perform renal / hepatic dosing optimization for medications commonly used in the inpatient care depending on pertinent laboratory values
- Conduct at least 8 patient interview to obtain drug histories (refer to activities schedules for due dates)
- Write a 2 page (maximum) SOAP note. Three per 4 week rotation
- Submit 4 adverse drug events and 8 medication errors identified during the 4 week rotation. Submit timely projects and student portfolios assigned by the preceptor

**The following description further explains the core activities to be performed.**

**The student shall attend Conferences, Morning Report, Grand Rounds, and medical Resident's Conference. The student may attend other conference as determined by the preceptor.**

**Patient Care:**

1. Rounds: The student is expected to actively participate in rounds on a daily basis. Team members should feel free to ask the student questions and ask them to research drug-related issues within a reasonable time period. Any responses to drug information requests or recommendations should be made at this time unless it was more prudent to provide the information/recommendation earlier in the day. The student should document clinical interventions/drug information requests on the appropriate form.
2. Patient Monitoring: The student should be monitoring the drug therapy of all patients on their team at all times. A complete database on each patient should be kept. Students should obtain necessary patient data in the morning so patients can be reviewed with a preceptor in the afternoon.
3. Drug Information Requests: During the morning the student should work on any requests for drug information that they have received from the team. Responses should be reviewed with the preceptor during the afternoon meeting or earlier if information is needed by the team prior to that time.

**Oral and Written Communication**

1. Case Presentations: Each day the student will informally present the patients currently assigned to their team.

2. **Written Drug Information:** If requested by the preceptor, the student will write a newsletter article or drug class summary. The preceptor will provide the student with a list of potential topics. The student may then choose the topic they would prefer to work on.
3. **Journal Club:** The student will review a recently published study that pertains to pediatric patients. The study will be presented at journal club. Attendees will include other pharmacy students, pharmacy faculty, and pharmacy staff members if available. The article should be approved by the preceptor. Please see the journal club information sheet for more details on choice of article and format for presentation format.

**Mode of delivery:**

Case presentation/morning session, patient chart review, attending medical wards, bed side teaching, journal club presentation, project work and seminar

**Mode of assessment:**

|      |                      |     |
|------|----------------------|-----|
| i.   | Case presentation    | 15% |
| ii.  | Seminar with pretest | 15% |
| iii. | Ward activities      | 20% |
| iv.  | Portfolio            | 10% |
| v.   | Internal oral exam   | 20% |
| vi.  | Written exam         | 20% |

**Clerkship Title: Hospital Pharmacy Clerkship**

**Clerkship Code: Phar5274**

**Clerkship EtCTS: 7 (4 weeks)**

**Pre-requisites: accomplishment of all didactic courses Year I – Year IV**

**Co-requisites: none**

**Clerkship Description:**

- Taking place in hospital-based pharmacy services, this experience emphasizes administrative and pharmacy service functions, to include: drug distribution and dispensing; D.U.E. [if any undergoing]; A.D.R. and Medication error reporting, formulary review and amendment [if any,]; cost containment outcome assessment; quality care assurance; inventory control; and maintenance of mandated hospital pharmacy records.
- The student Pharmacists are expected to provide hospital pharmacy Clerkship with integration to clinical pharmacy services in the above delivery sites.

### **Rotation task list/activities for Hospital Pharmacy Rotations**

#### **One week will be dedicated for hospital pharmacy related supply chain management activities**

A list of specific tasks has been established for each of the required clerkships to serve as a guide to the minimum requirements necessary to demonstrate competency. Provided below are samples of a few of the tasks defined for each specific core clerkships [drug distribution system, dispensing...]. Reasonable effort should be made to accomplish these tasks when possible).

The student will spend two weeks in a hospital pharmacy department. The student will experience pharmacy operations and services relating to systems for medication distribution, medication use and drug control, management of the department, scope of clinical services provided by the department, and department relationships within the institution and health systems

- As frequently as possible, if any of the following present at time of attachment otherwise we expect the preceptor to brief each of the students on the following issues, attend clinical/operational meetings: Pharmacy and Therapeutics Committee (PTC), PTC subcommittees, Patient Safety, Pharmacy Staff, Nursing/Pharmacy and others as identified by preceptor.
- As frequently as possible, attend educational meetings: staff in-services, Grand Rounds, and others as identified by preceptor.
- Participate in basic administrative activities needed to maintain a hospital pharmacy department
- Involve in purchasing, inventory control, and basic fiscal procedures
- Participate in drug security, storage, and control procedures and quality assurance works

- Describe the policies and procedures for maintaining quality assurance using JCAHO and ASHP standards [ these are international benchmarks for hospital pharmacy services]
- Participate in the intradepartmental and interdepartmental continuous quality improvement process and importance of project teams, if any
- Read about the role of technology and alternative distribution systems as they relate to expanded clinical services
- Assess and evaluate the financial impact of drug therapy
- Evaluate drug therapy costs, including costs of drugs as well as monitoring costs
- Evaluate financial consideration of alternative therapies
- Design the various means the pharmacy department seeks reimbursement for its distributive and cognitive services. The student should discuss the impact of pharmaceutical care programs on the health of patients and how they impact the cost of care
- Students should be given reading assignment on the following issues – effective utilization of automation / informatics - an assigned activity that provides insight into the benefits & challenges of automation / informatics (e.g. CPOE, decision support, distribution, compounding, drug information retrieval, etc.
- Participate in the drug dispensing process related to providing meds for new orders and ongoing supply (unit dose etc.) in order to understand system flow and verification that meds dispensed are correctly prepared.
- Under the supervision of a pharmacist, process a broad variety of new medication orders. Processes to include: medication order assessment, order entry or order verification, evaluation and application of computer alerts, resolution of problems and over-sight of medication dispensing. Closely observe and simulate those processes from printed profile and demonstrate an understanding of workflow.
- [students should be given reading assignment on these areas since we don't have currently such system]- how to prepare a number and variety of sterile parenteral products sufficient to demonstrate acceptable competency. Syringe, small and large volume, etc. Participate in TPN and chemotherapy processing and preparation if feasible. Discuss both patient and environmental safety issues.
- Develop/discuss/document resolution of 5 patient profile reviews or new medication order problems.

- Prepare at least **20** medication orders for the patient by evaluating the medication order and selecting the proper product.
- Identify drug-related problems (minimum of 5) and document in the work book or portfolio.
- Package and dispense multiple dosage forms including IV admixtures as assigned by the preceptor.
- Communicate therapeutic recommendations to other health care professionals.
- Perform at least **15** pharmaceutical calculations related to the medication order, including pediatric orders.
- Develop concise, applicable, comprehensive, and timely responses to requests for drug information from other health care providers in the hospital setting.
- Participate in the health system's formulary process/drug monograph, if any
- Perform prospective and retrospective financial and clinical outcomes analyses to support formulary recommendations and therapeutic guideline development
- Understand the relationship between medication distribution and clinical pharmacy services, and identify barriers between the two components.

**Mode of delivery:** Case presentation/morning session, Patient Chart review, attending hospital Pharmacy, bed side teaching, Journal Club presentation, project work and seminar

**Mode of assessment:**

- |                                                                              |     |
|------------------------------------------------------------------------------|-----|
| • Attendance and on-site activities at hospital pharmacy by preceptors       | 15% |
| • Portfolio                                                                  | 20% |
| • Hospital pharmacy related presentation – journal club presentation/seminar | 15% |
| • Internal oral exam                                                         | 25% |
| • Written Examination                                                        | 25% |



**Clerkship Title: Pediatrics Clerkship Rotation Syllabus**

**Clerkship Code: Phar5281**

**Clerkship ECTS: 7 (4 weeks)**

**Pre-requisites:** accomplishment of all didactic and Laboratory based courses Year I – Year IV

**Co-requisites: None**

**Clerkship Description:**

- **The purpose of Pediatrics clerkship is to train student to provide pharmaceutical care to pediatric patients**

#### Specific Goals

- To provide the student with an understanding of pediatric patient care in the clinical setting.
- To develop a knowledge base of common pediatric disease states and their therapy.
- To learn to apply therapeutic and pharmacokinetic principles to the pediatric patient.
- To develop an appreciation of the pharmacist's role in monitoring drug therapy in the pediatric patients.
- To be able to evaluate the appropriateness of drug therapy in pediatric illnesses.
- To be able to interact with the pediatric care system as a participant.
- To develop an understanding of drug dosage and formulation problems unique to pediatric drug therapy.

#### **Clerkship Objectives:**

**After completing this rotation, the student will be able to:**

##### **Provide Direct Patient Care**

1. Collect and organize all patient-specific information needed to detect and resolve drug related problems and to make appropriate drug therapy decisions in pediatric patients
  - Identify the types of information needed to detect and resolve problems
  - Discuss signs and symptoms, epidemiology, risk factors, pathogenesis, pathophysiology, natural history of disease, clinical course, etiology, and treatment of common diseases in pediatric patients
  - Discuss the mechanism of action, pharmacokinetics, pharmacodynamics, usual regimen (dose, schedule, form, route, and method of administration), indications, contraindications, interactions, adverse reactions, relative efficacies, and pharmacoeconomics of drugs used in pediatric patients

- Identify the differences that may occur in the pharmacokinetics and pharmacodynamics of drugs due to the developmental stage of a pediatric patient
- 2. Determine the presence of any of the following problems or concerns related to a patient's current drug therapy
  - a. Drugs used with no medical indication
  - b. Medical conditions for which there is no drug prescribed
  - c. Drugs prescribed inappropriately for a particular medical condition
  - d. Anything inappropriate with the current regimen (dose, schedule, route of administration, method of administration)
  - e. Presence of therapeutic duplication
  - f. Drugs to which a patient is allergic
  - g. Presence or potential for adverse drug reactions
  - h. Presence of clinically significant drug-drug, drug-disease, drug-nutrient, or drug laboratory test interactions
- 3. Design pharmacotherapeutic regimens
  - Using an organized collection of patient-specific information develop a problem list
  - Specify pharmacotherapeutic goals for a patient that integrate patient-specific, disease-specific and drug-specific information and economic, ethical, and QOL considerations
  - Design a regimen, including modifications to existing therapy, which meet pharmacotherapeutic goals established for a patient
- 4. Design monitoring plans for drug therapy regimens
  - Determine parameters to monitor that will measure achievement of pharmacotherapeutic goals for a regimen
  - Define a desirable value range for each selected parameter taking into account patient-specific information
- 5. Recommend regimens and corresponding monitoring plans to a prescriber in a way that is systematic and logical and secures consensus from the prescriber.
- 6. Modify a plan as necessary based upon evaluation of monitoring data
- 7. Document all pharmaceutical care activities appropriately
- 8. Participate effectively in patient care rounds
  - Formulate appropriate responses to drug information requests and drug policy questions occurring during patient care rounds

- Demonstrate a commitment to maintaining a database to support participation in patient care rounds

## 9. Provide Information Services

Provide concise, applicable, and timely responses to requests for information from members of the health care team.

- Appropriately elicit all background information necessary to respond to a request for drug information
- Identify the most appropriate sources of information on the use of drugs in pediatric patients
- Formulate a systematic, efficient, and thorough procedure for retrieving drug information.
- Efficiently utilize both manual and computerized sources of drug information
- Evaluate the quality of literature gathered
- Effectively communicate responses to the requestor and/or team

## 10. Prepare written information on drug therapy

- Write a newsletter article on a topic that is timely and is of interest to the intended audience.
  - Prepare a drug class review in a table format.
1. Prepare and orally disseminate information related to drug therapy in pediatric patients.
    - Effectively present cases informally and formally to preceptors and other students.
    - Effectively present an article as part of journal club.

## **Clerkship Task lists/ Responsibilities**

The responsibilities of the student during this rotation shall include the following:

- Attend daily patient work rounds in addition to attending rounds and other educational pediatric conferences as may be scheduled.
- Provide drug information when appropriate to other members of the health care team.
- Monitor patients' drug therapy for therapeutic effect, adverse drug reactions, and drug interactions.
- Provide patient counseling and education concerning drug therapy during hospitalization and at discharge, where possible.
- Participate in formal and informal consultations, including detailed literature searches.
- Provide in-service education for health professionals, when appropriate.
- Attend conferences that pertain to drug therapy and patient care.

- Participate in emergency situations, when possible, including the preparation of emergency drugs and provision of drug information.
- Obtaining complete and accurate medication records for each patient.
- Presenting patient-related information completely, succinctly, and accurately during rounds with preceptor as well as other health care professionals when queried.
- Evaluate the literature on a specific topic pertaining to a patient's therapy. Completing and presenting one literature searches relating to a pediatric drug therapy.
- Submit 2 written pharmaceutical care service given with a maximum of 2 pages using SOAP format
- Submit 4 written drug information responses that the student gave during the rotation
- Submit 4 adverse drug events and 8 drug therapy problems/interventions encountered during the rotation that comprises portfolio requirements in pediatric rotations

### **Core Activities**

**The following activities are described in detail as a guide for more focused student activity in pediatrics ward.**

#### **Patient Care:**

- **Rounds:** The student is expected to actively participate in rounds on a daily basis. Team members should feel free to ask the student questions and ask them to research drug-related issues within a reasonable time period. Any responses to drug information requests or recommendations should be made at this time unless it was more prudent to provide the information/recommendation earlier in the day. The student should document clinical interventions/drug information requests on the appropriate form.

**Patient Monitoring:** The student should be monitoring the drug therapy of all patients on their team at all times. A complete database on each patient should be kept. Students should obtain necessary patient data in the morning so patients can be reviewed with a preceptor in the afternoon.

**Drug Information Requests:** During the morning the student should work on any requests for drug information that they have received from the team. Responses should be reviewed with the preceptor during the afternoon meeting or earlier if information is needed by the team prior to that time.

Oral and Written Communication

Case Presentations: Each day the student will informally present the patients currently assigned to their team.

Written Drug Information: If requested by the preceptor, the student will write a newsletter article or drug class summary. The preceptor will provide the student with a list of potential topics. The student may then choose the topic they would prefer to work on.

Journal Club: The student will review a recently published study that pertains to pediatric patients. The study will be presented at journal club. Attendees will include other pharmacy students, pharmacy faculty, and pharmacy staff members if available. The article should be approved by the preceptor. Please see the journal club information sheet for more details on choice of article and format for presentation format.

Seminar presentation: The student will present selected topics preferably most common pediatrics cases encountered in the attachment site individually or in a group. With more focus on current management updates of the cases and local or national guidelines about that specific seminar topics.

Pharmacy only round: All students have to present highlight case presentation of each patients allocated and respond to specific pharmaceutical care concerns of the patient

#### Student Readings

During the rotation the student is expected to complete the readings to be suggested by preceptors. Articles on the following diseases states are required: Developmental Pharmacology, Asthma, Meningitis, Pneumonia, Antimicrobial Therapy for Infants and Children, GERD, Seizure disorders , Bone and soft tissue infections, Rheumatic heart disease, Fluid and electrolytes (diarrhea, dehydration, calculating fluid requirements), Cystic fibrosis , UTI, Diabetes, Nutritional disorders and neonatal respiratory distress syndromes.

**Mode of delivery:** Case presentation/morning session, Patient Chart review, attending pediatrics care team, bed side teaching, Journal Club presentation, project work and seminar

#### Mode of assessment:

|      |                      |     |
|------|----------------------|-----|
| i.   | Case presentation    | 15% |
| ii.  | Seminar with pretest | 15% |
| iii. | Ward activities      | 20% |
| iv.  | Portfolio            | 10% |
| v.   | Internal oral exam   | 20% |
| vi.  | Written exam         | 20% |

**Clerkship Title: Gynecology/Obstetric Clerkship Syllabus**

**Clerkship Code: Phar5282**

**Clerkship EtCTS: 3 (2 weeks)**

**Pre-requisites: accomplishment of all didactic courses Year I – Year IV**

**Co-requisites: None**

**Clerkship Description:**

- This rotation will be conducted in Gynecology and Obstetrics wards under department of gynecology and obstetrics. The student should collaborate with other health care providers for the pharmaceutical care needs of the patients.

**Clerkship Goals**

- To prepare the students with the necessary knowledge and problem-solving skills in an obstetrics/gynecology environment for provision of pharmaceutical care.
- To be responsible for providing accurate and timely drug information to gynecologic and obstetric patients and other health care professionals

**Clerkship Objectives:** At the completion of this rotation, the student should be able to:

- Identify and discuss different pharmacologic agents that may be harmful to the fetus or mother in different trimesters of pregnancy and also during breast feeding.
- Discuss pharmacotherapeutic modalities in reproductive system, gynecological and obstetrics pharmacotherapy.
- Consistently obtain complete and accurate drug histories including previous adverse reactions to medications.
- Present well-organized and accurate patient case histories including subsequent problem-plan management.
- Effectively and appropriately communicate with both patients and other health care professionals.
- Participate in the drug decision-making process with health care professionals, and patients
- Describe the clinical presentation and treatment for all encountered medical illnesses during pregnancy and also includes menstrual disorders, hormonal replacement therapy and contraception.

- Design a management therapeutic plan for different compelling diseases associated with pregnancy like ectopic pregnancy, miscarriage /bleeding, premenstrual syndrome, menopause, drugs used during labor and caesarian births.
- Discuss the problem-solving process to determine the risk vs. benefit of using drug therapy during pregnancy and lactation.
- Describe the standard medical care of a pregnant woman.
- Describe commonly encountered bacterial and fungal gynecologic diseases like gonorrhea, syphilis and their treatment, monitoring, and follow up.
- Design a pharmaceutical care plan for patients with pregnancy or other gynecological abnormalities.
- List drugs according to their pregnancy categories and discuss their toxic and beneficial potential.

### **Clerkship Task Lists**

The following **task list/ responsibilities** will include but are not limited to the following:

2. Punctual attendance at daily morning rounds with the assigned surgery team.
3. Punctual attendance at clinical pharmacy conferences.
4. Completely, accurately, and promptly responding to questions from health care professionals.
5. Initiating interactions with other health care professionals.
6. Filling out drug reaction forms for any undesired or unintended effect of a medication.
7. Completing and presenting two literature searches relating to a surgery topic.
8. Obtaining complete and accurate medication records for each patient.
9. Presenting patient-related information completely, succinctly, and accurately during rounds with preceptor as well as other health care professionals when queried.
10. Deciding which medications to use and how to use them with reference to pharmacokinetics and pharmacodynamics.
11. Completely and accurately evaluating the medications of all assigned patients on the surgery service.

**Mode of delivery:** Case presentation/morning session, Patient Chart review, attending surgical wards care team, bed side teaching, Journal Club presentation, project work and seminar

**Mode of assessment:**

|      |                      |     |
|------|----------------------|-----|
| i.   | Case presentation    | 15% |
| ii.  | Seminar with pretest | 15% |
| iii. | Ward activities      | 20% |
| iv.  | Portfolio            | 10% |
| v.   | Internal oral exam   | 20% |
| vi.  | Written exam         | 20% |



**Clerkship Title: Pharmaceutical Manufacturing Clerkship Syllabus**

**Clerkship Code: Phar5283**

**Clerkship ECTS: 5 (3 weeks)**

**Pre-requisites: accomplishment of all didactic courses Year I – Year IV**

**Co-requisites: None**

**Clerkship Description:**

This clerkship will be conducted in a selected pharmaceutical manufacturing industry. Students will get an opportunity to appreciate and be part of the day to day activities in a pharmaceutical industry.

The details of the clerkship activities will be communicated.

**Clerkship Title: Community Pharmacy Clerkship****Clerkship Code: Phar5284****Clerkship EtCTS: 5 (3 weeks)****Pre-requisites: accomplishment of all didactic courses Year I – Year IV****Co-requisites: none****Clerkship Description:**

- Taking place in community pharmacy services, this experience emphasizes a wide range of exposures, to include: prescription dispensing and OTC selection; patient communication and education; communication with prescribers and other health care providers; and pertinent management activities.
- This community pharmacy practice clerkship provides students with experience in providing Medication Therapy Management to patients in an outpatient pharmacy setting. The student's learning goal for this experience is to develop the essential skills necessary to provide patient- specific care in the community pharmacy setting.

**Clerkship Goals:**

- Provide medication therapy management; review profile and perform medication history to create a personal medication record; and design medication action plan for a patient.
- Communicate and collaborate verbally and in writing with patients, caregivers, health care providers, and others to improve patient care.
- Assess patients and recommend over-the-counter medication, non-drug therapy, medical goods, and complementary therapies beneficial for patient care.
- Triage patients to appropriate health care providers and social service agencies.
- Provide public health and wellness services and educational materials tailored to the needs of patients and caregivers in the community practice setting.
- Demonstrate professional conduct and demeanor that is ethical and responsible displaying integrity, compassion, empathy, and respect.

**Clerkship Objectives**

Upon successful completion of the experiential in community pharmacy, the student will be able to:

- **Process a prescription order**
  - Receive and evaluate the original prescription or refill
  - Review for accuracy, completeness, validity, and appropriateness
  - Differentiate between the prescription drop-off interview and the dispensing/exit interview
  - Prepare prescription for dispensing
  - Evaluate and monitor prescription refills
  - Determine therapeutic indications

- Help select correct medication and appropriate dosage forms
- Discuss the appropriateness of dosage, frequency, and route of administration with patients and other health care providers
- **Retrieve and interpret patient specific data**
  - Retrieve information from the prescription and/or patient, if possible patient medical record
  - Construct and maintain accurate patient profiles
  - Construct and maintain patient problem lists
  - Interpret patient data in regards to specific disease states and patient complaints.

#### **Identify patient specific drug related problems**

- Identify drug related problems through drug regimen reviews, clinical assessment of a patient, reviewing the patient profile and through patient consultations.
- Utilize available technology and patient interviewing techniques to:
- identify drug-drug/drug-disease/drug-nutritional and drug-allergy interactions
  - Identify appropriate drug therapy
  - Assess patient compliance
  - Assess patient understanding of their disease states and current medications
- **Demonstrate knowledge of country pharmacy laws and regulations**
  - Utilize professional and ethical judgment in the interpretation of laws and regulations of EFDA
  - Assure the medication order conforms to the federal regulations/EFDA regulations, including Laws/policies regarding controlled substances and generic/therapeutic substitution
  - **Demonstrate knowledge of management skills needed to maintain a pharmacy department**
  - Utilize various methods of purchasing and demonstrate knowledge as to the advantages and disadvantages of each method
  - List technologies that can be used in pharmacy operations
  - Utilize interpersonal skills in working with health care providers and pharmacy staff
  - Determine methods to improve a pharmacy operating
  - Apply licensing, regulatory, and accreditation standards which are necessary for the operation of a hospital pharmacy

- **Assess and evaluate the financial impact of drug therapy**
  - Evaluate drug therapy costs, including costs of drugs as well as monitoring costs
  - Evaluate financial consideration of alternative therapies
- **Provide patient counseling and disease state education**
  - ✓ Effectively conduct a patient interview
  - ✓ Review medication information with the patient to insure appropriate use and compliance of drug therapy
  - ✓ Utilize patient education materials to assist patients in understanding their roles in effective medication use
  - ✓ Counsel patients with respect to assessment of adverse effects of their medications and how to care for these effects
  - ✓ Counsel patients with respect to non-prescription medications
  - ✓ Counsel patients with respect to common community-related disease states
  - ✓ Counsel and train patients on the appropriate use of disease state monitoring tools (blood-pressure cuffs, blood glucose monitors, peak-flow monitors, etc.)
  - ✓ Document patient interventions and patient care appropriately
- **Conduct patient evaluations**
  - Recognize the need for pharmacist interventions
  - Refer patients to appropriate medical personnel when necessary
- **Provide drug information**
  - Effectively retrieve and evaluate medical information for patients and health care providers
  - Describe the pharmacist's role in providing health care information within the community
  - Demonstrate effective communication skills, written and verbal, to preceptors, patients, physicians, and other health professionals
  - Obtain necessary background information to accurately answer drug information questions.
  - Effectively evaluate, interpret, and summarize pharmaceutical and medical literature
  - Identify and utilize, both efficiently and effectively, appropriate drug information sources

- **Demonstrate professional attitude and conduct**

- Exhibit neatness and professionalism in appearance and work
- Accept constructive criticism, demonstrate receptiveness to feedback
- Demonstrate dependability, punctuality, courteousness, and tactfulness when dealing with patients and members of the health care team
- Maintain professional and ethical standards- compliance with laws and regulations, good professional judgment, reliability, and credibility when dealing with colleagues, patients, and other health care professionals
- Display self-directed (independent) learning, conduct self-assessment, develop a personal learning plan, and pursue knowledge independently
- Demonstrate competency in organizing and planning, establish management skills, set meaningful and attainable goals and be consistently well prepared
- Maintain confidentiality
- Display a patient and empathetic attitude towards patients including appropriately body language showing genuine interest in the well-being of the patient
- Respond to assignments in a timely manner and is consistently on time and ready for work upon arrival, with no unexcused absences.

**Mode of delivery:**

Prescription review, evaluation of community pharmacy establishments, journal club presentation, project work and seminar

**Mode of assessment:**

- |                                                                               |     |
|-------------------------------------------------------------------------------|-----|
| • Attendance and on-site activities at community pharmacy                     | 15% |
| • Portfolio                                                                   | 20% |
| • Community pharmacy related presentation – journal club presentation/seminar | 15% |
| • Internal oral exam                                                          | 25% |
| • Written Examination                                                         | 25% |

**Task List for Community Rotations**

A list of specific tasks has been established for each of the required rotations to serve as a guide to the minimum requirements necessary to demonstrate competency. Provided below are samples of a few of the tasks defined for each specific core experiential rotation. Reasonable effort should be made to accomplish these tasks when possible.)

- Process Prescriptions (> 60) of various types.

- Evaluate patient medication profiles. Discuss and document the resolution of 6 patient-profile (Clinical) of Drug related problems.
- Provide a minimum of two written patient assessments and pharmaceutical care plans (MTM services) of assigned patients.
- Develop/Discuss/Document resolution of 8 patient- specific financial problems in purchasing medications for chronic diseases
- Discuss the handling of the following scenarios with preceptor: suspected RX forged, discovery of an error, prescriber consensus challenges, notifying a patient that a medication error has occurred, and dealing with difficult patient / MD.
- Counsel patients on the use of medications (> 20 pts) including: insulin, statins, opiates (acute/chronic), inhalers, warfarin, pediatric dose measuring/administration, or others selected by preceptor. The student should report calculations done in the rotation. Students will be required to complete and document three counseling sessions and associated follow-up/interventions.

**Module 29: Professional elective clerkship****Module name:** Professional elective clerkship**Module category:** Elective**Module code:** Phar-M5292**Module Number:** 29**Module weight in ECTS:** = 5**Prerequisite:** successful accomplishment of all core modules courses Year I – Year IV  
**Courses:**

| Clerkship name                        | Clerkship code | ECTS           |
|---------------------------------------|----------------|----------------|
| Psychiatry clerkship                  | Phar5291       | 5 ECTS (3 wks) |
| Surgery clerkship                     | Phar5292       | 5 ECTS (3 wks) |
| Oncology & Hematology clerkship       | Phar5293       | 5 ECTS (3 wks) |
| Ophthalmology & ENT clerkship         | Phar5294       | 5 ECTS (3 wks) |
| Emergency Medicine Clerkship          | Phar5295       | 5 ECTS (3 wks) |
| Dermatology clerkship                 | Phar5296       | 5 ECTS (3 wks) |
| Pharmaceutical quality control        | Phar5297       | 5 ECTS (3 wks) |
| Pharmaceutical regulatory affairs     | Phar5298       | 5 ECTS (3 wks) |
| Pharmaceutical whole sale & promotion | Phar5299       | 5 ECTS (3 wks) |

**Module description:** This Module contained courses which students are to choose one among them. This will help the students delineate their future carrier, i.e., it is designed in such a way as to suit for the interest and competence of students who are interested to pursue their future career in one of the following courses: pharmaceutical industry, Pharmaceuticals quality control, Pharmaceutical regulatory, Pharmaceutical whole sale & promotion, Nuclear pharmacy, oncology & hematology clerkship, Ophthalmology & ENT clerkship, Dermatology clerkship, and emergency medicine.

**Module objective:** course specific

**Module competency:** the students should develop this competencies based on their choice of course

- Organize and control the manufacturing, compounding and packaging of pharmaceutical products
- Provide pharmacist-initiated care to patients and ensure the optimal use of medicines

**Mode of delivery: Parallel**

**Mode of Assessment:** The assessment criteria are based on continuous assessment of class activities, individual and group assignment, attachments and report writing, test and final exams.

**Module learning teaching methods**

Case presentation/morning session, patient chart review; ward rounds, journal club & seminar, project work, bedside teaching, visit to industries, whole sale, & regulatory bodies.



**Clerkship Title: Psychiatry Clerkship Syllabus**

**Clerkship Code: Phar5291**

**Clerkship EtCTS: 5 (3 weeks)**

**Pre-requisites: accomplishment of all didactic courses Year I – Year IV**

**Co-requisites: None**

**Clerkship Description:**

- The rotation will be conducted in Psychiatry Wards and/or Clinics under the department of Psychiatry. The students should team up with the psychiatry team for the patient care activities in the psychiatry unit attached.

**Clerkship Goals**

1. To be familiarized with Pharmaceutical care services for Psychiatric patients.
2. To provide appropriate and timely drug related information to Psychiatry team and other health care professionals working in the department.
3. To gain a general understanding of the Psychotropic medications used in different types of acute and chronic Psychiatric disorders such as antipsychotics, mood stabilizers, antidepressants, drugs for sleep disorders, anxiolytics and substance related disorders.

**Clerkship Objectives:** At the completion of this rotation, the students should be able to:

- Demonstrate an understanding of the pharmacotherapy of the most common acute and chronic Psychiatric disorders
- Provide Pharmaceutical care service for patients with psychiatric disorders
- Retrieve and interpret patient chart information
- Perform psychiatric patient interview
- Obtain a complete medication history including prescribed medications, over the-counters, complementary medications (herbals, supplements, etc.), previous adverse reactions and drug allergies. Include dose, route of administration, and frequency of use. Document in the appropriate manner
- Obtain information regarding behavioral issues such as tobacco, alcohol, and illicit drug use, diet, and exercise
- Provide rational drug therapy recommendations based on information obtained from the patient interview, physical assessment/examination, laboratory data, medical record, etc.

- Design medication regimens that are convenient, affordable, and will produce optimal patient outcomes (e.g. side effect profile, dosage form, etc.)
- Follow up and evaluation of drug regimen to evaluate outcome, contraindications, drug-drug and food-drug interactions utilizing knowledge of medicinal chemistry, biochemistry, pharmacokinetics, herbal/nutritional supplements/over-the-counter medicines
- Evaluate the primary literature and national treatment guidelines and its utility in meeting patient needs (including case reports, if appropriate) to make a reasonable decision based on available information
- Identify goals of therapy, including effects on quality of life (QOL)
- Present well-organized and accurate patient case histories with subsequent therapeutic plan.
- Understand the pathophysiology and the psychopharmacology of different psychiatric disorders with special emphasis on psychotropic medications, substance related disorders and no-pharmacologic therapies including: electroconvulsive therapy and psychotherapies.
- Communicate effectively with patients, health care professionals and other stakeholders
- Establish and maintain collaborative working relationship with psychiatric patients, health care professionals and other stakeholders.
- Apply the clinical therapeutic management principles of psychiatric disorders in light of specific patient factors, national and international guidelines.
- Demonstrate the ability to identify specific drug-related problems for psychiatric patients with acute and/or chronic illness.
- Discuss the therapeutic plans of the patients encountered on the psychiatric ward and OPD
- Identify monitoring parameters for monitoring patients' treatment outcomes.
- Conduct monitoring of psychiatric patient's drug therapy.
- Write a complete and formal drug information response
- Work with other health care professionals to ensure the safe and cost-effective drug therapy.
- Provide patient education regarding the optimal use, expected outcomes, and adverse effects of psychotropic drugs.
- Demonstrate ability to perform discharge/follow-up medication teaching and/or instruct patients in medication education classes
- Provide adherence support for Psychiatric patients on Psychotropic medications.
- Contribute in prevention and management of substance related disorders.
- Provide compassionate, respectful and caring service.
- Adherence to ethical principles and legal practice

## Clerkship Task Lists

The following **task list/ responsibilities** will be included but are not limited to the following:

1. Attend daily ward rounds, morning sessions and other educational conferences as scheduled.
2. Conduct Admission medication history taking, medication reconciliations and discharge medication use counselling
3. Provide patient counseling and education concerning drug therapy during hospitalization
4. Provide drug information when appropriate to other members of the Psychiatry team.
5. Monitor patients' drug therapy outcomes, and drug interactions.
6. Participate in formal and informal consultations, including detailed literature searches.
7. Participate in psychiatry emergency situations, when possible, including the preparation of emergency drugs and provision of drug information.
8. Obtaining complete and accurate medication records for each patient.
9. Presenting patient-related information completely during rounds with preceptor as well as other health care professionals when queried.
10. Evaluate the literature on a specific topic pertaining to a patient's therapy. Completing and presenting one literature searches relating to drug therapy on psychiatry.
11. Submit 2 written pharmaceutical care service given using SOAP format, 4 written drug information responses that the student gave during the rotation, and drug therapy problems/interventions encountered during the rotation that comprises portfolio requirements in psychiatry rotations
12. Punctuality for all activities.

**Mode of delivery:** Case presentation/morning session, patient chart review, attending psychiatric wards care team round, bed side teaching, journal club presentation, project work and seminar

### Mode of assessment:

- |      |                      |     |
|------|----------------------|-----|
| i.   | Case presentation    | 15% |
| ii.  | Seminar with pretest | 15% |
| iii. | Ward activities      | 20% |
| iv.  | Portfolio            | 10% |

|     |                    |     |
|-----|--------------------|-----|
| v.  | Internal oral exam | 20% |
| vi. | Written exam       | 20% |

**Clerkship Title: Surgery Clerkship Syllabus****Clerkship Code: Phar5292****Clerkship ECTS: 5 (3 weeks)****Pre-requisites: accomplishment of all didactic courses Year I – Year IV****Co-requisites: None****Clerkship Description:**

- This rotation will be conducted in surgical wards under department of surgery. The student should collaborate with other health care providers for the pharmaceutical care needs of the patients.

**Clerkship Goals**

4. To become familiar with the provision of clinical pharmacy services to surgery patients.
5. To be responsible for providing accurate and timely drug information to surgical team and other health care professionals related to a surgery service.
6. To gain a general understanding of the medications involved with a surgery service with emphasis on antibiotic therapy, nutritional support, and pain management.

**Clerkship Objectives:** At the completion of this rotation, the student should be able to:

- Consistently obtain complete and accurate drug histories including previous adverse reactions to medications.
- Present well-organized and accurate patient case histories including subsequent problem-plan management.
- Effectively and appropriately communicate with both patients and other health care professionals.
- Punctually and thoroughly complete and present the results of a literature search related to a disease, medication, or surgery performed on a surgery service.
- Spend the necessary amount of time in the patient care area to complete all service responsibilities related to his/her patients.
- Initiate interactions with other health care professionals.
- Understand the pathophysiology and pharmacology of the medications used on a surgery service with special emphasis on antibiotics and pain management.
- Understand the fundamental principles of acid-base balance, and fluid and electrolyte therapy.

- Have a general understanding of the role of parenteral and enteral nutrition in the surgery patient.
- Recognize and monitor adverse effects of medications and make a rational decision for their prevention and/or treatment.
- Understand the controversies involved with deciding which drug is the most effective and least toxic drug in specific clinical situations.
- Calculate a measured creatinine clearance given the appropriate data.
- Discuss the effects of renal and/or liver disease on the appropriate dosing of medications.
- Quickly and accurately calculate doses for medications used on the surgery service with emphasis on antibiotics and pain relievers.
- Realize that medication-related questions from health care professionals should only be answered when all of the relevant facts are known and that accuracy is of highest priority.
- Promptly and accurately search and answer questions from health care professionals when the answer is not initially known.
- Realize the importance of patient confidentiality.
- Know how to decide the order of priority for monitoring patient medications when on a busy surgery service.

### **Clerkship Task Lists**

The following **task list/ responsibilities** will include but are not limited to the following:

12. Punctual attendance at daily morning rounds with the assigned surgery team.
13. Punctual attendance at clinical pharmacy conferences.
14. Completely, accurately, and promptly responding to questions from health care professionals.
15. Initiating interactions with other health care professionals.
16. Filling out drug reaction forms for any undesired or unintended effect of a medication.
17. Completing and presenting two literature searches relating to a surgery topic.
18. Obtaining complete and accurate medication records for each patient.
19. Presenting patient-related information completely, succinctly, and accurately during rounds with preceptor as well as other health care professionals when queried.
20. Deciding which medications to use and how to use them with reference to pharmacokinetics and pharmacodynamics.

21. Completely and accurately evaluating the medications of all assigned patients on the surgery service.

**Mode of delivery:** Case presentation/morning session, Patient Chart review, attending surgical wards care team, bed side teaching, Journal Club presentation, project work and seminar

**Mode of assessment:**

|      |                      |     |
|------|----------------------|-----|
| i.   | Case presentation    | 15% |
| ii.  | Seminar with pretest | 15% |
| iii. | Ward activities      | 20% |
| iv.  | Portfolio            | 10% |
| v.   | Internal oral exam   | 20% |
| vi.  | Written exam         | 20% |

**Clerkship Title:** Oncology and Hematology Clerkship Syllabus

**Clerkship Code:** Phar5293

**Clerkship ECTS:** 5 (3 weeks)

**Pre-requisites:** accomplishment of all didactic and Laboratory based courses Year I – Year IV

**Co-requisites:** None

**Clerkship Description:**

- This clerkship program will be conducted at oncology center of a hospital to integrate pharmaceutical care and Clinical Pharmacy concepts and oncology patient care through patient care activities, case discussions, and others. The student is expected to provide pharmaceutical care to oncology patients.

**Clerkship Objectives:** At the completion of this clerkship, the student should be able to:

### **1. Neoplastic Diseases**

- Describe the general course of the disease including the clinical findings, diagnosis, complications of disease, and prognosis.
- Describe the histopathologic classification and staging of the malignancy.
- Define the goals and rational treatment programs including surgical, radiological, pharmacological, and immunological.
- Identify the agent or combination of agents of choice including rationale, dose, schedule, and potential toxicities.

### **2. Chemotherapeutic Agents**

- Discuss the pharmacology including mechanism of action, adverse reactions, dose ranges, and pharmacokinetics of the common cancer chemotherapeutic agents, hormonal and immunotherapeutic agents, and monitor their use in patients with cancer.
- Recommend dosage adjustments based on renal function tests, liver function tests, and hematologic or other indices.
- Discuss cancer chemotherapy with respect to cell kinetics.
- Discuss the rationale for induction, consolidation, maintenance, and adjuvant chemotherapy.



- Describe the agents implicated, time course, reversibility, symptoms, and predisposing factors for chemotherapy-induced adverse reactions including pulmonary disease, nephrotoxicity, cardiac toxicity, neurotoxicity, hepatotoxicity, radiation recall, gastrointestinal toxicity, hematologic complications, metabolic toxicity, and secondary malignancy.

### **3. Radiation Therapy and Surgery**

- Discuss the role of diagnostic, palliative, and curative radiation therapy and surgery in cancer management.
- Understand the monitoring and management of the complications associated with radiation therapy and surgery.

### **4. Nausea and Vomiting, Pain control**

- Recommend appropriate therapy with consideration of the proper drug, dose, and regimen.
- Know the relative onset, duration, and severity of nausea and vomiting with different chemotherapeutic agents.
- Describe nondrug methods for antiemetic control.

### **5. Transfusion Therapy**

- Discuss the use of blood products in cancer therapy including red blood cells, white blood cells, and platelets.
- Suggest premedication regimens to help reduce hypersensitivity reactions to blood products.
- Know the complications associated with the use of transfusion products.

### **6. Infectious Disease**

- Describe the types of bacterial and nonbacterial infections seen in patients with cancer.
- Recommend and monitor appropriate antibiotic therapy in patients with cancer with respect to patient specific factors.

### **7. Chemotherapy Extravasations**

- List drugs that are vesicants.
- Comprehend the management of extravasations.

### **Clerkship Activities:**

- Round with the multidisciplinary team and demonstrate appropriate interpersonal, leadership and collaboration skills. In addition, the student will provide pharmacotherapeutic recommendations

- Evaluate drug therapy regimens for appropriateness of drug, dose, and dosage regimen, route/method of administration, compliance, therapeutic duplications, therapeutic outcomes, cost, adverse drug reactions, and interactions.
- Design effective therapeutic regimens when therapy is initiated to best address patient specific goals and outcomes. Regimens should be guided by evidence based medicine.
- Design monitoring plans to achieve appropriate efficacy outcomes and avoid unwarranted adverse events/side effects with commonly used antineoplastic agents.
- Meet daily with preceptor to review patient care issues to ensure items have been addressed.
- Provide timely responses to drug information requests from the team, nursing, pharmacists, preceptor, and other health care providers.
- Counsel patients/caregivers when newly diagnosed or changes are made to routine home medications.
- Participate in preparing cytotoxic medications
- Document clinical activities and interventions and report ADEs and Medication error

**Mode of delivery:** Case presentation/morning session, Patient Chart review, attending surgical wards care team, bed side teaching, Journal Club presentation, project work and seminar

**Mode of assessment**

|      |                      |     |
|------|----------------------|-----|
| i.   | Case presentation    | 15% |
| ii.  | Seminar with pretest | 15% |
| iii. | Ward activities      | 20% |
| iv.  | Portfolio            | 10% |
| v.   | Internal oral exam   | 20% |
| vi.  | Written exam         | 20% |

**Clerkship Title: Ophthalmology and ENT Clerkship syllabus**

**Clerkship Code:** Phar5294

**Clerkship EtCTS: 5 (3 weeks)**

**Pre-requisites:** accomplishment of all didactic and Laboratory based courses Year I – Year IV

**Co-requisites:** None

**Clerkship Description:**

- The primary purpose of this clerkship is to develop an understanding of pharmacotherapy for the patients followed by Ophthalmology and ENT ward. The main focus will be providing pharmaceutical care for patients with glaucoma and infectious disease of the eye, ear, nose and throat

**Clerkship Objectives:** At the completion of this rotation, the student should be able to:

- Authoritatively discuss the pathophysiology and pharmacotherapy disease states seen.
- Demonstrate the ability to identify specific drug-related problems for patients with acute and/or chronic illness.
- Authoritatively discuss the therapeutic plans of the patients encountered on the wards
- Identify optimal variables for monitoring patients including the identification of adverse drug reactions.
- Conduct monitoring of patient's drug therapy.
- Write a complete and formal drug information response
- Provide patient education regarding the optimal use, expected outcomes, and adverse effects of drug therapy regimens.
- Work with other health care professionals to ensure the safe, correct, and cost-efficient administration of medications.

**Clerkship Activities:** The responsibilities of the students during this rotation shall include the following:

- Round with the multidisciplinary team and demonstrate appropriate interpersonal, leadership and collaboration skills for admitted patients at the ward.
- Closely follow patients from a medical team in the ward as assigned by the preceptor.
- Closely review all the drug therapy of each patient being followed (for both outpatient and inpatient) including pharmacology, toxicology, pharmacokinetics, drug interactions, and monitoring parameters. From these data, a therapeutic plan will be developed.
- Provide timely responses to drug information requests from the team, nursing, pharmacists, preceptor, and other health care providers.

- Counsel patients/caregivers when newly diagnosed or changes are made to routine home medications up on discharge. Emphasize on counseling administration technique and handling of glaucoma medication/eye drops, at least complete 5 patient counseling sessions on administration and adherence to glaucoma medications.
- Complete at least 2 drug related problems identified and the care plan
- Document all patient care activities in manner outlined by the preceptor.
- Prepare and give a formal presentation.

**Mode of delivery:** Case presentation/morning session, Patient Chart review, attending surgical wards care team, bed side teaching, Journal Club presentation, project work and seminar

**Mode of assessment**

|      |                      |     |
|------|----------------------|-----|
| i.   | Case presentation    | 15% |
| ii.  | Seminar with pretest | 15% |
| iii. | Ward activities      | 20% |
| iv.  | Portfolio            | 10% |
| v.   | Internal oral exam   | 20% |
| vi.  | Written exam         | 20% |

**Clerkship Title: Emergency Medicine Clerkship****Clerkship code: Phar5295****Clerkship EtCTS: 5 (3 weeks)****Clerkship Description**

The purposes of this attachment is to promote understanding of the ways in which pharmacists contribute to care in the Emergency medicine and to suggest future directions for the role of pharmacists in providing that care.

- working with emergency physicians, emergency nurses, and other healthcare professionals to develop and monitor medication-use systems that promote safe and effective medication use in the Emergency medicine, especially for high-risk Emergency medicine patients and procedures;
- Collaborating with emergency physicians, emergency nurses, and other healthcare professionals to promote medication use in the Emergency medicine that is evidence-based and aligned with national quality indicators;
- Participating in the selection, implementation, and monitoring of technology utilized in the medication-use process;
- Providing direct patient care as part of the interdisciplinary emergency care team;
- Participating in or leading emergency preparedness efforts and quality improvement initiatives;

Educating patients, caregivers, and healthcare professionals about safe and effective medication use

**Clerkship Objectives**

At the completion of this rotation:

**Patient care:** Hospital-Based Emergency Care together with the Emergency medicine care team the student should be able to ensure appropriate fulfillment of patient medication needs thereby to reduce or eliminate medication errors, and to evaluate for cost-effective medication therapy for the patient and hospital. As part of the interdisciplinary Emergency medicine care team, pharmacists can provide care to critically ill patients by:

- participating in resuscitation efforts;
- providing consultative services that foster appropriate evidence-based medication selection;
- providing consultation on patient-specific medication dosage and dosage adjustments;

- providing drug information consultation to emergency physicians, emergency nurses, and other clinicians;
- monitoring for patient allergies and drug interactions;
- monitoring patient therapeutic responses (including laboratory values);
- continuously assessing for and managing adverse drug reactions; and
- Gathering or reviewing medication histories and reconciling patients' medications.

In addition to the above, students can provide care to ambulatory patients in the Emergency medicine by:

- modifying medication regimens based on collaborative practice agreements for management of certain patient populations who return to ED;
- offering vaccination screening, referral, and administration;
- providing patient and caregiver education, including discharge counseling and follow-up; and
- Providing information on obtaining medications through patient assistance programs, care funds, and samples.

### **Emergency preparedness planning.**

Students expected to acquire basic knowledge to assertively exercise their responsibilities in preparing for and responding to disasters because treatment of disaster victims almost always involves the use of pharmacologic agents and ensuring the efficacy and safety of the medication-use process is a natural role for pharmacists.

**Quality improvement initiatives.** Students are expected to acquire basic knowledge about:

- guiding the development of evidence-based treatment protocols, algorithms, and/or clinical pathways that are congruent with nationally accepted practice guidelines and quality indicators;
- assisting in the development, implementation, and assessment of various technologies used throughout the ED medication-use process;
- conducting failure mode and effects analysis and root cause analysis on error-prone aspects of the medication-use process;
- participating in ED-based and hospital-wide committees (e.g., P&T, infection control, disaster) that impact medication use in the ED;
- Assisting in surveillance and reporting of adverse drug reactions.

**Education.** The will able to provide education and information to healthcare professionals, patients, and the public they come in contact with in the health systems' emergency service areas.

**Mode of Delivery**

- Ward round
- Bed side teaching
- Morning case presentation
- Journal club

**Mode of assessment**

|      |                      |     |
|------|----------------------|-----|
| i.   | Case presentation    | 15% |
| ii.  | Seminar with pretest | 15% |
| iii. | Ward activities      | 20% |
| iv.  | Portfolio            | 10% |
| v.   | Internal oral exam   | 20% |
| vi.  | Written exam         | 20% |

**Clerkship Title: Dermatology Clerkship Syllabus****Clerkship Code: Phar5296****Clerkship ECTS: 5 (3 weeks)****Pre-requisites: accomplishment of all didactic and Laboratory based courses Year I – Year IV****Co-requisites: None****Clerkship Description:**

- The primary goal of this rotation is to give the student an understanding of the recognition, diagnosis, and treatment of common dermatologic disorders and monitoring the safe and effective use of medications. Patients seen are out patients at dermatology clinic. The students shall have authoritative understanding of the different dermatologic preparation and their administration technique.

**Clerkship Activities**

- Conduct patient interview to organize and collect pertinent patient data including past medical history, past and current medication therapy
- List common drugs that are responsible for drug – induced skin reaction during the rotation
- Identify common drug – induced dermatological conditions
- Design and implement therapeutic plan for the conditions identified
- Design a monitoring plan for the care plan
- Write at least one detailed report of drug- induced skin reaction
- Provide timely responses to drug information requests from the team, nursing, pharmacists, preceptor, and other health care providers.
- Counsel patients/caregivers when newly diagnosed or changes are made to routine home medications up on discharge



**Mode of delivery:** Case presentation/morning session, Patient Chart review, attending surgical wards care team, bed side teaching, Journal Club presentation, project work and seminar

**Mode of assessment:**

|      |                      |     |
|------|----------------------|-----|
| i.   | Case presentation    | 15% |
| ii.  | Seminar with pretest | 15% |
| iii. | Ward activities      | 20% |
| iv.  | Portfolio            | 10% |
| v.   | Internal oral exam   | 20% |
| vi.  | Written exam         | 20% |

**Clerkship content:**

**A. ward activities**

1. Attend and actively participate in ward rounds and attending rounds according to the schedule of the service.
2. Each student will make a verbal presentation in fifteen minutes or less of any patient for which s/he is responsible. Each patient presentation should include the following elements;
  - a. Patient's name, age, sex
  - b. Reason for admission and chief complaint
  - c. History of present illness
  - d. Significant medical history
  - e. Present medication history
  - f. Summary of review of systems and physical examination
  - g. Admission of laboratory values
  - h. Pharmaceutical considerations
    - An evaluation of current therapy
    - Proposed alternative therapies
    - Monitoring parameters for therapeutic response and toxicity
    - Any pharmacokinetic parameters applicable endpoints of therapy
3. Interview and obtain medication histories from all patients assigned by the pharmacy preceptor.
4. Monitor drug therapy of all assigned patients.
5. Students will apply the information obtained through the interviewing and monitoring process to:

- A. evaluate current drug therapy
  - B. anticipates and identifies drug therapy problems
  - C. meets the objectives outlined in this syllabus
6. Provide patient specific drug information (verbally and/or in writing) to nurses, physicians, peers and pharmacy preceptors.
  7. Counsel all assigned patients about their drugs.

**B. Non-ward activities**

Students will prepare and make at least one in-service presentation (s) to the clinicians on their unit on a drug or pharmacy related topic. All students will attend and participate in morning meeting and journal club presentation. Presenters are assigned by pharmacy preceptors in turn.

**Clerkship Title:** Pharmaceutical quality control

**Clerkship Code:** Phar5297

**Clerkship ECTS:** 5 (3 weeks)

**Pre-requisites:** accomplishment of all didactic and Laboratory based courses Year I – Year IV

**Co-requisites:** None

**Clerkship Description:**

- This clerkship program will be conducted at the quality control units of big pharmaceutical industries and/or the quality control laboratory of the national regulatory authority.

**Clerkship Objectives:** At the completion of this clerkship, the student should be able to:

- Conduct basic quality control procedures for common pharmaceutical products in Ethiopia
- Identify the quality issues unique to the Ethiopian context

**Clerkship Title:** Pharmaceutical regulatory affairs

**Clerkship Code:** Phar5298

**Clerkship ECTS:** 5 (3 weeks)

**Pre-requisites:** accomplishment of all didactic and Laboratory based courses Year I – Year IV

**Co-requisites:** None

**Clerkship Description:**

- This clerkship program will be conducted at regional (zonal/wereda/city admin) and/or the national regulatory authority.

**Clerkship Objectives:** At the completion of this clerkship, the student should be able to:

- Conduct basic inspection of pharmaceutical retail outlets
- Participate in dossier evaluation
- Participate in product registration
- Participate in the registration and licensing of professionals

**Clerkship Title:** Pharmaceutical wholesale & promotion

**Clerkship Code:** Phar5299

**Clerkship ECTS:** 5 (3 weeks)

**Pre-requisites:** accomplishment of all didactic and Laboratory based courses Year I – Year IV

**Co-requisites:** None

**Clerkship Description:**

- This clerkship program will be conducted at local pharmaceutical import/wholesale and marketing institutions.

**Clerkship Objectives:** At the completion of this clerkship, the student should be able to:

- Participate in the import and wholesale of pharmaceutical products in the private sector
- Participate in promotion and marketing of pharmaceutical products

## Module 30 Pharmaceutical Research II

**Module Name:** Pharmaceutical Research II

**Module Category:** Core

**Module Code:** Phar-M4301

**Module Number:** 30

**Module Weight:** 5 ECTS

**Courses:**

| Course name    | Course code | ECTS |
|----------------|-------------|------|
| Directed study | Phar 5301   | 5    |

### Module description:

This module is intended to equip pharmacy students with a basic working knowledge of pharmaceutical research methods. It also gives the trainee an acquaintance with research proposal writing, critical appraisal of scientific paper and application of common statistical packages. The module includes hands on research experience in the form of a directed studies course, which offers the student to perform research which culminates in the submission of substantial research work in the form of a senior essay/directed studies report paper.

### Module objective:

After successful completion of this Module the students will be able to:

1. Differentiate the major types of study designs
2. Identify the main issues in the design, conduct and presentation of a research
3. Explain the major elements that need to be examined when making a critical assessment of a research paper.
4. Demonstrate how to deal with each of these elements with reference to a published paper
5. Demonstrate a basic understanding of common statistical packages useful for data processing and analysis
6. Explain the major components of research in the pharmaceutical sciences
7. Prepare a research protocol and conduct pharmaceutical research

### Module competencies:

Upon a successful completion of this module, students will be able to conduct with minimal supervision by a senior pharmacist (researcher) research in the different areas of pharmacy practice and/or pharmaceutical sciences..

**Mode of delivery (Parallel/Block):** Block

### Module teaching/learning method:

Learning Activities

Students are expected to undertake among others the following major learning activities:

- Engage in learning by doing (independent study, group assignments, presentation, report writing, etc. )
- Participation and note takings during class lectures and debates and discussions;
- Reviewing the literature
- Data collection/conducting experiment (as appropriate), analyse results, report write-up and presentaiton

Teaching Methods

The course facilitator is expected to:

- Lecture for introducing concepts and topics, and give references, facilitate discussions, ask questions, give and correct assessments, forming groups, encouraging students for peer learning
- Monitor student progress and provide feedback

**Module mode of assessment:**

- Quizzes
- Exam
- Assignments
- A directed studies report
- A directed studies presentation and defense

**Course syllabus**

**Course Name:** Directed study  
**Course code:** Phar5301  
**Module Name:** Pharmaceutical research II  
**Module Code:** **Phar-M 5301**  
**Course ECTS:** 5

**Totally required hours for the module: 135**

**Project work:** 133

**Presentation(s):** 2

**Year/Semester Course is offered:** Year V Semesters I and II

**Course prerequisite/s:** Research methods

**Course description:**

The module includes hands on and supervised research experience enabling the student to perform research, conduct experiments, collect data interpret the results, report in the form of formal scientific writing and present the findings to an audience of fellow students and instructors.

**Course objective:**

After completion of this course students will be able to:

- Prepare a research protocol and conduct research in the area of pharmaceutical sciences and or pharmacy practice

**Course mode of delivery:** Parallel

**Course learning and teaching methods**

- Mentoring and continuous monitoring of student progress

**Assessment techniques:**

- Continuous evaluation by the advisor
- Evaluation of the written report by the advisor
- Evaluation of the written report by the examiner
- Evaluation of the presentation by the examiner

**Teachers' and students' role****Roles of Instructors**

The instructor will be expected to:

- Allocation adequate consultation hours for the student
- Guide the student in the identification of plausible research topic
- Monitor the progress of the student



- Take remedial actions to ensure timely completion of the project

### **Roles of Students**

Students are expected to:

- Maintain regular contact with advisor
- Critically appraise available literature and identify research gaps in consultation with the advisor
- Develop appropriate data collection procedures and/or research protocol in consultation with the advisor
- Collect data interpret the results
- Act in compliance with all ethical and legal standards

### **References:**

- To be determined in consultation with assigned advisor

### **Course schedule**

- To be defined in consultation with the advisor. Students and advisors are encouraged to identify important milestones in the conduct of the research and accordingly monitor progress.

**Module 31 Team Training Program****Module Name: Team training program****Module Category:** Core**Module Code:** PUBHM5311**Module Number:** 31**Module Weight:** 7 ECTS**Courses:**

| S/N | Course name           | Course code | ECTS |
|-----|-----------------------|-------------|------|
| 1   | Team training program | ComH5311    | 7    |

**Module Description****Module Objective**

To produce competent health professional who went through an integrated series of learning experiences which will enable them to understand the health problems of a rural community and be able to solve them through evidence based innovative approaches.

The general objectives of the Rural Community Health Teaching Programme are:

1. To introduce students to the health problems and health service delivery system of the rural community as well as expose them to clinical service and research activities
2. To prepare health science students to perform effectively as competent health professionals who can effectively plan and deliver health services that responds to the priority needs of the community in a primary health care approach.

**Specific Objectives**

The specific objectives of the Rural Community Health Teaching Programme are,

1. To familiarize health science students to the context of the rural settings and the socio-demographic and health situation of people living in a rural area
2. To enable students conduct "community diagnosis" by defining socio-economic, political and environmental aspects of a given community (determinants of socio-economic status) and draw an action plan which would enable students to suggest appropriate intervention measures
3. To enable students obtain data pertaining to the socio-demographic and health status of a rural population that are useful to inform health service planning and implementation.
4. Organize intervention utilizing the concept components and strategies of community participation and multi-sectoral approach
5. To ensure health science students are capable of conducting mini research -collecting, analysing and writing up reports using data generated through quantitative and qualitative methods.
6. Familiarize students with public health importance of specific health problems such as

malnutrition, communicable diseases ... etc.

7. Learn tools how to identify and prioritize health problems, draw action plan and implement interventions and evaluate the program
8. To equip students with basic knowledge to evaluate the effectiveness and efficiency of community intervention.
9. Provide health education and health services in the health centres and surrounding communitiesDevelop the communication skill through interactive learning among team members, resident supervisors, supervisors and community
10. Understand and develop team spirit and create a solid foundation for their future work in a team with laboratory and pharmacy units.

Enrich sense of belongingness and develop positive attitude of graduates to work in a community where there is poor infrastructure

### **Module Competency**

At the completion of the program students are expected to:

- Acquire basic knowledge and skills of understanding and solving individual (clinical) and community health problems, as well as adopt positive and helping attitudes in patient care/handling, attitude on research, attitude on the importance of evidence for decision, plagiarism, etc.
- Moreover, students improve their oral and writing communication skills while interacting among themselves and with various community members including health and women development workers, community opinion leaders, faith leaders, school officials, students, local government officials, civil society organizations and many others. They shall also be involved in the discussion of community affairs and perhaps engaged in advocating basic health rights. Students shall also improve their writing communication skill through the preparation of study reports and communication aids, as well as presentation of findings in dissemination, learning and advocacy workshops
- Students will also improve skills in team building and team working as they will be assigned in groups throughout the programs. Through the process they will learn how individual behaves within a group, how to appreciate differences and consider as strength, handle conflict, etc.
- They will also be exposed to solve problems through adopting self-management of certain resources and opportunities such as serve their own food, timing for shared

resources including toilet and washing facilities etc., how to best conform to local community tradition and lifestyle.

### **Module Mode of Delivery**

As the program is competency and outcomes-based program it follows an innovative curriculum approach adopting problem solving and adult learning principles. In general the teaching learning process followed a principle of adult learning involved highly interactive discussions, case studies, group learning, problem based study in order to produce a medical/paramedical graduates who are problem solver inventively. The three domains of learning, namely, Knowledge, skill and attitude, will be given due attention in equal focus.

The following will be considered in the selection of the teaching and learning methods:

- **Student focused-** all learning methods should emphasize individual activities that students shall perform
- **Collaborative- learners should consider** collaboration and cooperation to enrich the delivery of teaching contents focused to specific learning outcome
- **Diversity** – learning teaching methods should be different depending on learning outcome

Each class room as well as field teaching activities should reflect the main curriculum of the different disciplines. Moreover, the teaching activities should help students to be imaginative, creative and innovative. Therefore, courses shall be properly designed and shall have alignment of the three components/elements (course objective, content and teaching methods) of the learning and teaching at course/unit/chapter levels.

The major teaching and learning methods and activities suggested to be used in the implementation of the curriculum are described below when and where.

- 1) Interactive lecture: class room teaching will be involved throughout the six week of the program, including the orientation week. This activity will help students improve their thinking, reasoning and discussion skills. The content of the lecture depends on the objectives of the teaching outcome. Lecture sections should involve brainstorming, dialogue, argument, case study, and encouraging individual reading.
- 2) actual conduct of community diagnosis that including field data collection using standard instrument, data processing and analysis, priority setting, drawing of intervention action plan;
- 3) mini research that includes
  - Problem identification, topic selection and development of proposal,
  - development of data collection instruments,
  - actual data collection (can be primary or secondary),

- data analysis, interpretation and recommendation of possible solutions for intervention
  - report writing;
  - Documentation and presentations of the research output
- 4) Class room seminars and case reports (clinical)
  - 5) Whole group session: this is to amalgamate and reflect on individual or group activities
  - 6) Health facility and community clinical services

### **Module mode of Assessment**

#### **Formative and Summative assessments**

- Quizzes
- Reports
- Seminar Presentations
- Assignments
- Final evaluations

## Broad schedule

### WEEK I& II

| DAY                            | RESPONSIBLE        | SPH & FAMILY MEDICINE                                                            | SPH, SoP& FAMILY MEDICINE                                                  |
|--------------------------------|--------------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------|
|                                | WEEK BRAND         | ORIENTATION & CommDx                                                             | INTERVENTION PLAN & Comm. Child health                                     |
|                                | TIME               | WEEK I                                                                           | WEEK II                                                                    |
| MONDAY                         | AM (8AM-12AM)      | Orientation, <b>lecture</b> -Census & MCH                                        | <b>Health Center SURVEY****<br/>Community opinion FOR PRIORITY SETTING</b> |
|                                | PM(2PM-4PM)        | <b>lecture</b> -Nutrition survey                                                 | <b>Data entry &amp; analysis</b>                                           |
|                                | EVENING(4:30-6:00) | <b>lecture</b> -School survey                                                    |                                                                            |
| TUSEDAY                        | AM (8AM-12AM)      | <b>lecture</b> -EH survey                                                        | <b>Hospital assessment</b>                                                 |
|                                | PM(2PM-4PM)        | <b>lecture</b> -Health F. survey                                                 |                                                                            |
|                                | EVENING(4:30-6:00) | <b>lecture</b> -priority setting                                                 | <b>Data entry &amp; analysis</b>                                           |
| WEDNSDAY                       | AM (8AM-12AM)      | <b>Census &amp; MCH SURVEY*</b>                                                  | <b>Data management &amp; analysis</b>                                      |
|                                | PM(2PM-4PM)        | <b>Data entry</b>                                                                | PRIORITY SETTING                                                           |
|                                | EVENING(4:30-6:00) | <b>Seminarby Family health</b>                                                   | PRELEMINARY REPORT                                                         |
| THURSDAY                       | AM (8AM-12AM)      | <b>Anthropometry/nutrition, U5 Morbidity SURVEY &amp;Comm. child health care</b> | <b>INTERVENTION PLAN</b>                                                   |
|                                | PM(2PM-4PM)        | <b>Data entry</b>                                                                |                                                                            |
|                                | EVENING(4:30-6:00) | <b>lecture</b> -Research proposal                                                | <b>Seminar on Proposal writing</b>                                         |
| FRIDAY                         | AM (8AM-12AM)      | <b>Env. Health SURVEY**</b>                                                      | <b>Individual Research proposal writing (RA)</b>                           |
|                                | PM(2PM-4PM)        | <b>School SURVEY***</b>                                                          |                                                                            |
|                                | EVENING(4:30-6:00) | <b>Data entry&amp;management</b>                                                 |                                                                            |
| SATURDAY                       | AM (8AM-12AM)      | <b>Data analysis practical training</b>                                          | <b>FINALIZE Draft research proposal</b>                                    |
|                                |                    |                                                                                  | <b>QUIZ</b>                                                                |
| <b>Deliverable by STUDENTS</b> |                    |                                                                                  | <b>Final Draft research Proposal</b>                                       |

### WEEK III

| DAY                     | RESPONSIBLE        | SPH; , SoP, PEDIATRICS & DERMATOLOGIST |    |                   |    |            |    |                   |    |
|-------------------------|--------------------|----------------------------------------|----|-------------------|----|------------|----|-------------------|----|
|                         | WEEK BRAND         | PEDIATRICS & DERMATOLOGY &INTERVENTION |    |                   |    |            |    |                   |    |
|                         | TIME               | WEEK III                               |    |                   |    |            |    |                   |    |
|                         |                    | A                                      | B  | C                 | D  | E          | F  | G                 | H  |
| MONDAY                  | AM (8AM-12AM)      | HC                                     | SH | RA                | RA | HC         | SH | RA                | RA |
|                         | PM(2PM-4PM)        | HC                                     | SH | RA                | RA | HC         | SH | RA                | RA |
|                         | EVENING(4:30-6:00) | SEMINAR-Ped                            |    |                   |    |            |    |                   |    |
| TUSEDAY                 | AM (8AM-12AM)      | RA                                     | HC | EH                | RA | RA         | HC | EH                | RA |
|                         | PM(2PM-4PM)        | RA                                     | HC | EH                | RA | RA         | HC | EH                | RA |
|                         | EVENING(4:30-6:00) | SEMINAR-Derma                          |    |                   |    |            |    |                   |    |
| WEDNSDAY                | AM (8AM-12AM)      | HP service                             |    | Community service |    | HP service |    | Community service |    |
|                         | PM(2PM-4PM)        | RA                                     |    |                   |    |            |    |                   |    |
|                         | EVENING(4:30-6:00) | SEMINAR-Ped                            |    |                   |    |            |    |                   |    |
| THURSDAY                | AM (8AM-12AM)      | RA                                     | RA | HC                | SH | RA         | RA | HC                | SH |
|                         | PM(2PM-4PM)        | RA                                     | RA | HC                | SH | RA         | RA | HC                | SH |
|                         | EVENING(4:30-6:00) | SEMINAR-Derma                          |    |                   |    |            |    |                   |    |
| FRIDAY                  | AM (8AM-12AM)      | EH                                     | RA | RA                | HC | EH         | RA | RA                | HC |
|                         | PM(2PM-4PM)        | EH                                     | RA | RA                | HC | EH         | RA | RA                | HC |
|                         | EVENING(4:30-6:00) | SEMINAR-Ped                            |    |                   |    |            |    |                   |    |
| SATURDAY                | AM (8AM-12AM)      | FOLLOWUP ON INTERVENTION               |    |                   |    |            |    |                   |    |
|                         |                    | QUIZ                                   |    |                   |    |            |    |                   |    |
| Deliverable by STUDENTS |                    | Start Data collection                  |    |                   |    |            |    |                   |    |

### WEEK IV

| DAY                     | RESPONSIBLE        | SHP, SoP, INTERNIST & OPHTHALMOLOGIST       |    |            |    |                   |    |            |    |
|-------------------------|--------------------|---------------------------------------------|----|------------|----|-------------------|----|------------|----|
|                         | WEEK BRAND         | INT. MEDICINE & OPHTHALMOLOGY &INTERVENTION |    |            |    |                   |    |            |    |
|                         | TIME               | WEEK IV                                     |    |            |    |                   |    |            |    |
|                         |                    | A                                           | B  | C          | D  | E                 | F  | G          | H  |
| MONDAY                  | AM (8AM-12AM)      | RA                                          | HC | SH         | RA | RA                | HC | SH         | RA |
|                         | PM(2PM-4PM)        | RA                                          | HC | SH         | RA | RA                | HC | SH         | RA |
|                         | EVENING(4:30-6:00) | SEMINAR-Med                                 |    |            |    |                   |    |            |    |
| TUSEDAY                 | AM (8AM-12AM)      | RA                                          | RA | HC         | EH | RA                | RA | HC         | EH |
|                         | PM(2PM-4PM)        | RA                                          | RA | HC         | EH | RA                | RA | HC         | EH |
|                         | EVENING(4:30-6:00) | SEMINAR-Ophta                               |    |            |    |                   |    |            |    |
| WEDNSDAY                | AM (8AM-12AM)      | Community service                           |    | HP service |    | Community service |    | HP service |    |
|                         | PM(2PM-4PM)        | RA                                          |    |            |    |                   |    |            |    |
|                         | EVENING(4:30-6:00) | SEMINAR- Med                                |    |            |    |                   |    |            |    |
| THURSDAY                | AM (8AM-12AM)      | SH                                          | RA | RA         | HC | SH                | RA | RA         | HC |
|                         | PM(2PM-4PM)        | SH                                          | RA | RA         | HC | SH                | RA | RA         | HC |
|                         | EVENING(4:30-6:00) | SEMINAR-Ophta                               |    |            |    |                   |    |            |    |
| FRIDAY                  | AM (8AM-12AM)      | HC                                          | EH | RA         | RA | HC                | EH | RA         | RA |
|                         | PM(2PM-4PM)        | HC                                          | EH | RA         | RA | HC                | EH | RA         | RA |
|                         | EVENING(4:30-6:00) | SEMINAR- Med                                |    |            |    |                   |    |            |    |
| SATURDAY                | AM (8AM-12AM)      | FOLLOWUP ON INTERVENTION                    |    |            |    |                   |    |            |    |
|                         |                    | QUIZ                                        |    |            |    |                   |    |            |    |
| Deliverable by STUDENTS |                    | Data analysis                               |    |            |    |                   |    |            |    |

#### WEEK V

| DAY                     | RESPONSIBLE        | SPH, SoP, OBSTATRICIAN & DERMATOLOGIST |    |            |    |                   |    |            |    |
|-------------------------|--------------------|----------------------------------------|----|------------|----|-------------------|----|------------|----|
|                         | WEEK BRAND         | GYN-OBS & DERMATOLOGY &INTERVENTION    |    |            |    |                   |    |            |    |
|                         | TIME               | WEEK V                                 |    |            |    |                   |    |            |    |
|                         |                    | A                                      | B  | C          | D  | E                 | F  | G          | H  |
| MONDAY                  | AM (8AM-12AM)      | RA                                     | RA | SH         | HC | RA                | RA | SH         | HC |
|                         | PM(2PM-4PM)        | RA                                     | RA | SH         | HC | RA                | RA | SH         | HC |
|                         | EVENING(4:30-6:00) | SEMINAR-GYN                            |    |            |    |                   |    |            |    |
| TUSEDAY                 | AM (8AM-12AM)      | EH                                     | RA | HC         | RA | EH                | RA | HC         | RA |
|                         | PM(2PM-4PM)        | EH                                     | RA | HC         | RA | EH                | RA | HC         | RA |
|                         | EVENING(4:30-6:00) | SEMINAR-GYN                            |    |            |    |                   |    |            |    |
| WEDNSDAY                | AM (8AM-12AM)      | Community service                      |    | HP service |    | Community service |    | HP service |    |
|                         | PM(2PM-4PM)        | RA                                     |    |            |    |                   |    |            |    |
|                         | EVENING(4:30-6:00) | SEMINAR-Pharma                         |    |            |    |                   |    |            |    |
| THURSDAY                | AM (8AM-12AM)      | HC                                     | SH | RA         | RA | HC                | SH | RA         | RA |
|                         | PM(2PM-4PM)        | HC                                     | SH | RA         | RA | HC                | SH | RA         | RA |
|                         | EVENING(4:30-6:00) | SEMINAR-Derma                          |    |            |    |                   |    |            |    |
| FRIDAY                  | AM (8AM-12AM)      | RA                                     | HC | RA         | EH | RA                | HC | RA         | EH |
|                         | PM(2PM-4PM)        | RA                                     | HC | RA         | EH | RA                | HC | RA         | EH |
|                         | EVENING(4:30-6:00) | SEMINAR-GYN                            |    |            |    |                   |    |            |    |
| SATURDAY                | AM (8AM-12AM)      | FOLLOWUP ON INTERVENTION               |    |            |    |                   |    |            |    |
|                         |                    | QUIZ                                   |    |            |    |                   |    |            |    |
| Deliverable by STUDENTS |                    | Start write up                         |    |            |    |                   |    |            |    |

#### WEEK VI

| DAY     | RESPONSIBLE        | SPH, SoP, & SURGEON               |    |    |    |    |    |    |    |
|---------|--------------------|-----------------------------------|----|----|----|----|----|----|----|
|         | WEEK BRAND         | SURGERY & INTERVENTION EVALUATION |    |    |    |    |    |    |    |
|         | TIME               | WEEK VI                           |    |    |    |    |    |    |    |
|         |                    | A                                 | B  | C  | D  | E  | F  | G  | H  |
| MONDAY  | AM (8AM-12AM)      | HC                                | HC | EH | SH | HC | HC | EH | SH |
|         | PM(2PM-4PM)        |                                   |    |    |    |    |    |    |    |
|         | EVENING(4:30-6:00) | SEMINAR-Sugery                    |    |    |    |    |    |    |    |
| TUESDAY | AM (8AM-12AM)      | EH                                | SH | HC | HC | EH | SH | HC | HC |

|                                |                    |                            |
|--------------------------------|--------------------|----------------------------|
|                                | PM(2PM-4PM)        | <b>RA</b>                  |
|                                | EVENING(4:30-6:00) | <b>SEMINAR-Surgery</b>     |
| <b>WEDNSDAY</b>                | AM (8AM-12AM)      | <b>REPORT PREPARATION</b>  |
|                                | PM(2PM-4PM)        |                            |
|                                | EVENING(4:30-6:00) | <b>SEMINAR-Surgery</b>     |
|                                |                    |                            |
| <b>THURSDAY</b>                | AM (8AM-12AM)      | <b>REPORT FINALIZATION</b> |
|                                | PM(2PM-4PM)        |                            |
|                                | EVENING(4:30-6:00) | <b>SEMINAR-Surgery</b>     |
|                                |                    |                            |
| <b>FRIDAY</b>                  | AM (8AM-12AM)      | <b>REPORT PRESENTATION</b> |
|                                | PM(2PM-4PM)        |                            |
|                                | EVENING(4:30-6:00) |                            |
|                                |                    |                            |
| <b>SATURDAY</b>                |                    |                            |
|                                | AM (8AM-12AM)      | <b>BACK TO CAMPUS</b>      |
| <b>Deliverable by STUDENTS</b> |                    | <b>Start write up</b>      |



**Module 32 Comprehensive Exit Exam****Module Name:** Comprehensive Exit Exam**Module Category:** Core**Module Code:** Phar-M5321**Module Number:** 32**Module Weight:** Non-credited**Courses:**

| S/N | Course name             | Course code | ECTS |
|-----|-------------------------|-------------|------|
| 1   | Comprehensive Exit Exam | Phar5321    | NA   |

**Module Description**

This module is a non-credited module that evaluates whether students have acquired the necessary knowledge, attitude and skills in their stay in the program. Questions will be prepared by all units in the school/department and will assess the major competency areas in the profession of pharmacy.

**Module Objective:**

- To assess the students' competence to practice, as an entry level pharmacist, in the different pharmacy practice settings

**Module Competency:** NA**Module Mode of Delivery:** NA**Module mode of Assessment:**

- Written comprehensive exam
- Evaluated as Pass or Fail